Parity equidistribution of nested floor powers, with descent applications to the Juggler map

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Abstract

For odd n set $m = \lfloor n^{3/2} \rfloor$, $v = \lfloor m^{3/2} \rfloor$, and so on: the integer chains obtained by iterating $x \mapsto \lfloor x^{3/2} \rfloor$ and $x \mapsto \lfloor \sqrt{x} \rfloor$ in a prescribed pattern. We prove complete depth-4 parity equidistribution for odd-rooted itineraries, with power savings: every pattern of depth at most four over odd starts receives its expected share. The obstacle at depth two and beyond is that the naive expansion of $m^{3/2}$ leaves the sawtooth $\{n^{3/2}\}$ with an amplitude that grows like $n^{3/4}$, which defeats the classical van der Corput method. The proof reorganizes the phase by an exact linearization — $m^{3/2} = \frac{3}{2} mn^{3/4} - \frac{1}{2} n^{9/4} + O(n^{-3/4})$ with a one-signed remainder, a Taylor rewrite that keeps the integer m linear with a smooth coefficient — and funnels the deeper patterns into a single *kernel*: the exponential sum of the level-2 floor defect $\{\lfloor n^{3/2} \rfloor^{3/2}\}$ against the monomial weight $c = \frac{3k}{4} n^{9/8}$. The fundamental obstruction inside that kernel is the exact level-2 wave $e(q\lfloor n^{3/2} \rfloor^{3/2})$ riding a frozen floor; Lemma 5.2 bounds these mixed pieces by $|q|^{-1/6} P^{23/24+\varepsilon}$, uniformly enough for depth four, and the central result (Theorem 5.3), $K_c \ll P^{1-1/96+\varepsilon}$, follows by double Weyl differencing over an exact carry-branch decomposition and master identity. All thresholds are effective: no divisor or gcd average occurs anywhere in the argument, so every ε is a power of $\log P$ — Theorem 5.3 holds in the form $K_c \ll P^{1-1/96} \log^{3/4} P$ — and the thirty-eight threshold inequalities of the proof are solved individually in Appendix A, giving $P_0 = 3.6 \cdot 10^{13}$. The depth-≤ 3 discrepancy estimates (Theorems 4.4 and 4.7) are also proved on sub-dyadic intervals of length $\geq P^{1/2}$, with a slowly varying twist attached (Section 3.5), which is the form a companion paper needs. The kernel theorem itself remains a dyadic-block statement.

The sequences arise as the itineraries of the Juggler map $J(n) = \lfloor \sqrt{n} \rfloor$ (n even), $\lfloor n^{3/2} \rfloor$ (n odd), whose finite exact theory is developed in a companion manuscript [22]. As a corollary, the class of starts carrying a uniform power-envelope descent certificate of length at most five has certificate density 7/8; the four-step subclass has certificate density 13/16. An unconditional counting argument shows that parity equidistribution at *all* depths would give density one to the set of starts with some finite descent certificate. None of these is a density of starts that reach 1, and no statement about the Juggler conjecture itself is claimed. What is solved here is the first genuinely nested layers of the parity process, not the infinite-depth problem: the remaining obstacle is the level-3 kernel, where the weight scale $n^{27/16}$ exceeds n; we state it as an open problem and record precisely what blocks every method of this paper, and what the depth-4 results do and do not buy for the termination problem.

1. Introduction

Equidistribution questions for a *single* floor of a smooth function — the Piatetski–Shapiro tradition — are classical and well developed. This paper concerns the *composition* of floors:

$$m = \lfloor n^{3/2} \rfloor, \qquad v = \lfloor m^{3/2} \rfloor, \qquad w = \lfloor m^{1/2} \rfloor, \qquad z = \lfloor v^{3/2} \rfloor, \ \ldots$$

and, specifically, the joint distribution of the *parities* along such chains. Fix a pattern (an *itinerary word*) $w_1w_2\cdots w_d$ over the alphabet $\{E, O\}$, where an O applies $x \mapsto \lfloor x^{3/2}\rfloor$ and an E applies $x \mapsto \lfloor\sqrt{x}\rfloor$. The question is whether the 2^d parity classes of the resulting chain equidistribute over odd starts $n \le N$, with a power saving. Odd starts have $w_1 = O$; the theorems of this paper concern the O-rooted words. (Even starts are a separate and easier problem: one square root drops the state to scale $\sqrt{N}$, and we make no claims about them.)

The difficulty is genuinely one of composition. Writing $\theta = \{n^{3/2}\}$, the naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ leaves a sawtooth of *growing* amplitude $n^{3/4}$ inside the phase. No estimate for exponential sums with smooth monomial phases applies; the phase is not smooth, and the sawtooth cannot be discarded. All results in the single-floor literature stop before this point (Section 1.2).

Two ideas carry the paper.

1. **Exact linearization** (Lemma 4.3). The identity $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + E(n)$ with $0 \le E(n) \le \frac{1}{2}n^{-3/4}$ eliminates the inner floor *exactly*: the integer m enters linearly, multiplied by a smooth coefficient, and the only non-smooth object left is m itself, which is handled by differencing and by an exact gap-cell decomposition. The identity is a Taylor rewrite, not a discovery; what carries the paper is the package built on it — one-signed remainders at every level, gap cells, the carry-branch decomposition, and a kernel with weight $n^{9/8}$. Iterated along the chain, this is what makes depth ≥ 2 accessible.

2. **The level-2 wave and the kernel theorem** (Lemma 5.2, Theorem 5.3). Every reorganization of the depth-4 pattern $OOO*$ funnels into one object, the exponential sum of the level-2 floor defect
$$K_c(P) = \sum_{\substack{n\sim P\\ n \text{ odd}}} e(c(n)\,\{\lfloor n^{3/2}\rfloor^{3/2}\}), \qquad c(n) = \tfrac{3k}{4}n^{9/8}.$$
After two Weyl differencings over an exact carry-branch decomposition, the master identity (Lemma 5.1) leaves no growing smooth part; what remains is bounded carries and, hardest, exact level-2 waves $e(qY)$, $Y = \lfloor n^{3/2}\rfloor^{3/2}$, possibly riding a frozen floor. The depth-2 nested-floor wave is the fundamental obstruction of the whole paper, and Lemma 5.2 is the result that overcomes it: a bound $|q|^{-1/6}P^{23/24+\varepsilon}$ for these mixed pieces by a targeted third differencing, exactly depth-2 strength, which is what pins the kernel saving at 1/96. We then prove $K_c(P) \ll P^{1-1/96+\varepsilon}$ for $c = \frac{3k}{4}n^{9/8}$, uniformly for $1 \le k \le P^{1/24}$ (Theorem 5.3). This is the hardest result of the paper, and Section 4 is written at full length — every estimate displayed with its constant — so that it can be checked without reference to anything outside this manuscript.

With the kernel theorem, depth-4 parity equidistribution for odd-rooted itineraries is complete (Theorem 6.1): the eight O-rooted length-4 words each receive their expected share with a power saving. The same estimates, applied to one further letter, count the two length-five contractors *OOOEE* and *OOEOE* (Theorem 6.3), so the certified descent class has certificate density 7/8 (Corollary 6.4). The leftover eighth is the expanding length-five tree $OOEOO \cup OOOEO \cup OOOO*$: the first two of those itineraries are counted and do not contract, and $OOOO*$ is the level-3 kernel. Section 3.5 proves the depth-≤ 3 theorems on sub-dyadic intervals of length $\ge P^{1/2}$ with a slowly varying twist attached, the form in which the companion paper [24] uses them. Theorem 5.5 localizes Theorem 5.3 to intervals of length $\ge P^{29/48+\delta}$, as a separate short-interval theorem. Its threshold does not reach the $P^{37/64}$ broad inverse scale of the proposed *OOOEEE* and *OOEOEE* productions, so no production is deduced from it here.

The logical dependence of the counting theorems is

Lemma 5.2(i) $\implies$ Lemma 5.2(ii) $\implies$ Theorem 5.3 $\implies$ Theorem 6.1 $\implies$ Theorem 6.3 and Corollary 6.4.

The localization used by the companion [24] is a separate chain:

$$\text{Theorems 4.11–4.12, Cor. 4.13, Thm. 5.5} \implies \text{companion [24]}.$$

The chains are not chosen at random: they are the itineraries of the Juggler map
$$J(n) = \begin{cases} \lfloor\sqrt{n}\rfloor, & n \text{ even},\\ \lfloor n^{3/2}\rfloor, & n \text{ odd},\end{cases}$$

introduced by Pickover [1] (OEIS A094683 [2]); universal arrival at 1 is open. The map is niche; the nested-floor sums must stand on their own, and the paper is organized so that they do. The exact finite-itinerary calculus of J — the power envelope $J^{|w|}(n)^{2^{|w|}} \le n^{3^{\#O(w)}}$, the defect identity, and a small-cycle census — is a companion manuscript [22]; here we use only the contraction criterion (Proposition 3.1), whose short induction is written out below. (The envelope's exponent, $3^{\#O(w)}/2^{|w|}$, is the scale exponent of $J^{|w|}$ that Section 7 uses to price the frontier: the envelope is the statement that flooring never raises an iterate above it.) The dynamical payoff of the counting theorems is a pair of *certified-descent densities*: the set of starts guaranteed to drop below their starting value within four steps has certificate density 13/16 (Corollary 4.9), and the two length-five contractors raise that class to certificate density 7/8 (Corollary 6.4). Equidistribution at all depths would give the set of starts with *some* finite descent certificate density one (Proposition 7.1). We state plainly what these corollaries are not: they are not densities of starts that reach 1, and they do not touch the Juggler conjecture, whose analogue of Terras's almost-all theorem for Collatz [4, 5, 6] remains open; see Lagarias [3] for the Collatz survey. A second companion paper [24] shows what the termination problem needs from parity statistics — control at depth of order $\log\log n$, not at any fixed depth — and Section 8 records what the fixed-depth results of this paper do and do not contribute to it.

Section 7 states the frontier precisely. The uncounted piece of the leftover eighth is the *OOOO*∗ split, one nesting deeper than Theorem 5.3, where the same kernel reappears with weight scale $n^{27/16} > n$ (Conjecture 7.3). Every method of this paper stops below that scale. What survives is a clean model problem — the amplitude-product sums $\sum e(A(t)\{B(t)\})$ with $1 \ll A' \ll A$ — for which a direct L^2 computation in the shift of the fractional argument gives square-root cancellation, up to a $\sqrt{\log}$ factor, for almost every shift (Proposition 7.4). That average says nothing about the single deterministic shift; the deterministic instance (Conjecture 7.5) is open, and we record in three sentences why the obvious shortcuts fail.

1.1 Verification

Every theorem below is a human proof from the stated classical inequalities. No numerical computation is a proof step. A companion repository records machine checks of the exact identities and a row-by-row audit ledger (paper_b_audit_ledger.md); nothing in this paper depends on it. Certificate densities are never densities of starts that reach 1. The six-stage proof of Lemma 5.2(i) is the author's reconstruction. The reduction of Lemma 5.2(ii) from (i), and the preservation of the classes (D1), (D2), (D3) under the third differencing, are written as a self-contained argument (Claims A–H in the proof of Lemma 5.2).

The boundary between those three kinds of warrant is worth stating once rather than reconstructing from the citations. *Human proof* means a proof in this paper; *Lean* lists the identifiers the repository declares, which are checks of identities, constants and thresholds, not of any estimate; *classical* names the external result used as given.

statement	human proof	Lean	classical input
Prop. 3.1 power envelope	companion [22]	in the companion, not here	—
Lem. 3.3 second-derivative test	quoted	—	van der Corput [11]
Lem. 3.4 discrepancy	quoted	—	Erdős–Turán [8]
Lem. 3.5 sawtooth expansion	quoted	—	Vaaler [10]
Lem. 3.7 shifted window	this paper	—	Lem. 3.5
Lem. 3.8 two-term test	this paper	`c6_eleven_eighths_five_fourths` (+`_attained`)	Lem. 3.3
Lem. 3.9 three-term sublevel	this paper	`step5b_curvature_inverse`, `step5b_curvature_norm`, `step5b_vector_transfer`, `step5b_c7_printed`	Lem. 3.3, 3.8

statement	human proof	Lean	classical input
Lem. 4.3 exact linearization	this paper	`lemma43_closed_form`, `lemma43_nonneg`, `lemma43_upper`, `lemma43_remainder_of_sqrt`, `carry_identity`, `carry_mem_zero_one`	Taylor
Thm. 4.4 nested discrepancy	this paper	—	Lem. 3.3–3.5, 4.3
Thm. 4.7, 4.8 depth three	this paper	—	Thm. 4.4
Thm. 4.11–4.13 localization	this paper	—	Thm. 4.4, 4.7
Lem. 5.1 master identity	this paper	`lemma51_i_identity`, `lemma51_i_closed_form`, `lemma51_i_nonneg`, `lemma51_i_upper`, `lemma51_double_gap`, `lemma51_brackets_le_two`, `lemma51_master`, `carry_as_sawtooth`, `double_difference_product`, `fract_diff_level2`, `mvt_cube_explicit`, `mvt_sqrt_diff_explicit`, `second_difference_exists_xi`, `second_difference_two_sided`	mean value theorem
Lem. 5.2(i), (ii), (iii)	**this paper**	**none**	Lem. 3.3, 3.5, 3.7, 3.8
Lem. 5.2b interpolant	this paper	`interpolant_assembly`, `interpolant_step_i`, `interpolant_step_ii_constant`, `gap_error_le_one`, `gap_error_one_attained`, `gap_error_not_halved_by_recentring`	—
Thm. 5.3 kernel cancellation	**this paper**	`step5b_curvature_norm`, `sublevel_raised_threshold` (Step 5b constants only; **no part of the assembly**)	Lem. 5.1, 5.2, 5.2b, 3.9
Lem. 5.4 transfer	this paper	—	—
Thm. 5.5 localized kernel	**this paper**	**none**	Thm. 5.3, Lem. 4.10
Thm. 6.1 depth four	this paper	—	Thm. 5.3
Thm. 6.3, Cor. 6.4 depth five	this paper	—	Thm. 5.3
Threshold P_0 (App. A)	computation, not a proof step	`row_5b_binding`, `step5b_c2_ceiling`, `step5b_c2_optimum_feasible`, `step5b_uniform_saturates`	rational arithmetic
Prop. 7.1 reduction	this paper	—	Hoeffding
Prop. 7.4 shift average	this paper	—	—

Three rows stand on the proof alone and are the ones to read first. Lemma 5.2 is proved here and has **no** machine check of any kind; Theorem 5.3 has two, and both are constants inside Step 5b rather than any step of the assembly; Theorem 5.5 has none, and what it rests on is stated in its own subsection — the completeness of an inventory of costs, which the repository checks by extraction from these proofs but which no Lean file certifies. Everything the repository verifies is an identity, a constant, or a threshold — it has checked no estimate in this paper, and a reader who treats the Lean column as corroboration of the analysis will be misled. The remaining rows are stated so that this one is unmistakable.

Two conventions in that column. Every identifier listed is declared under `formal/Problems/` and reachable from `Problems/JugglerParityPaper.lean`, a barrel importing exactly this paper's six modules — `BranchFreeze`, `MasterIdentity`, `MeanValues`, `MonomialSplitting`, `PaperBAssembly` and `ThresholdCertificate` — so the formal side can be built on its own with `lake build Problems.JugglerParityPaper`. It shares no module with Paper A's barrel. And three backticked names here are not theorems: `ring`, the tactic that discharges an inversion, and `sorry` and `native_decide`, named just below as the two ways a declaration can be green without being proved.

Declared and reachable is still not proved. A `sorry` anywhere in a declaration's dependency graph, or a `native_decide`, leaves the name declared, the module reachable and `lake build` green while this column asserts something false. So the third convention is that every identifier listed depends on Mathlib's three axioms and nothing else. `formal/AxiomCheckPaperB.lean` prints the axiom dependencies of all forty-seven, and every line of its output reads `[propext, Classical.choice, Quot.sound]`; the output is recorded beside it. That is what makes the column a claim about proofs rather than about names, and it is the check a reader should run first.

A column of identifiers says which theorems exist; it does not say that they are about this paper's constants. Two tables now say that. Appendix A pairs each of the thirty-eight threshold rows with its theorem, its substitution and a rational witness. `tools/lean_numeral_audit.py` pairs every numeral in the statement of every *other* Paper B theorem — three hundred and seventeen of them across `BranchFreeze`, `MonomialSplitting` and `PaperBAssembly` — either with the quantity it implements, checked by value against the constant this paper carries, or with the algebra it is a coefficient of. Sixty-two are of the first kind and the rest of the second; none is unclassified, which is the condition the check enforces.

The second table exists because of one failure it now prevents, recorded at the erratum in Lemma 5.2b: three theorems there proved that lemma's *superseded* chain, and the sentence citing them sat three lines under the corrected display. A check that the numerals appear somewhere in the manuscript would not have found it — both of them appear, inside the erratum's own list of what replaced them. Only a check by value does.

The same table checks the Lean files' *prose* about this paper, which goes stale by the same mechanism and usually in the opposite direction: a remark a Lean file makes about a gap here, once the gap is closed, falsifies its own framing. Two of `BranchFreeze`'s did. Nine such claims are now anchored to a sentence in Lean and a predicate on this manuscript, and a reworded sentence retires its row rather than quietly passing.

Locating a statement. Numbered statements run one ahead of the section headings until Section 6. Lemmas 3.x are in Section 2, Theorems 4.x in Section 3, Lemma 5.1, Lemma 5.2 and Theorem 5.3 in Section 4, and Theorem 6.1 in Section 5; from Section 6 onward the two agree. Subsection numbers are not offset — Section 3.5 and Appendices A.1–A.6 are where they say they are — and every cross-reference below is to a section heading, never to a statement's first digit.

1.2 Related work

Single floors. The distribution of $\lfloor n^c \rfloor$ ($c > 1$, $c \notin \mathbb{N}$) in arithmetic progressions and among primes goes back to Piatetski–Shapiro [12]; Leitmann [13] treated general smooth $\lfloor f(n) \rfloor$, and the prime exponent for $\lfloor n^c \rfloor$ has a long refinement history, e.g. Rivat–Sargos [14] and Rivat–Wu [15]. All of these concern a single floor of a smooth function; the analytic core is an exponential sum with smooth monomial phase, estimated by van der Corput's method in the form presented by Graham–Kolesnik [11].

Digital and parity properties of $\lfloor n^c \rfloor$. Mauduit–Rivat [16] established q-multiplicative equidistribution along $\lfloor n^c \rfloor$ for c in an explicit range, Morgenbesser [17] proved the analogous statements for the sum of digits of $\lfloor n^c \rfloor$ in residue classes, and Müllner–Spiegelhofer [23] proved normality of the Thue–Morse sequence along

$\lfloor n^c \rfloor$ for $1 < c < 3/2$. All of these estimate sums of the shape $\sum e(f(\lfloor n^c \rfloor))$ where f is a *digital* (automatic or q-additive) function: the outer function is handled by its own carry structure, and the analytic core is again a smooth-phase sum over the single floor. Parity of $\lfloor n^{3/2} \rfloor$ — our depth-1 statement, Theorem 4.1 — is the simplest instance of this circle and is classical; we include the short proof to fix constants and notation. Our outer function is not digital: it is a second convex power, and its fractional defect enters later phases with polynomially growing amplitude, which is a different regime.

Compositions and generalized polynomials. Two adjacent theories compose floors and do not cover our sums. For compositions of *Beatty* sequences $\lfloor \alpha \lfloor \beta n \rfloor \rfloor$, the inner floor error is bounded and the composition is again a bounded-perturbation linear sequence, so the linear theory applies. Bergelson–Leibman's generalized polynomials [7] — expressions built from polynomials by iterated floors, sums, and products — admit a complete distributional theory via nilmanifolds; but that theory is confined to maps generated by *polynomials*, and $n^{3/2}$ is not one: no nilsystem models $\{f(\lfloor g(n) \rfloor)\}$ for convex non-polynomial powers f, g, and the growing-amplitude sawtooth that drives Sections 3–6 has no analogue there. We are unaware of a published power-saving estimate for $\sum e(\xi\{f(\lfloor g(n) \rfloor)\})$-type sums, or for the fractional parts $\{f(\lfloor g(n) \rfloor)\}$, with f, g convex powers and growing ξ.

Arithmetic of Piatetski–Shapiro sequences. Baker, Banks, Brüdern, Shparlinski, and Weingartner [18] study squarefree values, prime factors, and Carmichael numbers along $\lfloor n^c \rfloor$; Glasscock [21] studies linear equations whose unknowns range over Piatetski–Shapiro sequences. These results intersect a Piatetski–Shapiro sequence with other arithmetic sets; none composes two floors.

Beatty sequences. For the linear case $\lfloor \alpha n \rfloor$ the discrepancy theory is governed by the continued fraction of α (three-distance/Ostrowski structure); see Abercrombie [19] and, for character sums along Beatty sequences, Banks–Shparlinski [20]. The linear case admits exact self-similar structure that the convex case $n^{3/2}$ does not.

What is not covered. We are unaware of a published equidistribution or parity result for *nested* floor powers $\lfloor \lfloor n^c \rfloor^d \rfloor$ with $c, d > 1$, or of a treatment of the level-2 defect sums $K_c(P)$ of Section 4. We do not make the novelty of this paper depend on that survey. What is new here, independently of the literature, is a specific object and a specific bound: the level-2 defect kernel $K_c(P) = \sum_{n \sim P \text{ odd}} e(c(n)\{\lfloor n^{3/2} \rfloor^{3/2}\})$ with smooth weights of scale $kP^{9/8}$, and the estimate $K_c(P) \ll P^{1-1/96+\varepsilon}$ uniformly for $k \leq P^{1/24}$ (Theorem 5.3), resting on the mixed-piece bound for exact level-2 waves (Lemma 5.2). Both are falsifiable statements about explicit sums. The obstruction they address is structural, not incremental: after one floor the argument of the second floor is an integer sequence, not a smooth function, and its fractional defect enters later phases with growing amplitude. The individual devices are not new — the linearization of Lemma 4.3 is a Taylor rewrite, and the carry bookkeeping of Lemma 5.1 will be recognized by anyone who has differenced $\lfloor f(n) \rfloor$; what we believe is new is the package (one-signed remainders, gap cells, the master identity, and the kernel bound) and the results it yields. A literature check was last refreshed in September 2026; we would welcome pointers to anything missed.

Dynamical context. For the Collatz map, Terras [4] and Everett [5] proved that almost every start has finite stopping time, and Tao [6] proved that almost all orbits attain almost bounded values; Lagarias [3] surveys the problem. These are methodological cousins — density statements through parity-word counting — and motivate Proposition 7.1; they prove nothing about J, whose branches are floor powers rather than affine maps.

Standard tools. The van der Corput estimates are used in the form presented by Graham–Kolesnik [11]; see also Iwaniec–Kowalski [9, Ch. 8]. The indicator expansions use Vaaler's extremal functions [10], and the discrepancy bridge is the Erdős–Turán inequality in the form of Kuipers–Niederreiter [8].

2. Exact linearization and the parity bridge

Throughout, n denotes an odd integer, $e(t) = e^{2\pi i t}$, $\{x\}$ the fractional part, and $\psi(x) = (-1)^{\lfloor x \rfloor}$. On a dyadic block $n \in (P, 2P]$ we write $n \sim P$. For the chains,

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m; \qquad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v;$$

and on even m, $U = m^{1/2}$, $w = \lfloor U \rfloor$, $\theta_w = U - w$. The same three letters are reused at later even states (Theorem 4.8 at m, Lemma 6.2 at v). An itinerary word of depth d at n is the sequence of parities $\mathrm{word}_d(n)$ of $n, J(n), \dots, J^{d-1}(n)$. We write $\mathrm{word}(n)$ for the infinite itinerary, and say it has prefix w when $\mathrm{word}_{|w|}(n) = w$.

Proposition 3.1 (power-envelope contraction; companion [22]). If the itinerary w is realized at $n \geq 2$ (the orbit's parities are the letters of w) and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Write $k = |w|$ and $m = J^k(n)$. We first prove the envelope $m^{2^k} \leq n^{3^{\#O(w)}}$, by induction on prefix length. The empty prefix is $n \leq n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \leq n^{3^o}$, and the next letter is realized. If that letter is even, then $J(x)^2 \leq x$, so $J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{2^\ell} \leq n^{3^o}$. If it is odd, then $J(x)^2 \leq x^3$, so $J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}$. This is the envelope at w. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, hence $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. □

The companion [22] develops the same envelope as a finite-itinerary identity with an exact defect; only the contraction criterion is used below. The induction is recorded here so that the criterion does not depend on an unpublished text.

An itinerary w with $3^{\#O(w)} < 2^{|w|}$ is *contracting*; realizing a contracting itinerary is a *descent certificate* of length $|w|$. Contracting words used below: E, OE, $OOEE$, and the two length-five words $OOOEE$ and $OOEOE$. Every even start realizes E, and every odd start with even image realizes OE; those two certificates cover all starts except the odd-to-odd class, which is where the counting problem lives. Longer contracting itineraries exist (length seven and eight); this paper makes no counting claims about them.

Lemma 3.2 (parity bridge). $\lfloor x \rfloor$ is odd if and only if $\{x/2\} \geq \frac{1}{2}$.

Proof. Write $x = \lfloor x \rfloor + \{x\}$ and split by the parity of $\lfloor x \rfloor$. □

Lemma 3.3 (van der Corput, second-derivative form [11]). Let f be twice differentiable on an interval of length M, with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then

$$\Big|\sum e(f)\Big| \ll_\alpha M\lambda^{1/2} + \lambda^{-1/2},$$

the sum over the integers of the interval.

The Weyl A-process used in Theorems 4.4 and 5.3 and in Lemma 5.2 is the following form, applied to a unimodular sequence on the odd integers of a block of length $\asymp P$ (equivalently, in the r-variable of Lemma 3.10, with shift h in r equal to shift $2h$ in n). For $1 \leq H \leq P$,

$$\Big|\sum a_n\Big|^2 \leq \frac{2P^2}{H} + \frac{4P}{H} \sum_{1 \leq h < H} \Big|\sum a_{n+2h}\overline{a_n}\Big|.$$

This is the classical inequality $|\sum a_n|^2 \leq \frac{P+2H}{H} \sum_{|h|<H} (1 - \frac{|h|}{H}) \sum a_{n+h}\overline{a_n}$ with the even-shift restriction and with the weights $1 - |h|/H \leq 1$ absorbed in the absolute constants. Every later appeal to "the classical inequality" is this display.

Lemma 3.4 (Erdős–Turán [8]). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies, for every $H \geq 1$,

$$D \ll \frac{R}{H} + \sum_{h=1}^{H} \frac{1}{h} \Big|\sum_{j \leq R} e(hx_j)\Big|.$$

Lemma 3.5 (Vaaler [10]). For every $J \geq 1$ there are trigonometric polynomials $V_J(t) = \sum_{0<|q|\leq J} a_q e(qt)$ with $|a_q| \leq \min(1, \frac{2}{|q|})$ and a nonnegative trigonometric polynomial $\Delta_J \geq 0$ of degree J with constant term and coefficients at most $1/(J+1)$, such that the period-2 square wave satisfies $\psi(x) = V_J(x/2) + O(\Delta_J(x/2))$. For products, $|\psi_1\psi_2 - V^{(1)}V^{(2)}| \leq \Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)}\Delta^{(2)}$, and each error term is again a nonnegative trigonometric polynomial of the same shape.

The class indicators below are products of half-wave factors $\frac{1}{2}(1\pm\psi)$ evaluated along *formal* chains — chains whose branches are dictated by the target itinerary, not by the orbit. The next lemma is the exact identity that justifies this; it replaces any appeal to sampled verification.

Lemma 3.6 (branch consistency). Let $d \ge 1$ and let $w = w_1 \cdots w_d$ be an itinerary with $w_1 = O$. (The hypothesis is necessary, not a convenience: the product below runs over $t \le d-1$ and so tests letters $2, \dots, d$ only, letter one being carried by the restriction to odd n. For an E-rooted w at odd n the two sides disagree outright, which is one reason Section 5 sets those starts aside.) For odd n, define the formal chain $x_1 = n$ and, for $1 \le t < d$,

$$\xi_t = \begin{cases} x_t^{3/2}, & w_t = O, \\ x_t^{1/2}, & w_t = E, \end{cases} \qquad x_{t+1} = \lfloor \xi_t \rfloor,$$

and set $\sigma_t = +1$ if $w_t = E$, $\sigma_t = -1$ if $w_t = O$. Then, for every odd n,

$$\prod_{t=1}^{d-1} \tfrac{1}{2}\big(1 + \sigma_{t+1}\, \psi(\xi_t)\big) = [\mathrm{word}_d(n) = w].$$

Proof. Induct on d. For $d = 1$ both sides are 1. Let Π_{d-1} be the product over $t \le d-2$; by induction $\Pi_{d-1} = [\mathrm{word}_{d-1}(n) = w_1 \cdots w_{d-1}]$. If $\Pi_{d-1} = 0$, both sides vanish, since every factor lies in $[0,1]$. If $\Pi_{d-1} = 1$, then the orbit realizes $w_1 \cdots w_{d-1}$, and a second induction along the chain gives $x_t = J^{t-1}(n)$ for $t \le d-1$: assuming $x_t = J^{t-1}(n)$, the true parity of x_t is w_t, so J applies exactly the w_t-branch and $J^t(n) = \lfloor \xi_t \rfloor = x_{t+1}$. Hence ξ_{d-1} is the true real image with $\lfloor \xi_{d-1} \rfloor = J^{d-1}(n)$, and the last factor $\frac{1}{2}(1 + \sigma_d \psi(\xi_{d-1}))$ equals 1 exactly when the parity of $J^{d-1}(n)$ is w_d, else 0. $\square$

Each $\psi(\xi_t)$ is defined for *every* odd n, which is what permits the unconditional Vaaler expansion of the product: off the cylinder, the vanishing of an earlier factor kills the term, and on the cylinder every factor computes the true letter.

The last three preliminaries are the remaining analytic tools of Section 4: a finite Fourier expansion for sawtooth phases whose coefficient may exceed 1; a second-derivative test for two-term monomial phases in which the curvature transition — the short set where the two curvatures cancel — receives a trivial bound; and a sublevel splitting for three-term monomial phases, again with the transition set destined for a trivial bound. No third- or fourth-derivative exponential-sum test is used anywhere in this paper: on the short cells of Section 4 those tests do not sum, and the transition sets are cheaper to measure than to oscillate over. A final remark fixes the parity reindexing once and for all.

Lemma 3.7 (finite Fourier expansion with a shifted window). Let $B \in \mathbb{R}$, and let T, J be integers with $J \ge 1$ and $T \ge 8(1 + |B|)$. There exist coefficients $(b_u)_{|u| \le T}$ with

$$|b_u| \le \min\Big(2, \tfrac{1}{\pi|u+B|}\Big) + \min\Big(2, \tfrac{1}{\pi|u|}\Big), \qquad \sum_{|u| \le T} |b_u| \le 8 + 2\log(2 + |B| + T),$$

and coefficients $(v_q)_{0<|q|\le J}$ with $|v_q| \le \min(2, 2\pi|B|)/|q|$, such that for every $t \in \mathbb{R}$

$$\Big| e(-B\{t\}) - \sum_{|u| \le T} b_u e(ut) - \sum_{0<|q|\le J} v_q e(qt) \Big| \le \min(2, 2\pi|B|)\, \Delta_J(t) + \frac{8(1+|B|)}{T},$$

where $\Delta_J \ge 0$ is a trigonometric polynomial of degree J with constant term and coefficients at most $1/(J+1)$.

A note on reading the coefficient bound. Both minima are written with a division whose denominator vanishes somewhere in range: the second at $u = 0$, which is always summed over, and the first at the shifted mode $u = -B$ when that is an integer. Here they are read with the convention $1/0 = +\infty$, so that min selects the constant branch 2 and the bound is the intended one. A formalization cannot read them that way — in Lean $1/0 = 0$, which makes min select 0 and turns the display into the false assertion that the coefficient vanishes at the one mode where it is largest. The safe statement is multiplicative: $|b_u| \le 2$ together with $\pi|u + B|\,|b_u| \le 1$, which is the same bound at every other mode and correct at these. `formal/Problems/Juggler/DividedBounds.lean` records the failure, the equivalence off it, and the convention itself. That file is a repository utility and is deliberately outside the Paper B barrel: it says nothing about the Juggler map, and the barrel's claim is that everything reachable from it is an identity, a constant or a threshold. The same discipline is why the window-mass hypotheses are better stated over a weight than over a minimum — after the two Weyl differencings of Theorem 5.3 the inner sums have shift-dependent lengths, and a fixed-length min cannot be applied to them.

Proof. Write $f(t) = e(-B\{t\})$ and $\sigma(t) = \frac{1}{2} - \{t\}$, the sawtooth with jump $+1$ at the integers. The jump of f at the integers is $1 - e(-B)$, so $g := f - (1 - e(-B))\,\sigma$ is continuous, 1-periodic, and piecewise C^1. Since $|f| = 1$, $|\sigma| \le \frac{1}{2}$, and $|1 - e(-B)| = 2|\sin \pi B| \le \min(2, 2\pi|B|) \le 2$, we have $|g| \le 2$, hence $|\hat{g}(0)| \le 2$. For $u \ne 0$, direct integration gives $\hat{f}(u) = (1 - e(-B))/(2\pi i(u + B))$ when $u + B \ne 0$ and $\hat{\sigma}(u) = 1/(2\pi i u)$, so

$$\hat{g}(u) = \frac{1 - e(-B)}{2\pi i}\Big(\frac{1}{u + B} - \frac{1}{u}\Big) = -\frac{(1 - e(-B))\,B}{2\pi i\,u(u + B)}.$$

Two bounds follow. If $\pi|u+B| \ge \frac{1}{2}$, the first form gives $|\hat{g}(u)| \le \frac{2}{2\pi}\big(\frac{1}{|u+B|} + \frac{1}{|u|}\big) = \frac{1}{\pi|u+B|} + \frac{1}{\pi|u|}$; if $\pi|u+B| < \frac{1}{2}$, then $|\hat{g}(u)| \le |\hat{f}(u)| + |1 - e(-B)||\hat{\sigma}(u)| \le 1 + \frac{1}{\pi|u|} \le 2 + \frac{1}{\pi|u|}$. In both cases (and when $u + B = 0$, where $|\hat{f}(u)| \le 1$ directly)

$$|\hat{g}(u)| \le \min\Big(2, \tfrac{1}{\pi|u+B|}\Big) + \min\Big(2, \tfrac{1}{\pi|u|}\Big),$$

and summing this over $|u| \le T$ — at most four terms hit each of the two caps, and the harmonic tails integrate to logarithms — gives $\sum_{|u|\le T} |\hat{g}(u)| \le 8 + 2\log(2 + |B| + T)$. The second form gives, for $|u| > T \ge 8(1 + |B|)$ (so $|u + B| \ge |u| - |B| \ge |u|/2$),

$$|\hat{g}(u)| \le \frac{2|B|}{2\pi}\cdot\frac{2}{u^2} = \frac{2|B|}{\pi u^2}, \qquad \sum_{|u|>T} |\hat{g}(u)| \le \frac{4|B|}{\pi T} \le \frac{8(1 + |B|)}{T}.$$

The full series $\sum_u |\hat{g}(u)|$ therefore converges, and since g is continuous its Fourier series converges to g everywhere; take $b_u = \hat{g}(u)$ for $|u| \le T$ and charge the tail to the flat error. For the sawtooth part, Vaaler's theorem [10] provides a trigonometric polynomial $V_J^*(t) = \sum_{0<|q|\le J} a_q^* e(qt)$ with $|a_q^*| \le 1/|q|$ and $|\sigma(t) - V_J^*(t)| \le \Delta_J(t)$ with Δ_J as stated; multiply by $1 - e(-B)$ and set $v_q = (1 - e(-B))a_q^*$. □

The mass this lemma costs, at every site that uses it. The bound carries a coefficient mass $\sum_u |b_u| + \sum_q |v_q| \le 8 + 2\log(2{+}|B|{+}T) + 4H_J$, which the paper absorbs into P^ε everywhere and prints nowhere. Since $|B|$, T and J are powers of P at every site, that is $O(\log P)$ at every site; the coefficient is $2\max(\beta, \tau) + 4\iota$ where P^β, P^τ, P^ι are $|B|, T, J$. Tabulated (`decoration_budget.lemma37_site_masses`):

site	$\lvert B\rvert$	T	J	coeff
Thm 4.1 St.3(s1)	$2.25P^{-1/16}$	$P^{1/2}$	R_0	9/4
Thm 4.1 St.3(s2)	$2.25P^{1/4}$	$P^{1/2}$	–	1
Thm 4.1 St.6(D1)	$O(1)$	$P^{1/2}$	–	1
Thm 4.1 St.6(D2)	$1.85khP^{1/8}$	$P^{1/2}$	R_0	9/4
Thm 5.3 St.3(a)	$\frac{15}{8}kh_2P^{1/8}$	$P^{1/2}/(2h_1)$	$P^{1/4}$	47/24
Thm 5.3 St.3(b)	$1.85kh_1P^{1/8}$	$P^{1/2}/(2h_2)$	–	11/12
Thm 5.3 St.5b (j=0)	≤ 6	$P^{1/2}$	–	1
Thm 6.1 Step E	≤ 6	$P^{1/2}$	–	1
Thm 6.3 depth five	$2P^{19/96}$	R_0	–	5/8
Lemma 5.2(iii)	$5P^{1/4}$	$P^{1/2}$	–	1

so the absorption is sound, and the coefficient never exceeds $\frac{9}{4}$. The two sites that reach it are exactly the two carrying Stage 2's truncation $J = R_0 = P^{5/16}$, where the v-mass $4 \cdot \frac{5}{16} = \frac{5}{4}$ outweighs the b-mass 1.

One thing in that table is worth reading twice. Theorem 6.3 carries the largest log power in the paper's final bounds, $\log^{15/4} P$, and it has the *thinnest* Lemma 3.7 site of the ten, at $\frac{5}{8}$ — its window parameter is

$R_0 = P^{5/16}$ rather than $P^{1/2}$. So that log power is not a fat mass at one site; it is the count of applications at depth five. The two are independent, and only the second grows with depth.

When the lemma is applied with $t = G(n)$ along a block, the Δ_J-term is a nonnegative majorant: its sum over the block is $P/(J+1)$ plus J-truncated mode sums of G with coefficients $\le 1/(J+1)$, and the flat term contributes $8(1+|B|)P/T$. Both costs are displayed at each application below.

Those applications quote the flat cost as $4P/J$ per majorant layer, with the layer count written out and the mode sums always shown separately or taken as the object of study. The 4 is a rounding rather than a derived value: the constant term contributes $P/(J+1)$ over the block, and $P/(2(J+1))$ over the odd n actually summed, so $4P/J$ carries a factor of at least eight. The slack is uniform and costs nothing — each of these majorants is weighed against a budget of the same exponent, so a sharper constant would move no exponent anywhere.

Lemma 3.8 (two-term monomial test with a trivial transition bound). Let $E \subset \mathbb{Q} \cap [-4, 4]$ be a fixed finite set disjoint from $\{0, 1, 2, 3\}$, let $\alpha \neq \beta$ lie in E, let $I \subseteq (P, 2P]$, and let

$$f(n) = a\, n^{\alpha} + b\, n^{\beta} + g(n), \qquad M := \max(|a|P^{\alpha-2},\, |b|P^{\beta-2}) \le 1,$$

where the perturbation satisfies, for every $n \in I$,

$$|g''(n)| \le \rho\,(|a|n^{\alpha-2} + |b|n^{\beta-2}), \qquad |g'''(n)| \le \rho\,(|a|n^{\alpha-3} + |b|n^{\beta-3}),$$

with $\rho \le \rho_0(E)$ sufficiently small in terms of E alone. Then

$$\Big|\sum_{n\in I} e(f(n))\Big| \ll_E |I|\, M^{1/2} + M^{-1/2} + (P/M)^{1/3}.$$

(If $M \le P^{-2}$ the third term exceeds $|I|$ and the bound is trivial; the content is the range $M > P^{-2}$.)

Proof. Write $A(n) = a\,\alpha(\alpha-1)\,n^{\alpha-2}$ and $B(n) = b\,\beta(\beta-1)\,n^{\beta-2}$, so that

$$f'' = A + B + g'', \qquad n\, f''' = (\alpha-2)A + (\beta-2)B + n\, g'''.$$

On the block, $|A(n)| \asymp_E |a|P^{\alpha-2}$ and $|B(n)| \asymp_E |b|P^{\beta-2}$. The ratio $|A/B| = |a\alpha(\alpha-1)/(b\beta(\beta-1))|\, n^{\alpha-\beta}$ is strictly monotone in n, so for any threshold $K = K(E) \ge 3$ the block splits into at most three consecutive intervals: I_1 where $|A/B| \ge K$, I_2 where $|A/B| \le 1/K$, and a middle I_0. (If a or b vanishes, only I_1 or I_2 is present and the argument shortens accordingly.)

On I_1: $|f''| \ge (1 - \frac{1}{K})|A| - |g''| \ge \frac{1}{2}|A| \gg_E M$-scale, with $\sup|f''|/\inf|f''| \le C(E)$; Lemma 3.3 gives $\ll_E |I|M^{1/2} + M^{-1/2}$. Symmetrically on I_2.

On I_0, $|A| \asymp_K |B|$. If A and B have the same sign there, $|f''| \ge |A| + |B| - |g''| \ge |A|$ and Lemma 3.3 applies as before. If they have opposite signs, write $B = -sA$ with $s = s(n) \in [1/K, K]$, monotone in n. Then

$$f'' = A\,(1-s) + g'', \qquad n\, f''' = A\,((\alpha-2) - s(\beta-2)) + n\, g'''.$$

The two affine functions $s \mapsto 1 - s$ and $s \mapsto (\alpha-2) - s(\beta-2)$ have distinct zeros: $s = 1$ and $s = (\alpha-2)/(\beta-2) \neq 1$ (here $\alpha \neq \beta$ and $\beta \neq 2$ are used). Hence

$$c_6(E) := \min_{s>0}\ \max(|1-s|,\ |(\alpha-2) - s(\beta-2)|) > 0.$$

Since $s(n)$ is monotone, the set where $|1-s(n)| \ge c_6$ is the complement of a single interval, so I_0 splits into at most three consecutive intervals: two on which $|f''| \ge \frac{1}{2}c_6|A|$ (after absorbing g by $\rho_0(E) \le c_6/8$) — Lemma 3.3 as before — and one transition interval J on which $|nf'''| \ge \frac{1}{2}c_6|A|$ throughout, i.e. $|f'''| \ge c_8(E)\, M/P$ in normalized scale. On J the second derivative is strictly monotone. Let $V \in (0, M]$ be a parameter: the set $\{n \in J : |f''(n)| < V\}$ is a single interval of length at most $2V/(c_8 M/P)$, which receives the trivial bound, and its at most two flanking intervals carry $|f''| \ge V$ of a single sign with $\sup|f''| \le C(E)M$, so Lemma 3.3 (applied at the piece's own infimum) gives $\ll_E |I|M^{1/2} + V^{-1/2}$ for each. Choosing $V = (c_8 M/P)^{2/3}$

equalizes $V^{-1/2}$ and the trivial cost at $\ll_E (P/M)^{1/3}$. Summing the $O_E(1)$ contributions proves the lemma. $\square$

The proof's only content is that the second and third derivatives of a two-term monomial phase cannot both be small on the same subinterval — the linear system with matrix $\left(\begin{smallmatrix} 1 & 1 \\ \alpha-2 & \beta-2 \end{smallmatrix}\right)$ is invertible — so the transition set where the second-derivative test fails is short, and *measuring* it costs less than any attempt to oscillate over it: crucially, the transition cost $(P/M)^{1/3}$ carries no factor $|I|$, so it sums over many short cells. All applications in Section 4 use exponent pairs from $E = \{\frac{3}{4}, \frac{5}{4}, \frac{11}{8}, \frac{3}{2}, \frac{15}{8}\}$.

For that E the constant c_6 is not merely positive but computable in closed form: with $p = \alpha - 2$, $q = \beta - 2$, the two V-shapes $|1 - s|$ and $|p - qs|$ have distinct zeros 1 and p/q, so the minimum of their maximum is attained at a crossing and is rational. Over the twenty ordered pairs of E:

$\alpha\backslash\beta$	$\frac{3}{4}$	$\frac{5}{4}$	$\frac{11}{8}$	$\frac{3}{2}$	$\frac{15}{8}$
$\frac{3}{4}$	—	$\frac{2}{7}$	$\frac{5}{13}$	$\frac{1}{2}$	1
$\frac{5}{4}$	$\frac{2}{9}$	—	$\frac{1}{13}$	$\frac{1}{6}$	$\frac{5}{9}$
$\frac{11}{8}$	$\frac{5}{18}$	$\mathbf{\frac{1}{14}}$	—	$\frac{1}{12}$	$\frac{4}{9}$
$\frac{3}{2}$	$\frac{1}{3}$	$\frac{1}{7}$	$\frac{1}{13}$	—	$\frac{1}{3}$
$\frac{15}{8}$	$\frac{1}{2}$	$\frac{5}{14}$	$\frac{4}{13}$	$\frac{1}{4}$	—

The table is not symmetric, and one might hope to exploit that: the proof is free to relabel so that the larger curvature is A, which makes $s = -B/A$ satisfy $|s| \leq 1$ and restricts the minimisation to $s \in (0, 1]$. That restriction does raise ten of the twenty entries — exactly those with $\alpha < \beta$, whose crossings lie beyond 1 — but every entry with $\alpha > \beta$ already has its crossing inside, and the minimum $\frac{1}{14}$ is attained at $(\frac{11}{8}, \frac{5}{4})$, which is one of those. So the normalisation is available and buys nothing: $c_6(E) = \frac{1}{14}$ and $\rho_0(E) \leq \frac{1}{112}$ either way.

Which constants carry numbers. Not all of them do, and it is worth saying which, so that a reader checking the paper knows where to spend the effort. Four are evaluated and used at their values: $c_6(E) = \frac{1}{14}$ and the perturbation gate $\rho_0(E) \leq \frac{1}{112}$ here; $c_7 = \frac{1}{232}$ of Lemma 3.9, which alone sets P_0; and the vector (c_2, c_3, c_4) that Lemma 3.9's proof actually needs. Two are not. The $c_8(E)$ of the proof above occurs three times, all within it, and is absorbed into the $\ll_E$ of the conclusion: choosing $V = (c_8 M/P)^{2/3}$ equalizes two costs whatever its value, so no closed form is called for. And $C(E)$ is never assigned a value anywhere in the paper.

The one place that matters is Step 5b of Theorem 5.3, where reading $C(E)P^{89/96} \log P$ against $P^{15/16}$ requires $C(E) \log P \leq P^{1/96}$. That comparison is not decided by the O_E that carries $C(E)$ everywhere else, which is exactly why Step 5b uses the ε-form rather than the sharp one, and why Appendix A.4 keeps the logarithmic thresholds out of P_0.

The minimum is $c_6(E) = \frac{1}{14}$, attained only at $(\alpha, \beta) = (\frac{11}{8}, \frac{5}{4})$ — there $p = -\frac{5}{8}$, $q = -\frac{3}{4}$ and the crossing is at $s = \frac{13}{14}$, where both V-shapes equal $\frac{1}{14}$. Consequently the admissible perturbation size of this section may be taken to be the explicit $\rho_0(E) = c_6/8 = \frac{1}{112}$, and every $\ll_E$ in Lemma 3.8 is a constant depending only on that number and on K. Lean: `c6_eleven_eighths_five_fourths` (the lower bound, for all s) and `c6_eleven_eighths_five_fourths_attained` (sharpness).

One corner of Section 4 (three curvature scales meeting on a zero-offset branch) needs the three-term analogue, which we state directly as a sublevel bound: there the transition set is measured once against a global smooth model, and only the second-derivative test is ever applied.

Lemma 3.9 (three-term monomial sublevel splitting). Let E be as in Lemma 3.8 and let $\alpha, \beta, \gamma \in E$ be pairwise distinct, $I \subseteq (P, 2P]$, and

$$f(n) = a\, n^{\alpha} + b\, n^{\beta} + c\, n^{\gamma} + g(n), \qquad S := \max(|a|P^{\alpha-2}, |b|P^{\beta-2}, |c|P^{\gamma-2}) > 0,$$

with g satisfying the analogues of the perturbation bounds of Lemma 3.8 for the second, third, *and fourth* derivatives, at some $\rho \leq \rho_0$. There are constants $c_7 = c_7(\alpha, \beta, \gamma) > 0$ and $C = C(\alpha, \beta, \gamma)$ such that for every V with $0 < V \leq c_7 S/2$:

(i) the sublevel set $\Omega_V = \{n \in I : |f''(n)| \le V\}$ is a union of at most $C(E)$ intervals of total length

$$|\Omega_V| \le C(E)\Big(\frac{PV}{S} + P\Big(\frac{V}{S}\Big)^{1/2}\Big);$$

(ii) $I \setminus \Omega_V$ is a union of at most $C(E)$ intervals, on each of which f'' is single-signed with $V \le |f''| \le C(E)\, S$. If one of a, b, c vanishes the claim reduces to Lemma 3.8; if two vanish, to Lemma 3.3. The application in Theorem 5.3, Step 5b, uses the three-term statement when a window mode is present and the two-term reduction when $w = 0$.

Proof. Write A, B, C for the three curvature terms $a\alpha(\alpha-1)n^{\alpha-2}$ etc., so that

$$f'' = A + B + C + g'', \quad nf''' = (\alpha{-}2)A + (\beta{-}2)B + (\gamma{-}2)C + ng''',$$

$$n^2 f'''' = (\alpha{-}2)(\alpha{-}3)A + (\beta{-}2)(\beta{-}3)B + (\gamma{-}2)(\gamma{-}3)C + n^2 g''''.$$

The coefficient matrix is the Vandermonde-type matrix in the distinct values $\alpha{-}2, \beta{-}2, \gamma{-}2$ (rows 1, x, $x(x{-}1)$), hence invertible with inverse bounded in terms of E: at every point of I,

$$\max(|f''|, |nf'''|, |n^2 f''''|) \ge c_7(E) \max(|A|, |B|, |C|) \ge c_7(E)\, \tilde{S}, \qquad \tilde{S} \asymp_E S,$$

after absorbing g by $\rho_0 \le c_7/8$ (zero coefficients only shorten the argument). The constant depends on the *triple*, not on the ambient set: $c_7 = 1/\|M^{-1}\|_\infty$, and since $\det M = \prod_{i<j}(x_j - x_i)$ with $x = \alpha - 2$ etc., it scales as the square of the exponent gap. For an equally spaced triple of gap δ about x_0, $\delta^2/c_7 = x_0^2 - 2x_0 + c$ with $c \in [1.75, 2]$ for $\delta \in [\frac{1}{8}, \frac{1}{2}]$. Writing $c_7(E)$ for a *uniform* constant over all triples of E would be weaker: over the exponent inventory of this paper the minimum is $1/259$, at $(\frac{9}{8}, \frac{5}{4}, \frac{11}{8})$. Each application below uses its own triple's value.

The proof needs less than a single scalar. What it actually uses is that the three tests are not *all* small, and that is a condition on a vector $c = (c_2, c_3, c_4)$, one constant per derivative order: if $|T_j| \le c_j \tilde{S}$ for $j = 2, 3, 4$ then $|A_i| \le \tilde{S} \sum_j |M^{-1}|_{ij} c_j$, so any $c \ge 0$ with $|M^{-1}|c \le 1$ rowwise will do (Lean `step5b_vector_transfer`). Only c_2 enters the hypothesis $V \le c_2 S/2$; c_3 and c_4 enter only through C. Appendix A.5 records what that buys, and what it costs. The ratios A/B, B/C, A/C are monotone, so I splits into $O_E(1)$ consecutive intervals on each of which one fixed derivative order $r \in \{2, 3, 4\}$ is good at full scale throughout. On an $r = 2$ piece, Ω_V is empty (as $V \le c_7 S/2$). On an $r = 3$ piece, $|f'''| \ge c_7 S/(2P)$ makes f'' strictly monotone, so Ω_V meets it in a single interval of length $\le 4PV/(c_7 S)$. On an $r = 4$ piece, $|f''''| \ge c_7 S/(4P^2)$ makes f''' strictly monotone, so f'' has at most one interior critical point and Ω_V meets the piece in at most two intervals; if $[x, y] \subseteq \Omega_V$, the second-difference identity $f''(x) - 2f''(\frac{x+y}{2}) + f''(y) = f''''(\xi)\,(y - x)^2/4$ forces $(y - x)^2 \le 16V \cdot 4P^2/(c_7 S)$, i.e. $y - x \le 8P\,(V/(c_7 S))^{1/2}$. Summing the $O_E(1)$ pieces gives (i). For (ii): on the complement, f'' is continuous with $|f''| \ge V$, hence single-signed on each maximal interval, and $|f''| \le |A| + |B| + |C| + |g''| \le C(E)S$. For the triple $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$ of Theorem 5.3, Step 5b, the inverse matrix is $\left(\begin{smallmatrix} 10 & 68 & 32 \\ -24 & -144 & -64 \\ 15 & 76 & 32 \end{smallmatrix}\right)$; the norm the argument needs — the ℓ^∞ operator norm, i.e. the maximal absolute row sum, since $(A, B, C)^\mathsf{T} = M^{-1}(f'', nf''', n^2 f'''')^\mathsf{T}$ — is 232, and its ℓ^1 operator norm (maximal absolute column sum) is 288. One may therefore take $c_7 = 1/232$; we keep the weaker value $c_7 = 1/288$ used in Step 5b, which remains valid. ☐

The Step-5b triple is in fact the extremal one, so $1/232$ serves as a single explicit constant for the whole section. Over the ten triples of E the norms $\|M^{-1}\|_\infty$ are $\frac{113}{10}$ $(\frac{3}{4}, \frac{5}{4}, \frac{15}{8})$ and $(\frac{3}{4}, \frac{11}{8}, \frac{15}{8})$, $\frac{113}{9}$, 35, $\frac{95}{3}$, $\frac{95}{2}$, 56, $\frac{149}{2}$, $\frac{181}{3}$, and 232 — the last being $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$. Hence $c_7(E) = \frac{1}{232}$ is admissible uniformly in the triple, and the absorption hypothesis of the lemma may be taken as the explicit $\rho_0 \le c_7/8 = \frac{1}{1856}$. Lean: `step5b_curvature_inverse` (the inversion, by `ring`) and `step5b_curvature_norm` (the ℓ^∞ bound $\max(|A|, |B|, |C|) \le 232 \max(|f''|, |nf'''|, |n^2 f''''|)$), with `step5b_c7_printed` recording that the manuscript's $1/288$ follows a fortiori. The step where the ℓ^∞/ℓ^1 confusion originally arose is therefore now machine-checked.

The lemma is a measure statement, not an exponential-sum estimate: in its one application (Theorem 5.3, Step 5b) the sublevel set receives the trivial bound and the complement the second-derivative test, with the parameter V chosen there.

Lemma 3.10 (parity reindexing). All counting sums below run over odd n. The substitution $n = 2r + 1$ maps a block interval of length L onto an interval of length $L/2$ and a phase f to $f^*(r) = f(2r + 1)$ with $(f^*)^{(k)}(r) = 2^k f^{(k)}(2r + 1)$. Consequently:

(a) any two terms of a composite phase have the same curvature *ratios* and the same *signs* in the r-variable as in the n-variable, so every dominance margin and single-sign check displayed below is invariant under the substitution;

(b) if $\lambda \le |f''| \le \alpha\lambda$ on an interval of length L, then Lemma 3.3 applied in the r-variable (length $L/2$, curvature window $[4\lambda, 4\alpha\lambda]$) gives

$$\Big|\sum_{n\in I,\ n \text{ odd}} e(f(n))\Big| \ll_\alpha \tfrac{L}{2}\,(4\lambda)^{1/2} + (4\lambda)^{-1/2} \le L\lambda^{1/2} + \lambda^{-1/2}\,:$$

the n-variable display dominates the true reindexed bound, and the same check for Lemma 3.8 replaces its conclusion by a smaller quantity;

(c) sublevel and transition sets (Lemmas 3.8–3.9) are measured in the n-line, and a set of total length Λ contains at most $\Lambda/2 + C(E)$ odd integers, so trivial bounds displayed in n-length dominate.

Two points the substitution needs, which (b) does not supply because (b) is about conclusions and these are about hypotheses. Lemma 3.8 asks for a two-term monomial, and $a(2r{+}1)^\alpha$ is not one. Writing it as $a2^\alpha r^\alpha\big(1{+}\tfrac{1}{2r}\big)^\alpha$, the correction has to be carried by that lemma's own perturbation g, and its second-derivative ratio

$$\Big|\big(1 + \tfrac{1}{2r}\big)^{\alpha-2} - 1\Big| \;\le\; \frac{|\alpha-2|}{2r}\Big(1 - \tfrac{1}{2r}\Big)^{-1}$$

clears $\rho_0(E) = \frac{1}{112}$ from $n \ge 143$, the worst case being $\alpha = \frac{3}{4}$. So the hypothesis does survive everywhere this paper works — but by absorption, not for free. And in (c) the domination $\Lambda/2 + C(E) \le \Lambda$ needs $\Lambda \ge 2C(E)$; on shorter sets both sides are $O_E(1)$, so the comparison there is vacuous rather than false.

Every application of Lemmas 3.3 and 3.7–3.9 in Sections 3–6 is to be read through this substitution; no displayed constant below needs adjustment.

3. Depths one to three

For odd n write $s(n) = \psi(n^{3/2}) = (-1)^m$ and $S_O(N) = \sum_{n\le N,\ n \text{ odd}} s(n)$. Let $M(N)$ be the number of odd $n \le N$ and $\mathrm{OO}(N) = \#\{n \le N : n \text{ odd},\ J(n) \text{ odd}\}$, so $S_O(N) = M(N) - 2\,\mathrm{OO}(N)$. By Lemma 3.2, with $g(r) = \frac{1}{2}(2r + 1)^{3/2}$, $s(2r + 1) = -1$ iff $\{g(r)\} \ge \frac{1}{2}$, and $S_O(N)$ is (twice) an interval discrepancy of the sequence $\{g(r)\}$.

Theorem 4.1 (depth one; classical). $|S_O(N)| \ll N^{5/6}$.

Proof. This is the standard van der Corput estimate for the monomial g, included for completeness. $g''(r) = \frac{3}{2}(2r + 1)^{-1/2}$ is positive and decreasing. Partition $1 \le r < R$ into dyadic blocks $[M, \min(2M, R))$; on a block $g'' \asymp M^{-1/2}$, so for the h-th mode $f = hg$ has $\lambda \asymp hM^{-1/2}$ and Lemma 3.3 gives $|\sum_{r\asymp M} e(hg(r))| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}$. Lemma 3.4 with $H = \lfloor M^{1/6}\rfloor$ bounds the block's discrepancy by $\ll M^{5/6}$, and the dyadic sum is $O(R^{5/6}) = O(N^{5/6})$. □

Corollary 4.2 (density of the odd-to-odd class). $\big|\mathrm{OO}(N) - N/4\big| \ll N^{5/6}$. Equivalently, the starts covered by the uniform one- or two-step certificates (E and OE) have natural density 3/4.

This is a density of a certificate class, not of starts that reach 1. Theorem 4.1 concerns consecutive source intervals; it supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step. The rest of the paper does not transfer it; it attacks the nested sums directly, by removing the inner floors exactly before any analytic step.

Lemma 4.3 (exact linearization and gap cells). (i) For every odd $n \ge 3$,

$$m^{3/2} = \tfrac{3}{2}\, m\, n^{3/4} - \tfrac{1}{2}\, n^{9/4} + E(n), \qquad 0 \le E(n) \le \tfrac{3}{8}\,(n^{3/2}-1)^{-1/2} \le \tfrac{1}{2}\, n^{-3/4}.$$

(ii) For $h \ge 1$ and odd n, let $g(n) = m(n+2h) - m(n)$ and $\delta(n) = (n+2h)^{3/2} - n^{3/2}$. Then

$$g(n) = \lfloor \delta(n) \rfloor + \kappa(n), \qquad \kappa(n) = [\{n^{3/2}\} \ge 1 - \{\delta(n)\}] \in \{0,1\}.$$

Proof. (i) Let $f(t) = (X-t)^{3/2}$ on $[0,\theta]$, so $f(\theta) = m^{3/2}$. Then $f'(t) = -\frac{3}{2}(X-t)^{1/2}$ and $f''(t) = \frac{3}{4}(X-t)^{-1/2}$. Taylor with Lagrange remainder at 0 gives $f(\theta) = f(0) + f'(0)\theta + \frac{1}{2}f''(\xi)\theta^2 = X^{3/2} - \frac{3}{2}X^{1/2}\theta + \frac{3}{8}(X-\xi)^{-1/2}\theta^2$ for some $\xi \in (0,\theta)$. Substituting $\theta = X - m$ in the linear term yields $-\frac{1}{2}X^{3/2} + \frac{3}{2}mX^{1/2} = -\frac{1}{2}n^{9/4} + \frac{3}{2}mn^{3/4}$. Since $\theta \in [0,1)$ and $f'' > 0$, the remainder $E(n) = \frac{3}{8}(X-\xi)^{-1/2}\theta^2$ lies in $[0, \frac{3}{8}(X-1)^{-1/2}]$, which is the bound of the statement; and $(X-1)^{-1/2} \le \frac{4}{3}X^{-1/2}$ already for $n \ge 3$, whence $E(n) \le \frac{1}{2}n^{-3/4}$. (ii) $g = \lfloor X + \delta \rfloor - \lfloor X \rfloor = \lfloor \delta \rfloor + \lfloor \{X\} + \{\delta\} \rfloor$, and the last floor is 1 precisely when $\{X\} + \{\delta\} \ge 1$ (Lean `carry_identity`, `carry_mem_zero_one`). $\square$

The remainder in closed form. The mean value ξ is avoidable: (i) is an algebraic identity in two square roots. Write $a = \sqrt{m}$ and $b = \sqrt{X} = n^{3/4}$, so that $m^{3/2} = a^3$, $mX^{1/2} = a^2 b$, $X^{3/2} = b^3$ and $\theta = b^2 - a^2$. Then

$$E = a^3 - \tfrac{3}{2}a^2 b + \tfrac{1}{2}b^3 = \tfrac{1}{2}(a-b)^2(2a+b),$$

identically. Both printed bounds are immediate from this: $E \ge 0$ because $a, b \ge 0$; and $Ea \le \frac{3}{8}\theta^2$ — i.e. $E \le \frac{3}{8}\theta^2 m^{-1/2}$, which at $\theta < 1$ is the stated $\frac{3}{8}(n^{3/2}-1)^{-1/2}$ — reduces after clearing $(a-b)^2$ to $4a(2a+b) \le 3(a+b)^2$, i.e. $(5a+3b)(a-b) \le 0$, which holds because $m \le X$. No Taylor expansion and no mean value are needed, and the resulting bound is sharper than the printed one at every n, not merely asymptotically. Machine-checked in `formal/Problems/Juggler/PaperBAssembly.lean` (`lemma43_closed_form`, `lemma43_nonneg`, `lemma43_upper`, `lemma43_remainder_of_sqrt`).

The point of (i): the non-smooth integer m enters the phase *linearly*, multiplied by a smooth coefficient; no fractional part of growing amplitude survives. The naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ instead leaves the sawtooth θ with the growing amplitude $n^{3/4}$, which defeats the van der Corput method directly. For (ii), since δ is smooth and increasing with $\delta' \asymp hn^{-1/2}$, the level sets of $\lfloor \delta \rfloor$ partition a dyadic block $n \in (P, 2P]$ into $O(hP^{1/2})$ cells of length $\asymp P^{1/2}/h$ on which the integer part of the gap is constant.

Theorem 4.4 (nested parity discrepancy). For every $\varepsilon > 0$ and every $(a,b) \in \{0,1\}^2 \setminus \{(0,0)\}$,

$$\Big| \sum_{n \le N,\ n \text{ odd}} \psi(n^{3/2})^a\, \psi(m(n)^{3/2})^b \Big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the four joint parity classes of $(m, \lfloor m^{3/2} \rfloor)$ on odd $n \le N$ has cardinality $\frac{N}{8} + O(N^{23/24+\varepsilon})$; in particular the odd-to-odd cylinder refines one letter deeper than Theorem 4.1.

Proof. Work on dyadic blocks $n \in (P, 2P]$, odd, with truncations $J_1 = J_2 = P^{1/24}$ (wave modes), $H = P^{1/12}$ (differencing), and $R = P^{1/4}$ (cell modes). All derivative tests are read through the parity reindexing of Lemma 3.10; every constant below already dominates the reindexed bound.

Step 1 (wave expansion). By Lemma 3.5 applied to each present factor, $\psi_1^a \psi_2^b$ differs from the product of the truncated polynomials $V^{(1)}V^{(2)}$ by at most $\Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)}\Delta^{(2)}$, and each error layer is a nonnegative trigonometric polynomial of degree J_1 resp. J_2 with constant term and coefficients $\le 1/(J+1)$. Summed over the block, a majorant layer costs its constant term $\le P/(2(J_1+1)) \le P^{23/24}$ plus mode sums $\sum_{0<r\le 2J_1} (J_1+1)^{-1} |\sum_n e(\frac{r}{2}n^{3/2})|$: each inner sum has $|(\frac{r}{2}X)''| \in [0.26, 0.75]\, rP^{-1/2}$, single sign, so Lemma 3.3 gives $\le 1.4 r^{1/2}P^{3/4} + 2P^{1/4}$, and the weighted total is $\le 2.4 J_1^{1/2} P^{3/4} + 2P^{1/4} \le 3P^{19/24}$. The theorem is thus reduced to bounds, uniform in the mode pair, for

$$S_{i,j}(P) = \sum_{n \in (P,2P],\ n \text{ odd}} e(\tfrac{i}{2}\, n^{3/2} + \tfrac{j}{2}\, m^{3/2}), \qquad |i| \le 2J_1,\ 1 \le |j| \le 2J_2,$$

with mode weights of total mass $O(\log^2 P)$; the $j = 0$ sums are the depth-1 sums of Theorem 4.1. Replacing the summand by its conjugate we may take $j \geq 1$.

Step 2 (linearization). By Lemma 4.3(i), replacing $\frac{j}{2}m^{3/2}$ by $\frac{3j}{4}mn^{3/4} - \frac{j}{4}n^{9/4}$ changes each summand by a phase of modulus $\leq \frac{j}{2}E(n) \leq \frac{j}{4}n^{-3/4}$, hence changes $S_{i,j}$ by at most $2\pi\frac{j}{4}\sum_{n\sim P} n^{-3/4} \leq 2jP^{1/4} \leq 4P^{7/24}$. Call the linearized phase Φ.

Step 3 (Weyl differencing). For $H = P^{1/12}$, the A-process recorded after Lemma 3.3 gives

$$|S_{i,j}|^2 \leq \frac{2P^2}{H} + \frac{4P}{H}\sum_{1\leq h<H}|T_h|, \qquad T_h = \sum_{\substack{n,\, n+2h\in(P,2P]\\ n \text{ odd}}} e\big(\Phi(n+2h) - \Phi(n)\big).$$

Step 4 (the exact differenced phase). Write $\Delta f = f(\cdot+2h) - f(\cdot)$, $g(n) = m(n+2h) - m(n)$, and $m = n^{3/2} - \theta_n$. Exactly,

$$\Phi(n+2h) - \Phi(n) = \tfrac{3j}{4}\, g(n)\,(n+2h)^{3/4} + \tfrac{j}{2}\, A_h(n) - B_h(n)\,\theta_n + \tfrac{i}{2}\,\Delta(n^{3/2}),$$

with $A_h = \frac{3}{2}n^{3/2}\Delta(n^{3/4}) - \frac{1}{2}\Delta(n^{9/4})$ and $B_h = \frac{3j}{4}\Delta(n^{3/4})$. Taylor with third-order remainders gives, for $n \sim P$ and $h \leq H$,

$$\Delta(n^{3/4}) = \tfrac{3}{2}hn^{-1/4} - \tfrac{3}{8}h^2n^{-5/4} + O(h^3P^{-9/4}), \qquad \Delta(n^{9/4}) = \tfrac{9}{2}hn^{5/4} + \tfrac{45}{8}h^2n^{1/4} + O(h^3P^{-3/4}),$$

so the two $n^{5/4}$-scale contributions of A_h cancel exactly at leading order (both equal $\frac{9}{4}hn^{5/4}$) and

$$A_h = -\tfrac{27}{8}\, h^2n^{1/4}(1 + O(hP^{-1})), \qquad |(\tfrac{j}{2}A_h)''| \leq 0.32\, jh^2P^{-7/4},$$

$$B_h \in [0.95,\ 1.13]\, jh\, P^{-1/4}, \qquad |(\tfrac{i}{2}\Delta(n^{3/2}))''| = \tfrac{i}{2}\, 2h|X'''(\xi)| \leq 0.38\, ihP^{-3/2}.$$

The sawtooth term is *discarded*: $|e(-B_h\theta_n) - 1| \leq 2\pi B_h \leq 7.1\, jhP^{-1/4}$ per term, so deleting it changes T_h by at most $3.6\, jhP^{3/4}$. This cost is charged in Step 7; it is admissible only because $jh \leq 2P^{1/24+1/12} = 2P^{1/8}$, which is the reason the truncation J_2 and the differencing length H cannot be enlarged in this argument.

Step 5 (cells and the exact shift device). Split T_h over the level sets of $\lfloor\delta\rfloor$, $\delta(n) = (n+2h)^{3/2} - n^{3/2}$: by (the argument of) Lemma 4.3(ii) and $\delta' \in (1.5, 2.2)\, hP^{-1/2}$, at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95]P^{1/2}/h$. On a cell $g = G + \kappa$ with $G = \lfloor\delta\rfloor$ frozen and $\kappa(n) = [\{n^{3/2}\} \geq \rho(n)]$, $\rho = 1 - \{\delta(n)\}$ smooth and monotone there. Vaaler-expand κ (one sawtooth layer, Lemma 3.5) with modes $e(rn^{3/2})$, $0 < |r| \leq R$, weights $\leq 1/|r|$, majorant cost $\leq P/(R+1) + (R\text{-mode sums at coefficients} \leq 1/(R+1)) \leq P^{3/4} + P^{7/8}$. The moving endpoint costs nothing: since $\rho(n) = 1 + G - \delta(n)$, the coefficient's n-dependent piece contributes the exact smooth phase $r\delta(n)$, and $e(rn^{3/2} + r\delta(n)) = e(r(n+2h)^{3/2})$ up to a unimodular constant, so every r-term falls into one of the two smooth families $e(rn^{3/2})$, $e(r(n+2h)^{3/2})$; the remaining in-cell coefficient variation is exactly 1 per cell and one Abel summation absorbs it at a factor $\leq 2+2\pi$.

Step 6 (second-derivative test per cell). After Steps 4–5 the phase on a cell is $\frac{3j}{4}G(n+2h)^{3/4} + \frac{j}{2}A_h + \frac{i}{2}\Delta(n^{3/2})$ plus, for the r-pieces, $rn^{3/2}$ or $r(n+2h)^{3/2}$.

For $r = 0$: the main curvature is the single monomial term

$$\left(\tfrac{3j}{4}G(n+2h)^{3/4}\right)'' = -\tfrac{9j}{64}\, G\,(n+2h)^{-5/4} \in -[0.15,\ 0.68]\, jh\, P^{-3/4},$$

using $G \in (3hP^{1/2}-1,\ 4.3hP^{1/2})$ from (the mean value bound for) δ and $(n+2h)^{-5/4} \in (2^{-5/4}, 1]P^{-5/4}$. Its competitors are dominated at displayed margins: $|(\frac{j}{2}A_h)''|/0.15jhP^{-3/4} \leq 2.2\, hP^{-1} \leq 2.2P^{-11/12}$ and $|(\frac{i}{2}\Delta(n^{3/2}))''|/0.15jhP^{-3/4} \leq 2.5\,(i/j)P^{-3/4} \leq 5P^{1/24-3/4}$, so the composite second derivative is single-signed with in-cell ratio ≤ 4.6. Lemma 3.3 per cell (through Lemma 3.10) and summation over cells give

$$\sum_{\text{cells}}(\ell_i\lambda^{1/2} + \lambda^{-1/2}) \leq 0.83\,(jh)^{1/2}P^{5/8} + 3.9\,(h/j)^{1/2}P^{7/8}.$$

For $r \neq 0$: the mode curvature $|(rX)''| \geq 0.53|r|P^{-1/2}$ dominates the full cell curvature $\leq 0.68jhP^{-3/4}$ at ratio $\geq 0.78\,|r|P^{1/4}/(jh) \geq 0.39\,P^{1/8} \geq 4$ for $P \geq P_0$ (here $jh \leq 2P^{1/8}$ is used again), so the composite keeps the mode's sign and, up to a factor $\frac{3}{4}$, its size. Lemma 3.3 per cell and the $1/|r|$ weights give

$$\sum_{0<|r|\leq R} \frac{1}{|r|}\Big(1.3|r|^{1/2}P^{3/4} + 1.5hP^{1/2}\cdot 1.6|r|^{-1/2}P^{1/4}\Big) \leq 5.2\,R^{1/2}P^{3/4} + 13\,hP^{3/4} \leq 6P^{7/8} + 13P^{5/6}.$$

Step 7 (totals and assembly). Collecting Steps 4–6,

$$|T_h| \leq C((jh)^{1/2}P^{5/8} + (h/j)^{1/2}P^{7/8} + P^{7/8} + jhP^{3/4})\log P \leq C'\,P^{7/8}(1+h^{1/2})\log P$$

uniformly in the modes (for the last step: $(jh)^{1/2}P^{5/8} \leq 2^{1/2}P^{1/16+5/8} \leq P^{3/4}$ and $jhP^{3/4} \leq 2h^{1/2}\cdot(jh^{1/2})P^{3/4} \leq 2h^{1/2}P^{1/24+1/24+3/4} \leq 2h^{1/2}P^{7/8}$, using $j \leq 2P^{1/24}$, $h^{1/2} \leq P^{1/24}$). Hence

$$|S_{i,j}|^2 \leq \frac{2P^2}{H} + \frac{4P}{H}\cdot C'\log P\sum_{h<H} P^{7/8}(1+h^{1/2}) \leq 2P^{23/12} + 6C'P^{15/8}H^{1/2}\log P \ll P^{23/12+\varepsilon}$$

at $H = P^{1/12}$, so $|S_{i,j}| \ll P^{23/24+\varepsilon}$. Summing mode weights ($O(\log^2 P)$), the majorant and truncation costs of Steps 1 and 5 ($\leq P^{23/24} + 3P^{19/24} + P^{7/8}$ per layer), and dyadic blocks gives the theorem. □

The exponent 23/24 is deliberately unoptimized; every stage has slack. The proof above is the undecorated instance of Lemma 5.2(i) below, which reruns the same six stages with three classes of decorations riding along; the reader who has checked this proof has checked the skeleton of that one.

Proposition 4.5 (OE-branch third letter). For $a \in \{0,1\}$,

$$\Big|\sum_{n\leq N,\ n\ \mathrm{odd}} \left((-1)^m\right)^a \psi(m^{1/2})\Big| \ll_\varepsilon N^{7/8+\varepsilon},$$

and consequently $\#\mathrm{OEO}(N),\ \#\mathrm{OEE}(N) = N/8 + O(N^{7/8+\varepsilon})$. Together with Theorem 4.4 this completes depth 3 for odd-rooted itineraries: each of *OOO*, *OOE*, *OEO*, *OEE* has density 1/8 among odd starts with a power saving.

Proof. Taylor expansion of $(X-\theta)^{1/2}$ with both correction terms one-signed gives

$$m^{1/2} = n^{3/4} + D_1(n), \qquad -\tfrac{1}{2}X^{-1/2} - \tfrac{1}{8}(X-1)^{-3/2} \leq D_1 \leq 0.$$

D_1 is decaying, so replacing $\frac{l}{2}m^{1/2}$ by $\frac{l}{2}n^{3/4}$ inside a Vaaler mode costs $\ll l\sum_{n\sim P} n^{-3/4} \ll lP^{1/4}$, absorbable at $l \leq 2P^{1/24}$. The mode sums are then pure: $\sum_{n\sim P} e(\frac{i}{2}n^{3/2} + \frac{l}{2}n^{3/4})$ with $l \neq 0$. For $i \neq 0$, $\varphi'' \asymp |i|P^{-1/2}$ is single-signed and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}P^{1/4}$; for $i = 0$, $\varphi'' \asymp lP^{-5/4}$ gives $\ll l^{1/2}P^{3/8} + l^{-1/2}P^{5/8}$. Vaaler majorants, truncation tails, and dyadic assembly are as in Theorem 4.4, and every bound is $\ll P^{7/8}$ after summing mode weights. Branch consistency is Lemma 3.6: the $(1+(-1)^m)/2$ factor vanishes exactly on odd m, where the true chain takes the 3/2-power branch. □

Lemma 4.6 (fourth-letter linearization). For every odd $n \geq 3$,

$$v^{1/2} = n^{9/8} + D(n), \qquad -\tfrac{3}{4}\,n^{-3/8} - n^{-9/8} \leq D(n) \leq 0.$$

Proof. Two applications of the Lemma 4.3(i) pattern. With $f(t) = (Y-t)^{1/2}$ on $[0,\theta_2]$: $v^{1/2} = m^{3/4} - \frac{1}{2}\theta_2 m^{-3/4} - E_2$, $0 \leq E_2 \leq \frac{1}{8}(Y-1)^{-3/2}$. With $f(t) = (X-t)^{3/4}$ on $[0,\theta]$: $m^{3/4} = n^{9/8} - \frac{3}{4}\theta n^{-3/8} - E_3$, $0 \leq E_3 \leq \frac{3}{32}(X-1)^{-5/4}$. Every term is nonpositive after the leading one, and $\frac{1}{2}m^{-3/4} + E_3 + E_2 \leq n^{-9/8}$ for $n \geq 3$. □

Unlike the depth-2 layer, *every* non-smooth term now has decaying amplitude: the cumulative phase cost of replacing $\frac{k}{2}v^{1/2}$ by the smooth $\frac{k}{2}n^{9/8}$ over a dyadic block is $\ll kP^{5/8}$, negligible against the target.

Theorem 4.7 (triple parity discrepancy). For every $\varepsilon > 0$ and every $(a, b, c) \in \{0,1\}^3 \setminus \{(0,0,0)\}$,

$$\Big|\sum_{n\le N,\ n\text{ odd}} \psi(n^{3/2})^a\, \psi(m^{3/2})^b\, \psi(v^{1/2})^c\Big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the eight sign classes of $(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$ on odd $n \le N$ has cardinality $\frac{N}{16} + O(N^{23/24+\varepsilon})$.

Proof. The $c = 0$ cases are Theorem 4.4 and Theorem 4.1. For $c = 1$, wave-expand all present sign factors as in Step 1 of Theorem 4.4, with truncations $J_1 = J_2 = J_3 = P^{1/24}$, and bound

$$S_{i,j,k}(P) = \sum_{n\in(P,2P],\ n\text{ odd}} e\big(\tfrac{i}{2}n^{3/2} + \tfrac{j}{2}m^{3/2} + \tfrac{k}{2}v^{1/2}\big)$$

uniformly for $|i| \le 2J_1$, $|j| \le 2J_2$, $1 \le |k| \le 2J_3$. By Lemma 4.6, replacing $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ costs $\ll |k|P^{5/8} \le P^{2/3}$, before any differencing.

If $j = 0$, the remaining sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8}$. Both curvature terms are positive for positive modes, and for mixed signs with $|i| \ge 1$ the first dominates by $P^{3/8-1/24}$; Lemma 3.3 gives $\ll P^{5/8}$ for $i = 0$ and $\ll |i|^{1/2}P^{3/4} + P^{1/4}$ otherwise, both $\ll P^{23/24}$.

If $j \ne 0$, the phase is exactly the Theorem 4.4 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. Steps 2–6 run verbatim; the passenger modifies the smooth part of the differenced phase by $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$, whose second derivative $\ll |k|hP^{-15/8}$ is smaller than the retained cell curvature $jhP^{-3/4}$ and the r-mode curvature $|r|P^{-1/2}$ by powers of P. Every sign-dominance check of Theorem 4.4 holds with these margins. $\square$

Theorem 4.8 (the OE splits).**

$$\#\mathrm{OEOE}(N),\ \#\mathrm{OEOO}(N) = \tfrac{N}{16} + O(N^{7/8+\varepsilon}), \qquad \#\mathrm{OEEE}(N),\ \#\mathrm{OEEO}(N) = \tfrac{N}{16} + O(N^{13/16+\varepsilon}).$$

Together with Theorems 4.4 and 4.7 this proves every depth-4 itinerary word class except $OOO*$.

Proof. On the OE branch m is even and $w = \lfloor U \rfloor$, $U = m^{1/2}$. The w-level linearization is Lemma 4.3(i) verbatim with base U: since $Um^{1/4} = m^{3/4}$ exactly,

$$w^{3/2} = m^{3/4} - \tfrac{3}{2}\, m^{1/4}\, \theta_w + E, \qquad 0 \le E \le \tfrac{3}{8}(U-1)^{-1/2},$$

and $m^{3/4} = n^{9/8}$ up to one-signed corrections of scale $n^{-3/8}$. For a Vaaler mode $k \ne 0$, the phase is therefore $\varphi_1(n) - B(n)\theta_w(n)$ up to absorbable errors, with $\varphi_1 = \frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4} + \frac{k}{2}n^{9/8}$ and $B = \frac{3k}{4}n^{3/8}(1 + O(n^{-3/2}))$: a *single growing sawtooth* of amplitude $\asymp kn^{3/8}$ riding $\theta_w = \{m^{1/2}\}$, whose underlying integer increments once every $\asymp \frac{4}{3}n^{1/4}$ steps. Truncations: $J_1 = J_2 = J_3 = P^{1/8}$.

Drift-1 intervals. $B' \asymp kn^{-5/8}$, so B drifts by at most 1 on intervals I of length $L_0 = P^{5/8}/k$; there are $\asymp kP^{3/8}$ of them. On I, expand $e(-B\{U\}) = \sum_r a_r(B)e(rU)$ (Fourier in U, Vaaler truncation $|r + B| \le T = P^{5/16}$); the coefficients satisfy $|a_r(B)| \le \min(1, |r + B|^{-1})$ with window mass $O(\log T)$, and their n-dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode.

Mode sums. Smoothing $rU \to rn^{3/4}$ costs $kP^{5/8}$ in total. Writing $r = -B_0 + t$, $|t| \le T$, the curvature of the mode phase is

$$\varphi'' = \tfrac{27k}{128}n^{-7/8} - \tfrac{3}{16}\big(t + \tfrac{j}{2}\big)n^{-5/4} + \tfrac{3i}{8}n^{-1/2}.$$

(The first coefficient collects two terms, which is worth saying because differentiating φ_1 alone does not produce it. $\frac{k}{2}n^{9/8}$ contributes $\frac{9}{128}k$; and the mode index is $r = -B_0 + t$, so the frozen $-B_0n^{3/4}$ contributes $\frac{3}{16}B_0n^{-5/4} \asymp \frac{9}{64}kn^{-7/8}$, landing on the same power because $B_0 \asymp n^{3/8}$. The sum is $\frac{9}{128} + \frac{18}{128} = \frac{27}{128}$, and only the t part of r is left in the middle term.)

For $i \ne 0$ the last term dominates and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}kP^{5/8}$ after summing intervals. For $i = 0$, $(T + J_2)P^{-5/4} \ll kP^{-7/8}$ because $T = P^{5/16} \ll kP^{3/8}$ for every $k \ge 1$, so $\lambda \asymp kn^{-7/8}$ is single-signed; Lemma 3.3 per interval and summation over $\asymp kP^{3/8}$ intervals give $S_k \ll k^{1/2}P^{13/16+\varepsilon}$. With Vaaler weights $1/k$, $\sum_{k\le J_3} k^{-1/2}P^{13/16} = J_3^{1/2}P^{13/16} = P^{7/8}$, balanced against the truncation error $P/J_3 = P^{7/8}$ at $J_3 = P^{1/8}$. Total $\ll P^{7/8+\varepsilon}$.

OEE branch. The fourth letter needs $\psi(w^{1/2})$, and $w^{1/2} = U^{1/2} - \frac{1}{2}\theta_w U^{-1/2} - \frac{1}{8}\theta_w^2(U-\xi)^{-3/2}$ has *decaying* amplitudes only, with $U^{1/2} = m^{1/4} \to n^{3/8}$ likewise. The pure modes $e(\frac{l}{2}n^{3/8})$ have $\lambda \asymp lP^{-13/8}$, giving $l^{1/2}P^{3/16} + l^{-1/2}P^{13/16}$; the mixed cases sit in the dominance hierarchy $iP^{-1/2} \gg jP^{-5/4} \gg lP^{-13/8}$ with all three second derivatives of the same sign. Total $\ll N^{13/16+\varepsilon}$. $\square$

Corollary 4.9 (certified-descent density $13/16$**).** The three uniform certificate classes

$$E, \qquad OE, \qquad OOEE$$

are disjoint, and

$$\#\{n \le N : \mathrm{word}(n) \text{ has prefix } E\} = \lfloor \tfrac{N}{2} \rfloor,$$

$$\big|\#\{n \le N : \mathrm{word}(n) \text{ has prefix } OE\} - \tfrac{N}{4}\big| \ll N^{5/6},$$

$$\big|\#\{n \le N : \mathrm{word}(n) \text{ has prefix } OOEE\} - \tfrac{N}{16}\big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Hence the class of starts with a certified descent within four steps has cardinality

$$\tfrac{N}{2} + \tfrac{N}{4} + \tfrac{N}{16} + O(N^{23/24+\varepsilon}) = \tfrac{13N}{16} + O(N^{23/24+\varepsilon}).$$

No other depth-≤ 4 word class is used. In particular this is not a census of E-rooted itineraries of length ≥ 2, and it is not a census of every O-rooted itinerary of length four (the classes $OOO*$ are Theorem 6.1, and they are non-contracting at this depth).

Proof. Every even start realizes E, so the first count is elementary. The OE count is Corollary 4.2. For $OOEE$, Lemma 3.6 gives $\#OOEE = \frac{1}{8}\sum_{n\le N \text{ odd}}(1-\psi_1)(1+\psi_2)(1+\psi_3)$ with $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$; expanding gives the main term $N/16$ and seven sign sums bounded by Theorem 4.7. Every $OOEE$ start descends within four steps by Proposition 3.1: $3^2 < 2^4$. The three classes are disjoint because they are distinct prefixes. $\square$

The certificate density 13/16 is the exact ceiling of this one-growing-layer machinery: an itinerary contracts iff $3^o < 2^\ell$, the method so far proves letters at positions 1–3 of any itinerary plus further letters along even branches only, and the contracting minimal words with all odd letters at positions ≤ 2 are exactly E, OE, and $OOEE$. Completing depth 4 over odd starts requires the $OOO*$ split — a second growing layer, where the fourth-letter phase coefficient $W \asymp kn^{9/8}$ crosses integers within single steps and no drift-1 interval exists. Sections 5 and 6 close that split. The $OOO*$ words are non-contracting at depth 4, so the four-step certified density remains 13/16; the next increment is the two length-five contractors of Theorem 6.3.

3.5 Localization to sub-dyadic intervals and slow twists

Theorems 4.4 and 4.7 are stated on dyadic blocks. Two applications outside this paper need them on shorter intervals and with a slowly varying phase attached: the contagion recursion of the companion paper [24] counts the $OOEEE$ starts landing in a prescribed even block, which is a statement about the interval $I(m') = [m'^{32/9}, (m'+1)^{32/9})$ of length $\asymp P^{23/32}$ at scale $P = m'^{32/9}$, and the fifth letter enters through a Fourier expansion whose modes $e(\frac{\ell}{2}n^{9/16})$ multiply the depth-4 sign products. This subsection proves both theorems in that generality. The proofs are the proofs of Theorems 4.4 and 4.7 with the number of summands Y in place of P wherever the number of summands enters, and the point of writing them out is to exhibit the terms that do *not* scale with Y — there are three, all of size at most $P^{7/16}$ — and to show that the twist is removed by one partial summation after the differencing.

Throughout, $I \subseteq (P, 2P]$ is an interval of length Y (so I contains $Y/2 + O(1)$ odd integers), and a *slow twist* is $g(n) = \frac{\ell}{2}n^{9/16}$ with an integer ℓ, $|\ell| \le P^{1/24}$. For $n \in (P, 2P]$,

$$|g'(n)| \le \tfrac{9}{32}|\ell|P^{-7/16}, \qquad |g''(n)| \le \tfrac{63}{512}|\ell|P^{-23/16} \le 0.13\, P^{1/24-23/16}. \tag{4.5}$$

Lemma 4.10 (removing a slow twist after differencing). Let a_n be complex numbers on an interval I and $w(n) = e(\gamma(n))$ with γ real and differentiable on I. Then

$$\Big|\sum_{n\in I} a_n w(n)\Big| \le (1 + 2\pi\,\mathrm{TV}_I(\gamma)) \max_{I'}\Big|\sum_{n\in I'} a_n\Big|,$$

the maximum over the initial sub-intervals I' of I, where $\mathrm{TV}_I(\gamma) = \int_I |\gamma'|$. In particular, for $\gamma = \Delta_{2h} g = g(\cdot + 2h) - g$ with g a slow twist, $h \le P^{1/12}$ and $|I| \le P$, $\mathrm{TV}_I(\gamma) \le 2h\,|I| \sup|g''| \le 0.26\,P^{1/24+1/12+1-23/16} = 0.26\,P^{-5/16}$.

Proof. Abel summation: with $A(x) = \sum_{n\in I,\, n\le x} a_n$, $\sum_{n\in I} a_n w(n) = \sum_n A(n)(w(n) - w(n+1)) + A(\max I) w(\max I)$, and $\sum_n |w(n) - w(n+1)| \le 2\pi \sum_n |\gamma(n+1) - \gamma(n)| \le 2\pi\,\mathrm{TV}_I(\gamma)$. The second claim is $|(\Delta_{2h} g)'| \le 2h \sup|g''|$ and (4.5). □

Theorem 4.11 (nested parity discrepancy on sub-dyadic intervals, with a slow twist). For every $\varepsilon > 0$ there is P_0 such that for all $P \ge P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \ge P^{1/2}$, every $(a,b) \in \{0,1\}^2 \setminus \{(0,0)\}$ and every slow twist g,

$$\Big|\sum_{n\in I,\ n \text{ odd}} \psi(n^{3/2})^a\, \psi(m^{3/2})^b\, e(g(n))\Big| \;\le\; Y\,P^{-1/24+\varepsilon}.$$

Proof. We run Steps 1–7 of Theorem 4.4 over I and record each cost as *proportional* (to the number of summands, hence to Y) or *absolute* (independent of it).

Step 1. The majorant layers are nonnegative and the twist is unimodular, so the expansion is unchanged: each layer costs its constant term $\le Y/(2(J_1+1)) \le \frac{1}{2} Y P^{-1/24}$ (proportional) plus the mode sums $\sum_{0<r\le 2J_1} (J_1+1)^{-1} |\sum_{n\in I} e(\frac{r}{2} n^{3/2})|$; Lemma 3.3 on I gives each inner sum $\le 1.4 r^{1/2} Y P^{-1/4} + 2P^{1/4}$, and the weighted total is $\le 2.6 J_1^{1/2} Y P^{-1/4} + 4P^{1/4} = 2.6\,Y P^{-11/48} + 4P^{1/4}$ (proportional plus absolute). The theorem reduces to the twisted mode sums

$$S^g_{i,j}(I) = \sum_{n\in I,\ n \text{ odd}} e\big(\tfrac{i}{2} n^{3/2} + \tfrac{j}{2} m^{3/2} + g(n)\big), \qquad |i| \le 2J_1,\ 1 \le |j| \le 2J_2,$$

with mode weights of total mass $O(\log^2 P)$; the $j = 0$ sums are the depth-1 sums $\sum_{n\in I} e(\frac{i}{2} n^{3/2} + g(n))$, $i \ne 0$, whose phase has curvature in $[0.53, 0.75]|i| P^{-1/2}$ up to the twist's $\le 0.13 P^{1/24-23/16}$ — a relative perturbation $\le 0.25 P^{1/24-15/16}$ — so Lemma 3.3 gives $\le 1.4|i|^{1/2} Y P^{-1/4} + 2P^{1/4}$.

Step 2. Linearization changes each summand by a phase of modulus $\le \frac{j}{4} n^{-3/4}$: total $\le 2\pi \cdot \frac{j}{4} \cdot \frac{Y}{2} P^{-3/4} \le jYP^{-3/4} \le 2YP^{1/24-3/4}$ (proportional).

Step 3. The A-process for a sum of $Y/2 + O(1)$ unimodular terms with $H = P^{1/12} \le Y$:

$$|S^g_{i,j}(I)|^2 \le \frac{2Y^2}{H} + \frac{4Y}{H} \sum_{1\le h<H} |T^g_h|, \qquad T^g_h = \sum_{\substack{n,\, n+2h \in I\\ n \text{ odd}}} e\big(\Phi(n+2h) - \Phi(n) + \Delta_{2h} g(n)\big).$$

By Lemma 4.10, $|T^g_h| \le (1 + 1.7P^{-5/16}) \max_{I'} |T_h(I')|$, where $T_h(I')$ is the *untwisted* differenced sum over an initial sub-interval I' of I. The twist has now been removed; it remains to bound $T_h(I')$ for every sub-interval $I' \subseteq I$ by an expression that is nondecreasing in $|I'|$, and to evaluate it at $|I'| = Y$.

Step 4. The exact differenced phase is unchanged. Discarding the sawtooth term costs $\le 7.1 jhP^{-1/4}$ per summand, i.e. $\le 3.6 jhY'P^{-1/4}$ on I' (proportional).

Step 5. The level sets of $\lfloor \delta \rfloor$ cut I' into at most $1.5hY'P^{-1/2} + 2$ cells: the full cells have length in $[\frac{2}{3}, 0.95] P^{1/2}/h$ as before, and at most two end cells are partial. The Vaaler layer for κ costs its constant term $\le Y'/(R+1) \le Y'P^{-1/4}$ plus the R-mode sums at coefficients $\le 1/(R+1)$; per cell and mode Lemma 3.3 gives $\ell \lambda_r^{1/2} + \lambda_r^{-1/2}$ with $\lambda_r \ge 0.53|r|P^{-1/2}$, and summing over cells and modes,

$$\le 1.2\, R^{1/2} Y' P^{-1/4} + (1.5hY'P^{-1/2} + 2) \cdot 5.5\, P^{1/4} R^{-1/2} = 1.2\, Y' P^{-1/8} + 8.3\, hY' P^{-3/8} + 11\, P^{1/8}$$

(two proportional terms and one absolute). The exact shift device is pointwise and unchanged.

Step 6. For $r = 0$ the per-cell test gives, over the cells meeting I',

$$\sum_{\text{cells}} (\ell_i \lambda^{1/2} + \lambda^{-1/2}) \le 0.83\,(jh)^{1/2} Y' P^{-3/8} + 3.9\,(h/j)^{1/2} Y' P^{-1/8} + 5.2\,(jh)^{-1/2} P^{3/8},$$

the last term being the two partial end cells. For $r \neq 0$ the same per-cell bounds with the mode curvature, summed against the $1/|r|$ weights, give $\le 5.2\,Y' P^{-1/8} + 13\,hY' P^{-1/4} + 17\,P^{1/4}$, the last term again from the end cells. All sign-dominance checks are pointwise and unchanged.

Step 7. Collecting, for every sub-interval I' of length Y',

$$|T_h(I')| \le C\,Y' P^{-1/8}(1 + h^{1/2}) \log P + C' P^{3/8},$$

nondecreasing in Y' (the absolute constant C' collects $5.2(jh)^{-1/2}P^{3/8}$, $17P^{1/4}$, $11P^{1/8}$ and the Step-1 term $4P^{1/4}$). Hence

$$|S^g_{i,j}(I)|^2 \le \frac{2Y^2}{H} + \frac{4Y}{H} \sum_{h<H} \Big(CYP^{-1/8}(1+h^{1/2}) \log P + C' P^{3/8} \Big)(1+1.7P^{-5/16}) \le Y^2 P^{-1/12}(3+8C\log P) + 5C' Y P^{3/8}.$$

For $Y \ge P^{1/2}$, $YP^{3/8} \le Y^2 P^{-1/8} \le Y^2 P^{-1/12}$, so $|S^g_{i,j}(I)| \ll Y P^{-1/24} \log^{1/2} P$. Summing the mode weights ($O(\log^2 P)$) and adding the proportional and absolute costs of Steps 1–2 gives the theorem, the absolute costs being $\le 4P^{1/4} \le YP^{-1/4}$. □

Theorem 4.12 (triple parity discrepancy on sub-dyadic intervals, with a slow twist). For every $\varepsilon > 0$ there is P_0 such that for all $P \ge P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \ge P^{1/2}$, every $(a,b,c) \in \{0,1\}^3 \setminus \{(0,0,0)\}$ and every slow twist g,

$$\Big| \sum_{n \in I,\ n \text{ odd}} \psi(n^{3/2})^a\, \psi(m^{3/2})^b\, \psi(v^{1/2})^c\, e(g(n)) \Big| \ \le\ Y\,P^{-1/24+\varepsilon}.$$

Proof. The $c = 0$ cases are Theorem 4.11 and its depth-1 case. For $c = 1$, wave-expand as in Theorem 4.7 and bound the twisted mode sums $S^g_{i,j,k}(I)$, $1 \le |k| \le 2J_3$. Lemma 4.6 replaces $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ at a per-summand cost $\le \frac{k}{2}(\frac{3}{4}n^{-3/8} + n^{-9/8})$, total $\le 2.4\,|k|YP^{-3/8} \le 5\,YP^{1/24-3/8}$ (proportional).

If $j = 0$ the sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8} + g(n)$. For $i \neq 0$ the first term's curvature $\ge 0.53|i|P^{-1/2}$ dominates the second ($\le 0.14|k|P^{-7/8} \le 0.14P^{1/24-7/8}$) and the twist ($\le 0.13P^{1/24-23/16}$) by powers of P, and Lemma 3.3 gives $\le 1.4|i|^{1/2}YP^{-1/4} + 2P^{1/4}$. For $i = 0$ the passenger's curvature $\ge 0.07|k|P^{-7/8}$ dominates the twist by $P^{-23/16+7/8-1/24} = P^{-7/12}$, and Lemma 3.3 gives $\le 0.27|k|^{1/2}YP^{-7/16} + 3.8|k|^{-1/2}P^{7/16}$: one proportional term and the absolute *pure passenger* term $3.8P^{7/16}$, which is $\le 3.8\,YP^{-1/16}$ for $Y \ge P^{1/2}$.

If $j \neq 0$, the phase is the Theorem 4.11 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. After the differencing of Step 3 and the removal of the twist by Lemma 4.10, the passenger contributes $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$ to the smooth part of the differenced phase, with second derivative $\le 0.15|k|hP^{-15/8}$, smaller than the retained cell curvature $0.15jhP^{-3/4}$ and the r-mode curvature $0.53|r|P^{-1/2}$ by the factors $P^{1/24-9/8}$ and $P^{1/24+1/12-11/8}$; every sign-dominance check of Steps 4–6 holds with these margins, and Step 7 is unchanged. □

The threshold $Y \ge P^{1/2}$ is where the three absolute terms — the two partial end cells ($5.2P^{3/8}$), the pure passenger ($3.8P^{7/16}$) and the majorant end cells ($17P^{1/4}$) — fall below the main saving $YP^{-1/24}$; nothing else in either proof uses the length of the block except through the number of summands. The twist enters only through Lemma 4.10 (after differencing) and through curvature comparisons in which it loses by a power of P; any g with $|g''| \le P^{-4/3}$ on the block would do, and the class $\frac{\ell}{2}n^{9/16}$, $|\ell| \le P^{1/24}$, is the one the application needs.

Corollary 4.13 (the *OOEEE* production on even blocks). For $m' \ge 2$ let $I(m') = [m'^{32/9}, (m'+1)^{32/9})$ and

$$\mathcal{O}(m') = \{ n \text{ odd} \in I(m') :\ \mathrm{word}_5(n) = OOEEE,\ J^5(n) = m' \},$$

where $\mathrm{word}_5(n)$ is the itinerary of the first five steps of n under J. Then every $n \in \mathcal{O}(m')$ has $J^4(n) \in [m'^2, (m'+1)^2)$ even, and

$$|\mathcal{O}(m')| = \tfrac{1}{16}\#\{n \text{ odd} \in I(m')\} + O_\varepsilon\big(|I(m')|\, m'^{-4/27+\varepsilon}\big).$$

Proof. Put $P = m'^{32/9}$ and $Y = |I(m')| = \frac{32}{9}m'^{23/9}(1 + O(1/m')) \asymp P^{23/32} \geq P^{1/2}$. Along $OOEEE$ the states are n (odd), m (odd), v (even), $\lfloor v^{1/2} \rfloor$ (even), $J^4(n) = \lfloor \sqrt{\lfloor v^{1/2} \rfloor} \rfloor = \lfloor v^{1/4} \rfloor$ (even), and $J^5(n) = \lfloor \sqrt{J^4(n)} \rfloor$; so $J^5(n) = m'$ forces $J^4(n) \in [m'^2, (m'+1)^2)$. Three elementary facts:

(a) *Nesting at the fifth letter.* For odd $n \geq 3$, $0 \leq n^{9/16} - v^{1/4} \leq n^{-15/16}$: the upper bound is $v \leq n^{9/4}$; the lower one follows from $v \geq (n^{3/2}-1)^{3/2} - 1 \geq n^{9/4} - \frac{3}{2}n^{3/4} - 1$ and the mean value theorem for $u \mapsto u^{1/4}$, using $v \geq \frac{1}{2}n^{9/4}$. Hence $\lfloor v^{1/4} \rfloor = \lfloor n^{9/16} \rfloor$ unless $\{n^{9/16}\} < n^{-15/16}$.

(b) *The exceptional set is small.* By Lemma 3.4 (Erdős–Turán) with $H = \frac{4}{9}P^{7/16}$ and the Kusmin–Landau bound $|\sum_{n \in I \text{ odd}} e(hn^{9/16})| \ll P^{7/16}/h$ (the phase $h(2k{+}1)^{9/16}$ has monotone derivative $\frac{9}{8}h(2k{+}1)^{-7/16} \in (0, \frac{1}{2}]$),

$$\#\{n \text{ odd} \in I :\ \{n^{9/16}\} < n^{-15/16}\} \ll YP^{-15/16} + YP^{-7/16} + P^{7/16} \ll YP^{-1/16}.$$

(c) *The fifth letter as a Fourier series.* With $J = \lfloor P^{1/24} \rfloor$, Lemma 3.5 gives coefficients b_q, $0 < |q| \leq J$, $|b_q| \ll 1/|q|$, and a majorant Δ_J of degree J with coefficients $\leq 1/(J+1)$, such that $|\psi(y) - \sum_q b_q e(qy/2)| \leq 2\Delta_J(y/2)$ for all real y; and by Kusmin–Landau, $|\sum_{n \in I \text{ odd}} e(\frac{q}{2}n^{9/16})| \ll P^{7/16}/|q|$ for $0 < |q| \leq J$, so $\sum_{n \in I} \Delta_J(\frac{1}{2}n^{9/16}) \ll YP^{-1/24} + P^{7/16}\log P$.

Now, off the exceptional set of (a), $n \in I(m')$ gives $\lfloor n^{9/16} \rfloor \in [m'^2, (m'+1)^2)$, so for such n with $\mathrm{word}_5(n) = OOEEE$ the value $J^5(n) = m'$ is automatic; therefore $|\mathcal{O}(m')|$ differs from $\#\{n \text{ odd} \in I : \ \mathrm{word}_5(n) = OOEEE\}$ by at most the exceptional count of (b). By Lemma 3.6 the indicator of $OOEEE$ on odd n is $\frac{1}{16}(1-\psi_1)(1+\psi_2)(1+\psi_3)(1+\psi_4)$ with $\psi_4 = \psi(v^{1/4})$, each factor enforcing the branch on which the next is the true letter. Expanding, the main term is $\frac{1}{16}\#\{n \text{ odd} \in I\}$; the seven sign sums without ψ_4 are Theorem 4.12 with $g = 0$; for the eight with ψ_4, replace $\psi(v^{1/4})$ by $\psi(n^{9/16})$ (they agree off the exceptional set, by (a)), expand by (c), and bound each twisted sum $\sum_n \psi_1^a \psi_2^b \psi_3^c\, e(\frac{q}{2}n^{9/16})$ by Theorem 4.12 when $(a,b,c) \neq 0$ and by the Kusmin–Landau bound of (c) when $(a,b,c) = 0$; the coefficient sums are $O(\log P)$. Every error is $\ll YP^{-1/24+\varepsilon} + P^{7/16}\log P \ll YP^{-1/24+\varepsilon}$, and $P^{-1/24} = m'^{-4/27}$. □

The corollary is what the contagion recursion of [24] consumes: for a backward-closed set $A \ni m'$, every member of $\mathcal{O}(m')$ lies in A, with log-mass at least $\frac{1}{9m'}(1 - O(m'^{-4/27+\varepsilon}))$, and the recursion gains the term $\frac{1}{9}g(9t/32)$, raising the contagion exponent from 0.4050 to 0.4922. Nothing in this subsection is used in Sections 5–7.

4. The level-2 wave and the kernel theorem

The reader should hold one object in mind through this section: the exact level-2 wave $e(qY)$, $Y = \lfloor n^{3/2} \rfloor^{3/2}$, a depth-2 nested-floor phase, possibly multiplied by a frozen floor of the same depth. It is the fundamental obstruction of the paper. The kernel theorem (Theorem 5.3) is organized so that, after two differencings and an exact bookkeeping of carries, *everything* that is not a standard second-derivative test is such a wave, and Lemma 5.2 — stated and proved first, as a standalone result with its own differencing — bounds it with exactly depth-2 strength, $|q|^{-1/6}P^{23/24+\varepsilon}$. The exponent $1 - 1/96$ of the kernel theorem is nothing but that depth-2 strength propagated through two differencings; any improvement of Lemma 5.2 improves the kernel theorem proportionally, and no other part of the argument is close to its limit.

Every reorganization of the $OOO*$ phase funnels into one object: for the monomial weight $c(n) = \frac{3k}{4}n^{9/8}$ on $n \sim P$,

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e\big(c(n)\,\{\lfloor n^{3/2} \rfloor^{3/2}\}\big).$$

This section proves the kernel bound at full length: every estimate that the proof uses is displayed, with its dependence on the mode k, the differencing shifts h_1, h_2, h_3, the branch offset j, and the expansion frequencies made explicit, and with numerical constants displayed at each of the sign-critical dominance checks. Two structural facts shape the proof and are worth stating in advance. First, the hardest pieces of the decomposition are exact level-2 waves $e(qY)$, possibly riding a frozen floor; a frozen-coefficient model $e(sX)$ with $s \asymp qP^{3/4}$ would silently discard the sawtooth $-\frac{3}{2}qX^{1/2}\theta$ of the same amplitude hidden in $Y = (X-\theta)^{3/2}$, so the waves are treated exactly — by the same exact linearization that proves Theorem 4.4 — and the honest bound is $q^{-1/6}P^{23/24+\varepsilon}$ (Lemma 5.2): the wave is a depth-2 object and receives exactly depth-2 strength, which is what pins the kernel saving at $\delta = \frac{1}{96}$. Second, the offset-branch pieces of Step 5 carry a factor $(k|j|)^{1/2}$, and $(k|j|)^{1/2}P^{15/16} \le P^{23/24}$ exactly on the standing range $k \le P^{1/24}$: the two bottlenecks meet at the endpoint, which is why the theorem is uniform in k on that range and why the range cannot be enlarged by this assembly.

The next lemma collects the exact reductions behind the treatment: the kernel phase is the level-2 local floor defect; the second difference of the level-2 integer obeys an exact floor identity; on an explicit carry-branch decomposition that second difference is a smooth function with a frozen floor; and the doubly differenced kernel phase decomposes exactly into four bounded pieces — the master identity — with no growing smooth part left to estimate.

Lemma 5.1 (level-2 defect, double gap, branch freeze, and the master identity). Fix shifts $d_1 = 2h_1$, $d_2 = 2h_2$ and write Δ_1, Δ_2, $\Delta\Delta$ for the corresponding difference operators in n, and $W = \Delta_1 Y$, $W' = \Delta_2 Y$.

(i) For odd $n \ge 3$,
$$\tfrac{3}{4}\, v^{1/2}\theta_2 = \tfrac{1}{2}(m^{9/4} - v^{3/2}) - R, \qquad 0 \le R \le \tfrac{3}{16}\, v^{-1/2}.$$
Hence for $c = \frac{3k}{4}v^{1/2}$ the kernel phase $c\,\theta_2$ equals $\frac{k}{2}(Y^{3/2} - v^{3/2})$ up to $kR \ll kP^{-9/8}$ per term: the kernel is the exponential sum of the level-2 local floor defect.

(ii) Exactly, with $g_2 = \Delta_1 v$,
$$\Delta_2 g_2 = \lfloor \Delta\Delta Y \rfloor + \kappa'' + \Delta_2\kappa_2, \qquad \kappa_2 = \big[\theta_2 \ge 1 - \{W\}\big], \quad \kappa'' = \big[\{W\} \ge 1 - \{\Delta\Delta Y\}\big],$$
and every carry is a difference of unit sawtooths: $\big[\{A\} + \{B\} \ge 1\big] = \{A\} + \{B\} - \{A{+}B\}$ for all reals A, B (both sides equal $\{A\} + \{B\} - \{A + B\}$: the left side because $\{A\} + \{B\} \in [0, 2)$ and $\{A + B\} = \{A\} + \{B\} - \lfloor \{A\} + \{B\} \rfloor$).

(iii) Write $b_1 = \lfloor \Delta_1 X \rfloor$, $b_2 = \lfloor \Delta_2 X \rfloor$, $b_{12} = \lfloor \Delta_{12} X \rfloor$ (shift $d_1{+}d_2$) — each constant on runs of length $\asymp P^{1/2}/h$ — and let $\kappa = (\kappa_1, \kappa_2, \kappa_{12}) \in \{0,1\}^3$ be the level-1 carries, so that the shifted values of m are $m + \beta_1$, $m + \beta_2$, $m + \beta_{12}$ with $\beta_i = b_i + \kappa_i$. On each b-run intersection and carry branch,
$$\Delta\Delta Y = F_\kappa(m), \qquad F_\kappa(m) = (m{+}\beta_{12})^{3/2} - (m{+}\beta_1)^{3/2} - (m{+}\beta_2)^{3/2} + m^{3/2}$$
exactly. The net offset $j = \beta_{12} - \beta_1 - \beta_2$ satisfies $-1 \le j \le 2$, hence $|j| \le 2$, for $h_1h_2 \le P^{1/2}/3$, both ends occurring, and F splits exactly into an offset term and a genuine second difference: for some ξ_1 between 0 and j and some $\xi_2 \in (0, \beta_1 + \beta_2)$,
$$F_\kappa(m) = \tfrac{3}{2}\, j\,(m + \beta_1 + \beta_2 + \xi_1)^{1/2} + \tfrac{3}{4}\,\beta_1\beta_2\,(m + \xi_2)^{-1/2}.$$
Consequently, with $G := F_\kappa \circ X$,
$$\tfrac{3}{2}|j|P^{3/4} \le \Big|\tfrac{3}{2}j(\cdot)^{1/2}\Big| \le 2.6\,|j|P^{3/4}, \qquad 1.4\,h_1h_2P^{1/4} \le \tfrac{3}{4}\beta_1\beta_2(\cdot)^{-1/2} \le 15\,h_1h_2P^{1/4},$$
$$|G'(n)| \le 2|j|P^{-1/4} + 20\,h_1h_2P^{-3/4} < 1, \qquad |G''(n)| \le 2|j|P^{-5/4} + 25\,h_1h_2P^{-7/4},$$
so $\lfloor G(n) \rfloor$ is constant on runs of length $\ge \frac{1}{22}\min(P^{1/4}/(|j|{+}1),\ P^{3/4}/(h_1h_2))$.

How much of the β-inventory above is attained. The offset erratum came from bounding two quantities separately that one variable determines; the rest of this lemma's inventory was scanned for the same thing. The β-product is clean: $\beta_1\beta_2/(h_1h_2P)$ attains $18 = (3\sqrt{2})^2$ exactly, and the Lean hypothesis pair $\beta_i \le 4.25h_iP^{1/2} + 1$ loses only the rounding $4.25^2 = 18.0625$, because both factors are extremal at the *same* $n = 2P$. Quantifying them separately costs nothing there.

The three bounds built on that product are another matter, because each pairs the product with a power of ν taken at the opposite end of the block. Measured exactly over $1 \le h_1, h_2 \le 7$ (`decoration_budget.beta_inventory_attained`; the β_i and their products are exact integers through $\lfloor n^{3/2}\rfloor = \lfloor\sqrt{n^3}\rfloor$, the ratios to the printed forms in floating point):

quantity	printed	attained
$\frac{3}{4}\beta_1\beta_2(m+\xi_2)^{-1/2}/(h_1h_2P^{1/4})$	[1.4, 15]	$[\frac{27}{4},\ \frac{27}{4}2^{1/4}] = [6.750,\ 8.027]$
β-part of $\lvert G'\rvert/(h_1h_2P^{-3/4})$	20	$\frac{81}{16} = 5.0625$
β-part of $\lvert G''\rvert/(h_1h_2P^{-7/4})$	25	$\frac{567}{64} = 8.8594$

So the printed interval in the first row is 13.6 wide where the attained one is 1.28, and the two derivative constants are 3.95 and 2.82 times what they bound. All three are the same arithmetic: $\beta_1\beta_2 \sim 9h_1h_2n$ and the accompanying power of n move together, and the printed forms take one at each end.

We do not sharpen them, and the reason is the same one that declined the widened constant in A.6: neither propagates anywhere that binds. The 20 enters the widened θ-coefficient only through the lower-order $20hP^{-1/4}$, so at 5.07 the collected constant $4 + \delta$ becomes valid from $7.6 \cdot 10^9$ rather than $2.95 \cdot 10^{11}$ — both far below P_0 — and the collected constant itself is unchanged. The 25 enters only through $25/0.35 \le 71.5$, whose certificate row moves from $8.2 \cdot 10^4$ to $1.0 \cdot 10^4$. And a rigorous sharpening would have to carry the $O(1)$ corrections in $\beta_i = \lfloor\Delta_i X + \theta\rfloor$ that the printed constants currently absorb. The measurement is recorded so that a later reader who needs one of these three knows what is there to be had. The branch indicator is a finite union of arcs in the single variable θ with slowly moving endpoints (drifts $\le 1.5hP^{-1/2}$ per step): exactly the moving-endpoint pattern of Step 4 of Theorem 4.4, absorbed by the same exact shift device at $O(\log P)$ mode-mass cost.

(iv) *(Master identity.)* With $\kappa_2 = [\theta_2 \ge 1-\{W\}]$, $\kappa_2' = [\theta_2 \ge 1-\{W'\}]$ the level-2 carries at the two shifts, and κ'', $\Delta_2\kappa_2$ as in (ii), the doubly differenced kernel phase decomposes exactly:

$$\Delta\Delta(c\,\theta_2) = (\Delta\Delta c)\,\theta_2 + (\Delta_2 c)(n+d_1)\,(\{W\}-\kappa_2) + (\Delta_1 c)(n+d_2)\,(\{W'\}-\kappa_2') + c_{11}\,(\{\Delta\Delta Y\}-\kappa''-\Delta_2\kappa_2),$$

where c_{11} is the doubly shifted weight. Every bracket is bounded by 2 in absolute value: no unbounded smooth part survives the differencing.

Proof. (i) Taylor of $(v+\theta_2)^{3/2}$ at v, with $Y^{3/2} = m^{9/4}$; as in Lemma 4.3(i) the mean value is avoidable, and the remainder has a closed form. Write $a = \sqrt{v}$, $b = \sqrt{Y}$, so $\theta_2 = b^2 - a^2$ and $m^{9/4} = b^3$. Then

$$\tfrac{3}{4}a\,(b^2 - a^2) = \tfrac{1}{2}(b^3 - a^3) - R, \qquad R = \tfrac{1}{4}(b-a)^2(2b+a),$$

identically; $R \ge 0$ by inspection, and $Ra \le \frac{3}{16}\theta_2^2$ reduces to $(a+3b)(a-b) \le 0$, which holds because $v \le Y$. At $\theta_2 < 1$ that is the printed $R \le \frac{3}{16}v^{-1/2}$, and it is nearly sharp: sampling odd $n \le 10^7$ gives $R\sqrt{v}$ up to 0.1867 against $\frac{3}{16} = 0.1875$ (Lean `lemma51_i_closed_form`, `lemma51_i_nonneg`, `lemma51_i_upper`, `lemma51_i_identity`). (ii) The gap identity of Lemma 4.3(ii) applied twice — to Y at shift d_1, then to the real sequence $n \mapsto W(n)$ at shift d_2, with $\Delta_2 W = \Delta\Delta Y$; the sawtooth form of the carry is the displayed elementary identity. (iii) The corner identities are Lemma 4.3(ii) at the three shifts, and the offset bound follows from $\lvert\Delta\Delta X\rvert \le 4h_1h_2\sup\lvert X''\rvert = 3h_1h_2P^{-1/2} < 1$. For the split, group $F = \big[(m+\beta_1+\beta_2+j)^{3/2} - (m+\beta_1+\beta_2)^{3/2}\big] + \big[(m+\beta_1+\beta_2)^{3/2} - (m+\beta_1)^{3/2} - (m+\beta_2)^{3/2} + m^{3/2}\big]$: the first bracket is $\frac{3}{2}j(\cdot)^{1/2}$ by the mean value theorem, the second is $\beta_1\beta_2\,f''(m+\xi_2)$ for $f = x^{3/2}$ by the double mean value theorem. The displayed numerical bounds use $m \in (P^{3/2}-1, 2^{3/2}P^{3/2}]$, $\beta_i \in [3h_iP^{1/2}-1,\ 3\sqrt{2}\,h_iP^{1/2}+1]$ (from $\Delta_i X = 2h_iX'(\xi)$

and (E1) below), and $G' = F'(X)X'$, $G'' = F''(X)X'^2 + F'(X)X''$ with $F' = \frac{3}{4}j(\cdot)^{-1/2} - \frac{3}{8}\beta_1\beta_2(\cdot)^{-3/2}$, $F'' = -\frac{3}{8}j(\cdot)^{-3/2} + \frac{9}{16}\beta_1\beta_2(\cdot)^{-5/2}$.

The bound on G'' is not term by term. Writing $n = s^4$, so that $X = s^6$, $X' = \frac{3}{2}s^2$, $X'' = \frac{3}{4}s^{-2}$, the two $\beta_1\beta_2$ contributions to $G'' = F''(X)X'^2 + F'(X)X''$ are

$$\tfrac{9}{16}\cdot\tfrac{9}{4} = \tfrac{81}{64} \quad \text{and} \quad -\tfrac{3}{8}\cdot\tfrac{3}{4} = -\tfrac{9}{32}, \qquad \tfrac{81}{64} - \tfrac{9}{32} = \tfrac{63}{64},$$

of *opposite sign.* With $\beta_1\beta_2 \le 19h_1h_2P$ the combined coefficient gives $\frac{63}{64}\cdot 19 = 18.7 \le 25$; bounding the two separately gives $\frac{99}{64}\cdot 19 = 29.4$, which exceeds the printed 25. The displayed bound is correct, but only through that cancellation, and the same happens for the j-terms ($-\frac{27}{32} + \frac{9}{16} = -\frac{9}{32}$, against the printed 2). Lean `Gsecond_beta_cancellation`, `Gsecond_naive_bound_fails`.

The run-length constant is $22 = 2 + 20$: with $M = \max((|j|+1)P^{-1/4}, h_1h_2P^{-3/4})$ the two parts of the G' bound are $\le 2M$ and $\le 20M$, so $|G'| \le 22M$ and the level sets of $\lfloor G\rfloor$ have length $\ge 1/(22M)$, which is the displayed minimum. The regrouping, the offset bound, the β-product bound and all four derivative estimates are machine-checked in `formal/Problems/Juggler/BranchFreeze.lean`, and the two mean values that produce ξ_1 and ξ_2 are supplied in `formal/Problems/Juggler/MeanValues.lean`, so (iii) is unconditional.

> *Erratum (the offset is ≤ 2, not ≤ 3).* The three level-1 carries are not free of the corner floors, and treating them as free is what produces the 3. Since $m = \lfloor X\rfloor$ and $\theta = X - m$, the shifted value is $\beta_i = m(n+d_i) - m(n) = \lfloor \Delta_i X + \theta\rfloor$ exactly — that is the content of $\kappa_i = \lfloor\{\Delta_i X\} + \theta\rfloor$ — and $\Delta_{12}X = \Delta_1 X + \Delta_2 X + \Delta\Delta X$, so the whole offset is one floor:
>
> $$j = \lfloor \{\Delta_1 X{+}\theta\} + \{\Delta_2 X{+}\theta\} - \theta + \Delta\Delta X\rfloor.$$
>
> Under the stated hypothesis $\Delta\Delta X = 3h_1h_2\xi^{-1/2} \in (0,1)$ for some $\xi > n$, and $\theta \in [0,1)$, so the argument lies in $(-1,3)$ and $j \in \{-1,0,1,2\}$.
>
> The printed 3 adds the corner-floor range $[-1,2]$ of $\lfloor\Delta_1 X{+}\Delta_2 X{+}\Delta\Delta X\rfloor - \lfloor\Delta_1 X\rfloor - \lfloor\Delta_2 X\rfloor$ to a carry vector $\kappa \in \{0,1\}^3$ as though the two could be chosen independently. They cannot: all three carries are $\lfloor\cdot{+}\theta\rfloor$ at the same θ, and folding θ in returns a single corner floor, at $\Delta\Delta X{-}\theta$ in place of $\Delta\Delta X$. What 3 is, is the bound at $h_1h_2 \le 2P^{1/2}/3$, where $\Delta\Delta X < 2$: the range gains one for each unit of $\Delta\Delta X$, and an exact census (`decoration_budget.branch_offset_ladder`, integer arithmetic through $\lfloor n^{3/2}\rfloor = \lfloor\sqrt{n^3}\rfloor$) finds $\max j = r+1$ and $\min j = -1$ at $h_1h_2 \le rP^{1/2}/3$ for $r = 1,2,3,6$. So the printed bound is the conclusion of a hypothesis twice the one stated beside it.
>
> Nothing in the lemma's displays changes: they are stated in $|j|$, and $|G'| \le 2|j|P^{-1/4} + 20h_1h_2P^{-3/4}$ is unaffected. What changes is every downstream use of $2|j| \le 6$, which becomes $2|j| \le 4$. Lean carries `offset_abs_le_two` beside the printed `offset_abs_le_three`, with both ends of $\{-1,0,1,2\}$ witnessed.

Two of the three ingredients need no analysis at all, because $x^{3/2}$ has *explicit* mean values in root coordinates. For the first, with $a = \sqrt{A}$ and $b = \sqrt{A+j}$,

$$b^3 - a^3 = \tfrac{3}{2}(b^2 - a^2)\,c, \qquad c = \tfrac{2}{3}\,\frac{a^2 + ab + b^2}{a+b},$$

and $a \le c \le b$ reduces to $(2b+a)(b-a) \ge 0$ and $(2a+b)(a-b) \le 0$; so $\xi_1 = c^2 - A$ is an exact witness (`mvt_cube_explicit`). The *inner* step of the second is the **arithmetic mean of the square roots**: $F'(A{+}B) - F'(A) = BF''(\eta)$ holds identically with $\sqrt{\eta} = \frac{1}{2}(\sqrt{A} + \sqrt{A+B})$ (`mvt_sqrt_diff_explicit`). Only the outer step uses a genuine mean value theorem, applied to $g(t) = F(t{+}\beta_2) - F(t)$; rationalising its increment gives the two-sided form the inventory actually uses,

$$\tfrac{3}{4}\,\frac{\beta_1\beta_2}{\sqrt{m{+}\beta_1{+}\beta_2}} \;\le\; \Delta\Delta \;\le\; \tfrac{3}{4}\,\frac{\beta_1\beta_2}{\sqrt{m}},$$

and hence ξ_2 itself, as $\sqrt{m+\xi_2} = \frac{3}{4}\beta_1\beta_2/\Delta\Delta$ (`second_difference_two_sided`, `second_difference_exists_xi`). Throughout, $x^{3/2}$ is written $x\sqrt{x}$, so no real-power machinery is needed. On 20,000 samples the witness ξ_2 sits between 0.32 and 0.52 of $\beta_1+\beta_2$, comfortably interior. The printed ranges are comfortable except the offset lower bound, which is *exactly* attained: the ratio to $|j|P^{3/4}$ is $\frac{3}{2}(m+\beta_1+\beta_2+\xi_1)^{1/2}P^{-3/4}$, and over $n \in (P, 2P]$ the bracket runs from $m \sim P^{3/2}$ to $m \sim (2P)^{3/2}$, so the ratio runs over

$$\left[\tfrac{3}{2},\ \tfrac{3}{2}\cdot 2^{3/4}\right] = [1.5000,\ 2.5227]$$

against the printed $[1.5, 2.6]$ — the lower end attained and the upper carrying three percent (`decoration_budget.offset_term_attained`). An earlier printing gave this from a 300-point sample as $[1.510, 2.514]$, which is the same statement with the endpoints missed by a sampling grid; the closed forms are what the block gives. (iv) By (ii) and the definitions: $\Delta_1\theta_2 = \Delta_1 Y - \Delta_1 v = W - (\lfloor W\rfloor + \kappa_2) = \{W\} - \kappa_2$ (Lemma 4.3(ii) applied to Y), likewise $\Delta_2\theta_2 = \{W'\} - \kappa_2'$, and $\Delta\Delta\theta_2 = \Delta\Delta Y - \Delta_2 g_2 = \{\Delta\Delta Y\} - \kappa'' - \Delta_2\kappa_2$ by (ii). Substitute these into the exact product rule

$$\Delta\Delta(cf) = c_{11}\,\Delta\Delta f + (\Delta_2 c)(n+d_1)\,\Delta_1 f + (\Delta_1 c)(n+d_2)\,\Delta_2 f + (\Delta\Delta c)\,f,$$

which is verified by expanding both sides over the four base points n, $n+d_1$, $n+d_2$, $n+d_1+d_2$.

All of (i), (ii) and (iv) are machine-checked in `formal/Problems/Juggler/MasterIdentity.lean`: the carry sawtooth (`carry_as_sawtooth`), the substitution $\{y+w\} - \{y\} = \{w\} - \kappa$ that drives every step of (iv) (`fract_diff_level2`), the double-gap identity of (ii) (`lemma51_double_gap`), the four-point product rule (`double_difference_product`), the master identity itself (`lemma51_master`), and the bracket bound (`lemma51_brackets_le_two`). These were previously supported only by the probe's 60-digit sampling on random odd n; they are exact, so they are now proved rather than sampled. □

One warning, essential to the organization: $\lfloor\Delta\Delta Y\rfloor$ itself is *not* frozen. The level-1 carry toggles at essentially every step and shifts $\Delta\Delta Y$ by $\frac{3}{2}j_1 m^{1/2}$, a jump of size $\asymp P^{3/4}$. The branch decomposition of (iii), which carries the flicker inside the indicator over the eight carry branches, is therefore forced: no decomposition conditioning on the *values* of the level-1 gaps produces sets on which the second difference is smooth.

Standing estimates

All estimates of this section take place on a dyadic block $n \in (P, 2P]$, n odd, under two standing constraints:

$$\text{(C3)}\quad 1 \le k \le P^{1/24}, \qquad \text{(C4)}\quad h_1,\, h_2 \le P^{1/24}.$$

Two consequences are cited often enough below to carry names of their own:

$$\text{(C1)}\quad kh_1h_2 \le P^{1/8}, \qquad \text{(C2)}\quad h_1h_2 \le P^{1/2}/3.$$

(C1) is the product of the three caps and is tight: equality at $k = h_1 = h_2 = P^{1/24}$. (C2) follows with room to spare — (C3) and (C4) give $h_1h_2 \le P^{1/12}$, and $P^{1/12} \le P^{1/2}/3$ once $P \ge 3^{12/5} = 14$ — and is in fact invoked nowhere below; it is recorded because the differencing steps are easier to read against a named bound on the shift product. So verifying the standing setup at an invocation means checking two inequalities, not four.

The individual caps are what the argument needs, not the products: (C1) and (C2) alone would permit one of h_1, h_2 to be as large as $\asymp P^{1/2}$, while the decoration class (D1) and the third-differencing reduction of Lemma 5.2(ii) need $h_1 + h_2 \le 2P^{1/24}$. Theorem 5.3 takes $H_1 = P^{1/48}$ and $H_2 = P^{1/24}$, inside (C4) with room on the first. Every displayed constant below is valid for $P \ge P_0$ with an absolute P_0. All sums run over odd n, and every derivative test below is read through the parity reindexing of Lemma 3.10: margins and signs are invariant, and each displayed test bound dominates its reindexed value, so no constant needs adjustment. The base derivatives are

$$\text{(E1)}\quad X' = \tfrac{3}{2}n^{1/2} \in (1.5, 2.13]\,P^{1/2}, \quad X'' = \tfrac{3}{4}n^{-1/2} \in [0.53, 0.75)\,P^{-1/2},$$

$$-X''' = \tfrac{3}{8}n^{-3/2} \in [0.13, 0.375)\,P^{-3/2}, \quad X'''' = \tfrac{9}{16}n^{-5/2} \le 0.57\,P^{-5/2},$$

and $m = X - \theta \in (P^{3/2} - 1,\, 2.83\, P^{3/2}]$. Every gap is a mean value: for any shift $2h$,

$$\text{(E2)} \quad \Delta X = 2hX'(\xi) \in (3hP^{1/2},\, 4.3hP^{1/2}), \qquad \beta := \lfloor \Delta X \rfloor + \kappa \in (3hP^{1/2} - 1,\, 4.3hP^{1/2} + 1),$$

so that every level-1 gap integer at shift $2h$ is $\asymp hP^{1/2}$ with the displayed constants. For the monomial weight $c = \frac{3k}{4} n^{9/8}$ of Theorem 5.3,

$$\text{(E3)} \quad c' = \tfrac{27}{32} kn^{1/8}, \quad c'' = \tfrac{27}{256} kn^{-7/8}, \quad c''' = -\tfrac{189}{2048} kn^{-15/8}, \quad c'''' = \tfrac{2835}{16384} kn^{-23/8},$$

$$\text{(E4)} \quad \Delta_i c = 2h_i c'(\xi) \in (1.68, 1.85)\, kh_i P^{1/8}, \qquad \Delta\Delta c = 4h_1 h_2 c''(\xi) \in (0.22, 0.43)\, kh_1 h_2 P^{-7/8}.$$

On the level-2 side, with β_i as in Lemma 5.1(iii),

$$\text{(E5)} \quad W = \Delta_1 Y \in (4.4, 11)\, h_1 P^{5/4}, \qquad \left|(W \circ \mathrm{cell})'\right| = \left|W_1'(m)\, X'\right| \in (2.2, 10.4)\, h_1 P^{1/4},$$

where $W_1(m) = (m + \beta_1)^{3/2} - m^{3/2}$, so $\{W\}$ is a *fast* sawtooth (its argument moves by $\gg 1$ per step), while θ, $\{\Delta_i X\}$, and $\{\Delta\Delta Y\}$ are slow. The branch bounds of Lemma 5.1(iii) give, per branch with offset j,

$$\text{(E6)} \quad \left|(cF)''\right| = \tfrac{945}{512}\, k|j|\, n^{-1/8}\, (1 + O(P^{-1/4})) + O(kh_1 h_2 P^{-5/8}).$$

The first summand is the offset monomial $c \cdot \frac{3}{2} j\, m^{1/2} = \frac{9}{8} kj\, \nu^{15/8}$, differentiated in ν with j frozen. The second is the zero-offset scale: on a cell the gaps β_i are frozen, and Lemma 5.2b computes the local curvature exactly as $-\frac{135}{1024}\, k\beta_1\beta_2 \nu^{-13/8}$ — read the erratum there: the anchor's value is $-\frac{27}{128}$ — which is $\asymp kh_1 h_2 \nu^{-5/8}$ and *negative*. (A model that differentiates the moving gaps $\Delta_i X(\nu)$ produces a different, positive leading coefficient and is not the local f'': $F_{\mathrm{sm}} = \frac{27}{4} h_1 h_2 \nu^{1/4}$ at leading order, so $cF_{\mathrm{sm}} = \frac{81k}{16} h_1 h_2 \nu^{11/8}$ and the foil is $\frac{81}{16} \cdot \frac{11}{8} \cdot \frac{3}{8} = \frac{2673}{1024}$, measured at $P = 10^8$ to seven figures. Earlier printings gave $\frac{243}{128}$ here, which is not this model's value — and is, by an unlucky coincidence, exactly the magnitude of the *corrected* frozen anchor $9 \cdot \frac{216}{1024}$, so the two errors concealed each other.) Finally, the run and cell inventories: level-1 gap cells at shift $2h$ number at most $1.5hP^{1/2} + 1$ with lengths in $[\frac{2}{3},\, 0.95]\, P^{1/2}/h$, and the frozen-floor runs of $\lfloor F_\kappa(X) \rfloor$ number at most $22(|j| + 1)P^{3/4}$ by the derivative bound of Lemma 5.1(iii).

Throughout, "A is dominated by B at margin $P^{-\rho}$" means $\sup|A|/\inf|B| \le CP^{-\rho}$ with the displayed C: the composite second derivative then keeps the sign and, up to the factor $1 + CP^{-\rho}$, the size of B, so Lemma 3.3 applies at B's scale.

The symbol P^ε is used only for factors of the form $C(\varepsilon)(\log P)^{O(1)}$. The number of Fourier/Vaaler expansion layers that produce those logarithms is three (the pieces M_2, M_3, M_4 of the master identity), independent of P and of k, h_1, h_2, t. Each layer contributes $O(\log P)$ mode mass; there is no further nesting of expansions whose depth would grow with P. Implied constants in every $\ll_\varepsilon$ of Sections 4–6 depend on ε and on (C3) and (C4) only: on those ranges they do not grow with k, h_i, or t. The threshold P_0 is independent of ε (Appendix A.3) and of k, h_i, t; it is carried by the Lemma 3.9 comparison $W \le c_7 S/2$.

Every numerical margin in this section is claimed only for $P \ge P_0$, and P_0 is **effective**: Appendix A solves each of the thirty-eight printed threshold inequalities of Sections 4–6 separately and takes the maximum,

$$P_0 = 3.6 \cdot 10^{13},$$

attained at the Lemma 3.9 hypothesis $W \le c_7 S/2$ in Theorem 5.3, Step 5b. Two features of that number should be said here. First, P_0 does not depend on ε. No divisor sum, gcd sum or large-sieve average occurs anywhere in Sections 3–6, so every $\ll_\varepsilon$ in those sections is a power of $\log P$ and nothing else; the mode masses are $O(\log^3 P)$ and the two Weyl steps halve the exponent twice, giving the ε-free form $K_c(P) \ll P^{1-1/96}(\log P)^{3/4}$ of Theorem 5.3 (Appendix A.3). That halving is the same one that takes Lemma 5.2(ii)'s saving $\frac{1}{24}$ to the kernel's $\frac{1}{96}$: each differencing squares the sum and then takes a square root, which halves a saving exponent and a logarithmic exponent alike. So $\frac{1}{24} \to \frac{1}{96}$ and $\log^3 \to \log^{3/4}$ are one quartering, not two coincidences. The threshold at which $\log^A P$ is absorbed into P^ε is a statement about ε, not about the proof, and is not part of P_0.

Second, the comparisons stratify sharply, and one of them is the whole story. The Lemma 3.7 / Lemma 5.2 window hypotheses first hold together on the printed majorants at a moderate threshold (of size $3 \cdot 10^5$); every inequality in the paper except four holds from $2.8 \cdot 10^{10}$ on. The four are the three Lemma 3.9 balance comparisons of Steps 5a and 5b, and the Step 5b(a) q'' curvature ratio, which alone sits at $3.0 \cdot 10^{11}$ — the price of $R_0 = P^{5/16}$, and still two and a half orders below P_0 (Appendix A.6). The comparison $W \le c_7 S/2$ then adds three and a half orders of magnitude on its own: until it holds, a three-term zero of Φ'' can keep the sublevel Ω_W of length $\Theta(P)$ on a dyadic block, and the printed length $P^{89/96}$ is the bound of Lemma 3.9, not a claim that the sublevel is short at small P. Its two ingredients are $c_7 = 1/232$, exactly $1/\|M^{-1}\|_\infty$ for the Vandermonde-type matrix M of Lemma 3.9 at the exponent triple of Step 5b and hence not improvable there, and the interpolant error E of Lemma 5.2b (Appendix A.5). The interpolant error of Step 5b is $O(P^{-5/6})$ with leading coefficient 0.11, and is not absorbed into a smaller constant. The constants are not asserted to be sharp.

The proof of the kernel theorem classifies the differenced pieces into four classes; three are handled by standard tests, and the fourth — an exact level-2 wave $e(qY)$, possibly riding a frozen floor — is the genuinely new difficulty. We isolate it as a standalone lemma with its own differencing, so that it can be checked independently. The checkable objects, in order, are: Lemma 5.1 (exact identities); Lemma 5.2(i) (differenced wave); Lemma 5.2(ii) from (i) (Claims A–H); Lemma 5.2b (frozen-shape interpolant: local f'' versus global Λ); Theorem 5.3 Step 5a (offset composite); Theorem 5.3 Step 5b (three regimes, the middle band citing Lemma 5.2b and Lemma 3.9); Theorem 6.1 Step E (frozen-shape total phase $\Delta\Delta(\frac{k}{2}m^{9/4}) - \Delta\Delta(c\theta_2)$, composites $\frac{243}{512}$ and $\frac{2187}{2048}$). No later section re-derives these.

Lemma 5.2 (level-2 waves: the mixed-piece bound). Assume (C3) and (C4), write $\mathcal{D} = \{0, d_1, d_2, d_1{+}d_2\}$, and call a *decoration* any sum ρ of at most nine terms of the classes

- (D1) $q'\,\Delta_{2h}\Delta_{2h'}Y(n{+}d')$ with $|q'| \le 4P^{1/24}$, $1 \le h' \le 2P^{1/24}$, $d' \in \mathcal{D}$;
- (D2) $-\Delta_{2h}(c\,\lfloor F_\kappa(X)\rfloor)(n)$, with F_κ a branch function of Lemma 5.1(iii), offset $|j| \le 2$;
- (D3) a smooth φ with $|\varphi''| \le 6kh_1h_2h\,P^{-13/8}$ and $|\varphi'''| \le 6kh_1h_2P^{-13/8}$, where $2h$ is the shift of part (i);

here $2h$ is the shift of part (i), and in part (ii) the decorations are read after the differencing there. Then, for sums over $n \in (P, 2P]$, n odd:

(i) *(differenced wave)* for all integers $u, h \ge 1$ with $h \le P^{1/8}$, $uh \le P^{1/2}$, and $d \in \mathcal{D}$,

$$V := \Big|\sum_n e\big(u\,\Delta_{2h}Y(n{+}d) + \rho(n)\big)\Big| \ll \Big((uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + P^{7/8} + P^{1/24}(uh)^{-1/2}P^{7/8} + R_0^{1/2}P^{3/4}\Big)P^{\varepsilon};$$

The fifth term is Stage 5's dominant-mode sum over the Stage-2 families, $3R_0^{1/2}P^{3/4}\log P$; it is the only term carrying the Stage-2 truncation, which is why Appendix A.6 weighs R_0 against $P^{23/24}$ and not against the other four. At $R_0 = P^{5/16}$ it is $P^{29/32}$; at $R_0 = P^{1/4}$ it would read $P^{7/8}$ and be absorbed by the third term, which is why the trade of Appendix A.6 makes it visible.

The absorption runs both ways, and at the truncation this paper uses it runs the other way. Over the admissible range $h \le P^{1/8}$, $uh \le P^{1/2}$ the first term is largest at $uh = P^{1/2}$, where it is $P^{7/8}$, and the third is $P^{7/8}$ throughout; both are therefore strictly below the fifth's $P^{29/32}$ everywhere. Only three of the five are ever the largest: the second, at $u = 1$, $h = P^{1/8}$, where it is $P^{15/16}$; the fourth, at $u = h = 1$, where it is $P^{11/12}$; and the fifth, on the rest of the range. All five are kept because Stage 6 cites the fourth by position and because the proof produces them separately, but a reader balancing the bound should know that at $R_0 = P^{5/16}$ it is a three-term bound.

(ii) *(waves)* for all integer coefficients $(q_d)_{d\in\mathcal{D}}$ with $|q_d| \le 4P^{1/24}$ and total frequency $t := \sum_{d\in\mathcal{D}} q_d$ obeying $0 < |t| \le 3P^{1/24}$, all $\varepsilon_0 \in \{0,1\}$, and smooth φ with $|\varphi'''| \le 3kh_1h_2P^{-13/8}$,

$$U := \Big|\sum_n e\Big(\sum_{d\in\mathcal{D}} q_d\,Y(n{+}d) - \varepsilon_0\,c(n)\lfloor F_\kappa(X(n))\rfloor + \varphi(n)\Big)\Big| \ll |t|^{-1/6}\,P^{23/24+\varepsilon}.$$

(iii) *(widened decoration budget)* the conclusion of (i) continues to hold when at most two of the (D1) terms of ρ carry, in place of $|q'| \le 4P^{1/24}$, the widened budget

$$|q'|\, h' \ \le\ P^{1/2}, \qquad h' \ \le\ P^{1/24}, \tag{D1$'$}$$

provided $uhh' \ge 46$ for each such term. For fixed u and h' the shifts violating that are $h < 46/(uh')$, at most 46 positive integers per widened term.

Part (ii) is the mixed-piece bound proper: level-2 waves $e(qY)$, possibly riding a frozen floor. Part (i) is the engine that (ii)'s targeted differencing feeds, and is also used directly for the $\{W\}$-content of the kernel proof. The exponent $\frac{23}{24}$ is not an accident: written exactly, the wave $e(qY)$ is the depth-2 object of Theorem 4.4, and part (ii) recovers exactly the depth-2 strength — no more. A model of U as $e(sX)$ with a frozen real coefficient $s \asymp tP^{3/4}$ would discard the sawtooth $-\frac{3}{2}tX^{1/2}\theta$ of amplitude $\asymp tP^{3/4}$ inside $tY = t(X-\theta)^{3/2}$; no such shortcut is taken anywhere below.

Proof of (ii) from (i). Eight claims. No appeal is made to "the same treatment", "absorbed", or "inside the budget" except as a pointer to a displayed comparison in the claim that uses it.

Claim A (conjugate). Replacing the summand by its conjugate replaces (q_d) by $(-q_d)$ and t by $-t$. Henceforth $t \ge 1$. The classes (D1)–(D3) are unchanged (they are absolute-value conditions).

Claim B (telescoping, with signs). Fix $e_1 \in \mathcal{D}$ with $q_{e_1} \ne 0$. For each $d \in \mathcal{D}$,

$$Y(n{+}d) = Y(n{+}e_1) + \sigma_{d,e_1}\,\Delta_{|d-e_1|}Y(n{+}\min(d,e_1)),$$

where $\sigma_{d,e_1} = +1$ if $d > e_1$, $\sigma_{d,e_1} = -1$ if $d < e_1$, and $\sigma_{e_1,e_1} = 0$. (If $d > e_1$ this is the definition of Δ_{d-e_1}; if $d < e_1$ then $Y(n{+}d) - Y(n{+}e_1) = -(Y(n{+}e_1) - Y(n{+}d)) = -\Delta_{e_1-d}Y(n{+}d)$.) Therefore, exactly,

$$\sum_{d\in\mathcal{D}} q_d\, Y(n{+}d) = t\, Y(n{+}e_1) + \sum_{d\ne e_1} q_d\sigma_{d,e_1}\,\Delta_{|d-e_1|}Y(n{+}\min(d,e_1)).$$

Each difference of elements of $\mathcal{D} = \{0, d_1, d_2, d_1{+}d_2\}$ lies in $\{\pm d_1, \pm d_2, \pm(d_1{+}d_2)\}$, so $|d - e_1| \in \{2h_1, 2h_2, 2(h_1{+}h_2)\}$ and $h' := |d-e_1|/2 \in \{h_1, h_2, h_1{+}h_2\}$. By (C4), $h' \le h_1{+}h_2 \le 2P^{1/24}$. Each $\min(d, e_1)$ lies in $\mathcal{D}$. Each coefficient $q' := q_d\sigma_{d,e_1}$ satisfies $|q'| = |q_d| \le 4P^{1/24}$. There are at most three nonzero seeds.

Claim C (the A-process). Let $H_3 := \lceil t^{1/3}P^{1/12}\rceil$. The recorded A-process after Lemma 3.3, applied to the unimodular sequence of odd $n \in (P, 2P]$ (at most P terms; using P in place of the true count $P/2 + O(1)$ only enlarges the right-hand side), gives

$$|U|^2 \le \frac{2P^2}{H_3} + \frac{4P}{H_3}\sum_{1\le h_3<H_3} |V_{h_3}|,$$

where

$$V_{h_3} = \sum_n e\big(t\,\Delta_{2h_3}Y(n{+}e_1) + \rho_{h_3}(n)\big)$$

and ρ_{h_3} is the increment of the remaining phase at shift $2h_3$, written out in Claim E. (The recorded display sums $1 \le h < H$; we have renamed the index h_3.)

Claim D (the parameters of (i) are admissible). The hypothesis gives $t \le 3P^{1/24}$. (The individual bound $|q_d| \le 4P^{1/24}$ would give only $16P^{1/24}$, and that is not a harmless slack: it would put the shift comparison below at $16^{12} = 2.8\cdot 10^{14}$ and make it the largest threshold in the paper. The total-frequency bound is the one to carry, and Step 4 of Theorem 5.3 — the only place (ii) is invoked — supplies it, because the wave modes come one per expansion layer from three layers at truncation $J_2 = P^{1/24}$, so $|t| \le 3J_2$.) Write $x := t^{1/3}P^{1/12}$, so $H_3 = \lceil x\rceil$. Claim C sums over $1 \le h_3 < H_3$, hence $h_3 \le \lceil x\rceil - 1 \le x$ (the integer $\lceil x\rceil - 1$ never exceeds x). In particular $h_3 \le t^{1/3}P^{1/12} \le 3^{1/3}P^{7/72}$ with $3^{1/3} < 1.45$. Two comparisons for the shift range of (i):

$$\frac{7}{72} < \frac{1}{8}, \qquad 1.45\, P^{7/72} \le P^{1/8} \quad \text{once} \quad P \ge 1.45^{36}.$$

The threshold is $1.45^{36} = 644537$, comfortably under P_0; A.1 rounds it up to $6.5 \cdot 10^5$, which is the direction that column rounds in. The exponent gap here is $\frac{1}{8} - \frac{7}{72} = \frac{1}{36}$, so whatever constant stands in front of $P^{7/72}$ is paid at the thirty-sixth power; that is why the sharp t matters. Thus every index in the Claim C sum satisfies $1 \le h_3 \le P^{1/8}$ for $P \ge P_0$. (The averaging length itself obeys $H_3 \le x + 1 \le 3P^{7/72}$ once $1 \le 0.48P^{7/72}$, i.e. once $P^{7/72} \ge 3$, which holds for $P \ge 3^{72/7} < 10^6$; Claim G uses only $H_3 \le 2x$, which follows from $x \ge 1$.) The product range of (i) is $th_3 \le tx = t^{4/3}P^{1/12}$. Substituting $t \le 3P^{1/24}$ gives $t^{4/3}P^{1/12} \le 3^{4/3}P^{1/18+1/12} = 4.33\,P^{5/36}$, and $4.33P^{5/36} \le P^{1/2}$ once $4.33 \le P^{13/36}$, i.e. once $P \ge 58$, again far under P_0. Hence every V_{h_3} in the Claim C sum is an instance of (i) with $(u, h) = (t, h_3)$, provided ρ_{h_3} is a legal decoration.

Claim E (class membership of ρ_{h_3}). Expanding the increment of Claim B, the frozen-floor term, and φ,

$$\rho_{h_3} = \sum_{d \ne e_1} q_d \sigma_{d,e_1}\, \Delta_{2h_3} \Delta_{|d-e_1|} Y(n+\min(d, e_1)) - \varepsilon_0\, \Delta_{2h_3}(c\lfloor F_\kappa(X) \rfloor) + \Delta_{2h_3}\varphi.$$

This is a sum of at most five terms (three of type (D1), one of type (D2), one of type (D3)). Step 4 of Theorem 5.3 adds at most two further (D1) terms, the leftover first-differenced modes of Lemma 5.2(iii), so seven is the largest decoration this paper forms; the budget of nine is a ceiling on the class, not a count that is reached, and no estimate below depends on its value.

- (D1). Each summand is $q'\, \Delta_{2h}\Delta_{2h'} Y(n+d')$ with $2h = 2h_3$ the shift of this invocation of (i), $q' = q_d\sigma_{d,e_1}$, $|q'| \le 4P^{1/24}$, $h' = |d - e_1|/2 \le 2P^{1/24}$, and $d' = \min(d, e_1) \in \mathcal{D}$. This is the printed class (D1).
- (D2). The middle term is exactly the printed class (D2) at the shift $2h$ of (i), with the same branch function F_κ and the same offset $|j| \le 2$. If $\varepsilon_0 = 0$ the term is absent.
- (D3). The input φ of (ii) is produced, in every application in this paper, with $|\varphi'''| \le 3kh_1h_2P^{-13/8}$ (the kernel's smooth remnants satisfy $|c''| \le 0.11kP^{-7/8}$ and $|(\Delta_i c)''| \le 0.19kh_iP^{-15/8}$, both far smaller). Then
$$\left|(\Delta_{2h_3}\varphi)''\right| \le 2h_3 \sup|\varphi'''| \le 6kh_1h_2h_3P^{-13/8}.$$
This is *exactly* the (D3) second-derivative budget at the shift $h = h_3$ of this invocation, and it is the sharp form Stage 6 needs. It must not be relaxed to a budget of the shape $3kh_1h_2P^{-5/8}$: that is larger by a factor $P/(2h_3)$ — at $h_3 \le P^{1/8}$, a factor $\ge \frac{1}{2}P^{7/8}$ — and Stage 6 cannot dominate it. Indeed $3kh_1h_2P^{-5/8}\big/(0.35uhP^{-3/4}) = 8.6\,kh_1h_2P^{1/8}/(uh)$, which by (C1) reaches $8.6P^{1/4}$ at $uh = 1$; at any $uh \le 8.6P^{1/4}$ such a φ'' could cancel the Stage-4 curvature outright, and Lemma 3.3 at that scale would not apply. The third derivative of the increment satisfies $\left|(\Delta_{2h_3}\varphi)'''\right| \le 2\sup|\varphi'''| \le 6kh_1h_2P^{-13/8}$, which is the printed (D3) third-derivative budget. (The printed 6 rather than 3 is exactly this factor of two: (D3) is closed under one difference of an input that starts at 3.) Thus $\Delta_{2h_3}\varphi$ is of class (D3).

No other terms appear. In particular this reduction does not introduce a large-u first-differenced wave; those, when they sit on the same piece as a $t \ne 0$ wave in Theorem 5.3, are treated in Step 4 of that theorem, after this lemma.

Claim F (invoke (i)). By Claims D and E, part (i) applies to each V_{h_3} and gives

$$|V_{h_3}| \ll \Big((th_3)^{1/2}P^{5/8} + (h_3/t)^{1/2}P^{7/8} + P^{7/8} + P^{1/24}(th_3)^{-1/2}P^{7/8}\Big)P^{\varepsilon}.$$

The fourth term is the (D1) run-boundary remainder of Stage 6; it is part of the printed bound of (i), not an extra cost smuggled through the average.

Claim G (the H_3-average). Substitute Claim F into Claim C. Write S_1, S_2, S_3, S_4 for the four contributions to $(4P/H_3)\sum_{h_3<H_3} |V_{h_3}|$, and use the crude bounds $\sum_{h_3<H_3} h_3^{1/2} \le H_3 \cdot H_3^{1/2} = H_3^{3/2}$ and $\sum_{h_3<H_3} h_3^{-1/2} \le H_3 \cdot 1 = H_3$ (the second is the worst case $h_3 = 1$ on every term; a tighter integral $2H_3^{1/2}$ is available and is

not needed). Then, with $H_3 \le 2t^{1/3}P^{1/12}$,

$$\begin{aligned}
\frac{2P^2}{H_3} &\le 2t^{-1/3}P^{2-1/12} = 2t^{-1/3}P^{23/12},\\
S_1 &\le 4t^{1/2}H_3^{1/2}P^{13/8}\,P^{\varepsilon} \le 6t^{2/3}P^{5/3}\,P^{\varepsilon} = 6\,(tP^{-1/4})\,t^{-1/3}P^{23/12}\,P^{\varepsilon}\\
&\le 96P^{1/24-1/4}\,t^{-1/3}P^{23/12}\,P^{\varepsilon},\\
S_2 &\le 4t^{-1/2}H_3^{1/2}P^{15/8}\,P^{\varepsilon} \le 6t^{-1/3}P^{23/12}\,P^{\varepsilon},\\
S_3 &= 4P^{15/8}\,P^{\varepsilon} = 4\,(t^{1/3}P^{-1/24})\,t^{-1/3}P^{23/12}\,P^{\varepsilon}\\
&\le 11P^{1/72-1/24}\,t^{-1/3}P^{23/12}\,P^{\varepsilon},\\
S_4 &\le 4P^{1/24}t^{-1/2}P^{15/8}\,P^{\varepsilon} = 4t^{-1/2}P^{23/12}\,P^{\varepsilon} = 4t^{-1/6}\cdot t^{-1/3}P^{23/12}\,P^{\varepsilon}.
\end{aligned}$$

The identities used here are $1/24+13/8 = 5/3$, $5/3-23/12 = -1/4$, $1/24+15/8 = 23/12$, $15/8+1/24 = 23/12$, and $1/24+15/8 = 23/12$. The exponent contributed by $H_3^{1/2}$ is $1/24$, not $1/12$: the bound $H_3 \le 2t^{1/3}P^{1/12}$ enters each of S_1, S_2 under a square root, giving $H_3^{1/2} \le \sqrt{2}\,t^{1/6}P^{1/24}$. Each prefactor is $O(1)$ on $t \ge 1$, $P \ge P_0$: $96P^{-5/24} \to 0$, $P^{1/72-1/24} = P^{-1/36} \to 0$, and $4t^{-1/6} \le 4$. Hence $|U|^2 \ll t^{-1/3}P^{23/12+\varepsilon}$.

Only two of the five contributions survive with an $O(1)$ prefactor — the A-process head $2P^2/H_3$ and S_2, of constants 2 and $4\sqrt{2} \le 6$ — so the balance is exact and explicit:

$$|U|^2 \le (8+o(1))\,t^{-1/3}P^{23/12+\varepsilon}, \qquad |U| \le (2.83+o(1))\,t^{-1/6}P^{23/24+\varepsilon}.$$

The choice $H_3 = \lceil t^{1/3}P^{1/12}\rceil$ is exactly the one that equalizes those two terms; any other exponent in H_3 makes one of them dominate and loses the 23/24.

Claim H (square root). Taking square roots, $|U| \ll t^{-1/6}P^{23/24+\varepsilon}$, which is (ii). The implied constant depends on ε and on (C3) and (C4) only. ☐ of the reduction.

The six stages below prove (i). They are used as a black box by Claims F–H; the only decoration information those claims need is Claim E together with the printed fourth term of (i).

Proof of (i). Six stages; write $\nu = n + d$ (a shift of the block by at most $d_1{+}d_2 \le P^{1/2}$, harmless in every estimate) and $\Delta = \Delta_{2h}$.

Stage 1 (the m-linear form). By Lemma 4.3(i) at the bases ν and $\nu{+}2h$,

$$u\,\Delta Y = u\,A_h(\nu) + \tfrac{3u}{2}\,g(\nu)\,(\nu{+}2h)^{3/4} - \tfrac{3u}{2}\,\theta_\nu\,\Delta(\nu^{3/4}) + u\,\Delta E,$$

with $g = m(\nu{+}2h) - m(\nu)$, $|u\Delta E| \le uP^{-3/4}$ (total cost over the block $\le 2\pi uP^{1/4} \le 2\pi P^{3/4}$), and

$$A_h(\nu) = \tfrac{3}{2}\nu^{3/2}\Delta(\nu^{3/4}) - \tfrac{1}{2}\Delta(\nu^{9/4}) = -\tfrac{27}{8}h^2\nu^{1/4}(1 + O(hP^{-1})):$$

the two $\nu^{5/4}$-scale contributions cancel exactly at leading order (both equal $\frac{9}{4}h\nu^{5/4}$), leaving $|uA_h''| \le 0.64\,uh^2P^{-7/4}$. The sawtooth coefficient is

$$B(\nu) := \tfrac{3u}{2}\,\Delta(\nu^{3/4}) = \tfrac{9}{4}\,uh\,\xi^{-1/4} \in (1.89, 2.25]\,uhP^{-1/4}.$$

Stage 2 (gap cells and the exact shift device). Split into the level sets of $\lfloor\delta_h\rfloor$, $\delta_h(\nu) = (\nu{+}2h)^{3/2} - \nu^{3/2}$: at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95]\,P^{1/2}/h$ (E1). On a cell $g = G + \kappa$ with $G = \lfloor\delta_h\rfloor$ frozen and $\kappa = [\theta_\nu \ge 1 - \{\delta_h\}]$; Vaaler-expand κ at truncation $R_0 = P^{5/16}$ (majorant cost $\le 4P/R_0 = 4P^{11/16}$). The exponent 5/16 is not free choice: Stage 5 below pays $R_0^{1/2}$ for it and Step 5b(a) pays R_0, while the fifth-letter window of Theorem 6.3 needs R_0 *large*. Appendix A.6 solves five competing sites; 5/16 is the smallest value at which all five hold below P_0, and 1/4 is not — indeed the fifth, Lemma 5.2(iii)'s widened mode index, fails at 1/4 for every P. R_0 is the truncation of *this* expansion. It is reused as the Lemma 3.7 mode truncation J in Stage 3(s1) below, and again in Step 5b(a), where the modes counted are Stage

2's own and the identification $J = R_0$ is deliberate; the per-window Lemma 3.7 truncation of Step 3a is a separate parameter, chosen there as $P^{1/4}$, and is not affected by the value of R_0. The moving endpoint costs nothing: as in Theorem 4.4, Step 4, the endpoint's smooth part contributes the exact phase $r\delta_h(\nu)$, and $e(r\nu^{3/2}+r\delta_h(\nu)) = e(r(\nu+2h)^{3/2})$, so every r-term falls into the two smooth families $e(r\nu^{3/2})$, $e(r(\nu+2h)^{3/2})$, $0 < |r| \le R_0$, with weights $\le 1/|r|$; the in-cell variation of the coefficient is exactly 1 and one Abel summation absorbs it at a factor $\le 2+2\pi$.

Stage 3 (the θ-sawtooth). Two regimes. (s1) If $uh \le P^{3/16}$ then $|B| \le 2.25P^{-1/16} < \frac{1}{2}$ for $P \ge P_0$: Lemma 3.7 with $T = P^{1/2}$, $J = R_0$ (hypothesis $T \ge 8(1 + |B|)$: $P^{1/2} \ge 9$ for $P \ge P_0$) expands $e(-B_0\{\nu^{3/2}\}\dots)$ per window (here one window: the total drift of B is $\le 0.6uhP^{-1/4} \le 0.6P^{-1/16} < 1$) into the same two mode families at coefficient factor $\min(2, 2\pi|B|) \le 14.2P^{-1/16}$ (the constant is $2\pi \cdot 2.25 = 14.14$, not 14), mass $O(\log P)$, flat cost $\le 12P^{1/2}$, majorant $\le 14.2P^{-1/16} \cdot 4P^{11/16}$. (s2) If $P^{3/16} < uh \le P^{1/2}$ then $|B| \le 2.25P^{1/4}$ and B drifts by at most 1 on windows of length $\ge P^{5/4}/(0.6uh) \ge 1.2P^{3/4}$: at most $0.6P^{1/4} + 1$ windows. Per window Lemma 3.7 at the centre B_0 ($T = P^{1/2} \ge 8(1 + 2.25P^{1/4})$ for $P \ge P_0$, since $P^{1/4} \ge 19$; flat cost $\le 8(1+2.25P^{1/4})P^{1/2} \le 19P^{3/4}$ in total, once $P \ge 4096$) produces modes $e(w\nu^{3/2})$ whose coefficients decay as $\min(2, \frac{1}{\pi|w+B_0|}) + \min(2, \frac{1}{\pi|w|})$; the window-boundary cost is, using $uh > P^{3/16}$, $\le (0.6P^{1/4}{+}1)\,(0.35\,uh)^{-1/2}P^{3/8} \le 1.1\,P^{1/4-3/32+3/8} = 1.1\,P^{17/32} \le P^{5/8}$. Modes with $|w|$ in the collision window are treated in Stage 5.

Stage 4 (the main curvature, $r = w = 0$). On a cell the remaining phase is $\frac{3u}{2}G(\nu{+}2h)^{3/4} + uA_h +$ (smooth decorations), with second derivative

$$-\tfrac{9}{32}\,uG\,(\nu{+}2h)^{-5/4} \in -[0.35,\ 1.20]\,uh\,P^{-3/4} \qquad \text{(E2)},$$

single-signed with in-cell ratio sup / inf $\le$ 3.5: the second derivative of $\frac{3u}{2}G(\nu{+}2h)^{3/4}$ is the single term $\frac{3u}{2}G \cdot \frac{3}{4} \cdot (-\frac{1}{4})(\nu{+}2h)^{-5/4}$, with G frozen and bounded by (E2). The competitor uA_h'' has ratio $\le 0.64uh^2P^{-7/4}/(0.35uhP^{-3/4}) \le 1.9hP^{-1} \le 1.9P^{-7/8}$; decoration competitors are bounded in Stage 6. Lemma 3.3 per cell and summation give

$$\sum_{\text{cells}}\Big(\ell_i\lambda^{1/2} + \lambda^{-1/2}\Big) \le 1.1\,(uh)^{1/2}P^{5/8} + 2.6\,(h/u)^{1/2}P^{7/8},$$

the first because the cells partition the block, so $\sum_i \ell_i\lambda_i^{1/2} \le P\lambda_{\max}^{1/2} = (1.20)^{1/2}(uh)^{1/2}P^{5/8} = 1.096\,(uh)^{1/2}P^{5/8}$, and the second because there are at most $1.5hP^{1/2}+1$ cells, giving $1.5\cdot(0.35)^{-1/2} = 2.536$ plus a lower-order term. These are the two main terms of (i).

Stage 5 (nonzero modes and collisions). A mode $w \ne 0$ (or $r \ne 0$; identical treatment with weight $1/|r|$) adds the phase $w\nu^{3/2}$ of curvature $\ge 0.53|w|P^{-1/2}$. If $0.53|w|P^{-1/2} \ge 4\cdot 1.20uhP^{-3/4}$, i.e. $|w| \ge 9.1\,uhP^{-1/4}$, the mode curvature dominates at margin $\ge$ 4: Lemma 3.3 over the full block gives $\le 1.4|w|^{1/2}P^{3/4} + 1.4|w|^{-1/2}P^{1/4}$, and the coefficient-weighted sums are $\le 3R_0^{1/2}P^{3/4}\log P = 3P^{29/32}\log P$ (families of Stage 2, regime (s1)) and $\le CP^{7/8}\log P$ (regime (s2) tails). If $|w| \le 0.11\,uhP^{-1/4}$ the main curvature dominates at margin ≥ 4 and the mode rides along at Stage 4's scale, weight summable to $O(\log P)$. In the remaining collision window $0.11\,uhP^{-1/4} \le |w| \le 9.1\,uhP^{-1/4}$ (nonempty only in regime (s2)), the two curvatures can cancel: there the phase is the two-term monomial $a\nu^{3/4} + w\nu^{3/2}$ with $a = \frac{3u}{2}G \asymp uhP^{1/2}$ plus perturbations already shown ρ_0-small, and Lemma 3.8 with $(\alpha, \beta) = (\frac{3}{4}, \frac{3}{2})$ applies per window. The range of M is pinned from below by the $\nu^{3/4}$ scale alone, which never degenerates: by (E1), $\delta_h = 3h\xi^{1/2} \in (3hP^{1/2},\, 4.25hP^{1/2}]$, so $G = \lfloor\delta_h\rfloor > 3hP^{1/2} - 1$ and

$$|a|P^{-5/4} = \tfrac{3}{2}uG\,P^{-5/4} > uhP^{-3/4}\Big(4.5 - \tfrac{1.5}{hP^{1/2}}\Big) \ge 4.4\,uhP^{-3/4} \qquad (P \ge P_0),$$

while $|a|P^{-5/4} \le 6.37\,uhP^{-3/4}$ and $|w|P^{-1/2} \in [0.11, 9.1]uhP^{-3/4}$ on the band. Hence

$$M \in [4.4,\ 9.1]\,uhP^{-3/4}, \qquad M \le 9.1\,P^{-1/4} \le 1 \quad (P \ge 7000),$$

using $uh \le P^{1/2}$ for the last two. (The $\nu^{3/4}$ curvature carries a factor $G \asymp hP^{1/2}$, so M cannot be small on this band.) Then:

$$\le C_E(|I_w|M^{1/2} + M^{-1/2} + (P/M)^{1/3}) \quad \text{per window.}$$

Summing over the at most $0.6P^{1/4}{+}1$ windows — the transition cost $(P/M)^{1/3}$ carries no window-length factor, which is why it sums — with the $O(\log P)$ collision-band coefficient mass:

$$\sum_w |I_w| M^{1/2} = P\, M^{1/2} \le 3.1\,(uh)^{1/2} P^{5/8}, \qquad \sum_w M^{-1/2} \le 0.77P^{1/4}{\cdot}0.48\,(uh)^{-1/2} P^{3/8} \le 0.37\,(uh)^{-1/2} P^{5/8},$$

$$\sum_w (P/M)^{1/3} \le 0.77P^{1/4} \cdot 0.611\,(uh)^{-1/3} P^{7/12} = 0.47\,(uh)^{-1/3} P^{5/6} \le 0.47\,P^{5/6-1/16} = 0.47\,P^{37/48},$$

the windows numbering at most $0.6P^{1/4} + 1 \le 0.77P^{1/4}$ ($P \ge 1200$), the first sum using that the windows partition the block, and the last using $uh > P^{3/16}$ in regime (s2). The collision-band total is $\le C((uh)^{1/2}P^{5/8} + P^{37/48}) \log P \le CP^{7/8} \log P$, with $\frac{37}{48} < \frac{7}{8}$.

Stage 6 (decorations). Each class is dominated at a displayed margin against the Stage-4 curvature $\ge 0.35uhP^{-3/4}$.

- (D1). On the branch decomposition of Lemma 5.1(iii) at the shift pair (h_3-role h, h'): the arcs are absorbed by the shift device (log mass); the b-run boundaries add $\le 1.5(h{+}2h')P^{1/2}{+}2$ cells, at boundary cost $\le 1.5(h{+}2h')P^{1/2}(0.35uh)^{-1/2}P^{3/8} \le 2.6(h/u)^{1/2}P^{7/8} + 5.1\,h'(uh)^{-1/2}P^{7/8}$. The second term is the printed fourth term of (i): since $h' \le 2P^{1/24}$, it is $\le 11\,P^{1/24}(uh)^{-1/2}P^{7/8}$. When (i) is invoked from (ii), Claim G averages this term at the worst-case $h_3 = 1$, giving $S_4 \le 4t^{-1/6}\,t^{-1/3}P^{23/12}P^{\varepsilon}$. The θ-coefficient of the decoration is $\le |q'|(2|j'|P^{-1/4}{+}20hh'P^{-3/4}) \le 16P^{1/24-1/4}{+}160P^{1/24+1/8+1/24-3/4} = 16P^{-5/24} + 160P^{-13/24}$. This is $O(P^{-5/24})$, hence $|B| \le 1$ for $P \ge P_0$; Lemma 3.7 at $T = P^{1/2}$ applies either way (even a constant-size leftover would satisfy $T \ge 8(1 + |B|)$). The modes are expanded in the Stage-2 families. Its smooth curvature has ratio $\le |q'|(2|j'|P^{-5/4} + 25hh'P^{-7/4})/(0.35uhP^{-3/4}) \le 46P^{1/24-1/2} + 572P^{1/24+1/24-1} \le P^{-1/4}$: dominated.
- (D2). Write $\Delta(c\lfloor F\rceil) = (\Delta c)\lfloor F\rceil + c_+\Delta\lfloor F\rceil$ with $c_+ = c(\cdot{+}2h)$.
  (a) $(\Delta c)\lfloor F\rceil = (\Delta c)F - (\Delta c)\{F\}$: the smooth part has curvature $\le \big((\Delta c)F\big)'' \le 7kh|j|P^{-9/8} + 28khh_1h_2P^{-13/8}$, ratio to the main curvature $\le 20k|j|P^{-3/8}/u \le 60P^{1/24-3/8}$: dominated. The sawtooth part has coefficient $\Delta c \in (1.68, 1.85)\,khP^{1/8}$, handled per windows of drift ≤ 1: at most $2khP^{1/8}{+}1$ windows, at boundary cost $\le 2khP^{1/8}(0.35\,uh)^{-1/2}P^{3/8} \le 3.4\,k\,(h/u)^{1/2}P^{1/2}$. Per window, Lemma 3.7 at $T = P^{1/2}$ yields modes $e(q''(F \circ X))$ whose index obeys $|q''| \le |B_0| + J$ with the *truncation* $J = R_0 = P^{5/16}$ of Stage 2 — not the window parameter T — so that $|q''| \le 1.85khP^{1/8} + P^{5/16} \le 2.85P^{5/16}$ by (C3) and $h \le P^{1/8}$ (here R_0 is the larger of the two, which it was not at $R_0 = P^{1/4}$), and the curvature is

  $$\le (1.85khP^{1/8}{+}R_0) \cdot 3|j|P^{-5/4} \le 2.85P^{5/16} \cdot 6P^{-5/4} = 17.1P^{-15/16},$$

  ratio to the main curvature $\le 48.9P^{-3/16}$, which is 0.14 at P_0 against the margin $\frac{1}{4}$. Weakening it to $49P^{-1/16}$ is true but useless — a constant of 49 on a gap of $\frac{1}{16}$ does not clear $\frac{1}{4}$ until 10^{36}. In the other direction $48.9P^{-3/16}$ is itself not the sharpest form available, because merging $1.85khP^{1/8} + R_0$ into $2.85P^{5/16}$ raises $P^{7/24}$ to $P^{5/16}$ and loses $P^{1/48}$ — a factor 1.45 at P_0. Kept apart, the ratio is

  $$\frac{(1.85khP^{1/8} + R_0) \cdot 6P^{-5/4}}{0.35\,uh\,P^{-3/4}} \le \frac{6(1.85P^{7/24} + P^{5/16})}{0.35} P^{-1/2},$$

  which clears $\frac{1}{4}$ from $2.98 \cdot 10^{11}$ rather than from $1.66 \cdot 10^{12}$. Both are far below P_0, so nothing in Sections 4–6 changes; the two-term form matters only where this site is itself the binding one, which is the floor of Appendix A.5. (Substituting T for J here would give only $9P^{-3/4}$, whose ratio $26/(uh)$ is not $o(1)$ at $uh = O(1)$; $17.1/0.35 = 48.9$ fixes the constant either way. The 0.35 here is the Stage-4 curvature $0.35\,uh\,P^{-3/4}$ of Theorem 4.1, not Lemma 5.2b's λ_0 floor, which the erratum there moved to 0.56; the two constants share a value and nothing else.) The flat cost is, per point, $8(1{+}B_0)/T$, so in total $\le 8(1{+}1.85khP^{1/8})P^{1/2} \le 8P^{1/2} + 15\,khP^{5/8} \le 23\,P^{1/24+1/8+5/8} = 23\,P^{19/24}$, using $h \le P^{1/8}$ and (C3); this sits inside the Step 6 budget $P^{23/24}$ from $P \ge 23^6 = 1.5 \cdot 10^8$. (It is *not* below $P^{7/8}$: the gap $\frac{7}{8} - \frac{19}{24} = \frac{1}{12}$ would make that claim wait until $23^{12} = 2.2 \cdot 10^{16}$.)

(b) $c_+\Delta\lfloor F\rfloor$: by the gap identity applied to the sequence $F\circ X$, $\Delta\lfloor F\rfloor=\lfloor\Delta F\rfloor+\kappa_F$. ΔF has total drift $\le P\cdot 2h\sup|(F\circ X)''|\le P\cdot 2h(2|j|P^{-5/4}+25h_1h_2P^{-7/4})\le 13hP^{-1/4}+50h\,h_1h_2P^{-3/4}<1$, so $\lfloor\Delta F\rfloor$ takes at most two values, on at most two intervals; per value the phase $-q_0c_+$ ($|q_0|\le 2+6hP^{-1/4}\le 3$) has curvature $\le 0.4kP^{-7/8}$, ratio $\le 1.2kP^{-1/8}/(uh)\le 1.2P^{1/24-1/8}$: dominated. The carry $\kappa_F=\{F\}+\{\Delta F\}-\{F{+}\Delta F\}$ is a sum of unit sawtooths in slow variables (all drifts <1 per step): Lemma 3.5 at truncation $P^{1/8}$ (majorant $\le 3\cdot 4P^{7/8}$) gives modes $e(q''(F\circ X))$-type of curvature $\le P^{1/8}\cdot 3|j|P^{-5/4}\le 9P^{-9/8}$, ratio $\le 26P^{-3/8}$: dominated.

- (D3). The class budget is $|\varphi''|\le 6kh_1h_2h\,P^{-13/8}$, which is what Claim E delivers for $\Delta_{2h_3}\varphi$. The ratio to the Stage-4 curvature is

$$\frac{6kh_1h_2h\,P^{-13/8}}{0.35uhP^{-3/4}}=\frac{18\,kh_1h_2P^{-7/8}}{u}\le 18P^{1/8-7/8}=18P^{-3/4},$$

using (C1) in the form $kh_1h_2\le P^{1/8}$ and $u\ge 1$: dominated. Only the second-derivative budget is used; the third-derivative budget is carried so that (D3) is closed under one further difference. It is essential that the budget be the differenced one: a second derivative of size $3kh_1h_2P^{-5/8}$ — the shape an *undifferenced* smooth remnant would have — is larger by $P/(2h)$ and is *not* dominated (Claim E).

The wide (D3) class: what is and is not covered. The narrowing of (D3) above is not cosmetic, and it is worth recording exactly where the wider class — $|\varphi''|\le 3kh_1h_2P^{-5/8}$, the shape an *undifferenced* smooth remnant would have — stands. Write $\Phi_2=3kh_1h_2P^{-5/8}$.

Regime A: $\Phi_2\le\frac14\cdot 0.35uhP^{-3/4}$, i.e. $uh\ge 34.3\,kh_1h_2P^{1/8}$. Here the decoration is dominated and Stage 6 applies verbatim; nothing changes.

Regime B: $uh<34.3\,kh_1h_2P^{1/8}\le 34.3P^{1/4}$ by (C1). Here φ'' may cancel the Stage-4 curvature and φ''' may exceed λ'_{main}, so neither the second- nor the third-derivative test is available on a cell; and Φ is not a monomial combination, so Lemmas 3.8–3.9 do not apply either. What replaces them is the *frozen* G. Since δ_h is continuous and increasing, $G=\lfloor\delta_h\rfloor$ increases by exactly 1 at each cell boundary, so f'' carries a sawtooth of amplitude $\frac{9}{32}u(\nu{+}2h)^{-5/4}\in[0.118,0.282]\,uP^{-5/4}$ that no continuous φ'' can follow, and f' jumps by $\frac98u(\nu{+}2h)^{-1/4}\in[0.946,1.125]\,uP^{-1/4}$ at each boundary. Two consequences, both elementary:

- the cells are *flat*: at the sawtooth curvature scale the phase departs from linear by $\le 0.282\,uP^{-5/4}\ell^2\le 0.26\,uP^{-1/4}h^{-2}$, which is <1 throughout regime B once $h\ge 3$; so the second-derivative test is the wrong tool there and Kusmin–Landau is the right one;
- the cell frequencies are *spread*: across the $\ge 1.24hP^{1/2}$ cells, f' sweeps at least $1.17\,uhP^{1/4}$ full periods.

If the $\|f_i'\|$ inherit that spread, then $\sum_i\min(\ell,(2\|f_i'\|)^{-1})\ll hP^{1/2}\log P\le P^{5/8}\log P$ by $h\le P^{1/8}$ — comfortably inside the third printed term $P^{7/8}$ of (i). Numerically this is what happens. Taking $\varphi=\frac{27}{10}\nu^{5/4}$, which lies in the wide class at $u=h=k=h_1=h_2=1$ and cancels the smooth part of the Stage-4 curvature exactly, the block sums at $P=2\cdot10^3,8\cdot10^3,3.2\cdot10^4,1.28\cdot10^5,5.12\cdot10^5$ are 36, 12, 220, 479, 385, against $P^{1/2}=45,\dots,716$ and a $P^{7/8}$ larger by a further factor $P^{3/8}$; the cells are flat ($\mathrm{amp}\cdot\ell^2\approx 3\cdot10^{-3}$) and f' sweeps 19 to 37 periods.

Closing regime B. Write $t=\{\delta_h(\nu)\}\in[0,1)$ on a cell and $q(\nu)=\frac98u(\nu{+}2h)^{-1/4}\in[0.946,1.125]\,uP^{-1/4}$, so that

$$f'(\nu)=\Psi(\nu)-q(\nu)\,t,\qquad \Psi:=\tfrac{27}{8}uh\nu^{1/4}+uA_h'+\varphi'.$$

The decoration enters only through Ψ; the term $-qt$ comes from the frozen G and no choice of φ can remove it. Since t sweeps $[0,1)$ across each cell, f' sweeps an interval of length $\asymp q$ there unless Ψ' cancels the sweep, and that dichotomy is the whole argument. Put $D_i:=\Psi'-q/\ell$ on cell i.

(a) $|D_i|\ge q/(2\ell)$. Then f' is monotone on the cell and sweeps an interval of length $\ge q/2$. Split the cell by dyadic $\|f'\|$: the j-th piece has length $\ll 2^{-j}\ell$ and Kusmin–Landau bounds its sum by $\ll 2^j/q$; balancing at $2^j=(\ell q)^{1/2}$ gives a cell total $\ll(\ell/q)^{1/2}\le 1.01\,(uh)^{-1/2}P^{3/8}$. Summing over the $\le 1.5hP^{1/2}{+}1$ cells,

$$\ll 1.6\,(h/u)^{1/2}P^{7/8},$$

which is exactly the second printed term of (i). Case (a) is closed.

(b) $|D_i| < q/(2\ell)$. Then $\Psi' \asymp q/\ell \asymp uhP^{-3/4}$, so φ'' must sit within $O(uhP^{-3/4})$ of $\frac{1}{2}uhP^{-3/4}$, which by $|\varphi''| \le \Phi_2$ confines this case to $uh \le 6\,kh_1h_2P^{1/8}$. Here f' is nearly constant on each cell, equal to $\alpha_i := \Psi(\nu_i)$ up to $O(q)$, and $\alpha_{i+1} - \alpha_i = \Psi'\ell \asymp q$: the cell frequencies advance by $\asymp uP^{-1/4}$ per cell. The cell sums are $\ll \min(\ell, \|\alpha_i\|^{-1})$, and Ψ is monotone here, sweeping $V \asymp uhP^{1/4}$ periods over the block. Counting the samples in each period,

$$\sum_i \min(\ell, \|\alpha_i\|^{-1}) \ll hP^{1/2}\log P + V\ell \ll P^{5/8}\log P + 1.1\,uP^{3/4},$$

using $h \le P^{1/8}$. The first term is inside the third printed term of (i) with room. The second is $\le P^{7/8}$ precisely when $u \le 0.9P^{1/8}$.

So regime B closes outright for $u \le P^{1/8}$, and case (a) closes for every u. What remains is the sliver

$$P^{1/8} < u \le 6\,kh_1h_2P^{1/8}/h \quad \text{inside case (b)},$$

where the argument above yields only the crude $V\ell$: it charges a full cell to each of the V crossings of Ψ through $\mathbb{Z}$. That is certainly lossy — those V cells carry different constant terms and should not add coherently — but making the saving explicit needs a bound on how often $\|\alpha_i\|$ is small, i.e. the classical $\sum_{i<N}\min(\ell, \|iq\|^{-1})$ estimate for the *slowly varying* difference q, and that is not carried out here.

The sliver is a narrow-class problem. Case (b) looks like the wide class but is not. Its defining condition $|D_i| < q/(2\ell)$ reads $\Psi' \in (\frac{q}{2\ell}, \frac{3q}{2\ell})$, and with $q/\ell = \frac{27}{16}uh\nu^{-3/4}$ and $\Psi' = \frac{27}{32}uh\nu^{-3/4} + \varphi''$ this pins

$$0 \;<\; \varphi'' \;<\; \tfrac{27}{16}\,uh\nu^{-3/4} \qquad \text{throughout case (b)}.$$

So in the only case left open, $|\varphi''|$ is at most 4.8 times the Stage-4 curvature $0.35uhP^{-3/4}$: the wide budget $\Phi_2 = 3kh_1h_2P^{-5/8}$, larger by $P/(2h)$, is *never* attained there. The wide class collapses to a narrow one exactly where it mattered.

Two consequences follow at once. First, Ψ is strictly increasing with $\Psi' \in [0.84, 2.53]uh\nu^{-3/4}$, so the cell frequencies $\alpha_i = \Psi(\nu_i)$ form a strictly increasing sequence with gaps

$$\alpha_{i+1} - \alpha_i \in [0.562,\ 1.688]\,uP^{-1/4},$$

of bounded ratio 3. Second, in the extreme sub-case $\Psi' \equiv q/\ell$ one has $\Psi' = q\,\delta_h'$, hence $\alpha_i = \Psi(\nu_i) \approx qG_i$+const with G_i consecutive integers: an approximate arithmetic progression of difference $q \asymp uP^{-1/4}$. The residual is therefore exactly the classical sum $\sum_{i<N}\min(\ell, \|iq\|^{-1})$, for a difference that drifts by 19% across the block (because $q = \frac{9}{8}u(\nu{+}2h)^{-1/4}$); that drift is what prevents the progression from locking onto a rational, and it is the saving the crude $V\ell$ throws away.

The sliver, measured. Maximising over the free linear term as before, at $h = 1$ and u up to the top of the sliver:

P	u	case (b)	case (a)	printed (i)	(b)/(i)
$8 \cdot 10^3$	1	192	397	5478	0.035
$8 \cdot 10^3$	13	234	193	4315	0.054
$3.2 \cdot 10^4$	1	475	1035	18154	0.026
$3.2 \cdot 10^4$	7	546	403	13788	0.040
$3.2 \cdot 10^4$	13	544	413	13536	0.040

The ratio to the printed bound stays near 0.04 across the whole sliver and falls with P. At $P = 3.2 \cdot 10^4$, $u = 13$ the crude $V\ell$ is 34 214 against an actual 544: the period-counting step is lossy there by a factor 63, which is the whole of the remaining gap. *Numerical check of the whole regime.* The decoration is free to carry a linear term, so the honest test is to maximise over it; on the natural grid that maximum is a discrete Fourier transform of $e(f(n))$. At $u = h = k = h_1 = h_2 = 1$, the optimum over all class-(D3) linear shifts, for the undecorated phase and for the two extremal decorations $\varphi = \pm\frac{27}{10}\nu^{5/4}$ (which cancel the Stage-4 curvature and the drift of f' respectively), is

P	none	curvature	drift	$P^{7/8}$	worst$/P^{7/8}$
$8 \cdot 10^3$	169	192	397	2601	0.152
$3.2 \cdot 10^4$	383	475	1035	8750	0.118
$1.28 \cdot 10^5$	792	958	2536	29431	0.086

The worst case grows like $P^{0.67}$ and its ratio to $P^{7/8}$ *decreases*; the drift-cancelling decoration is the stronger of the two, as the analysis above predicts, and neither approaches the printed bound.

Totals. Stages 1–6 bound V by

$$C\Big((uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + P^{7/8} + P^{1/24}(uh)^{-1/2}P^{7/8} + k\,(h/u)^{1/2}P^{1/2}\Big)\log^3 P.$$

The last term is $\leq P^{1/24}(h/u)^{1/2}P^{1/2}$, and $P^{1/24+1/2} = P^{13/24} < P^{7/8} = P^{21/24}$, so it is absorbed by the second printed term of (i). The (D1) run-boundary term $P^{1/24}(uh)^{-1/2}P^{7/8}$ is kept in the printed bound of (i); it is $\leq P^{11/12}$ for $uh \geq 1$, and $11/12 < 23/24$. This is (i).

Costs collected. Each class contributes to V as follows, all times P^{ε}.

Class	Bound contributed to V	Absorbed by
Stage 1 remainder $u\Delta E$	$P^{3/4}$	$P^{7/8}$
Stage 2 majorant / shift device	$P^{3/4}$	$P^{7/8}$
Stage 3 (s1) sawtooth	$P^{3/4}$	$P^{7/8}$
Stage 3 (s2) windows / collision	$(uh)^{1/2}P^{5/8} + P^{7/8}$	(i)
Stage 4, $r = w = 0$	$(uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8}$	(i)
Stage 5, non-collision modes	$P^{7/8}$	(i)
(D1) curvature	dominated at $P^{-1/4}$	Stage 4
(D1) run boundaries	$(h/u)^{1/2}P^{7/8} + h'(uh)^{-1/2}P^{7/8}$	printed (i); Claim G
(D2)(a) smooth / sawtooth	dominated; $P^{7/8}$ flat	Stage 4; (i)
(D2)(b) gap / carry	dominated; $P^{7/8}$ majorant	Stage 4; (i)
(D3)	dominated at $P^{-3/4}$	Stage 4

No decoration class is used later unless it appears in this table.

Proof of (iii). The budget $|q'| \leq 4P^{1/24}$ enters Stages 1–6 in exactly two places, both in Stage 6's (D1) bullet: the smooth curvature ratio and the θ-coefficient. Everything else there depends on h' and not on $|q'|$ — in particular the run-boundary cost $2.6(h/u)^{1/2}P^{7/8} + 5.1h'(uh)^{-1/2}P^{7/8}$, whose second term is the fourth printed term of (i) and is unchanged because $h' \leq P^{1/24}$ still holds under (D1′).

Curvature ratio. Against the Stage-4 curvature $0.35uhP^{-3/4}$, using $2|j'| \leq 4$, $4/0.35 \leq 11.5$, $25/0.35 \leq 71.5$ and $|q'|h' \leq P^{1/2}$ in both summands,

$$\frac{|q'|(2|j'|P^{-5/4} + 25hh'P^{-7/4})}{0.35uhP^{-3/4}} \;\leq\; \frac{11.5}{uhh'} + \frac{72}{u}\,P^{-1/2}.$$

The second summand is $o(1)$ for $P \geq P_0$; the first is $\leq \frac{1}{4}$ once $uhh' \geq 46$, which is the stated hypothesis. Stage 6 then dominates at margin ≥ 4, as in the printed case.

θ-coefficient, and the window inventory it forces. By (D1′), $|j'| \leq 2$, $h' \geq 1$ and $h \leq P^{1/8}$,

$$|q'|(2|j'|P^{-1/4} + 20hh'P^{-3/4}) \;\leq\; \frac{4P^{1/4}}{h'} + 20hP^{-1/4} \;\leq\; 5P^{1/4}.$$

This is not $O(P^{-5/24})$, so the decoration's sawtooth is expanded by the large-B treatment of Stage 3(s2) and not by the small-B one of (s1). Each line below is that treatment's with 2.25 replaced by 5. The hypothesis $T = P^{1/2} \geq 8(1 + |B|)$ reads $P^{1/4} \geq 40.20$, i.e. $P \geq 2.6 \cdot 10^6$. There are at most $5P^{1/4} + 1$ windows on which B moves by ≤ 1.

> *Erratum (the count is right; one route to it is not).* The window count does not follow from monotonicity. "B is monotone on the dyadic block, so its drift is at most $\sup|B|$" is the natural one-line argument here, and neither half of it survives inspection. Write the coefficient as $q'(2j'f_1 + hh'f_2)$, with $f_1 \leq P^{-1/4}$ and $f_2 \leq 20P^{-3/4}$ the two smooth profiles of the display

above. j' is the branch offset of Lemma 5.1(iii): it is frozen on each b-run and jumps between runs, so B is not monotone on the block; and it takes both signs, so even a monotone B would give $2\sup|B|$ and not $\sup|B|$. What is true is that f_1 and f_2 are monotone and single-signed on the whole block, and total variation is additive over a partition, so with $|j'| \le 2$ on each run

$$\sum_{\text{runs}} \operatorname{Var}(B) \;\le\; 4|q'|\operatorname{Var}(f_1) + |q'|hh'\operatorname{Var}(f_2) \;\le\; 4|q'|\sup f_1 + |q'|hh'\sup f_2 \;\le\; 5P^{1/4},$$

which is the printed bound, reached without knowing either sign. The jumps across runs are the b-run boundaries already inventoried in the first bullet, at the same Stage-4 curvature, so they cost nothing further. Both readings give $5P^{1/4}+1$; only one of them is a proof.

The boundary charge at the Stage-4 curvature is

$$(5P^{1/4}{+}1)(0.35uh)^{-1/2}P^{3/8} \;\le\; 8.5\,(uh)^{-1/2}P^{5/8},$$

which needs no lower bound on uh: Stage 3(s2) may simplify its own using $uh > P^{3/16}$ because (s2) *is* that regime, whereas a widened decoration is large independently of the main mode and can occur while the main sawtooth sits in (s1). It is dominated by the fourth printed term of (i), since $\frac{5}{8} < \frac{1}{24} + \frac{7}{8}$. The flat cost is $8(1{+}|B|)P^{1/2} \le 48P^{3/4}$ in total. The modes are the (s2) families, with weights $\min(2, \frac{1}{\pi|w+B_0|}) + \min(2, \frac{1}{\pi|w|})$ and index $|w| \le |B_0| + R_0 \le 2R_0$ for $P \ge P_0$, because $5P^{1/4} \le P^{5/16}$ once $P \ge 5^{16} = 1.53 \cdot 10^{11}$ — a certificate row in its own right, and the one that pins R_0 from below (A.6); at the printed offset bound it read 7^{16} and was the largest row not mentioning c_7, which is what A.5's floor turns on; Stage 5 therefore pays at most $\sqrt{2}$ more on its dominant sum and a factor $5/0.6 \le 9$ on its (s2) tails, both absorbed by P^{ε}. $\square$

Checklist for (ii) from (i). Claims A–H are the verification. In order: conjugate (A); telescoping with signs σ_{d,e_1} (B); the recorded A-process $2P^2/H_3 + 4P/H_3\sum|V|$ (C); admissible range $h_3 \le P^{1/8}$ and $th_3 \le P^{1/2}$ (D); class membership of ρ_{h_3}, five terms, (D3) closed under one difference from an input with $|\varphi'''| \le 3kh_1h_2P^{-13/8}$ (E); invoke the printed (i), fourth term included (F); average, five displayed comparisons, worst-case $h_3 = 1$ on the (D1) remainder (G); square root (H). Collision windows of (i) are only the band $0.11\,uhP^{-1/4} \le |w| \le 9.1\,uhP^{-1/4}$ of Stage 5, and only in regime (s2). On a cell the integer $G = \lfloor \delta_h \rfloor$ is held frozen; the smooth remainder A_h does not differentiate a frozen gap. Lemma 3.9 is not used in (i) or (ii); it is used only on the global interpolant Φ of Lemma 5.2b. Every k, h_1, h_2 dependence passes through (C3) and (C4). $\square$

The balance to keep in mind: with the trivial bound $|V_{h_3}| \le P$, part (ii)'s differencing returns nothing beyond the choice of H_3; the content of the lemma is that the floor, coefficient, and carry bookkeeping of Stages 3–6 survives the targeted differencing with the Stage-4 curvature intact. What is *not* available is the frozen-coefficient shortcut: per frozen run the run length $P^{1/4}$ is shorter than $\lambda^{-1/2} \asymp (uh)^{-1/2}P^{3/8}$ for small uh, and the run-boundary cost $(h/u)^{1/2}P^{7/8}$, summed against the H_3-average, is exactly what pins the final exponent at $\frac{23}{24}$.

The remaining object that Theorem 5.3 must not confuse with Lemma 5.2 is the *zero-offset anchor*: the phase $c(G_F - J_F)$ on a branch with $j = 0$. On a cell the integers β_1, β_2, J_F are frozen, so $(cG_F)'' = c''G_F + 2c'G_F' + cG_F''$ is computed with those integers held constant while $X(\nu)$ and $c(\nu)$ move. The interpolant used on the whole block is the same frozen-shape formula with the smooth gap values $\Delta_i X(\nu)$ substituted as numbers, not differentiated. The next lemma records that computation, so that Step 5b is only a classification of scales.

Lemma 5.2b (frozen-shape interpolant on a zero-offset piece). Assume (C3) and (C4), $j = 0$, and the *middle-band wave bound*

$$u \;\le\; 300\,kh_2P^{1/8}, \qquad u' \;\le\; 300\,kh_1P^{1/8}. \tag{C5}$$

(C5) is not implied by (C3) and (C4): Lemma 5.2(i) admits $uh_1 \le P^{1/2}$, hence u as large as $P^{1/2}$, and at that size the first error term below is $\frac{9}{16}P^{-3/4}$, larger than the stated bound by a factor $P^{7/24}$. It *is* the middle-band condition of Step 5b: $\mu \le 60\lambda_0$ with $\mu = 0.84\max(uh_1, u'h_2)P^{-3/4}$ and $\lambda_0 \le 4.2\,kh_1h_2P^{-5/8}$ gives $\max(uh_1, u'h_2) \le \frac{60\cdot 4.2}{0.84}kh_1h_2P^{1/8} = 300\,kh_1h_2P^{1/8}$, with no opening needed. Keeping the shifts visible, rather than passing to the cruder $u, u' \le 300P^{5/24}$ through $kh_i \le P^{1/12}$, is what lets the two error terms below combine before they are converted. Work on a common-refinement piece of the gap cells of both shifts, the frozen-floor runs of $\lfloor F_\kappa(X)\rfloor$, and the sawtooth windows already counted, so that the integers G_1, G_2, β_1, β_2, and $J_F = \lfloor F_\kappa(X)\rfloor$ are constant. Write $G_F = F_\kappa \circ X$. The remaining phase on the piece has second derivative

$$f'' = -\tfrac{9}{32}\Big(uG_1(\nu{+}2h_1)^{-5/4} + u'G_2(\nu{+}2h_2)^{-5/4}\Big) + 2c'G_F' + c\,G_F'' + c''(G_F - J_F) + wX'' + O(\rho_0),$$

where the $O(\rho_0)$ collects decorations already shown small against the scales of Lemma 3.8, and the chain rule uses *frozen* β_i:

$$G_F' = F'(X)\,X', \qquad G_F'' = F''(X)\,(X')^2 + F'(X)\,X'',$$

with, from Lemma 5.1(iii) at $j = 0$,

$$F'(m) = -\tfrac{3}{8}\beta_1\beta_2\,m^{-3/2}, \qquad F''(m) = \tfrac{9}{16}\beta_1\beta_2\,m^{-5/2}$$

up to the relative $O(\beta/m) = O(P^{-1})$ of the mean-value ξ. Define the *frozen-shape interpolant* Λ by substituting the smooth gap values $\widetilde{\beta}_i(\nu) = \Delta_i X(\nu)$ and $\delta_{h_i}(\nu) = (\nu{+}2h_i)^{3/2} - \nu^{3/2}$ for those frozen integers *as values only*. The ν-derivatives of $\widetilde{\beta}_i$ are not introduced:

$$\begin{aligned}\Lambda(\nu) = {} & -\tfrac{9}{32}u\,\delta_{h_1}(\nu)\,(\nu{+}2h_1)^{-5/4} - \tfrac{9}{32}u'\,\delta_{h_2}(\nu)\,(\nu{+}2h_2)^{-5/4}\\ & + 2c'(\nu)\,\widetilde{F}'(X(\nu))\,X'(\nu) + c(\nu)\,\widetilde{G}''(\nu) + \tfrac{1}{2}c''(\nu) + wX''(\nu),\end{aligned}$$

where $\widetilde{F}'(m) = -\frac{3}{8}\tilde{\beta}_1\tilde{\beta}_2\,m^{-3/2}$ and $\widetilde{G}''$ is the frozen-shape formula for G_F'' evaluated at $\tilde{\beta}_i$. Then, for $P \ge P_0$,

$$|f''{-}\Lambda| \le \tfrac{9}{32}(u{+}u')P^{-5/4}{+}|c''|{+}0.91\,k(h_1{+}h_2)P^{-9/8} \le 85.3\,k(h_1{+}h_2)P^{-9/8}{+}0.11P^{-5/6} \le 170.6\,P^{-25/24}{+}0.11P^{-5/6}.$$

The second term is the leading interpolant error and is not absorbed into a smaller multiple of $P^{-5/6}$. Moreover $\Lambda = \Phi'' + r$, where

$$\Phi(\nu) = a\,\nu^{5/4} + b\,\nu^{11/8} + w\,\nu^{3/2}, \qquad a = -\tfrac{27}{10}(uh_1 + u'h_2), \quad b = -\tfrac{81}{22}\,kh_1h_2,$$

the exponents $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$ lie in E and are pairwise distinct, and r obeys the $\rho_0(E)$ bounds of Lemma 3.9. If $w = 0$, drop the third term. The local frozen curvature of the anchor itself is

$$2c'G_F' + c\,G_F'' = -\tfrac{27}{128}\,k\beta_1\beta_2\,\nu^{-13/8}(1 + O(P^{-1/4})),$$

and therefore

$$\lambda_0 := \big|\big(c(G_F - J_F)\big)''\big| \in [0.56,\ 4.2]\,kh_1h_2P^{-5/8}$$

on the standing range: the summand $c''(G_F - J_F)$ is $O(kP^{-7/8})$ and sits in the $O(P^{-1/4})$ relative error.

Erratum (constants only; the exponent does not move and P_0 falls). Earlier printings gave $-\frac{135}{1024}$ here, which is $(cG_F)''$ and not $2c'G_F'{+}cG_F''$. The three chain-rule terms $c''G_F = \frac{81}{1024}$, $2c'G_F' = -\frac{972}{1024}$, $cG_F'' = \frac{756}{1024}$ do sum to $-\frac{135}{1024}$; but the anchor is $c(G_F - J_F)$, and with J_F frozen its first term is $c''(G_F - J_F) = O(kP^{-7/8})$, not $c''G_F$. Dropping the term the phase does not carry leaves $-\frac{216}{1024} = -\frac{27}{128}$, confirmed against $(c(G_F - J_F))''$ at $P = 10^8$ to six figures (0.210938; the old constant is what $(cG_F)''$ measures, 0.131836). Step 5a always made this subtraction on the offset branch, $\frac{945}{512} - \frac{81}{512} = \frac{864}{512}$, and so does Theorem 6.1 Step E at zero offset; only this display did not.

The factor is $\frac{216}{135} = \frac{8}{5}$, and it runs through: $-\frac{1215}{1024} \to -\frac{243}{128}$, $b = -\frac{405}{176} \to -\frac{81}{22}$, the range $[0.38, 2.44] \to [0.62, 3.90]$ (opened $[0.35, 2.6] \to [0.56, 4.2]$), the (C5) cap $186 \to 300$, $0.567 \to 0.907$, $52.9 \to 85.3$ and E's $106 \to 170.6$. It was never cosmetic: under the old b the residue r keeps a leading term of size $\frac{729}{1024}kh_1h_2\nu^{-5/8}$, comparable to S itself, and the $\rho_0(E)$ ratios of Lemma 3.9 fail outright.

The correction *lowers* the threshold. S rises by $\frac{8}{5}$ while $V = \kappa S^{1/2}P^{-11/24}$ rises only by $\sqrt{8/5}$, which more than pays for E: P_0 falls from $8.95 \cdot 10^{13}$ to $3.6 \cdot 10^{13}$ and the non-vacuity point P_1 from $5.04 \cdot 10^{19}$ to $9.9 \cdot 10^{18}$. The optimal κ is unchanged at $\frac{1}{12}$, and so is the exponent $1 - \frac{1}{96}$. Every figure below is the corrected one.

Proof. There are three replacements of a frozen integer by a smooth gap of the same scale, and each moves its argument by at most 1.

(i) By the gap identity $G_i = \lfloor \delta_{h_i} \rfloor + \kappa_i$ with $\kappa_i \in \{0,1\}$, so $|G_i - \delta_{h_i}| = |\kappa_i - \{\delta_{h_i}\}| \le 1$ — *not* merely < 2, which would double the constant below. The bound 1 is sharp and cannot be halved by recentring Λ (Lean `gap_error_le_one`, `gap_error_one_attained`, `gap_error_not_halved_by_recentring`): $\kappa_i = 1$ exactly when $\{\nu^{3/2}\} + \{\delta_{h_i}\} \ge 1$, so both values of κ_i occur, and a global shift of the interpolant that halves the error for one doubles it for the other. Hence the wave terms differ from Λ by at most $\frac{9}{32}(u{+}u')P^{-5/4}$, and by (C5) this is $\le \frac{9}{32} \cdot 300\, k(h_1{+}h_2)P^{1/8-5/4} = 84.375\, k(h_1{+}h_2)P^{-9/8}$ — the *same shape* as the third term, which is why the two are added before converting. Together they are $85.2820 \le 85.3\, k(h_1{+}h_2)P^{-9/8}$, i.e. $\le 170.6\, P^{-25/24}$ by $k(h_1{+}h_2) \le 2P^{1/12}$ from (C3) and (C4).

(ii) $|\beta_i - \tilde{\beta}_i| \le 1$, so $|\beta_1\beta_2 - \tilde{\beta}_1\tilde{\beta}_2| \le |\beta_1| + |\tilde{\beta}_2| \le 4.3(h_1{+}h_2)P^{1/2} + 1$. The frozen-shape second derivative is linear in that product with coefficient $\frac{27}{128}k\nu^{-13/8}$, so the difference is at most $\frac{27}{128} \cdot 4.3\ k(h_1{+}h_2)P^{1/2}P^{-13/8} = 0.9070\, k(h_1{+}h_2)P^{-9/8}$, printed as 0.91 — not 0.95, which would push the sum in (i) past 85.3: this is the third displayed term.

(iii) $|G_F - J_F| < 1$, and Λ replaces this fractional part by $\frac{1}{2}$ only in the already-expanded c''-term. The difference is at most $|c''| \le 0.11kP^{-7/8} \le 0.11P^{-5/6}$. This replacement is *not* made in the phase $c(G_F - J_F)$: substituting $J_F \mapsto G_F - \frac{1}{2}$ there would collapse the anchor to $c/2$ and destroy the curvature.

Steps (i)–(iii) and the assembly are machine-checked in `formal/Problems/Juggler/PaperBAssembly.lean` (`interpolant_step_i`, `interpolant_step_ii_constant`, `interpolant_assembly`), with the power identities $P^{1/8}P^{-5/4} = P^{-9/8}$ and $P^{1/12}P^{-9/8} = P^{-25/24}$ as hypotheses — at the cap 300, the constants 84.38 and 0.91, and the conclusion 170.6, which are the corrected figures above and not the ones the erratum replaced.

> *Erratum (what the machine check was checking).* Until this revision those three theorems carried the pre-correction chain — cap 186, constants 52.32 and 0.57, conclusion 106 — while the display above them carried the corrected one. The sentence claiming them was therefore true of a statement the erratum at this lemma had already replaced, which is the one thing the trust table of Section 1.1 exists to prevent. Regenerated; the superseded chain is retained under `interpolant_step_i_precorrection`, `interpolant_step_ii_precorrection` and `interpolant_assembly_precorrection`, as `step5b_c7_printed` retains the weaker $c_7 = 1/288$, so that a reader checking the erratum's list $186 \to 300$, $0.567 \to 0.907$, $106 \to 170.6$ finds both ends of it in Lean. `interpolant_gain` now records $219/170.6 = 1.2837$, the factor Appendix A.5 claims, beside the $219/106 > 2$ of the superseded chain.
>
> Nothing else moves: the threshold certificate was already solved against 170.6 (`p0_certificate.interpolant_error`), so P_0 and P_1 are unchanged. What was wrong was the corroboration, not the arithmetic.

These are all the differences between f'' and Λ. In particular Λ is not $(cF_{\mathrm{sm}})''$ for $F_{\mathrm{sm}} = \frac{3}{4}(\Delta_1 X)(\Delta_2 X)X^{-1/2}$. That composite differentiates the moving gaps and is a different function.

It remains to match leading monomials. Mean-value expansion gives $\delta_h(\nu) = 3h\,\xi^{1/2}$ with $\xi \in (\nu, \nu{+}2h)$, hence $\delta_h(\nu)\,(\nu{+}2h)^{-5/4} = 3h\,\nu^{-3/4}(1 + O(hP^{-1}))$ and

$$-\tfrac{9}{32}u\,\delta_{h_1}(\nu)\,(\nu{+}2h_1)^{-5/4} = -\tfrac{27}{32}uh_1\,\nu^{-3/4}(1 + O(h_1P^{-1})).$$

This is Φ'' of $a\nu^{5/4}$, because $a \cdot \frac{5}{4} \cdot \frac{1}{4} = -\frac{27}{32}$.

For the anchor, keep β_1, β_2 frozen and expand the chain rule at $c = \frac{3k}{4}\nu^{9/8}$, $X = \nu^{3/2}$. Then $G_F = \frac{3}{4}\beta_1\beta_2X^{-1/2} = \frac{3}{4}\beta_1\beta_2\nu^{-3/4}$ at leading order, $G_F' = -\frac{9}{16}\beta_1\beta_2\nu^{-7/4}$, and $G_F'' = \frac{63}{64}\beta_1\beta_2\nu^{-11/4}$, so

$$\begin{aligned}
c''G_F &= \tfrac{81}{1024}\,k\beta_1\beta_2\nu^{-13/8},\\
2c'G_F' &= -\tfrac{972}{1024}\,k\beta_1\beta_2\nu^{-13/8},\\
c\,G_F'' &= \tfrac{756}{1024}\,k\beta_1\beta_2\nu^{-13/8}.
\end{aligned}$$

The sum of the three is $-\frac{135}{1024}\,k\beta_1\beta_2\nu^{-13/8}$, which is $(cG_F)''$; the anchor is $c(G_F - J_F)$ and drops the first of them, leaving $-\frac{216}{1024} = -\frac{27}{128}$ (erratum above). (The term $\frac{1}{2}c''$ in Λ is $O(kP^{-7/8})$ and is absorbed in r.) Substituting the interpolating values $\tilde{\beta}_i = 3h_i\nu^{1/2}(1+O(hP^{-1}))$ converts the product $\beta_1\beta_2$ into $9h_1h_2\nu$, and the curvature becomes $-\frac{243}{128}kh_1h_2\nu^{-5/8}$. This is Φ'' of $b\nu^{11/8}$, because

$$b\cdot\tfrac{11}{8}\cdot\tfrac{3}{8} = -\tfrac{81}{22}\cdot\tfrac{33}{64} = -\tfrac{243}{128}.$$

The window mode is exactly $wX'' = \frac{3}{4}w\nu^{-1/2}$. Write $r = \Lambda - \Phi''$, so that Lemma 3.9 is applied to $\Phi + g$ with $g'' = r$. The remainder r is the sum of the mean-value errors $O(hP^{-1})$ on the two leading monomials and the leftover $\frac{1}{2}c''$. In the middle band of Step 5b one has $S \geq 0.56\,kh_1h_2P^{-5/8}$ (and $S \geq 0.56P^{-5/8}$ whenever $kh_1h_2 \geq 1$). The three ratios Lemma 3.9 needs are then, for $P \geq P_0$,

$$\frac{|r|}{S} \leq C_1P^{-1/4}, \qquad \frac{P|r'|}{S} \leq C_2P^{-1/4}, \qquad \frac{P^2|r''|}{S} \leq C_3P^{-1/4},$$

with absolute C_1, C_2, C_3 coming from $|\frac{1}{2}c''| \leq 0.053kP^{-7/8}$, $|\frac{1}{2}c'''| \leq 0.047kP^{-15/8}$, $|\frac{1}{2}c''''| \leq 0.044kP^{-23/8}$, and the wave remainders $O(uh^2\nu^{-7/4})$ which are $O(P^{-35/24})$ on the printed inventory. Each ratio is therefore $\leq \rho_0(E) = c_7/8 = 1/2304$ for $P \geq P_0$. (The worst standing cell is $kh_1h_2 = 1$; larger products only enlarge S.)

The range of λ_0 is the same leading term against $\beta_i \in (3h_iP^{1/2} - 1,\ 4.3h_iP^{1/2} + 1)$ and $\nu \in (P, 2P]$: the product $\beta_1\beta_2\nu^{-13/8}$ runs through $[2.92, 18.5]\,h_1h_2P^{-5/8}$, and $\frac{27}{128} \approx 0.211$ converts this to $[0.62, 3.90]$, opened to $[0.56, 4.2]$ for the $O(P^{-1/4})$ and the ± 1 in the β-bounds. □

The global monomial $\nu^{11/8}$ appears only after $\tilde{\beta}_1\tilde{\beta}_2 \sim \nu$ is substituted. On a single frozen run the curvature is a multiple of $\nu^{-13/8}$, i.e. the second derivative of a $\nu^{3/8}$ phase. Lemma 3.9 is applied to the *global* model Φ, whose second derivative tracks Λ; the local f'' stays within the interpolant error of that model, which is the only comparison the sublevel argument uses.

Architecture of the proof of Theorem 5.3

The proof is long, and its length is bookkeeping rather than depth: one reduction, one identity, one hard lemma, and a classification. This table is the map. Nothing in it is proved here; every row points at the statement that does the work.

#	component	input	output	consumed by
—	Lemma 5.1(i)	exact Taylor of $m^{9/4}$ at v	kernel phase = level-2 local floor defect, error $kR \ll kP^{-9/8}$	the definition of K_c
—	Lemma 5.1(ii)	$[\{A\}+\{B\} \geq 1] = \{A\}+\{B\}-\{A+B\}$	every carry is a difference of unit sawtooths	Step 3(3b), (3d)
—	Lemma 5.1(iii)	branch functions F_κ, offset $\lvert j\rvert \leq 2$	branch decomposition; smooth-per-branch, frozen floor J_F	Steps 3(3e), 5
—	Lemma 5.1(iv)	(i)–(iii)	master identity $\varphi_2 = M_1+M_2+M_3+M_4$	Step 2
1	Step 1	A-process twice, $H_1 = P^{1/48}$, $H_2 = P^{1/24}$	K_c reduced to T_2; (C3), (C4) hold, so (C1) does with room $P^{-1/48}$	Steps 2–6

#	component	input	output	consumed by
2	Step 2	Lemma 5.1(iv), estimate (E4)	M_1 deleted at cost $2.7P^{1/4}$; M_2, M_3, M_4 remain	Step 3
3	Step 3	Lemma 3.7 (sawtooth windows), Lemma 3.5 at $J_2 = P^{1/24}$ (carries)	every piece = anchor + waves + differenced waves + (D3)-smooth	Steps 4–5
4	Step 4	pieces with total wave frequency $t \neq 0$, $\lvert t\rvert \leq 3P^{1/24}$	$\ll \lvert t\rvert^{-1/6}P^{23/24+\varepsilon}$; $t = 0$ collapses exactly to (D1)	Step 6; $t = 0$ to Step 5
—	Lemma 5.2(i)	$u, h \geq 1$, $h \leq P^{1/8}$, $uh \leq P^{1/2}$, decoration ρ	V-bound $\ll ((uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + \cdots)P^{\varepsilon}$	Lemma 5.2(ii); Steps 3(3a), 4, 5b
—	Lemma 5.2(ii)	$\lvert q_d\rvert \leq 4P^{1/24}$, $0 < \lvert t\rvert \leq 3P^{1/24}$	$\ll \lvert t\rvert^{-1/6}P^{23/24+\varepsilon}$	Step 4
—	Lemma 5.2(iii)	up to two (D1) terms with $\lvert q'\rvert h' \leq P^{1/2}$, and $uhh' \geq 46$	the bound of (i) at the widened budget	Step 4, leftover modes
5a	Step 5a	offset branches $j \neq 0$	$\leq 1.8P^{23/24+\varepsilon}$; anchor curvature $\lambda_a \in [1.30, 1.43]k\lvert j\rvert P^{-1/8}$	Step 6
5b	Step 5b	zero-offset branches $j = 0$	$\ll P^{15/16+\varepsilon}$	Step 6
6	Step 6	all of the above	$\lvert T_2\rvert \ll P^{23/24+\varepsilon}$, hence $\lvert K_c\rvert \ll P^{1-1/96+\varepsilon}$	Theorem 6.1

Two structural facts the table is meant to make visible.

Where the exponent comes from. The saving $\frac{1}{96} = \frac{1}{4} \cdot \frac{1}{24}$ is inherited, not produced: it is the depth-2 strength $P^{23/24}$ of Lemma 5.2(ii) passed back through the two differencings of Step 1. Every other estimate in the proof is arranged to reach that exponent and no further; improving the wave bound improves δ proportionally, and improving anything else improves nothing.

Where the threshold comes from. P_0 is not distributed across the proof. Of the thirty-eight displayed inequalities, thirty-three hold from $2.8 \cdot 10^{10}$ or below. Five do not: the widened mode index of Lemma 5.2(iii) at $1.5 \cdot 10^{11}$, the Step 5b(a) q'' curvature ratio at $3.0 \cdot 10^{11}$, and the three Lemma 3.9 balance comparisons of Steps 5a and 5b, the largest of which — $W \leq c_7 S/2$ inside Step 5b's middle band — is P_0 itself (Appendix A). That single row is why Lemma 3.9's constant $c_7 = 1/232$ is proof-critical and why the middle band is the part of the argument to check first; the other thirty-three constants are bookkeeping, in the exact sense that moving any of them leaves P_0 where it is.

The mode-index row is bookkeeping in that sense too, and it very nearly was not. Read with the printed offset bound $\lvert j'\rvert \leq 3$ it stands at $7^{16} = 3.3 \cdot 10^{13}$, 8% under P_0 and the largest row not mentioning c_7 — which would make it, and not the q'' ratio, the ceiling on every improvement to the middle band. The erratum at Lemma 5.1(iii) takes it to $5^{16} = 1.53 \cdot 10^{11}$, back under the q'' row, and A.5's floor is the q'' ratio after all. Two constants had to be right for the printed 120 to be the right answer, and only one of them was.

Step 5b in detail. It is the only step that splits, and the only one where several curvature scales meet. Write $\lambda_0 \in [0.56, 4.2]\, kh_1h_2P^{-5/8}$ for the frozen anchor curvature and $\mu = 0.84 \max(uh_1, u'h_2)P^{-3/4}$ for the strongest differenced-wave scale present.

regime	condition	tool	output
anchor-dominant	$60\mu \le \lambda_0$	Lemma 3.3 per run at λ_0	$\le 1.7(kh_1h_2)^{1/2}P^{11/16} + 40(h_1h_2/k)^{1/2}P^{9/16}$
mode-dominant	$\mu \ge 60\lambda_0$	Lemma 5.2(i), anchor as decoration	$\ll ((uh_1)^{1/2}P^{5/8} + (h_1/u)^{1/2}P^{7/8} + P^{7/8})P^{\varepsilon}$
middle band	$\frac{1}{60} \le \mu/\lambda_0 \le 60$	Lemma 5.2b, then Lemma 3.9	transition set measured once, trivial bound; **binds** P_0

The middle band is the only place in the paper where the composite second derivative can cross zero inside a cell, and the only consumer of the three-term form of Lemma 3.9. It is also where condition (C5) of Lemma 5.2b is discharged: (C5) does *not* follow from (C3) and (C4), and the middle-band inequality $\mu \le 60\lambda_0$ is exactly what supplies it.

Theorem 5.3 (kernel cancellation). Let $c(n) = \frac{3k}{4}n^{9/8}$ with $1 \le k \le P^{1/24}$. Then

$$K_c(P) \ll P^{1-1/96+\varepsilon},$$

uniformly in k. The cancellation in Step 5a uses the exact ratio of this monomial's derivatives; a two-sided scale $c^{(r)} \asymp kP^{9/8-r}$ would not determine the composite.

Proof. Work on a dyadic block $n \sim P$, odd.

Step 1 (double Weyl differencing). Set $H_1 = P^{1/48}$, $H_2 = P^{1/24}$. The A-process recorded after Lemma 3.3, applied twice, gives

$$|K_c|^2 \le \frac{2P^2}{H_1} + \frac{4P}{H_1}\sum_{h_1=1}^{H_1}|T_1(h_1)|, \qquad |T_1|^2 \le \frac{2P^2}{H_2} + \frac{4P}{H_2}\sum_{h_2=1}^{H_2}|T_2(h_1,h_2)|,$$

with $T_2 = \sum_n e(\varphi_2)$, $\varphi_2 = \Delta\Delta(c\,\theta_2)$. For all $h_1 \le H_1$, $h_2 \le H_2$, $k \le P^{1/24}$: $kh_1h_2 \le P^{1/24+1/48+1/24} = P^{5/48} \le P^{1/8}$, so (C3) and (C4) hold, so (C1) does too, with room $P^{-1/48}$ on it (and (C4) is $P^{1/48}, P^{1/24} \le P^{1/24}$). We prove

$$|T_2| \ll P^{23/24+\varepsilon} \quad \text{uniformly in } h_1, h_2, k;$$

then $|T_1| \ll P^{1-1/48+\varepsilon}$ and $|K_c| \ll P^{1-1/96+\varepsilon}$, the three savings balancing exactly at the chosen H_1, H_2.

Step 2 (master identity; the trivial piece). By Lemma 5.1(iv), exactly,

$$\varphi_2 = \underbrace{(\Delta\Delta c)\,\theta_2}_{M_1} + \underbrace{(\Delta_2 c)(n{+}d_1)(\{W\} - \kappa_2)}_{M_2} + \underbrace{(\Delta_1 c)(n{+}d_2)(\{W'\} - \kappa_2')}_{M_3} + \underbrace{c_{11}(\{\Delta\Delta Y\} - \kappa'' - \Delta_2\kappa_2)}_{M_4}.$$

By (E4), $|M_1| \le 0.43\,kh_1h_2P^{-7/8}$, so $|e(M_1) - 1| \le 2.7\,kh_1h_2P^{-7/8}$ and deleting M_1 changes T_2 by at most $2.7\,kh_1h_2P^{1/8} \le 2.7\,P^{1/4}$ by (C1). No growing smooth part remains: every other bracket is bounded by 2.

Step 3 (the expansion inventory). The three remaining pieces are expanded as follows; all truncations, masses, and error costs are displayed here once and summed in Step 6.

(3a) *M_2, sawtooth part* $(\Delta_2 c)(n{+}d_1)\{W\}$. The coefficient $B(n) = \Delta_2 c(n{+}d_1) \in (1.68, 1.85)kh_2P^{1/8}$ drifts by $|B'| \le 0.22kh_2P^{-7/8}$ per step: freeze it on at most $2kh_2P^{1/4}{+}1$ windows on which it moves by $\le P^{-1/8}$ (residual cost $\sum_n |e((B{-}B_0)\{W\}) - 1| \le 6.3P^{-1/8} \cdot P = 6.3P^{7/8}$). Per window, Lemma 3.7 at the centre B_0 with $T = P^{1/2}/(2h_1)$, $J = P^{1/4}$ (hypothesis $T \ge 8(1 + |B|)$: $P^{1/2}/(2h_1) \ge \frac{1}{2}P^{23/48}$ and $8(1 + |B|) \le 15\,kh_2P^{1/8} \le 15P^{10/48}$, so the inequality holds for $P \ge P_0$): flat cost in total $\le 8(1{+}1.85kh_2P^{1/8}) \cdot 2h_1P^{1/2} \le$

$16h_1P^{1/2} + 30\,kh_1h_2P^{5/8} \le 46P^{3/4}$ by (C1); the majorant Δ_J-part costs $\le 2\cdot 4P/J = 8P^{3/4}$ plus J-mode sums at coefficients $\le 1/(J{+}1)$; modes $e(uW)$ with mass $\le C\log P$ and $u \le 1.85kh_2P^{1/8} + P^{1/2}/(2h_1)$, so that $uh_1 \le 1.85P^{1/4} + P^{1/2}/2 \le P^{1/2}$: every mode is a Lemma 5.2(i) object at shift $h_1 \le P^{1/48}$. Window boundaries cost $\le 2kh_2P^{1/4}\cdot 3.4P^{3/8} \le 7P^{17/24}$.

(3b) *M_2, carry part $(\Delta_2 c)(n{+}d_1)\kappa_2$.* Here $\kappa_2 \in \{0,1\}$ is an integer, so exactly $e(-(\Delta_2 c)\kappa_2) = 1 + \kappa_2(e(-\Delta_2 c) - 1)$: the large coefficient never multiplies a sawtooth inside an exponential. The factor $e(-\Delta_2 c(n{+}d_1))$ is a smooth phase with $|(\Delta_2 c)''| \le 2h_2 \sup|c'''| \le 0.19kh_2P^{-15/8}$: class (D3). The weight $\kappa_2 = \{Y\} + \{W\} - \{Y{+}W\}$ (Lemma 5.1(ii)) is expanded *additively* as real numbers, each unit sawtooth by finite Fourier (Lemma 3.5) at truncation $J_2 = P^{1/24}$, coefficients $\le 1/(\pi|q|)$, majorant cost $\le 4P/J_2 = 4P^{23/24}$ per layer plus J_2-mode averages. Since $Y{+}W = Y(n{+}d_1)$ exactly, the resulting modes are $e(qY(n))$, $e(qY(n{+}d_1))$ — Lemma 5.2(ii) objects — and $e(qW)$ — Lemma 5.2(i) objects.

(3c) *M_3, sawtooth and carry.* The same two expansions as (3a)–(3b), with the roles of the shifts exchanged: $B(n) = \Delta_1 c(n{+}d_2) \in (1.68, 1.85)\,kh_1P^{1/8}$ and Lemma 3.7 at $T = P^{1/2}/(2h_2)$. The window hypothesis is $T \ge \frac{1}{2}P^{11/24} = \frac{1}{2}P^{22/48}$ (since $h_2 \le P^{1/24}$) against $8(1+|B|) \le 15\,kh_1P^{1/8} \le 15P^{9/48}$, so it holds for $P \ge P_0$ with more room than (3a). Modes $e(uW')$ satisfy $uh_2 \le 1.85\,kh_1h_2P^{1/8} + P^{1/2}/2 \le P^{1/2}$ by (C1) and are Lemma 5.2(i) objects at shift $h_2 \le P^{1/24}$. Window boundaries cost $\le 2kh_1P^{1/4}\cdot 3.4P^{3/8} \le 7P^{11/16}$. The carry expansion is identical to (3b) at truncation $J_2 = P^{1/24}$, and produces $e(qY(n{+}d_2))$ and $e(qW')$.

(3d) *M_4, carries.* $\kappa'' = \{W\} + \{\Delta\Delta Y\} - \{W{+}\Delta\Delta Y\}$ with $W{+}\Delta\Delta Y = W(n{+}d_2)$ exactly, and $\Delta_2\kappa_2 = \kappa_2(n{+}d_2) - \kappa_2(n)$ with each κ_2 expanded as in (3b) at its base. The smooth factors $e(\mp c_{11})$ have $|c''| \le 0.11kP^{-7/8}$: class (D3). New mode species: $e(q\{\Delta\Delta Y\}\text{-content})$ — slow modes, treated in Step 5 — and $e(qY(n{+}d))$ for $d \in \{0, d_2, d_1{+}d_2\}$, $e(qW(n{+}d_2))$: Lemma 5.2 objects again.

(3e) *M_4, the anchor.* The factor $e(c_{11}\{\Delta\Delta Y\})$ is never expanded away. On the branch decomposition of Lemma 5.1(iii) (arcs absorbed by the exact shift device at $O(\log P)$ mass and $4P^{11/16}$ majorant, as in Theorem 4.4, Step 4), $\{\Delta\Delta Y\} = G - J_F$ with $G = F_\kappa(X{-}\theta)$ and $J_F = \lfloor G\rfloor$ frozen on the runs of (E6). The anchor phase $c_{11}(G - J_F)$ is present in **every** final piece; its curvature scale (E6) is what orders the whole classification of Step 5.

After (3a)–(3e), every final piece of T_2 has the exact form

$$e\Big(\underbrace{c_{11}(G - J_F)}_{\text{anchor}} + \underbrace{\textstyle\sum_i q_iY(n{+}e_i)}_{\text{waves},\ e_i\in\mathcal{D}} + \underbrace{u\,W\text{- and } u'W'\text{-modes}}_{\text{differenced waves}} + \underbrace{\text{slow modes} + \varphi}_{\text{(D3)-smooth}}\Big)$$

with at most one mode from each expansion layer (three layers), and piece weights whose total mass is $O(\log^3 P)$ beyond the displayed majorant and flat costs.

Step 4 (pieces with wave content: Lemma 5.2(ii)). Let $t = \sum_i q_i$ be the total wave frequency, $|t| \le 3J_2 \le 3P^{1/24}$, inside the coefficient budget of Lemma 5.2(ii).

If $t \ne 0$: split the phase of the piece into the printed Lemma 5.2(ii) content and the leftover first-differenced modes.

Printed content. The Y-wave, the frozen-floor anchor, and the (D3) content are exactly an instance of Lemma 5.2(ii). Claims A–H give $\ll |t|^{-1/6}P^{23/24+\varepsilon}$ for a piece that carries only that content. The carry-expansion modes of (3b)–(3d) have $|q| \le J_2 = P^{1/24}$ and become printed (D1) decorations after the differencing of Claim C ($|q'| \le P^{1/24} \le 4P^{1/24}$, $h' \in \{h_1, h_2\} \le P^{1/24}$ by (C4)); they are already covered by Claims E–H.

Leftover modes. The same piece may also carry first-differenced modes uW and $u'W'$ from Steps (3a) and (3c). Those modes are not in the printed statement of (ii), and their coefficients need not obey $|q'| \le 4P^{1/24}$. After the A-process of Claim C they become

$$u\,\Delta_{2h_3}W = u\,\Delta_{2h_3}\Delta_{d_1}Y, \qquad u'\,\Delta_{2h_3}W' = u'\,\Delta_{2h_3}\Delta_{d_2}Y,$$

which have the algebraic shape of (D1) decorations with coefficients $q' = u, u'$ and second shifts $h' = h_1, h_2$, and which Lemma 5.2(iii) covers. From (3a), $u \le 1.85kh_2P^{1/8} + P^{1/2}/(2h_1)$ and $uh_1 \le P^{1/2}$; likewise $u'h_2 \le P^{1/2}$ from (3c). The second-shift bounds $h_1, h_2 \le P^{1/24}$ still hold, so both satisfy (D1′), and Lemma

5.2(iii) applies with main coefficient t and shift h_3: its good-shift condition reads $th_3h_i \geq 46$, and it supplies the Stage-6 comparisons, the $(s2)$ window inventory for the enlarged sawtooth, and the mode accounting.

The *bad* set for the u-mode is $h_3 < 46/(th_1)$. It contains at most 46 positive integers (since $t \geq 1$ and $h_1 \geq 1$). The bad set for the u'-mode likewise contains at most 46 integers. Their union has at most 92 elements. On those h_3 use the trivial bound $|V_{h_3}| \leq P$. The recorded A-process charges

$$\frac{4P}{H_3} \cdot 92 \cdot P = 368\,\frac{P^2}{H_3} \leq 368\,t^{-1/3}P^{23/12},$$

which is a constant multiple of the target of Claim G (the same first-term comparison $P^2/H_3 \leq t^{-1/3}P^{23/12}$). On the complementary good set Lemma 5.2(iii) gives the bound of (i), and Claims F–H average it exactly as for a printed decoration.

Thus every $t \neq 0$ piece, with or without leftover first-differenced modes, is $\ll |t|^{-1/6}P^{23/24+\varepsilon}$. The weight sum is

$$\sum_{t\neq 0}\ \sum_{q_1+q_2+q_3=t} \frac{1}{\pi^3 \max(1,|q_1|)\max(1,|q_2|)\max(1,|q_3|)}\,|t|^{-1/6} \ll \sum_{t\geq 1}\frac{\log^2(2+t)}{t^{7/6}} \ll 1,$$

so the total wave-piece contribution is $\ll P^{23/24+\varepsilon}$.

The saving in t is what makes that converge, and it is the only thing that does: the inner weight sum is $\asymp \log^2(2+t)/t$, so with no t-saving at all the total reads $\sum \log^2(2+t)/t$, which diverges. The exponent $\frac{1}{6}$ is not consumed, though. Any fixed $\delta > 0$ in its place leaves $\sum_t \log^2(2+t)\,t^{-1-\delta} < \infty$, so this step goes through with $|t|^{-\delta}P^{23/24+\varepsilon}$ for any positive δ, and a different route to the wave pieces need only produce *some* power saving in t, however small. The contrast with the other exponent is exact: $P^{23/24}$ is consumed in full, being where $\frac{1}{96} = \frac{1}{4}\cdot\frac{1}{24}$ comes from, while $\frac{1}{6}$ is spent on convergence and nothing else.

If $t = 0$: the wave content collapses exactly to differenced waves — e.g. $q(Y(n+d_2) - Y(n)) = q\,W'(n)$ — i.e. to (D1)-type resonant decorations of the remaining piece; these are handled in Step 5. This exactness is the reason no near-resonant analysis is ever needed for the wave frequencies: the resonant combination is an identity, not an approximation.

Step 5 (pieces without wave content: the anchor classes). The remaining pieces carry the anchor, at most two differenced-wave modes $u\,W$, $u'W'$ (including the resonant remnants of Step 4, which by Lemma 5.1(iii) are smooth-per-branch with θ-coefficients $O(P^{-5/24})$), slow modes, and (D3)-smooth content. Split by the anchor branch type.

(5a) Offset branches ($j \neq 0$). The anchor curvature is, by (E6) and the window-centre composite,

$$\lambda_a = \tfrac{729}{512}\,k|j|\,n^{-1/8}\,(1 + O(P^{-1/8})) \ \in\ [1.30,\,1.43]\,k|j|P^{-1/8},$$

where $\frac{729}{512} = \frac{945}{512} - \frac{27}{64}$, by the following algebra. The smooth part of the anchor on an offset branch is $cF = \frac{3k}{4}\nu^{9/8}\cdot\frac{3}{2}j(m+\beta_1+\beta_2+\xi_1)^{1/2} = \frac{9}{8}\,kj\,\nu^{15/8}\,(1+O(P^{-1/4}))$ (Lemma 5.1(iii) and $m = \nu^{3/2}-\theta$), with second derivative $\frac{9}{8}\cdot\frac{15}{8}\cdot\frac{7}{8}\,kj\,\nu^{-1/8} = \frac{945}{512}\,kj\,\nu^{-1/8}$ up to the same relative error. The window-centre mode carries $u = -B(n_0)$, where $B = c\,\partial_m F = \frac{3k}{4}\nu^{9/8}\cdot\frac{3}{4}j\,m^{-1/2} = \frac{9}{16}\,kj\,\nu^{3/8}\,(1+O(P^{-1/4}))$ is the θ-sawtooth coefficient of the anchor (the θ-linear term of $cF(X-\theta)$), so its curvature contribution is $uX'' = -\frac{9}{16}\cdot\frac{3}{4}\,kj\,\nu^{-1/8} = -\frac{27}{64}\,kj\,\nu^{-1/8}$. The two terms have ratio $\frac{945}{512} : \frac{27}{64} = 4.375$, so the composite is single-signed at the displayed scale. Every competitor is dominated at a displayed margin: differenced-wave modes $\leq 0.84\,uh_1P^{-3/4} \leq 0.51P^{-1/4}$ (since $uh_1 \leq 0.6P^{1/2}$), ratio $\leq 0.34P^{-1/8}$ against $\lambda_a \geq 1.30P^{-1/8}$; resonant (D1) content $\leq 6\,|q'|P^{-5/4}$ with $|q'| \leq 4P^{1/24}$, ratio $\leq 20P^{1/24-5/4+1/8} = 20P^{-13/12}$; slow modes $\leq 3J_2|j|P^{-5/4}$, ratio $\leq 8P^{1/24-9/8}$; (D3) content, ratio $\leq 3h_1h_2P^{-1/2} \leq P^{-1/4}$. The θ-sawtooth of the anchor has coefficient $\frac{9}{16}k|j|P^{3/8}$-scale with per-step drift $\frac{27}{128}k|j|\nu^{-5/8} \leq 0.22k|j|P^{-5/8} < 1$. Its total drift over the block is $\frac{9}{16}(2^{3/8}-1)k|j|P^{3/8} \leq 0.17k|j|P^{3/8}$, so there are at most $0.17k|j|P^{3/8}+1$ windows, each of length at least $1/B'(P) = \frac{128}{27}P^{5/8}/(k|j|) \geq 4.7P^{5/8}/(k|j|)$, at window-boundary cost $\leq 0.17k|j|P^{3/8}\cdot 0.88(k|j|)^{-1/2}P^{1/16} \leq 0.15(k|j|)^{1/2}P^{7/16}$. Off the collision band, window modes w are dominated at margin ≥ 4 either way and ride along or are estimated by Lemma 3.3

at their own scale with $1/|w{+}B_0|$ weights ($\ll P^{7/8}\log P$ in total); on the collision band $|wX''|\in[\frac14,4]\lambda_a$, Lemma 3.8 with $(\alpha,\beta)=(\frac{15}{8},\frac32)$ and $M\in[1.30,\,5.75]\,k|j|P^{-1/8}$ gives $\le C_E(|I_w|M^{1/2}+M^{-1/2}+(P/M)^{1/3})$ per window; summed over the at most $0.17k|j|P^{3/8}{+}1$ windows with the $O(\log P)$ band mass,

$$\sum_w |I_w|M^{1/2}=P\,M^{1/2}\le 2.4\,(k|j|)^{1/2}P^{15/16},\qquad \sum_w M^{-1/2}\le 0.15\,(k|j|)^{1/2}P^{7/16},$$

$$\sum_w (P/M)^{1/3}\le 0.17k|j|P^{3/8}\cdot 0.92\,(k|j|)^{-1/3}P^{3/8}=0.16\,(k|j|)^{2/3}P^{3/4}\le 0.33\,P^{1/36+3/4},$$

so the band total is $\ll (k|j|)^{1/2}P^{15/16}\log P$. The main estimate is Lemma 3.3 per frozen run at scale λ_a: run lengths $\ge \frac{1}{22}P^{1/4}/(|j|{+}1)$ (Lemma 5.1(iii)), and $\lambda_a^{-1/2}\le 0.88(k|j|)^{-1/2}P^{1/16}\ll$ run length, so

$$\sum_{\text{runs}}(\ell\lambda_a^{1/2}+\lambda_a^{-1/2})\le 1.2\,(k|j|)^{1/2}P^{15/16}+20\,(|j|{+}1)|j|^{-1/2}k^{-1/2}P^{13/16}.$$

Summed over the eight carry branches and $|j|\le 2$ with the $O(\log^3 P)$ piece masses: $\ll (k)^{1/2}P^{15/16+\varepsilon}\le 1.8\,P^{1/48}P^{15/16+\varepsilon}=1.8\,P^{23/24+\varepsilon}$ at $k\le P^{1/24}$ — the absorbed $(k|j|)^{1/2}$ loss, exactly at the bottleneck.

(5b) Zero-offset branches ($j=0$). Lemma 5.2b gives the local frozen curvature $\lambda_0\in[0.56,\,4.2]\,kh_1h_2P^{-5/8}$ (leading coefficient $\frac{27}{128}$ in $k\beta_1\beta_2\nu^{-13/8}$, converted by $\beta_i\asymp h_iP^{1/2}$). Runs of length $\ge \frac{1}{22}P^{3/4}/(h_1h_2)$. Let $\mu=0.84\max(uh_1,u'h_2)P^{-3/4}$ be the strongest differenced-wave scale present. Three regimes.

- *Anchor-dominant* ($60\mu\le\lambda_0$): Lemma 3.3 per run at λ_0, all else dominated at margin ≥ 20: $\le 1.7(kh_1h_2)^{1/2}P^{11/16}+40\,(h_1h_2/k)^{1/2}P^{9/16}$.
- *Mode-dominant* ($\mu\ge 60\lambda_0$, i.e. $uh_1\ge 60\,kh_1h_2P^{1/8}$-form): Lemma 5.2(i) with the undifferenced anchor as decoration: its run boundaries number $\le 22h_1h_2P^{1/4}\le 22P^{5/16}$, which uses the Theorem 5.3 shift caps $h_1\le H_1=P^{1/48}$, $h_2\le H_2=P^{1/24}$ (so $h_1h_2\le P^{1/16}$) and *not* (C4) alone, under which $h_1h_2\le P^{1/12}$ would give only $22P^{1/3}$, cost $\le 22P^{5/16}\cdot 3.4(uh_1)^{-1/2}P^{3/8}\le 75P^{11/16}$; its smooth part is dominated at margin ≥ 20 by hypothesis. On a zero-offset branch the anchor's θ-coefficient is $B=c\,\partial_mF=-\frac38 c\,\beta_1\beta_2\,m^{-3/2}=-\frac{9}{32}k\beta_1\beta_2\nu^{-9/8}(1{+}O(P^{-1/4}))$. With $\beta_i\in(3h_iP^{1/2}{-}1,\,4.3h_iP^{1/2}{+}1)$ this is $|B|\le 5.3\,kh_1h_2P^{-1/8}\le 5.3$ by (C1), opened to $|B|\le 6$. The sawtooth is of constant size, not sub-unit. Lemma 3.7 still applies in a single window at $T=P^{1/2}$, because $T\ge 8(1+|B|)$ is $P^{1/2}\ge 56$ for $P\ge P_0$; there is no large-B window inventory. (The offset-scale amplitude $kh_1h_2P^{3/8}$ does not occur at $j=0$.) Centre modes w are treated as in Lemma 5.2(i), Stage 5. Total $\ll ((uh_1)^{1/2}P^{5/8}+(h_1/u)^{1/2}P^{7/8}+P^{7/8})P^{\varepsilon}$.
- *Middle band* ($\frac{1}{60}\le\mu/\lambda_0\le 60$; coefficient mass $O(1)$ per layer): here up to three curvature scales meet, and this is the one place in the paper where the composite second derivative can cross zero *inside* cells. Since the cells are short ($\asymp P^{1/2}/h$) while the third-derivative scale is tiny, no inverse-power van der Corput term may be summed per cell; instead the transition set is measured once, against a global smooth model, and receives the trivial bound (Lemma 3.9).

  Inventory. In this band $0.84\,uh_1P^{-3/4}=\mu\le 60\lambda_0\le 252\,kh_1h_2P^{-5/8}$ forces $u\le 300\,kh_2P^{1/8}\le 300\,P^{5/24}$, and likewise $u'\le 300\,kh_1P^{1/8}\le 300\,P^{5/24}$. The phase's second derivative f'' is smooth on the common refinement of the gap cells of both shifts (at most $1.5(h_1{+}h_2)P^{1/2}+2\le 3.1P^{13/24}$), the anchor runs ($\le 22h_1h_2P^{1/4}\le 22P^{3/8}$), and the sawtooth windows already counted: in total at most $N\le 3.5P^{13/24}$ pieces, for $P\ge P_0$. On each piece,

  $$f''=-\tfrac{9}{32}\Big(uG_1(\nu{+}2h_1)^{-5/4}+u'G_2(\nu{+}2h_2)^{-5/4}\Big)+\big(c_{11}(G_F-J_F)\big)''+wX''+O(\rho_0\text{-small}),$$

  with G_1,G_2,J_F (and the β_i inside G_F) frozen integers. This is the local f'' of Lemma 5.2b.

  Interpolant. Invoke Lemma 5.2b: the frozen-shape interpolant Λ (values of Δ_iX substituted, not differentiated) satisfies $|f''-\Lambda|\le 170.6P^{-25/24}+0.11P^{-5/6}=:E$ and $\Lambda=\Phi''+r$ with $a=-\frac{27}{10}(uh_1+u'h_2)$ and $b=-\frac{81}{22}\,kh_1h_2$. The raised threshold $W=V+E$ below is read from that two-term majorant.

The scale is

$$S = \max(|uh_1 + u'h_2|P^{-3/4},\ kh_1h_2P^{-5/8},\ |w|P^{-1/2}),$$

and the middle-band constraints give $0.56P^{-5/8} \le S \le 610\,P^{-1/2}$: the lower bound is λ_0 when $kh_1h_2 \ge 1$, and for the upper bound, $|uh_1{+}u'h_2| \le 2\max(uh_1, u'h_2) = 2\mu P^{3/4}/0.84$ with $\mu \le 60\lambda_0 \le 60 \cdot 4.2\,kh_1h_2P^{-5/8}$, so the first entry of S is at most $2 \cdot 252/0.84 = 600\,kh_1h_2P^{-5/8} \le 600\,P^{-1/2}$ by $kh_1h_2 \le P^{1/8}$ from (C1) (opened to 610); the third entry is the collision-band restriction $|wX''| \ll P^{-1/2}$. (The factor 2 from the sum and the 1/0.84 from the definition of μ are both needed: 252 alone would give 300, which does not cover the sum.) If $w = 0$, drop the third term of Φ and apply Lemma 3.8 (or Lemma 3.3 if only one of a, b is present).

Splitting. Choose $V := \frac{1}{12}S^{1/2}P^{-11/24}$, so that $V/S \le 0.12P^{-7/48}$ and, at the lower end $S \ge 0.56P^{-5/8}$, $V \ge 0.062P^{-37/48}$. Lemma 3.9 is applied not at V but at the *raised* threshold

$$W := V + E, \qquad E := \sup|f'' - \Lambda| \le 170.6P^{-25/24} + 0.11P^{-5/6},$$

whose only hypothesis is $W \le c_7S/2$. Off $\Omega_W = \{|\Lambda| \le W\}$ one has $|\Lambda| \ge W > E$, so f'' carries the sign of Λ there and $|f''| \ge W - E = V$ (Lean `sublevel_raised_threshold`). No separate comparison $V \ge 10|f'' - \Lambda|$ is needed. That factor 10 was margin, and demanding it was actively harmful: it forced V to be *large* exactly where $V \le c_7S/2$ wants it small, so the two comparisons pulled against each other and pinned κ near $\frac{1}{3}$. At the exact $c_7 = 1/232$ of Lemma 3.9 (the ℓ^∞ operator norm, Lean `step5b_curvature_norm`), $W \le c_7S/2$ holds from $P \ge 3.6 \cdot 10^{13}$; the interpolant error alone would allow $4.1 \cdot 10^{12}$, and the balance between the two is what now fixes κ. At that threshold V and E take 55% and 45% of the budget $c_7S/2$, and E itself splits 70:30 between its two terms. Both halves of E are therefore load-bearing — which they were not under the old comparison, where $V \ge 10E$ made E irrelevant to the binding inequality and the 219 cost nothing.

Writing $V = \kappa S^{1/2}P^{-11/24}$: the piece-boundary cost carries $\kappa^{-1/2}$, while $W \le c_7S/2$ pushes the threshold up as κ grows. With the raised threshold these no longer conflict — both the threshold and the non-vacuity point P_1 of Appendix A.5 fall together as κ decreases, until near $\kappa = \frac{1}{12}$ the boundary term turns P_1 around — and the turning point is where it is for geometric reasons, not numerical ones: correcting λ_0 by $\frac{8}{5}$ moved P_0 and P_1 but left the optimum at $\frac{1}{12}$. We take that value: it gives $P_0 = 3.6 \cdot 10^{13}$ and $P_1 = 9.9 \cdot 10^{18}$, against $1.2 \cdot 10^{16}$ and $2.8 \cdot 10^{20}$ at $\kappa = \frac{1}{3}$, and $2.7 \cdot 10^{22}$ at $\kappa = 3$. The exponent 89/96 does not depend on κ. Until $W \le c_7S/2$ holds, a three-term zero of Φ'' can keep Ω_W of length $\Theta(P)$ on a dyadic block. The length bound below is that of Lemma 3.9, not a claim that the sublevel is short at small P; interval counts remain $O_E(1)$. By Lemma 3.9 the set $\Omega_W = \{\nu : |\Lambda(\nu)| \le W\}$ is a union of at most $C(E)$ intervals of total length $\le C(E)\,P(W/S)^{1/2} \le 0.46\,C(E)\,P^{89/96}$ (using $W/S \le 0.21P^{-7/48}$ for $P \ge P_0$), and on its complement f'' is single-signed per interval with $V \le |f''| \le C(E)S + E$. The three costs:

$$\text{transition (trivial):} \quad \le 0.46\,C(E)\,P^{89/96};$$

$$\text{piece boundaries:} \quad \le (N{+}C(E))\,V^{-1/2} \le 3.5P^{13/24} \cdot 4.01P^{37/96} + O_E(P^{37/96}) \le 14.1\,P^{89/96};$$

$$\text{good pieces (Lemma 3.3):} \quad \sum \ell\,(1.1\,C(E)S)^{1/2} \le C'(E)\,P \cdot (610)^{1/2}P^{-1/4} \le 26\,C'(E)\,P^{3/4}.$$

The middle band therefore totals $\le C(E)\,P^{89/96}\log P = O_E(P^{89/96+\varepsilon})$, which is the form Step 6 uses. The sharper reading $\le P^{15/16}$ would need $C(E)\log P \le P^{1/96}$, i.e. $\ln P \ge 96\ln\ln P$, which first holds near $P = 10^{274}$; at $P_0 = 3.6 \cdot 10^{13}$ one has $\ln P = 31.2$ against $P^{1/96} = 1.38$, so that reading is **not** available at P_0. Nothing downstream depends on it: Step 6 carries P^ε and $\frac{89}{96} < \frac{15}{16}$.

In all three regimes the $j = 0$ pieces total $\ll P^{15/16+\varepsilon} \ll P^{23/24}$.

Step 6 (assembly). Additive costs: the M_1 deletion ($2.7P^{1/4}$); majorants (≤ 3 sawtooth layers at $4P/J_2 = 4P^{23/24}$ each, plus $8P^{3/4}$, $46P^{3/4}$, and the displayed window residuals $\le 7P^{7/8}$); flat costs ($\le 46P^{3/4}$ per layer). Multiplicative costs: mode masses over at most three expansion layers plus the shift devices,

$O(\log^3 P) = P^{\varepsilon}$. Piece totals: Step 4, $\ll P^{23/24+\varepsilon}$; Step 5a, $\leq 1.8P^{23/24+\varepsilon}$; Step 5b, $\ll P^{15/16+\varepsilon}$; slow modes and (D3) remnants, $\ll P^{7/8+\varepsilon}$. Hence $|T_2| \ll P^{23/24+\varepsilon}$, and by Step 1

$$|T_1|^2 \leq 2P^{2-1/24} + CP^{1+23/24+\varepsilon} \Rightarrow |T_1| \ll P^{1-1/48+\varepsilon}, \qquad |K_c|^2 \leq 2P^{2-1/48} + CP^{2-1/48+\varepsilon},$$

so $|K_c| \ll P^{1-1/96+\varepsilon}$. Exponents deliberately unoptimized. □

Two remarks on the constants. First, the k-range is sharp for this assembly: the class-(5a) loss $(k|j|)^{1/2}P^{15/16}$ meets the wave-piece bottleneck $P^{23/24}$ exactly at $k = P^{1/24}$, which is why the theorem is uniform on the original range with no k-explicit statement needed. Second, the exponent $\frac{1}{96} = \frac{1}{4} \cdot \frac{1}{24}$ traces entirely to the depth-2 strength $P^{23/24}$ of the exact level-2 waves (Lemma 5.2(ii)); any improvement of the wave bound improves δ proportionally.

Localization of the kernel theorem

Theorem 5.3 is stated on a dyadic block. This subsection asks how far its proof localizes. An earlier application identified the relevant companion intervals as $I(m') = [m'^{32/9}, (m'+1)^{32/9})$, of length $\asymp P^{23/32}$. Those are the *OOEEE* intervals of Corollary 4.13, whose landing exponent is 9/32, not the intervals for the proposed *OOOEEE* and *OOEOEE* productions. The latter words have landing exponent 27/64, hence broad inverse scale $P^{37/64}$, below the threshold proved here. The result below is therefore a standalone short-interval theorem, not a production theorem for [24]. As in Section 3.5 the proof is the proof already given, with the number of summands in place of P wherever the number of summands enters, and the content is the inventory of terms that do *not* scale. Here one of them dominates the rest, and it is not the one the architecture suggests.

Two kinds of cost. Write a cost on an interval $I \subseteq (P, 2P]$ of length Y as $cY^{\alpha}P^{\beta}$; on the block it is $cP^{\alpha+\beta}$, the exponent Steps 2–6 print. Every displayed cost is of one of two kinds.

Proportional ($\alpha = 1$): a per-summand cost — a flat cost $8(1+|B|)N/T$, a majorant $4N/J$, a deletion, a residual — or a sum over gap cells, sawtooth windows, frozen runs or middle-band pieces. Those objects partition the interval, so their number is $|I|$ times a density plus $O(1)$, and a printed exponent e becomes YP^{e-1}.

A unit ($\alpha = 0$): the $O(1)$ additive term in such a count, paying one unit cost. A unit is a van der Corput $\lambda^{-1/2}$, or the transition term of Lemma 3.8; never a length.

Lemma 5.4 (transfer). Let $H = P^{\eta} \leq Y$ and suppose $|T_h| \leq C\big(YP^{p} + P^{a}\big)$ for every $h \leq H$ and every sub-interval, with the dyadic argument balancing $2P^2/H$ against $(4P/H)\sum_h |T_h|$ at the printed p. Then

$$|S| \;\leq\; C'\Big(YP^{p'} + P^{(y+a)/2}\Big), \qquad Y = P^{y},$$

with p' the printed half of the dyadic saving.

Proof. The A-process on I reads $|S|^2 \leq 2Y^2/H + (4Y/H)\sum_{h\leq H}|T_h| \leq 2Y^2P^{-\eta} + 4CY^2P^{p} + 4CYP^{a}$, and $\sqrt{x+y+z} \leq \sqrt{x} + \sqrt{y} + \sqrt{z}$. □

The first two terms are the dyadic balance, unchanged because both carry Y^2; the third is new. A unit is therefore not squared away but pushed to the geometric mean with the length, and three A-processes stand between a unit and K_c — Claim C inside Lemma 5.2(ii), then the two of Step 1 — so P^{a} arrives as $P^{(7y+a)/8}$.

The balance is against an average. The hypothesis of Lemma 5.4 is the one thing localization could break, and it does not. In Claim C the balance at $H_3 = t^{1/3}P^{1/12}$ is between $2P^2/H_3 = 2t^{-1/3}P^{23/12}$ and the h_3-*average* of Lemma 5.2(i), not its maximum: with $\sum_{h\leq H} h^{1/2} \sim \frac{2}{3}H^{3/2}$, the second printed term contributes

$$\frac{4P}{H_3} \cdot t^{-1/2} \cdot \tfrac{2}{3}H_3^{3/2} \cdot P^{7/8} = \tfrac{8}{3}\,t^{-1/3}P^{23/12},$$

the same quantity, while the first, third, fourth and fifth printed terms give $P^{5/3}$, $P^{15/8}$, $P^{15/8}$ and $P^{61/32}$, all strictly below $P^{23/12}$. So H_3 balances against one term and that term is proportional; an average of proportional quantities is proportional, both sides carry $(Y/P)^2$ on I, and H_3 is still the balancing choice. The same holds at H_1 and H_2. No parameter is re-optimized and the exponent $\frac{1}{96}$ is untouched.

The inventory of units. Collecting every count-times-unit cost in Lemma 5.2(i) and in Steps 2–6:

site	unit	exponent
Lemma 5.2(i), Stage 3(s2) window boundaries	$(0.35uh)^{-1/2}P^{3/8}$, $uh > P^{3/16}$	9/32
Lemma 5.2(i), Stage 4 gap-cell boundaries	$1.69(uh)^{-1/2}P^{3/8}$	3/8
Lemma 5.2(i), Stage 5 inverse-root term	$0.48(uh)^{-1/2}P^{3/8}$	9/32
Lemma 5.2(i), Stage 5 transition term	$0.611(uh)^{-1/3}P^{7/12}$, $uh > P^{3/16}$	**25/48**
Lemma 5.2(i), Stage 5 mode ends	$\lambda^{-1/2} \ll P^{1/4}$	1/4
Steps 3a, 3c window boundaries	$3.4P^{3/8}$	3/8
Step 5a run boundaries	$0.88(k\|j\|)^{-1/2}P^{1/16}$	1/16
Step 5a transition term	$0.92(k\|j\|)^{-1/3}P^{3/8}$	3/8
Step 5b anchor-dominant run boundaries	$1.34(kh_1h_2)^{-1/2}P^{5/16}$	5/16
Step 5b mode-dominant boundaries	$3.4(uh_1)^{-1/2}P^{3/8}$	3/8
Step 5b middle-band piece boundaries	$4.01P^{37/96}$	37/96

The largest is the Stage 5 transition term, and it is the one unit that is not an inverse root. Lemma 3.8's third term carries no window-length factor — which, as Stage 5 says, is exactly why it sums — so one window's worth of it survives as a unit, at $P^{7/12}$ before the regime-(s2) constraint $uh > P^{3/16}$ is used and $P^{25/48}$ after. That single cost, and nothing about the architecture, sets the threshold below.

Theorem 5.5 (localized kernel cancellation). Let $c(n) = \frac{3k}{4}n^{9/8}$ with $1 \le k \le P^{1/24}$, let $\delta > 0$, and let g be a slow twist. For every interval $I \subseteq (P, 2P]$ of length $Y \ge P^{29/48+\delta}$,

$$\Big|\sum_{\substack{n\in I\\ n\text{ odd}}} e(c(n)\{\lfloor n^{3/2}\rfloor^{3/2}\} + g(n))\Big| \ll Y\,P^{-1/96+\varepsilon},$$

uniformly in k, in g, and in the position of I.

Proof. The twist first. By Lemma 4.10 applied after the h_1 differencing of Step 1, with $\mathrm{TV}(\Delta_{2h_1}g) \le 2h_1|I|\sup|g''| \le 0.26\,YP^{1/48+1/24-23/16} = 0.26\,YP^{-11/8} \le 0.26\,P^{-3/8}$, the twist costs a factor $1 + o(1)$ and every sum below is untwisted; the maximum over initial sub-intervals it introduces is harmless because the bound proved is nondecreasing in $|I|$.

Now run Steps 2–6 over I. Each proportional cost of printed exponent e becomes YP^{e-1}, so the largest, the Step 3b majorant $4N/J_2$ and the three costs of Steps 4 and 5a that meet it, give $YP^{-1/24}$; each unit contributes its exponent, the largest being $P^{25/48}$ through Lemma 5.2(i). Lemma 5.2(ii) follows by Lemma 5.4 at H_3, and T_1, K_c by Lemma 5.4 at H_2, H_1. Hence

$$|K_c^g(I)| \ll Y\,P^{-1/96+\varepsilon} + P^{(7y+25/48)/8},$$

and the second term is at most the first exactly when $y \ge \frac{25}{48} + \frac{8}{96} = \frac{29}{48}$, with equality at the endpoint; $\delta > 0$ makes the inequality strict. □

At the admissible reference length $y = \frac{23}{32}$ the unit chain ends at $P^{533/768}$ against a target $P^{17/24} = P^{544/768}$, a margin of $P^{11/768}$. This is arithmetic inside the theorem's range, not the scale of the two proposed productions.

Three remarks. First, the theorem does **not** reach $P^{1/2}$, where Section 3.5 leaves Theorems 4.11 and 4.12: $\frac{29}{48} > \frac{1}{2}$, and the obstruction is the transition term rather than the nesting. The proposed productions lie at the shorter broad scale $P^{37/64}$. Formally substituting $y = 37/64$ into the two-term bookkeeping bound leaves a saving $(37/64 - 25/48)/8 = 11/1536$ below the trivial length, but that substitution lies outside the theorem just stated and does not transport the exact floor and parity restrictions of a production. It is a possible salvage proof obligation, not a result of this paper. A shorter localization would need that transport justified, with Lemma 3.8's third term retained rather than silently absorbed. Second, the loss is entirely in the admissible length; the saving is the printed $P^{-1/96}$ and the constants are the printed constants, because no step was replaced. Third, the earlier estimate in Section 8 that the per-window absolute costs are "at most $P^{7/16}$" was the depth-≤ 3 figure carried over; the kernel's largest unit is $P^{25/48}$, and $\frac{25}{48} > \frac{7}{16}$.

The bookkeeping is arithmetic and is recorded as such: the two cost tables, the transfer recursion, the threshold, the balance above and the twist are in `research.juggler_sequence.localized_kernel`, which also checks that the tables' maxima are the printed $P^{15/16}$ of Lemma 5.2(i) and $P^{23/24}$ of T_2, that setting $Y = P$ returns $P^{23/24}$, $P^{47/48}$, $P^{95/96}$, and that every P-exponent at or above $P^{1/4}$ displayed in either proof is either one of the tabulated costs or is named as a count, a length, a parameter, a hypothesis or an intermediate step. That last check is the one worth stating plainly: the completeness of the inventory is what the theorem rests on, and a missing unit larger than $P^{25/48}$ would raise the threshold. It would not touch the exponent.

5. Depth-four equidistribution

Theorem 6.1 (the OOO* splits; complete depth-4 parity equidistribution for odd-rooted itineraries). For $w \in \{OOOE, OOOO\}$,

$$\#\{n \le N \text{ odd}: \ \mathrm{word}_4(n) = w\} = \tfrac{N}{16} + O(N^{1-1/96+\varepsilon}).$$

Hence every length-4 itinerary word class of *odd* starts — the eight words with first letter O — satisfies $\#\{n \le N\} = 2^{-4}N + O(N^{1-\delta_w})$ with explicit $\delta_w > 0$: depth-4 parity equidistribution over odd starts is complete. (E-rooted words, i.e. even starts, are a different and easier problem — one square root drops the state to scale $\sqrt{N}$ — and are not treated in this paper.)

Proof. Step A (expansion; the mode ranges). The class indicator expands (Lemma 3.6, then Lemma 3.5 at truncation $J_3 = P^{1/96}$, majorant cost $4P/J_3 = 4P^{1-1/96}$ per layer) into mode sums

$$S_{ijk} = \sum_{\substack{n \sim P \\ n \text{ odd}}} e\big(\tfrac{i}{2}X + \tfrac{j}{2}Y + \tfrac{k}{2}v^{3/2}\big), \qquad |i| \le 2P^{1/96}, \quad |j| \le 2P^{1/96}, \quad 1 \le |k| \le 2P^{1/96},$$

with weights $\le \min(1, 2/|i|)\min(1, 2/|j|)\min(1, 2/|k|)$ of total mass $O(\log^3 P)$. The $k = 0$ sums are Theorems 4.1 and 4.4. Conjugating, take $k \ge 1$; then every displayed sign below is as written.

Step B (the depth-4 identity). Three applications of the Taylor pattern of Lemma 4.3(i), taken to second order, give the exact identity (odd $n \ge 5$)

$$v^{3/2} = -\tfrac{5}{64}n^{27/8} + \tfrac{9}{32}mn^{15/8} - \tfrac{45}{64}m^2n^{3/8} + \tfrac{15}{64}vn^{9/8} + \tfrac{45}{32}vmn^{-3/8} - \tfrac{9}{64}vm^2n^{-15/8} + \mathrm{err},$$

$|\mathrm{err}| \le \frac{3}{4}n^{-9/8}$ (discarding err costs $\le 2\pi \cdot \frac{k}{2} \cdot \frac{3}{4}P^{-9/8} \cdot P = \frac{3\pi k}{4}P^{-1/8} \le 4.8\,P^{-11/96} < 1$ for $P \ge 7.6 \cdot 10^5$): the fourth-letter phase is a polynomial of degree $(2,1)$ in the integer pair (m, v) with smooth coefficients. Substituting $m = X - \theta$ in the v-coefficients, the total smooth v-coefficient is *exactly*

$$c(\nu) := \tfrac{k}{2}\Big(\tfrac{15}{64} + \tfrac{45}{32} - \tfrac{9}{64}\Big)\nu^{9/8} = \tfrac{3k}{4}\,\nu^{9/8},$$

the monomial weight of Theorem 5.3, and the $v\theta$- and $v\theta^2$-cross coefficients are $\le \frac{45}{64}k\nu^{-3/8} + \frac{9}{32}k\nu^{-3/8} \le 1.2k\nu^{-3/8}$ and $\le \frac{9}{128}k\nu^{-15/8}$: sub-unit and decaying. The pure-m part is $\mu_1 m + \mu_2 m^2$ with $\mu_1 = \frac{9}{64}k\nu^{15/8}$, $\mu_2 = -\frac{45}{128}k\nu^{3/8}$, and the pure-smooth part $-\frac{5}{128}k\nu^{27/8} + \mu_1 X + \mu_2 X^2$.

Step C (double differencing; corner exactness). Apply Step 1 of Theorem 5.3 ($H_1 = P^{1/48}$, $H_2 = P^{1/24}$; now $kh_1h_2 \le 2P^{1/96+1/48+1/24} \le P^{1/8}$, so (C3) and (C4) hold) to the whole mode phase. That last comparison is where Theorem 5.3's exponent is actually spent, and it says how far an improvement would carry. Written for a general kernel saving δ, Step A truncates at $J_3 = P^{\delta}$ — the majorant $4P/J_3$ balancing $P^{1-\delta}$ — and the check here reads $kh_1h_2 \le 2P^{\delta+1/16} \le P^{1/8}$, since $\frac{1}{48} + \frac{1}{24} = \frac{1}{16}$. That comparison has room — $\frac{7}{96}$ against a budget of $\frac{12}{96}$ — and it is not what pins the exponent.

What pins it is (C4). The two balances of Step 1 fix $H_2 = P^{\delta_0}$ and $H_1 = P^{\delta_0/2}$ from Lemma 5.2(ii)'s own saving δ_0, returning the kernel saving $\delta_0/4$; and (C4) caps h_2 at $P^{1/24}$. So $H_2 = P^{\delta_0}$ sits *exactly* at that cap, with no slack at all, and δ_0 cannot exceed $\frac{1}{24}$ — which is why the kernel saving is $\frac{1}{96}$ and not something larger. The headroom along this route is zero.

(C1) is tight, but only at the corner of its own hypothesis box: the equality $k = h_1 = h_2 = P^{1/24}$ is never the configuration in force, because Theorem 6.1 enters with $k \le 2P^{1/96}$. Tight over the domain, slack at every invocation. And (C4) is not free either: the decoration class (D1) and the third-differencing reduction of Lemma 5.2(ii) are what require $h_1+h_2 \le 2P^{1/24}$. So $\frac{1}{96}$ traces back, without slack anywhere along the way, to the decoration budget — and improving the kernel means re-deriving that, not re-balancing anything.

The chain closes at (D1), tightly. Stage 6 bounds that class's boundary cost by $5.1\,h'(uh)^{-1/2}P^{7/8}$ and then puts h' at its cap $2P^{1/24}$; that substitution is where the printed fourth term of Lemma 5.2(i) comes from. And the fourth term is one of the three that ever lead — it is the largest at $u = h = 1$ — so the cap is spent at a configuration that occurs, not merely at a corner of the hypothesis box. Collecting the slack at each link:

link	at the operating point	slack
(D1) $h' \le 2P^{1/24}$, used at the cap	makes term four	0
term four against term five	leads at $u = h = 1$	—
(C4) $h_2 \le P^{1/24}$ against H_2	equality	0
(C1) $P^{1/8}$ against Step C's load	7/96 of 12/96	5/96

One exact relation falls out, worth recording because its two sides are fixed by unrelated criteria. At $u = h = 1$ the fourth term is $P^{11/12}$ and the fifth is $P^{29/32}$, and $\frac{11}{12} - \frac{29}{32} = \frac{1}{96}$: the fourth exceeds the fifth by exactly the kernel exponent. That is equivalent to $R_0 = 2(\frac{1}{24} + \frac{1}{8} - \frac{1}{96}) = \frac{5}{16}$, so the truncation Appendix A.6 selects — the left endpoint of the band five unrelated sites leave open — is also the one making that gap the kernel's own saving. We record the coincidence without claiming a mechanism for it.

Nothing after this theorem consumes the value at all — Corollary 6.4's density 7/8 needs the error to be $o(N)$, and Proposition 7.1 asks only for $\delta_d > 0$.

On each level-1 carry branch of Lemma 5.1(iii), the four corner values obey the exact relations

$$m(n{+}d) = m + \beta_d, \qquad \theta(n{+}d) = \theta + \delta_d(\nu) - \beta_d \qquad (d \in \mathcal{D}),$$

with β_d frozen integers and δ_d smooth: the doubly differenced phase on a branch is an exact function $\Phi(\nu,\theta)$ of ν and the *single* sawtooth $\theta = \{X\}$. Two consequences are used repeatedly. First, the θ^2-contents of the four corners cancel exactly in the $(+,-,-,+)$ pattern, so $|\partial_\theta^2\Phi| \le 3kh_1h_2P^{-13/8} + 4|\Delta\Delta\mu_2| \le P^{-1}$: no quadratic sawtooth survives, and $e(\Phi) = e(\Phi(\nu,0) + B(\nu)\theta)(1 + O(P^{-1}))$ with $B = \partial_\theta\Phi(\nu,0)$. Second, the wave and kernel contents are exactly those of Theorem 5.3 with $c = \frac{3k}{4}\nu^{9/8}$: the v-linear part contributes $\Delta\Delta(c\,\theta_2)$ (the master identity, Steps 2–3 of Theorem 5.3) plus $\Delta\Delta(cY)$-content whose wave frequencies join Step 4's bookkeeping.

Step D (passenger inventory). Every phase content beyond the bare kernel is listed here with its range and its check.

- *The $\frac{i}{2}X$-passenger*, $|i| \le 2P^{1/96}$: $\Delta\Delta(\frac{i}{2}X)$ is smooth with second derivative $\le \frac{i}{2} \cdot 4h_1h_2 \sup|X''''| \le 2.3\,ih_1h_2P^{-5/2} \le P^{-9/4}$: class (D3), dominated by every retained scale at margin $\le P^{-3/2}$.

- *The $\frac{j}{2}Y$-passenger*, $|j| \le 2P^{1/96}$: $\Delta\Delta(\frac{j}{2}Y)$ is exactly four waves $\pm\frac{j}{2}Y(n+d)$, $d \in \mathcal{D}$, with total frequency 0. In Step 4 of Theorem 5.3 it shifts each wave coefficient q_d by at most $P^{1/96}$, keeping $|q_d| \le 3P^{1/24} + P^{1/96} \le 4P^{1/24}$ — inside Lemma 5.2(ii)'s coefficient budget — and leaves t unchanged; its $t = 0$ remnants are differenced waves that shift the u-coefficients of Step 5 by $\le P^{1/96}$, harmless against the budget $uh_1 \le 0.6P^{1/2}$.
- *The pure-m passengers $\mu_1 m + \mu_2 m^2$*: on a branch, by corner exactness, $\Delta\Delta(\mu m) = (\Delta\Delta\mu)m + (\Delta_2\mu)\beta_1 + (\Delta_1\mu)\beta_2 + \mu_{11}j$ and $\Delta\Delta(\mu m^2) = (\Delta\Delta\mu)m^2 + (\Delta_2\mu)(2\beta_1 m+\beta_1^2) + (\Delta_1\mu)(2\beta_2 m+\beta_2^2) + \mu_{11}(2jm + \beta_{12}^2-\beta_1^2-\beta_2^2)$, j the branch offset. The θ-coefficients that arise from $m = X - \theta$ in these terms are $|\Delta\Delta\mu_1| \le 0.92kh_1h_2P^{-1/8} \le P^{-1/48}$, $|2(\Delta_2\mu_2)\beta_1| \le 2.3kh_1h_2P^{-1/8} \le 3P^{-1/48}$, and $|2j\mu_2|$-content, which joins $B(\nu)$ below; all sub-unit ones are expanded in the slow-mode families at $O(\log P)$ mass. The smooth parts join the two anchor families and are accounted in Step E.
- *The $v\theta$-crosses*: writing $v = Y - \theta_2$ and linearizing $Y = m^{3/2}$ once more by Lemma 4.3(i), the crosses decompose into $\nu^{15/8}$-scale θ-sawtooth content (joining $B(\nu)$), a $\theta\theta_2$-product with coefficient $\le 1.2k\nu^{-3/8}$ (deleted at cost $\le 31kP^{5/8} \le P^{2/3}$), and decaying remainders.

Step E (the two sign-critical composites, rerun). The passengers change the two composites of Step 5 of Theorem 5.3. Both are recomputed here from the *frozen-shape* model of Lemma 5.2b, not from the moving-(m, v) interpolant of $\frac{k}{2}\nu^{27/8}$. The identity that organises the difference is Lemma 5.1(i):

$$\tfrac{k}{2}\, v^{3/2} = \tfrac{k}{2}\, m^{9/4} - c\,\theta_2 - kR, \qquad c = \tfrac{3k}{4}\, v^{1/2},$$

with $|kR| \ll kP^{-9/8}$ discarded as in Step B. The differenced total phase is therefore $\Delta\Delta(\frac{k}{2}m^{9/4}) - \Delta\Delta(c\theta_2)$ plus the Step D passengers. On a piece the integers β_i, J_F are held frozen; $X(\nu)$ and the smooth weights move; the values of $\Delta_i X$ may be substituted for the β_i as *values only*.

Offset branches ($j \ne 0$). The frozen-shape offset content of $\Delta\Delta(\frac{k}{2}m^{9/4})$, with $m = X$ as a value, is the same monomial as in Theorem 5.3, $\frac{9}{8}kj\nu^{15/8}$, with curvature $\frac{945}{512}kj\nu^{-1/8}$. The frozen kernel anchor $(c(G_F - J_F))''$, now including the $-J_Fc''$ term that Lemma 5.2b isolates as part of $c''(G_F - J_F)$ only after the fractional part is taken, is $\frac{27}{16}kj\nu^{-1/8} = \frac{864}{512}kj\nu^{-1/8}$. The difference — the surviving total-phase offset — is

$$\left(\tfrac{945}{512} - \tfrac{864}{512}\right)kj\nu^{-1/8} = \tfrac{81}{512}\, kj\nu^{-1/8}.$$

This is J_Fc'' at leading order: after the master identity cancels $c_{11}F_\kappa$ against $\Delta\Delta(cY)$, what remains of the growing smooth offset is the frozen integer times c''. The total θ-coefficient on the piece is

$$B = \tfrac{27}{32}\, kj\, \nu^{3/8}\, (1 + O(P^{-1/4})),$$

one and a half times the bare-kernel value $\frac{9}{16}kj\nu^{3/8}$ (not the moving-gap value $\frac{45}{32}$). The window-centre mode therefore carries $u_0X'' = -\frac{27}{32} \cdot \frac{3}{4}\, kj\, \nu^{-1/8} = -\frac{81}{128}\, kj\, \nu^{-1/8} = -\frac{324}{512}\, kj\, \nu^{-1/8}$, and the composite is

$$\left(\tfrac{81}{512} - \tfrac{324}{512}\right)kj\, \nu^{-1/8} = -\tfrac{243}{512}\, kj\, \nu^{-1/8},$$

single-signed, with $\lambda_a' \in [0.40,\, 0.52]\, k|j|P^{-1/8}(1 + O(P^{-1/8}))$. The competitors of Step 5a, now read against $\lambda_a' \ge 0.40\, k|j|P^{-1/8} \ge 0.40P^{-1/8}$, remain dominated for $P \ge P_0$: differenced-wave modes $\le 0.84\, uh_1P^{-3/4} \le 0.51P^{-1/4}$, ratio $\le 1.3P^{-1/8}$; resonant (D1) content, ratio $\le 13P^{-9/16}$; slow modes $\le 3J_2|j|P^{-5/4}$, ratio $\le 9P^{-13/12}$; (D3) content, ratio $\le 4h_1h_2P^{-1/2}/(0.40P^{-1/8}) \le 3P^{-1/8}$. The window count grows by $\frac{27/32}{9/16} = \frac{3}{2}$, so at most $1.8\, k|j|P^{3/8} + 1$ windows; the collision-band Lemma 3.8 sum is then $\le 3.0\, (k|j|)^{1/2}P^{15/16}\log P$. Run lengths are unchanged, and

$$\sum_{\text{runs}} (\ell\lambda_a'^{1/2} + \lambda_a'^{-1/2}) \le 1.4\, (k|j|)^{1/2}P^{15/16} + 34\, (|j|{+}1)|j|^{-1/2}k^{-1/2}P^{13/16}.$$

Summed over the eight branches and $|j| \le 2$ with the $O(\log^3 P)$ piece masses: $\ll k^{1/2}P^{15/16+\varepsilon} \le P^{1/48}P^{15/16+\varepsilon} = P^{23/24+\varepsilon}$ already at the kernel's $k \le P^{1/24}$; the actual range $k \le 2P^{1/96}$ sits strictly inside. The inverse-power term remains $P^{13/16}$.

Zero-offset branches ($j = 0$). The same frozen-shape difference, in two halves. With β_1, β_2 frozen the smooth double difference is $\Delta\Delta f(X) = \beta_1\beta_2 f''(X)$ up to the mean-value error, so for $f(Z) = \frac{k}{2}Z^{9/4}$, $f'' = \frac{45k}{32}Z^{1/4}$ and $\Delta\Delta(\frac{k}{2}m^{9/4}) = \frac{45k}{32}\beta_1\beta_2\nu^{3/8}$, whose ν-curvature is $\frac{45}{32}\cdot\frac{3}{8}\cdot(-\frac{5}{8}) = -\frac{675}{2048}\,k\beta_1\beta_2\nu^{-13/8}$. The anchor $\Delta\Delta(c\theta_2)$ is the $(c(G_F - J_F))''$ of Lemma 5.2b, $-\frac{216}{1024} = -\frac{432}{2048}$ in the same units — the *corrected* value of the erratum there, with the $-J_F c''$ taken, as this step takes it at offset. Subtracting,

$$\lambda_0' = (\tfrac{675}{2048} - \tfrac{432}{2048})k\beta_1\beta_2\nu^{-13/8} = \tfrac{243}{2048}\,k\beta_1\beta_2\nu^{-13/8} = \tfrac{2187}{2048}\,kh_1h_2\,\nu^{-5/8} \in [0.60,\,1.25]\,kh_1h_2P^{-5/8}$$

after $\beta_1\beta_2 \sim 9h_1h_2\nu$, measured against the true integer gaps at $P = 10^8$ to seven figures on each half separately. Equivalently $\lambda_0' = \frac{9}{16}\lambda_0$ exactly.

> **Erratum (constants only; nothing downstream moves).** Earlier printings gave $\frac{1095}{1024} = \frac{2190}{2048}$ here and $b' = -\frac{365}{176} = -\frac{730}{352}$ below. The exact values are $\frac{2187}{2048} = \frac{3^7}{2^{11}}$ and $-\frac{729}{352} = -\frac{3^6}{352}$: a single slip $729 \to 730$, carried into b' by the propagation the audit ledger had already noticed. The printed bracket $[0.60, 1.25]$ still holds — the true range is $(0.6924, 1.0679]$ against $(0.6934, 1.0694]$ — so no threshold and no appendix row moves. Note that $\frac{2190}{2048}$ is already the value the *corrected* Lemma 5.2b anchor produces: with that lemma's printed $-\frac{135}{1024}$ the difference would be $-\frac{3645}{2048}$, off by a factor $\frac{5}{3}$. This step computed the anchor correctly and Lemma 5.2b displayed it wrongly.

The three-regime split of Step 5b is read at this scale. The thresholds (factor 60) are scale-free. In the anchor-dominant regime, Lemma 3.3 per run gives $\le 2.0(kh_1h_2)^{1/2}P^{11/16} + 30\,(h_1h_2/k)^{1/2}P^{9/16}$. In the mode-dominant regime the undifferenced-anchor decoration is the same as in Step 5b. The $j = 0$ θ-coefficient of the *total* phase is $O(1)$, of the same species as the kernel's zero-offset B (constant size, $|B| \le 6$): Lemma 3.7 applies in a single window at $T = P^{1/2}$ (hypothesis $T \ge 8(1 + |B|)$ holds for $|B| \le 6$). In the middle band the frozen-shape interpolant of Lemma 5.2b is reused with the kernel-anchor leading $-\frac{243}{128}kh_1h_2\nu^{-5/8}$ replaced by $-\frac{2187}{2048}kh_1h_2\nu^{-5/8}$, so $\Phi = a\nu^{5/4} + b'\nu^{11/8} + w\nu^{3/2}$ with $a = -\frac{27}{10}(uh_1 + u'h_2)$ and $b' = -\frac{729}{352}\,kh_1h_2$, which is $-\frac{2187}{2048}\cdot\frac{64}{33}$ and is $\frac{9}{16}$ of the kernel's own $b = -\frac{81}{22}$: the interpolant scales with the anchor it is built from, now by a ratio in lowest terms. The $\rho_0(E)$ bounds are unchanged, and

$$S \le \max(60\lambda_0',\ \lambda_0',\ |w|P^{-1/2}) \le 80\,P^{-1/2}$$

by (C1) and $|w| \le 2$. The same choice $V = \frac{1}{12}S^{1/2}P^{-11/24}$ and the same raised threshold $W = V + E$ satisfy $W \le c_7S/2$ at $c_7 = 1/232$ from $P \ge 1.6\cdot 10^{13}$ – a lower threshold than Step 5b's, because S is larger here: at the lower end $S \ge 0.60P^{-5/8}$ one has $V/S \le 0.11P^{-7/48}$ and $V \ge 0.064P^{-37/48}$, against the interpolant error $106P^{-25/24} + 0.11P^{-5/6}$. Transition length and piece-boundary costs are as in Step 5b (the same exponents; the smaller S enlarges V/S by a constant, still $O(P^{-7/48})$). Good pieces cost $\le P\cdot(80)^{1/2}P^{-1/4} \le 9P^{3/4}$. The middle band therefore remains $\ll P^{15/16+\varepsilon}$.

The passengers of Step D are inside these estimates: the $\frac{i}{2}X$-term is (D3) as listed; the $\frac{j}{2}Y$-term shifts each q_d by at most $P^{1/96}$, keeping $|q_d| \le 4P^{1/24}$ and $uh_1 \le 0.6P^{1/2}$; the pure-m smooth pieces are already in $\Delta\Delta(\frac{k}{2}m^{9/4})$ and do not add a further relative $O(1)$ to λ_a' or λ_0'; the sub-unit θ-coefficients of those pieces are among the slow modes already bounded.

Step F (assembly). With the inventory of Step D and the composites of Step E, Steps 2–6 of Theorem 5.3 give $|T_2| \ll P^{23/24+\varepsilon}$ uniformly in (i, j, k) on the stated ranges, hence $S_{ijk} \ll P^{1-1/96+\varepsilon}$. Summing the mode weights ($O(\log^3 P)$), the majorant costs ($\le 4P^{1-1/96}$ per layer, three layers), and dyadic blocks gives the theorem. □

The four-step certified class remains 13/16: *OOO*∗ does not contract ($3^3 > 2^4$). The next two contracting itineraries are *OOOEE* and *OOEOE* ($3^3 < 2^5$). Neither is a third growing layer. After *OOOE* the fifth letter is a decaying nest plus one slow sawtooth of coefficient $n^{3/16} < n$; after *OOEO* it is a single growing sawtooth of coefficient $n^{9/16} < n$, of the same species as Theorem 4.8. Theorem 6.3 counts both cylinders.

6. Depth-five contractors

Lemma 6.2 (fifth-letter identities). Let $n \ge 5$ be odd, and write $z = \lfloor v^{3/2}\rfloor$, $U = v^{1/2}$, $w = \lfloor U\rfloor$, and $\theta_w = \{U\}$ (the same letters as in Theorem 4.8, now at the v-level).

(i) *(OOOE∗ smoothing.)*

$$z^{1/2} = n^{27/16} - \tfrac{9}{8} n^{3/16}\,\theta + D_5, \qquad |D_5| \le \tfrac{3}{4}\,m^{-3/8} + \tfrac{1}{2}\,v^{-3/4} + \tfrac{9}{128}\,(X-1)^{-7/8} + \tfrac{3}{32}(Y-1)^{-5/4} + \tfrac{1}{8}(v^{3/2}-1)^{-3/2}.$$

The second term is $O(n^{-27/16})$, and the last three are $O(n^{-21/16})$, $O(n^{-45/16})$ and $O(n^{-81/16})$; the leading remainder is $O(n^{-9/16})$.

(ii) *(OOEO∗ linearization.)*

$$w^{3/2} = n^{27/16} - \tfrac{9}{8} n^{3/16}\,\theta - \tfrac{3}{2}\,v^{1/4}\,\theta_w + D_5',$$

with $|D_5'| \le \tfrac{3}{4}\,m^{-3/8} + \tfrac{3}{8}(U-1)^{-1/2} + \tfrac{9}{128}(X-1)^{-7/8} + \tfrac{3}{32}(Y-1)^{-5/4}$.

Proof. (i) Three applications of the Lemma 4.3(i) pattern. With $f(t) = (v^{3/2}-t)^{1/2}$ on $[0,\theta_z]$, $\theta_z = \{v^{3/2}\}$: $z^{1/2} = v^{3/4} - \frac{1}{2}\theta_z v^{-3/4} - E_z$ and $0 \le E_z \le \frac{1}{8}(v^{3/2}-1)^{-3/2}$. With $f(t) = (m^{3/2}-t)^{3/4}$ on $[0,\theta_2]$: $v^{3/4} = m^{9/8} - \frac{3}{4}\theta_2 m^{-3/8} - E_2$ and $0 \le E_2 \le \frac{3}{32}(Y-1)^{-5/4}$. With $f(t) = (X-t)^{9/8}$ on $[0,\theta]$: $m^{9/8} = n^{27/16} - \frac{9}{8}\theta n^{3/16} + \frac{9}{128}\theta^2(X-\xi)^{-7/8}$ for some $\xi \in (0,\theta)$. Hence $D_5 = \frac{9}{128}\theta^2(X-\xi)^{-7/8} - \frac{3}{4}\theta_2 m^{-3/8} - E_2 - \frac{1}{2}\theta_z v^{-3/4} - E_z$, and the triangle inequality with $\theta, \theta_2, \theta_z < 1$ gives the displayed bound term by term. (The two Lagrange remainders E_2, E_z are of orders $n^{-45/16}$ and $n^{-81/16}$; they are displayed rather than absorbed into the coefficients $\frac{3}{4}$ and $\frac{1}{2}$, which have no slack when θ_2 or θ_z is close to 1.) The leftover sawtooth has coefficient $n^{3/16}$.

(ii) Lemma 4.3(i) at base U, using $U\,v^{1/4} = v^{3/4}$ exactly: $w^{3/2} = v^{3/4} - \frac{3}{2}v^{1/4}\theta_w + E$ with $0 \le E \le \frac{3}{8}(U-1)^{-1/2}$. The chain $v^{3/4} \to n^{27/16}$ is the second and third steps of (i), so $D_5' = E - \frac{3}{4}\theta_2 m^{-3/8} - E_2 + \frac{9}{128}\theta^2(X-\xi)^{-7/8}$, and the bound follows term by term. Two sawtooths remain, of coefficients $n^{3/16}$ on θ and $n^{9/16}$ on θ_w. □

Theorem 6.3 (two length-five splits; not a census of depth five).

$$\#\mathrm{OOOEE}(N),\ \#\mathrm{OOOEO}(N) = \tfrac{N}{32} + O(N^{1-1/96+\varepsilon}), \qquad \#\mathrm{OOEOE}(N),\ \#\mathrm{OOEOO}(N) = \tfrac{N}{32} + O(N^{47/48+\varepsilon}).$$

Of these four words only *OOOEE* and *OOEOE* contract. The expanding tree *OOOO∗* is not estimated: it is the level-3 kernel of Conjecture 7.3. All estimates below are for $P \ge P_0 = 3.6 \cdot 10^{13}$, the effective threshold of Appendix A — but only because Stage 2 of Theorem 5.3 is run at $R_0 = P^{5/16}$. The fifth-letter window below is opened at $T = R_0$ against a sawtooth coefficient $|C| \le 1.30\,P^{19/96}$, and both of its requirements — the Lemma 3.7 hypothesis $T \ge 8(1+|C|)$ and the flat cost $8(1+|C|)/T \le P^{-1/96}$ per point — have a margin of only $T/P^{19/96}$. At $R_0 = P^{1/4}$ that margin is $P^{5/96}$, and the two requirements first hold at $2.55 \cdot 10^{19}$ and $1.81 \cdot 10^{24}$: both far above P_0, the second by ten orders. At $R_0 = P^{5/16}$ the margin is $P^{11/96}$ and they hold from $7.5 \cdot 10^8$ and $5.5 \cdot 10^9$. Appendix A.6 records the trade in full.

Proof. OOOE∗. By Lemma 3.6 the class indicators are the *OOOE* indicator of Theorem 6.1 times $\frac{1}{2}(1 \pm \psi(z^{1/2}))$. Vaaler-expand the fifth wave at truncation $J_5 = 2P^{1/96}$ (majorant $4P/J_5 = 2P^{1-1/96}$ per block). Lemma 6.2(i) replaces $\frac{l}{2}z^{1/2}$ by $\frac{l}{2}n^{27/16} - \frac{9l}{16}n^{3/16}\theta$. The remainder costs $|l| \cdot P \cdot P^{-9/16} \ll P^{1/96+7/16} = P^{43/96}$, inside $P^{1-1/96}$. Write $C = \frac{9l}{16}n^{3/16}$, so $|C| \le 2P^{19/96}$ on $|l| \le 2P^{1/96}$. Lemma 3.7 with $T = R_0 = P^{5/16}$ satisfies $T \ge 8(1+|C|)$ for $P \ge P_0$, since $8|C|/T \le 11P^{-11/96}$, which is below 1 from $P \ge 7.5 \cdot 10^8$. The produced modes are $e(uX)$ with $|u| \le P^{5/16}$. These are first-letter monomials of the shape already estimated in the proof of Lemma 5.2(i) — the families $e(r\nu^{3/2})$ with truncation $R_0 = P^{5/16}$, which is exactly why Stage 2 is run at that truncation — and are not decorations of class (D1). (The coefficient budget $|q'| \le 4P^{1/24}$ of Lemma 5.2(ii) is the kernel's own Y-wave range after differencing; it is not a slot for X-modes.)

Write I_{tot} for the combined first-letter index: Theorem 6.1's own $|i| \le 2P^{1/96}$ plus the fifth-letter $|u| \le P^{5/16}$. Then $|I_{\mathrm{tot}}| \le 2P^{5/16}$. Theorem 6.1 Steps D–E apply at this range.

- *The $\frac{i}{2}X$-passenger.* $\Delta\Delta(\frac{i}{2}X)$ has second derivative $\le 2.3\,|i| h_1 h_2 P^{-5/2}$. At $|i| \le 2P^{5/16}$ and $h_1 h_2 \le P^{1/16}$ this is $O(P^{-34/16})$, inside class (D3) ($|\varphi''| \le 6kh_1h_2h\,P^{-13/8}$) by $P^{-34/16}/P^{-26/16} = P^{-1/2}$. The fifth-letter chirp $\frac{l}{2}n^{27/16}$, after the same double difference, is likewise (D3): $h_1h_2 \cdot l \cdot P^{27/16-4} = O(P^{1/16+1/96-37/16})$.
- *The $\frac{j}{2}Y$-passenger.* The fifth letter does not enlarge j or the Y-wave frequencies. The bound $|q_d| \le 3P^{1/24} + P^{1/96} \le 4P^{1/24}$ is unchanged.

- *The sign-critical composites.* An X-mode is smooth, so it does not enter the frozen θ-coefficient B. The offset composite $\frac{243}{512}$ and the zero-offset curvature $\frac{2187}{2048}kh_1h_2\nu^{-5/8}$ are therefore the values of Theorem 6.1 Step E. The (D3) curvature of the new modes is dominated at the ratios already displayed there ($\leq 3P^{-1/8}$ against $\lambda'_a \geq 0.40P^{-1/8}$).
- *The Lemma 5.2(i) budget.* Modes with $|u| \leq P^{5/16}$ are a subset of the family already bounded by $3R_0^{1/2}P^{3/4}\log P = 3P^{29/32}\log P$, which is inside $P^{23/24}$ with $P^{5/96}$ to spare. Collision-band mass stays $O(\log P)$.
- *The flat cost of Lemma 3.7.* Per point $8(1+|C|)/T \leq 11P^{19/96-5/16} = 11P^{-11/96}$; over a block $\leq 11P^{1-11/96}$, inside Theorem 6.1's $P^{1-1/96}$ budget from $P \geq 5.5\cdot 10^9$. This is the requirement that binds hardest on R_0: at $R_0 = P^{1/4}$ the same line reads $11P^{1-5/96}$, which does not clear $P^{1-1/96}$ until $1.8\cdot 10^{24}$.

Theorem 6.1 therefore applies to every fifth-letter decorated *OOOE$*$* mode sum, uniformly in $|l| \leq 2P^{1/96}$. Hence $\#\mathrm{OOOEE}(N),\ \#\mathrm{OOOEO}(N) = N/32 + O(N^{1-1/96+\varepsilon})$.

OOEO$*$. By Lemma 3.6, the two class indicators are

$$\tfrac{1}{16}(1-\psi(X))(1+\psi(Y))(1-\psi(U))(1\pm\psi(w^{3/2})).$$

Vaaler-expand the four waves. The fifth-letter $k=0$ sums are the depth-4 class *OOEO*, bounded by Theorem 4.7. For $k \neq 0$, Lemma 6.2(ii) writes the fifth-letter phase as $\frac{k}{2}n^{27/16} - C\theta - B\theta_w$ with $B = \frac{3k}{4}v^{1/4} \asymp kn^{9/16}$ and $C = \frac{9k}{16}n^{3/16}$, up to the decaying D'_5. The remainder costs $|k|\cdot P\cdot P^{-9/16} \ll P^{1/48+7/16} = P^{11/24}$ at the fifth-letter truncation $P^{1/48}$ used below, inside $P^{47/48}$. The leading sawtooth is of the same species as Theorem 4.8, now riding $\theta_w = \{v^{1/2}\}$.

Drift-1 intervals. $B' \asymp kn^{-7/16}$, so B drifts by at most 1 on intervals I of length $L_B = P^{7/16}/k$; there are $\asymp kP^{9/16}$ of them. On I, expand $e(-B\{U\})$ by the same shifted-window Fourier expansion as in Theorem 4.8 (Fourier in U, Vaaler truncation $|r+B| \leq T_w$ with $T_w = P^{1/4}$): the coefficients satisfy $|a_r(B)| \leq \min(1, |r+B|^{-1})$ with window mass $O(\log T_w)$, and their n-dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode. The window is legal for the sign check below: $T_w = P^{1/4} \ll kP^{9/16}$ for every $k \geq 1$; the majorant is $P/T_w = P^{3/4}$.

At frequency $\ell = -B+t$, $|t| \leq T_w$, the combined phase is $-\frac{3k}{4}v^{3/4} + t\,v^{1/2}$ plus the original $\frac{k}{2}n^{27/16} - C\theta$. Linearizing $v^{3/4}$ and $v^{1/2}$ by Lemma 6.2, the θ-coefficients from the two expansions cancel at the window centre up to a residual $C_{\mathrm{net}} = \frac{9k}{32}n^{3/16}$: the contribution of $-\frac{3k}{4}v^{3/4}$ is $\frac{27k}{32}n^{3/16}$, and $\frac{27k}{32} - \frac{9k}{16} = \frac{9k}{32}$. One slow θ-sawtooth remains. Expand $e(C_{\mathrm{net}}\theta)$ on the same intervals, again by the shifted-window device of Theorem 4.8 (C_{net} drifts by $P^{-3/8} \ll 1$ on each I; window $T_\theta = P^{1/8} \ll kP^{3/16}$; majorant $P^{7/8}$). The window-centre X-mode of this last expansion has curvature $\frac{27k}{128}n^{-5/16}$. The leading $n^{27/16}$ coefficient of the combined phase is $\frac{k}{2} - \frac{3k}{4} = -\frac{k}{4}$, of curvature $-\frac{297k}{1024}n^{-5/16}$. Hence

$$\lambda_2 = \Big(-\tfrac{297k}{1024} + \tfrac{27k}{128} + O(T_w n^{-9/16}) + O(J_* n^{-3/16})\Big)n^{-5/16},$$

with leading combination $-\frac{297}{1024} + \frac{216}{1024} = -\frac{81}{1024} \neq 0$. The t-error is $O(P^{1/4-9/16}) = O(P^{-5/16})$ and the J_*-error, at the truncation $J_* = P^{1/48}$ used below, is $O(P^{1/48-3/16}) = O(P^{-1/6})$. Both are smaller than the leading coefficient, so λ_2 is single-signed and $\lambda_2 \asymp kn^{-5/16}$. By Lemma 3.10(a) the curvature ratios and signs are invariant under $n = 2r+1$; Lemma 3.10(b) says the n-variable van der Corput display dominates the reindexed bound.

Lemma 3.3 on each I: $L_B\lambda_2^{1/2} + \lambda_2^{-1/2} \ll k^{-1/2}P^{9/32} + k^{-1/2}P^{5/32}$. The drift pieces have length $Z \asymp P^{7/16}/|k|$, and at most $1 + |k|YP^{-7/16}$ of them meet a source interval of length Y. With the unabsorbed estimate of Theorem 4.12 on a piece, $|S_D| \ll_\varepsilon P^\varepsilon(ZP^{-1/24} + Z^{1/2}P^{3/16})$, summing the pieces gives

$$P^\varepsilon\Big(YP^{-1/24} + |k|^{1/2}YP^{-1/32} + |k|^{-1/2}P^{13/32} + |k|^{-1}P^{19/48}\Big).$$

Against the $1/|k|$ Vaaler weights, truncation at $J_* = P^a$ costs YP^{-a} while the second term costs $YP^{-1/32+a/2}$; the two balance at $a = \frac{1}{48}$, which is the fifth-letter truncation used above. Hence $O_\varepsilon(YP^{-1/48+\varepsilon})$, and dyadic blocks sum to $N^{47/48+\varepsilon}$. The third and fourth terms are $P^{13/32}$ and $P^{19/48}$, both far inside this, and the two window majorants cost $YP^{-1/8}+P^{5/16}$ and $YP^{-1/8}+P^{3/16}$. $\square$

Erratum (the omitted mixed mode, and the exponent it costs). Until this revision the second half of Theorem 6.3 was printed at $N^{43/48+\varepsilon}$, on a proof with two defects.

First, the four-wave product carries $\frac{j}{2}Y+\frac{k}{2}w^{3/2}$ with $Y=\lfloor n^{3/2}\rfloor^{3/2}$ and $j,k\neq 0$; the λ_2 calculation retained the k-wave and its window modes and dropped $jY/2$. That term is not a passenger — it is the nested phase for which Theorem 4.4 spends one Weyl A-process — and there is no reduction to $j=0$. Second, a centered Fourier frequency was written dynamically as $r=-B(n)+t$; a frequency must be frozen before differentiation.

Both are repairable. For the window, put $B=N+\beta$ with $N=\lfloor B\rfloor$ and $0\le\beta<1$, so that $e(-B\{t\})=e(-Nt)\,e(-\beta\{t\})$: apply Lemma 3.7 to the bounded β and shift every produced frequency by the fixed integer $-N$, splitting once more on each piece where B drifts by one. The apparent singularity at $u+\beta=0$ is removable, and the two centered expansions cost $O(\log^2 P)$ under Abel summation with fixed supports.

For the omitted mode, do not replace RU by $Rn^{9/8}$. Use $U=\sqrt{\lfloor m^{3/2}\rfloor}=m^{3/4}+O(P^{-9/8})$, whose mode error is $O(|k|P^{-9/16})$ per summand, and after the A-process write the existing first-level gap as $g=G+\kappa$, $\kappa\in\{0,1\}$, with $D_\kappa(n)=(X+G+\kappa)^{3/4}-X^{3/4}$ and $X=n^{3/2}$. Then $\Delta(m^{3/4})=D_\kappa(n)+O(hP^{-11/8})$, $D_\kappa''\ll hP^{-15/8}$ and $(RD_\kappa)''\ll|k|hP^{-21/16}$, which at the fifth-letter truncation $J_*=P^{1/48}$ sits below the retained $|j|hP^{-3/4}$ curvature by $P^{-13/24}$. Splitting the carry that is already there introduces no new nesting.

What the repair costs is the exponent. The surviving balance is $a=\frac{1}{48}$ rather than the $J_*=P^{5/48}$ of the old display, so the saving is $P^{-1/48+\varepsilon}$ and not $P^{-5/48+\varepsilon}$. The stronger $P^{-1/24}$ localization does not follow from this architecture — the partial-end-cell term forces the $1/48$ balance. Corollary 6.4 is unaffected in form: $47/48=94/96$ is still below $1-\frac{1}{96}=95/96$, so the combined error remains $N^{1-1/96+\varepsilon}$, now with one ninety-sixth of room rather than nine.

Corollary 6.4 (certified-descent density 7/8). The five uniform certificate classes

$$E,\qquad OE,\qquad OOEE,\qquad OOOEE,\qquad OOEOE$$

are disjoint, and

$$\big|\#\{n\le N:\mathrm{word}(n)\text{ has prefix }OOOEE\}-\tfrac{N}{32}\big|\ll_\varepsilon N^{1-1/96+\varepsilon},$$

$$\big|\#\{n\le N:\mathrm{word}(n)\text{ has prefix }OOEOE\}-\tfrac{N}{32}\big|\ll_\varepsilon N^{47/48+\varepsilon}.$$

Hence the class of starts with a certified descent within five steps has cardinality

$$\tfrac{N}{2}+\tfrac{N}{4}+\tfrac{N}{16}+\tfrac{N}{32}+\tfrac{N}{32}+O(N^{1-1/96+\varepsilon})=\tfrac{7N}{8}+O(N^{1-1/96+\varepsilon}).$$

No other depth-≤ 5 word class is used. In particular this is not a census of every O-rooted itinerary of length five: the classes $OOOO*$ remain open, and $OOEOO$, $OOOEO$ are counted by Theorem 6.3 but do not contract.

Proof. The first three counts are Corollary 4.9. The new cylinders are Theorem 6.3. Every $OOOEE$ or $OOEOE$ start descends within five steps by Proposition 3.1: $3^3<2^5$. The five classes are disjoint because they are distinct prefixes. The error is the worse of the two fifth-letter exponents. $\square$

7. The Terras-style reduction and the frontier

The depth-by-depth counting assembles into a conditional Terras-style statement, and the reduction is unconditional:

Proposition 7.1 (equidistribution implies density-one descent). Let $d\ge 1$, let N_d be the number of length-d words with no contracting prefix, and suppose that each of *those* words satisfies the one-sided bound

$$\#\{n\le N:\mathrm{word}_d(n)=w\}\ \le\ 2^{-d}N+E_d(N).$$

Then the starts with no contracting prefix of length $\le d$ number at most

$$\frac{N_d}{2^d}\,N + N_d\,E_d(N),$$

and $N_d \le 2^d e^{-cd}$ with $c = 2\left(\frac{\log 2}{\log 3} - \frac{1}{2}\right)^2 > 0.0342$. Every other start $n \ge 2$ satisfies $J^t(n) < n$ for some $t \le d$ with the uniform power-envelope certificate of Proposition 3.1. Consequently, if $E_d(N) = O_d(N^{1-\delta_d})$ with $\delta_d > 0$ for every d, the set of starts admitting a finite descent certificate has natural density 1.

Proof. Every E-rooted word has a contracting prefix at length one ($3^0 < 2$), so the starts with no contracting prefix of length $\le d$ all realize an O-rooted itinerary of length d. An itinerary w of length d has a contracting prefix iff $3^{o_t} < 2^t$ for some $t \le d$, where o_t counts odd letters among the first t; so the words to be counted are exactly those whose lattice path (t, o_t) satisfies $3^{o_t} \ge 2^t$ throughout. That condition depends on nothing but (t, o_t), so N_d is a two-line dynamic program over the triangle $0 \le o \le t \le d$, exact in integers. Each of them has at most $2^{-d}N + E_d(N)$ members by hypothesis, and multiplying by N_d gives the count. For the closed form, drop every constraint but $t = d$: the endpoint condition is $o_d \ge d\log 2/\log 3$, and Hoeffding's inequality gives $\#\{o_d \ge d\log 2/\log 3\} \le 2^d e^{-cd}$ with c as above. The density-one statement follows by letting $d \to \infty$ slowly with N (any $d(N) \to \infty$ with $N_d E_d(N)/N \to 0$). □

The name of the proposition is historical. Its proof consumes neither equidistribution nor two-sided control: an upper bound suffices, and only on the words that survive. At depth five that is four words — *OOOOO*, *OOOOE*, *OOOEO*, *OOEOO* — against the sixteen O-rooted classes a full equidistribution statement covers, and the gap widens with depth: 19 against 128 at $d = 8$, 2114 against 32768 at $d = 16$, 286581 against 8388608 at $d = 24$. Whatever a prover establishes beyond that is not used here.

The exact count is worth carrying rather than the closed form, because Hoeffding is lossy in exactly the range the paper certifies:

d	N_d	endpoint only	2^d	$N_d/2^d$	e^{-cd}	certificate density
4	3	5	16	0.1875	0.8718	0.8125
5	4	6	32	0.1250	0.8425	0.8750
6	8	22	64	0.1250	0.8141	0.8750
8	19	37	256	0.0742	0.7601	0.9258
12	226	794	4096	0.0552	0.6627	0.9448
16	2114	6885	65536	0.0323	0.5778	0.9677
24	286581	1271626	16777216	0.0171	0.4392	0.9829

The ratio $e^{-cd}2^d/N_d$ is 6.7 at $d = 5$, 11.4 at $d = 10$, 43.6 at $d = 40$ and $1.13 \cdot 10^4$ at $d = 1600$. It grows without bound, and the reason is not the exponential rate. The rate is essentially untouched: the sharp value is $\rho = \min_\theta \frac{1}{2}((3/2)^\theta + 2^{-\theta}) = 0.965907$, i.e. $-\log\rho = 0.034688$ against Hoeffding's $c = 0.034285$, a difference of one part in eighty. What Hoeffding discards is a polynomial factor, and it is a large one:

$$\frac{N_d}{2^d} \;\sim\; C\,\rho^d\,d^{-3/2}, \qquad C \approx 11,$$

the $d^{-1/2}$ being the cost of staying nonnegative under the zero-drift tilt and the further d^{-1} the cost of the tilted endpoint, which sits at height $\asymp \sqrt{d}$ rather than at the origin. Over any range a depth-d theorem could occupy, that polynomial factor is the whole story: the *observed* per-letter rate $-d^{-1}\log(N_d/2^d)$ is 0.1696 at $d = 24$, 0.0635 at $d = 200$ and still 0.0401 at $d = 1600$, against an asymptote of 0.0347 it approaches only logarithmically slowly. (The value at $d = 200$, $N_{200}/2^{200} = 3.06 \cdot 10^{-6}$, is the figure recorded independently in the theorem ledger.) What changes in Proposition 7.1 is therefore the operative constant, in both terms: at $d = 5$ the error term carries $N_5 = 4$ rather than $2^5 = 32$, and at $d = 16$ it carries 2114 rather than 65536.

The table is exact and has been re-run. N_d is the two-line dynamic program described in the proof, so the whole table is recomputable: all seven rows agree, integers and decimals alike, and the program agrees with brute enumeration of all 2^d words for every $d \le 18$. The two analytic constants beside it check as well: $c = 2(\log 2/\log 3 - \frac{1}{2})^2 = 0.0342852$, printed 0.034285 — rounded *down*, which is the safe direction for a rate appearing as e^{-cd} in an upper bound — and $\rho = 0.9659066$, $-\log \rho = 0.0346882$. The constant $C \approx 11$ is approached from below: the fit gives 10.45 at $d = 800$ and 10.76 at $d = 1600$.

One figure did not survive. The loss ratio at $d = 1600$ was printed $1.3 \cdot 10^4$ and is $1.13 \cdot 10^4$, a fifteen per cent overstatement, where the three values quoted beside it — 6.7, 11.4, 43.6 — are right to under one per cent. It overstated how lossy Hoeffding is, which is the direction that flatters the argument being made, and nothing downstream depends on it: the sentence it supports is that the ratio grows without bound, which either figure shows.

The $d = 5$ row is a cross-check rather than a new number. The four surviving words are *OOOOO*, *OOOOE*, *OOOEO* and *OOEOO*, so the certificate density available at depth five is $1 - 4/32 = 7/8$ — which is Corollary 6.4's figure, reached here by counting words rather than by counting contractors. The $d = 6$ row is 7/8 again: all eight children of those four words survive, so depth six buys nothing without depth five first. The two words of the open *OOOO*$*$ split are half of the depth-five obstruction and, by the same table, half of the depth-six one.

Proposition 7.1b (the depth ceiling, and which depths pay). Write $\theta = \log 2/\log 3$, so that $\beta_* = 1-\theta$.

- (i) *(ceiling)* Once the N_d surviving classes are known to carry at least their *total* Bernoulli density $N_d/2^d$ — no class needs to be resolved on its own — no depth-d argument resting on Proposition 3.1 can certify a set of density above $1 - N_d/2^d$.
- (ii) *(attained at depths four and five)* Corollaries 4.9 and 6.4 attain it, unconditionally. At $d = 4$ the lower bounds are Theorem 6.1. At $d = 5$ the four survivors are *OOEOO*, *OOOEO*, *OOOOE* and *OOOOO*; Theorem 6.3 estimates the first two at $\frac{1}{32}$ each, and Theorem 6.1 gives the remaining pair — the children of the open *OOOO*$*$ split — through their *sum*, $\frac{1}{16}$. The total is $\frac{1}{32} + \frac{1}{32} + \frac{1}{16} = \frac{1}{8}$. Conjecture 7.3 is therefore not needed for the ceiling at depth five: the split it would resolve is one whose halves enter only added together.
- (iii) *(which depths pay)* The ceiling is unchanged from depth $d-1$ to depth d exactly when $\{(d-1)\theta\} \le \beta_*$. The stalling depths therefore have density β_*; they begin $3, 6, 9, 11, 14, 17, 19, 22, 25, 28, 30, \ldots$ and are not eventually periodic.
- (iv) *(what carries a gain)* At a paying depth the words leaving the surviving set are exactly the length-$(d-1)$ survivors with the *least* odd count $\lceil (d-1)\theta \rceil$ — those sitting on the contraction line rather than comfortably above it. Hence

$$\Big(1 - \frac{N_d}{2^d}\Big) - \Big(1 - \frac{N_{d-1}}{2^{d-1}}\Big) = \frac{L_{d-1}}{2^d},$$

  where L_t counts the length-t survivors with $o_t = \lceil t\theta \rceil$: $L_t = 1, 1, 1, 2, 3, 3, 7, 12, 12, 30$ for $t \le 10$.

Proof. (i) A start whose word has no contracting prefix of length $\le d$ satisfies $3^{o_t} \ge 2^t$ for every $t \le d$, so Proposition 3.1's envelope certifies no descent at any of those steps, whatever else is known about the start; the certified set omits the whole class. (ii) is the displayed arithmetic.

- (iii) A surviving word of length t with o odd letters extends by O to a surviving word always, since $3^{o+1} \ge 3 \cdot 2^t > 2^{t+1}$, and by E exactly when $3^o \ge 2^{t+1}$. So $N_d = 2N_{d-1}$ iff every survivor of length $d-1$ has that slack, and it suffices to test the leanest. The least odd count among survivors of length t is $\lceil t\theta \rceil$: the inequality $o_t \ge t\theta$ forces $o_t \ge \lceil t\theta \rceil$, and since $t \mapsto \lceil t\theta \rceil$ steps by 0 or 1 some word realizes it at every prefix at once. The test $3^{\lceil (d-1)\theta \rceil} \ge 2^d$ reads $\lceil (d-1)\theta \rceil \ge d\theta$; as θ is irrational, $\lceil (d-1)\theta \rceil = (d-1)\theta + 1 - \{(d-1)\theta\}$, and the test becomes $\{(d-1)\theta\} \le 1 - \theta$. Weyl's theorem on the equidistribution of $(d\theta)$ gives the density $1 - \theta$, and irrationality forbids eventual periodicity.
- (iv) By the same extension rule, the E-extension of a length-$(d-1)$ survivor with odd count o contracts exactly when $3^o < 2^d$, that is $o \le \lceil d\theta \rceil - 1$; survival already forces $o \ge \lceil (d-1)\theta \rceil$. At a paying depth $\lceil d\theta \rceil = \lceil (d-1)\theta \rceil + 1$, so the two constraints pin o to the single value $\lceil (d-1)\theta \rceil$, and the words removed are precisely the lean survivors. □

The constant is Proposition 7.7's. There β_* is the largest node-wise odd share still compatible with contraction; here it is the frequency with which one more letter of information is worth nothing. Both are the same remark about θ — that a step of the lattice path buys $\log 3$ against a cost of $\log 2$ — read once along the walk and once along the depth.

For this paper the consequence is concrete. Depths four and five are exhausted, depth six is a stalling depth, and the next increment is 115/128 at depth seven, worth 3/128 over Corollary 6.4. It requires the three contractors *OOEOOEE*, *OOOEOEE* and *OOOOEEE*, which descend from three *different* depth-five survivors — *OOEOO*, *OOOEO* and *OOOOE*. So the *OOOO*∗ split of Conjecture 7.3 does not by itself unlock depth seven; all of depth five does.

Part (iv) also says how that increment is priced. A run of k consecutive odd letters accumulates k nested 3/2-powers and an even letter square-roots the scale back down, so the longest odd run of a word is the kernel level its split needs, plus one: *OOOO*∗ is run four, the level-3 kernel of Conjecture 7.3, while Theorem 6.1 reaches run three. Sorting the removed words by that statistic prices each depth separately:

d	gain	run 2	run 3	run 4	run 5	run ≥ 6
4	1/16	1/16				
5	1/16	1/32	1/32			
7	3/128	1/128	1/128	1/128		
8	7/256	1/256	3/256	1/128	1/256	
10	3/256	1/1024	1/256	1/256	1/512	1/1024

The statistic has a mechanism under it, and the mechanism is the sharper object. Each iterate $J^t(n)$ sits at a scale P raised to an exponent obtained from the previous one by $x \mapsto \frac{3}{2}x$ or $x \mapsto \frac{1}{2}x$, as letter t is O or E; call it the *scale exponent* of that iterate. Let θ_s be the s-th floor defect, so that $J^s = (J^{s-1})^{3/2} - \theta_s$ or $(J^{s-1})^{1/2} - \theta_s$ according to letter s — the $\theta = \{n^{3/2}\}$ and $\theta_w = \{v^{1/2}\}$ of Theorem 6.3 are θ_1 and θ_3. Letter t constrains the parity of J^{t-1}, so its wave sits at that iterate's scale exponent, and linearizing the wave in an earlier defect θ_s gives θ_s a coefficient whose exponent is the *difference* of the two scale exponents, J^{t-1}'s less J^s's.

Those differences are the paper's own displayed constants. At the fifth letter of *OOEO*∗, where J^4 has scale exponent 27/16, they are 3/16, −9/16 and 9/16 for $s = 1, 2, 3$ — which are, in that order, the kept coefficient $\frac{9k}{16}n^{3/16}$ of θ_1; the discarded remainder $|k|P \cdot P^{-9/16}$; and the coefficient $\frac{3k}{4}v^{1/4} \asymp kn^{9/16}$ riding $\theta_3 = \{v^{1/2}\}$. At the fifth letter of *OOOE*∗ they are 3/16 and −9/16, the kept coefficient and the remainder of that proof. At the fourth letter of *OOO*∗, where J^3 has scale exponent 27/8, the $s = 2$ difference is 9/8: the coefficient $W \asymp kn^{9/8}$ that Section 3 names as the reason no drift-1 interval exists.

That last line is the criterion. A coefficient n^c has derivative $\asymp n^{c-1}$, so it drifts by less than 1 between consecutive integers exactly when $c < 1$; above the threshold the shifted-window device of Theorem 4.8 has no interval to run on and the letter must go through a kernel theorem instead. Counting the differences above the threshold prices each split:

The rule gives the constants too, not only the exponents. Letter t's wave is $e(kJ^{t-1}/2)$, and J^{t-1} feels θ_s through the chain, so the coefficient is

$$\frac{k}{2}\prod_{q=s+1}^{t-1} p_q \quad \text{at exponent } e_{t-1} - e_s, \qquad p_q = \tfrac{3}{2} \text{ or } \tfrac{1}{2},$$

and this returns, with no further input, every such constant the paper names: $\frac{3k}{4}n^{9/8}$, which is Theorem 5.3's kernel monomial verbatim; $\frac{9k}{16}n^{3/16}$ and $\frac{3k}{4}v^{1/4}$, the two kept coefficients of Theorem 6.3's *OOEO*∗ proof; the weight $\frac{3k}{4}n^{27/16}$ of the level-3 kernel, whose derivative $\asymp kn^{11/16}$ is the $\varrho' \asymp kP^{11/16}$ that Conjecture 7.3 quotes; and $\frac{9k}{8}n^{45/16}$, the family that same paragraph says the forced inner linearization trades θ_3 for.

There are two thresholds, and both are the paper's. Below 1 a coefficient drifts by less than one between consecutive integers and Theorem 4.8's shifted window applies. Above 9/4 — the figure Conjecture 7.3

names when it says the traded family at $kn^{45/16}$ sits *above the threshold where every method of this paper stops* — nothing applies. Between them lies kernel territory. And it is the *deepest* blocked defect that names the kernel. The shallower ones are not resolved elsewhere and not expanded either: their floors stay exact inside the kernel's argument, which is exactly why K_c is written over $\{\lfloor n^{3/2}\rfloor^{3/2}\}$ with the inner floor intact. A defect is expanded when its coefficient drifts slowly enough to window, and otherwise becomes part of what the kernel is a kernel *of*:

> **An open question about which defects the $9/4$ stop should screen.** The two sentences above say that only the deepest blocked defect names the kernel, and that shallower ones are not expanded but stay exact inside its argument. The table below, and the screen built from it, nevertheless apply the $9/4$ test to *every* blocked defect of a letter — the $OOOO{*}$ row lists $57/16$ and $45/16$ while its deepest sits at $27/16$. Both readings cannot be right, and the difference is not academic.
>
> Exactly one contractor on the whole frontier is barred by the $9/4$ stop and by nothing else: *OOOEOOEE*, at depth eight. Its letters ask for $\frac{3k}{4}n^{9/8}$ at level 2 — Theorem 5.3's own monomial — and $\frac{27k}{32}n^{33/32}$ at level 1, which are precisely the two the surviving words *OOOEOEE* and *OOOEOEOE* ask for and nothing more. Its letter 7 rides θ_5 at $81/64$, whose base $27/16$ has runs and whose $E<2$ holds. What bars it is θ_1, the *shallowest* defect, at $\frac{81k}{64}n^{147/64}$ — and $\frac{147}{64}$ exceeds $\frac{9}{4}=\frac{144}{64}$ by $\frac{3}{64}$.
>
> Under the printed reading the screen stops at $227/256$. Under the deepest-only reading *OOOEOOEE* joins, worth $\frac{1}{256}$, and the screen stops at $\frac{57}{64}$ — the same figure this section reaches only by way of a square-root level-3 kernel for *OOEOOEE*, a species with no theorem, no branch runs and the worst composite factor on the frontier. Two routes to one number, and one of them turns on whether a $3/64$ excess in an unexpanded coefficient is an obstruction.
>
> This paper does not settle it. The $9/4$ is the figure Conjecture 7.3 names when it places $kn^{45/16}$ beyond where every method here stops, and $45/16 = 2.8125$ is far above the line where $147/64 = 2.2969$ sits just over it. Whether the stop is about coefficients that must be handled — in which case an unexpanded θ_1 is not one — or about the linearisation as a whole is a question for the derivation of that threshold, which is not in this paper. The screen is left as printed, and this is recorded as the one place where a reading of a threshold, rather than a theorem, stands between $227/256$ and $\frac{57}{64}$.

split	deepest blocked	monomial	$>9/4$	status
OOEO$*$, letter 5	none	—	none	$N^{47/48}$, windows only
OOO$*$, letter 4	level 2, $3/2$	$\frac{3k}{4}n^{9/8}$	none	Thm 5.3
OOOO$*$, letter 5	level 3, $3/2$	$\frac{3k}{4}n^{27/16}$	$57/16$, $45/16$	open (Conj. 7.3)
OOOEOEE, letter 6	level 1, $3/2$	$\frac{27k}{32}n^{33/32}$	none	—
OOEOOEE, letter 6	level 3, $\surd$	$\frac{9k}{8}n^{45/32}$	none	—
OOOOEEE, letter 5	level 3, $3/2$	$\frac{3k}{4}n^{27/16}$	$57/16$, $45/16$	—

The first three rows are checks: the rule assigns *OOEO*$*$ no kernel and that proof uses none; it assigns *OOO*$*$ the level-2 kernel and returns Theorem 5.3's monomial; it assigns *OOOO*$*$ the level-3 kernel and returns the weight Conjecture 7.3 describes. The last three are the reading.

The criterion is not calibrated on depth seven. It reproduces the frontier of this paper wherever that frontier is known, and does so without exception. Over the eight O-rooted words of depth four, six carry no blocked defect at any letter — *OEEE*, *OEEO*, *OEOE*, *OEOO*, *OOEE*, *OOEO* — and those six are exactly the ones Theorems 4.4, 4.7 and 4.8 prove by drift-1 windows. The two that are blocked are *OOOE* and *OOOO*, which is the *OOO*$*$ split, the one word of depth four that needed Sections 5 and 6. So Theorem 4.8 stops where it does for the drift reason and for no other.

Depth five grades three ways, and so does the criterion:

word	blocked	the paper
OOEOE, *OOEOO*	none	Theorem 6.3, $N^{47/48}$, windows only
OOOEE, *OOOEO*	letter 4, level 2	Theorem 6.3, $N^{1-1/96}$, on Theorem 6.1's kernel
OOOOE, *OOOOO*	letters 4 and 5, levels 2 and 3	open

The unblocked pair still carries the better error exponent, 47/48 against $1-\frac{1}{96}$ — that is 94/96 against 95/96, a margin of one ninety-sixth, where the superseded 43/48 claimed nine. The direction is what an argument that never needs a kernel should give; the size of it is not evidence of anything. Two depths, sixteen words, three outcomes, no exceptions: that is the ground for reading the table above at depths where nothing is proved.

And it says something about the starts this paper excludes. Theorem 6.1 sets aside E-rooted words as "a different and easier problem", one square root having dropped the state to scale $\sqrt{N}$. At depth four the criterion agrees without exception: all eight E-rooted words are unblocked, against two of the eight O-rooted ones. Deeper the advantage in frequency persists — 2 of 16 blocked at depth five against 4, 4 of 32 against 12, 10 of 64 against 24 — but its character changes completely:

depth	E-rooted blocked	of square-root species	O-rooted blocked	of square-root species
4	0/8	0	2/8	0
5	2/16	2	4/16	0
6	4/32	4	12/32	2
7	10/64	10	24/64	6

Every blocked E-rooted word is blocked on a square-root defect, at every depth — necessarily, since its first letter makes $\theta_1 = \{n^{1/2}\}$ — and that is the species for which this paper has no kernel at any level. The first of them, *EOOOE* at letter five, carries $\frac{27k}{16}n^{19/16}$ on θ_1 with $E = \frac{27}{8}$, so it fails the linearisation test as well. The excluded problem is easier to enter and harder to finish, which is a better reason to set it aside than the one the parenthetical gives.

The square-root species is untouched, not overlooked. Screening everything this paper proves — every word of depth at most four, and Theorem 6.3's four of depth five — not one has a square-root defect at its deepest blocked position. Thirty-two words, none. The species first appears at depth five among E-rooted words, at *EOOOE* and *EOOOO*, and at depth six among O-rooted ones, at *OOEOOE* and *OOEOOO* — where it carries $\frac{9k}{8}n^{45/32}$ on θ_3 with $E = \frac{9}{4}$, failing the linearisation test in the same breath.

So the absence of a square-root kernel here is not a gap in the treatment: the treatment stops before the species occurs. And the first O-rooted instance of it is exactly *OOEOOE*, the prefix of the depth-seven word this section ranks last. What blocks *OOEOOEE* is not an unhandled case of something familiar but the first appearance, anywhere in the problem, of a kind of defect this paper never meets.

The same is true of the level, and it applies to the other target too. All four blocked defects among those thirty-two proved words sit at level 2; level 1 never occurs either. The first level-1 blockages are the E-rooted pair at depth five, and the first of 3/2 species are four words of depth six — *OOOEOE*, *OOOEOO*, *OOOOEE*, *OOOOEO* — all carrying the same $\frac{27k}{32}n^{33/32}$ with $E = \frac{27}{16}$. So *OOOEOEE*'s kernel is a first appearance as well, of a level rather than a species, and "one theorem closes both depth seven and depth eight" should not be read as one routine theorem.

What separates the two targets is not that one is precedented. It is the barrier this paper names for itself: after one floor the argument of the second is an integer sequence rather than a smooth function. Level 1's argument is $n^{3/2}$, still smooth, and sits below that barrier; the level-3 square-root object sits above it. Both are new; one is new on the side the paper's machinery was built to reach.

Is the level-2 reading forced by the depth? No, and it is not a coincidence either. Sorting every word of depths four and five by its blocked profile gives four classes:

blocked (level, species)	words	proved here
none	40	16
(2, 3/2)	4	4
(1, $\sqrt{\ }$)	2	0
(2, 3/2), (3, 3/2)	2	0

Among the words blocked at all, "proved" and "blocked only at level 2 of 3/2 species" coincide exactly — four words with that profile, all proved; four with any other, none proved. So the characterisation is neither forced by the depth nor accidental: it is the reach of the level-2 kernel, and the proved set is exactly what that kernel covers.

The unblocked rows deserve a plain word. Twenty-four words of depth five carry no obstruction of any kind and are still not proved here, and the reason is that they are worth nothing: Corollary 6.4 already attains the depth-five ceiling of 7/8, so no further depth-five class can raise the certified density at all. Theorem 6.3 says in its own title that it is not a census of depth five; the twenty-four are what it declines to census, and declining costs nothing.

Why the two counts behave differently. Blocking and contraction are conditions on the same lattice path (t, o_t), with $e_t = 3^{o_t}/2^t$, but they are not the same kind of condition. Contraction asks $3^{o_t} \geq 2^t$, which depends on (t, o_t) and nothing else — a pointwise condition, which is why N_d is a two-line dynamic program with the closed asymptotic $C\rho^d d^{-3/2}$ recorded above. Blocking asks $e_u - \min_{s<u} e_s > 1$, which couples two positions of the path. There is no closed form for the count, and none should be expected from a condition of that shape; but carrying the running minimum in the state restores a dynamic program, which counts $0, 2, 6, 16, 34, 82, 164, 368, \ldots$ blocked words at depths $3, 4, 5, \ldots$ in a few hundred states rather than 2^d words. The blocked fraction is 0.40 at depth 16 and 0.43 at depth 28, still climbing.

One more check, on the *form* of those monomials. Writing the kernel's coefficient as a power of n drops the floors kept exact beneath it, at a cost equal to its sensitivity to each, of exponent $(e_{t-1} - e_{s^*}) - e_s$ for the deepest blocked s^*. For Theorem 5.3 that is $-3/8$: negative, so the coefficient really is the monomial $\frac{3k}{4} n^{9/8}$ its statement fixes, to within $o(1)$. For the level-3 kernel it is $+3/16$, and the weight cannot be written as a monomial in n at all — a structural difference between the two levels beyond the derivative count, and one more reason the level-3 case is not the level-2 case with larger exponents. *OOEOOEE*'s level-3 kernel, by contrast, has sensitivities $-3/32$ and $-27/32$, so its weight *is* a clean monomial: it is a tidier level-3 problem than Conjecture 7.3, though of the wrong species.

And *OOOEOEE* has no inner floor at all. Its kernel is $\sum_{n\sim P} e\big(\frac{27k}{32} n^{33/32} \{n^{3/2}\}\big)$, whose argument $n^{3/2}$ is a smooth function of the summation variable. That is below the barrier this paper locates at level two — *after one floor the argument of the second floor is an integer sequence, not a smooth function* — and it is the barrier, not the exponent, that the whole apparatus of Sections 4 and 5 exists to cross. Note also that Theorem 4.7 does not cover this sum: it bounds products of sawtooths after Vaaler expansion at frequencies $|i|, |j|, |k| \leq P^{1/24}$, whereas here the sawtooth sits inside the exponential against an amplitude $\asymp P^{33/32}$. The level-1 kernel is a smaller problem than the level-2 one, but it is still a kernel and this paper does not contain it.

What the two differencings cost. How much smaller is settled by an accounting that never looks at the weight. Step 1 of Theorem 5.3 balances $|K|^2 \le 2P^2/H + (4P/H)\sum_{h\le H}|\cdot|$ twice; if the doubly-differenced sum is $\ll P^{1-\delta}$, the two balances force $H_2 = P^{\delta}$ and $H_1 = P^{\delta/2}$ — so that $H_2 = H_1^2$ — and return $K \ll P^{1-\delta/4}$. Each differencing halves the saving, and that is the whole content of the paper's $\frac{1}{96} = \frac{1}{4}\cdot\frac{1}{24}$: Lemma 5.2(ii) supplies $\delta = \frac{1}{24}$ and the two differencings quarter it. At that value the formula returns $H_1 = P^{1/48}$ and $H_2 = P^{1/24}$, which are the ranges Step 1 uses.

Since the chain never sees the weight's exponent, 9/8 and 33/32 enter it identically: a level-1 kernel matching $P^{1-1/96}$ needs its doubly-differenced sum to reach the same $P^{23/24}$, and no amplitude in the range considered here changes that arithmetic. What the amplitude can change is whether the differenced sum reaches the bar at all — but any $\delta > 0$ whatever already gives a nontrivial kernel bound, at a quarter of the saving. So the question left for *OOOEOEE* is not how much the amplitude costs in the bookkeeping, where it costs nothing, but whether the doubly-differenced level-1 sum admits any power saving.

Where the branch decomposition comes from. That question has a structural answer, and it is one the paper already contains without stating it as a criterion. A level-ℓ kernel takes its branches from $\lfloor \Delta_1 J^{\ell-1}\rfloor$; if $J^{\ell-1}$ sits at scale exponent e, that difference has derivative $\asymp hP^{e-1}$, so the floor is constant on runs of length $\asymp P^{2-e}/h$ and the decomposition exists exactly when $e < 2$. At $\ell = 2$ this is $e = 3/2$ and the runs have length $P^{1/2}/h$ — the b-runs of Lemma 5.1(iii), with that length. At $\ell = 3$ it is $e = 9/4$, the exponent is negative, and there are no runs at all: this is Conjecture 7.3's own complaint that v jumps by $\asymp n^{5/4}$ per step and Lemma 5.1(iii) has no analogue at the v-level. So $e < 2$ is a third threshold, alongside the drift-1 condition on coefficients and the 9/4 stop, and it is the one that separates Theorem 5.3 from Conjecture 7.3.

The drift threshold is graded, and the grading is already in the paper's own $\frac{1}{96}$. Being above it is not a yes-or-no fact about a coefficient but a count. Weyl differencing lowers a weight's exponent by exactly one — $\Delta_h c \asymp \alpha\, khn^{\alpha-1}$ — so a coefficient at α is $\lceil\alpha\rceil - 1$ differencings from the near side of the threshold. And each differencing splits the phase in two: a branch that loses a *level*, because the defect's outermost floor is exposed, and a branch that keeps the level and loses an *exponent*. Both must bottom out, so a level-ℓ defect at coefficient exponent α costs

$$d = \max(\ell,\ \lceil\alpha\rceil - 1)$$

differencings, and the chain halves a saving each time: a factor 2^{-d}.

At level two with $\alpha = \frac{9}{8}$ that is $d = \max(2,1) = 2$, and the saving $\frac{1}{24}$ of Lemma 5.2(ii) becomes $\frac{1}{96} = \frac{1}{4}\cdot\frac{1}{24}$ — the paper's own headline constant, read off the grading. At level one with $\alpha = \frac{33}{32}$ it is $d = \max(1,1) = 1$, a single halving, which is what the level-1 analysis below spends.

The grading separates the frontier where the binary reading cannot. Over the 26 663 blocked sites carried by contractors of depth at most thirteen, d runs from 1 to 128: 7% sit at $d = 1$, a fifth at $d \le 2$, under a third at $d \le 3$, and the tail reaches $\alpha = \frac{525297}{4096}$, where 2^{-128} is not a saving in any useful sense. Which count binds is close to even — the level for 40% of sites, the drift depth for 45%, and they tie for 16%.

Among the targets that matter the level binds, and only once do they tie:

target	α	level	$\lceil\alpha\rceil - 1$	d	cost
Theorem 5.3	9/8	2	1	2	$\frac{1}{4}$
OOOEOEE, letter 6	33/32	1	1	1	$\frac{1}{2}$
OOEOOEE, letter 6	45/32	3	1	3	$\frac{1}{8}$
Conjecture 7.3	27/16	3	1	3	$\frac{1}{8}$

So the drift threshold, which this section treats as the obstruction for these words, costs one differencing on all four; it is the level that costs two and three. And *OOOEOEE* is the tractable one for a sharper reason than being a level below Theorem 5.3: it is the only target where the two counts coincide, so one chain serves both purposes at once. That is exactly what the identity below exhibits.

The grading has a domain, and the domain is the branch-run condition. A cost of 2^{-d} is only payable if the chain can be organised into runs at the level it passes through, which is $e < 2$. So $e < 2$ is not a fourth axis beside the grading; it is the grading's precondition, and it makes the primary split: Theorem 5.3 and *OOOEOEE* have runs, *OOEOOEE* and Conjecture 7.3 do not. Inside the second pair the grading is blind — both cost 2^{-3} — and it is species and the composite factor that separate them, both isolating *OOEOOEE*, while the 9/4 stop isolates Conjecture 7.3 the other way.

axis	partition of the four
differencing cost d	1: *OOOEOEE*; 2: Thm 5.3; 3: the other two
branch runs $e < 2$	Thm 5.3, *OOOEOEE* | *OOEOOEE*, Conj 7.3
above 9/4	Conj 7.3 | the other three
species	*OOEOOEE* | the other three
worst κ	*OOEOOEE* | the other three

And level three is run-less everywhere, not just at OOOO∗. The statement above is made for that split, but it is forced. A contractor must satisfy $3^{o_2} \geq 4$ at $t = 2$, and $3^1 = 3 < 4$, so $o_2 = 2$: **every contractor begins** *OO*, and therefore every level-3 defect on the frontier branches on $e_2 = \frac{9}{4}$. Over the 26 663 blocked sites of depth at most thirteen, all 3910 at level three lack runs, without exception. Conjecture 7.3's complaint is not a fact about *OOOO∗*; it is a theorem about level three.

Levels beyond it are not uniformly barred, which the 9/4 reading does not suggest. The base at level s is e_{s-1}, and survival pins it only from below, so a prefix carrying an even letter early brings it back under 2: at level four the base is $\frac{9}{8}$ for *OOE∗* and $\frac{27}{8}$ for *OOO∗*, and 746 of the 3167 level-four sites have runs. Level five recovers 1156 of 2546, and runs reappear at levels seven, eight and ten. They vanish again at six, nine and eleven. Overall 51% of blocked sites have runs and the pattern in the level is not monotone — which is worth recording only because the reading that "level three and above has no runs" would predict none of it.

At $\ell = 1$ the base is n itself, $e = 1$, and $\Delta_1 n = d_1$ is constant outright: the runs fill the block and the branch set is a single point. So the carry bookkeeping does have a level-1 form and it is degenerate — there is nothing to branch on. Read the other way, Lemma 5.1(iii) is already a level-1 construction: its $b_i = \lfloor \Delta_i X \rfloor$ are floors of differences of $X = n^{3/2}$, and it calls its own κ the *level-1 carries*. What a level-1 kernel would not need is the layer built on top of them — parts (i) and (ii) of that lemma, the level-2 defect identity and the double-gap carry algebra.

What the level-1 kernel actually is. It is not a new species. For integer r, $e(r\{x\}) = e(rx)$, since $r\lfloor x \rfloor$ is an integer; so Fourier-expanding $e(c(n)\{n^{3/2}\})$ in the sawtooth produces exactly the monomial waves $e(rn^{3/2})$ that Theorems 4.4 and 4.7 estimate, with coefficients $|a_r(c)| \leq \min(1, 1/(\pi|c - r|))$ concentrating the mass at $r \approx c(n)$. The whole difference is the range: those theorems work at $|r| \leq P^{1/24}$, and the kernel needs $|r| \lesssim kP^{33/32}$ — a gap of $P^{95/96}$, and nothing else.

That also says what the drift threshold means, which is more than that an interval fails to exist. The window on which c moves by less than 1 has length $\asymp 1/c' \asymp P^{-1/32}/k$, shorter than the spacing 2 of the summation variable, so it holds *at most one* odd integer. A coefficient exponent above 1 is exactly the statement that the shifted window is finer than the lattice it is supposed to sit on, and that is why no amount of care with Lemma 3.7 recovers it: what that lemma expands is a sum over a window, and a window with one term is not a sum.

> *Erratum (at most one, not none).* This read "so it contains no integer at all", which is stronger than the length gives and stronger than the argument needs. A window of length $1/c' < 2$ laid on a lattice of spacing 2 holds one point with density $1/(2c')$ and none otherwise — 0.43 at $P = 10^4$, 0.37 at 10^6, 0.22 at P_0 — and the density falls only like $n^{-1/32}$, so it is never zero. Tiling the block and counting reproduces $1/(2c')$ to four figures

(`decoration_budget.level1_drift_window_occupancy`). The conclusion is unaffected: at most one summand per window is already fatal to Lemma 3.7, and that is what "finer than the lattice" says.

And "fatal" can be priced, which says why care does not help. Lemma 3.7 expands $e(-c\{t\})$ into modes with total coefficient mass $\sum|b_u| + \sum|v_q| \le 8 + 2\log(2{+}c{+}U) + 4H_J$, under $U \ge 8(1{+}c)$ — the lemma's frozen coefficient is this section's c, and U is its mode cutoff. On a window of W terms the trivial bound is W, and the expansion beats it only if the individual mode sums do. At $W = 1$ every mode sum is a single unimodular term, of modulus exactly 1, so the expansion returns the whole mass against a trivial bound of 1. With the frozen $c \sim \frac{27k}{32}P^{33/32}$ the kernel would need, and $J = R_0$:

P	b-mass	v-mass	total
10^6	40.5	19.6	60.1
P_0	76.4	41.3	117.7
10^{24}	126.0	71.4	197.4

so expanding is worse than not expanding by two orders, not merely no better (`decoration_budget.lemma37_one_term_`
There are two edges and not one: at the hypothesis boundary $U = 8(1{+}c)$ the flat error $8(1{+}c)/U$ is exactly 1 on its own, the trivial bound, before a single mode is counted; taking U larger kills that term and raises the b-mass, which carries $\log U$.

The two failures have one cause, which is the point. The window is short because c is large, $1/c' \asymp P^{-1/32}/k$; the mass is large because c is large, $2\log c \asymp \frac{33}{16}\log P$. The paper needs only "no better than trivial" and has it; what the price adds is that the two ends of the lemma are driven by the same quantity, so there is no setting of U or J that trades one against the other.

And nothing printed depends on how much larger than 1 *the drift is.* The condition enters as the binary $2c' > 1$, which is window-length against lattice-spacing; no constant anywhere in Sections 4–7 carries c' beyond that. What grows with c' is the occupancy $1/(2c')$, which describes the situation rather than bounding anything. Worth stating because the margin is thin and slow: $2c' = \frac{891k}{512}n^{1/32}$ is 2.32 at 10^4, 4.62 at P_0, and reaches 10 only near $2\cdot 10^{24}$. A reader who assumed the drift condition was comfortable at P_0 because the exponent exceeds 1 would be assuming the wrong thing; it exceeds it by a factor under five, and the argument is built so that this does not matter.

And the same drift is what makes the sum cancel. The two readings of $c' \gg 1$ point opposite ways, and only the first is stated above. Consecutive summands of K_1 see values of $c\theta_1$ that differ by more than a whole period, so they are decorrelated — which is what a shifted window is for, and what its absence therefore costs the method rather than the object. That is a claim about the sum, and it can be measured. Splitting one pass over $n \sim P$ into blocks and fitting the root-mean-square block sum against block length (`paper_b_audit.level1_kernel_block_scaling`; the instrument reads $0.497{\pm}0.043$ at its default 200 trials on data whose exponent is exactly $\frac{1}{2}$) gives, over $P \in \{10^4, 3\cdot 10^4, 10^5, 10^6\}$ and $k \in \{1,2,4\}$, twelve exponents with mean 0.487 (`decoration_budget.level1_exponent_sweep`, which is that ladder). The instrument's own 90% interval on data that is exactly $\frac{1}{2}$ is $[0.4268, 0.5684]$, and exactly one of the twelve lies outside it: $P = 10^4$, $k = 1$, at 0.3866. The other eleven run over $[0.446, 0.535]$, and dropping the smallest P alone lifts the mean to 0.505.

The ratio rms$/\sqrt{L}$ over the five fitted block lengths is flat in L and sits in $[0.94, 1.12]$ at $P = 10^6$. That is a statement about the top of the ladder and not about $P \ge 10^5$: at $P = 10^5$ the same ratio reaches 0.874 at $k = 4$ and 1.150 at $k = 2$, so over $P \ge 10^5$ the honest interval is $[0.87, 1.15]$. Both readings say the same thing about the object — the block sums grow like $\sqrt{L}$ — and only the narrower one is a statement about a single P.

The control on the same pass is the unweighted defect, $e(\{n^{3/2}\}) = e(n^{3/2})$. It behaves the other way. At $P = 10^6$ its rms$/\sqrt{L}$ climbs $0.99, 1.07, 1.48, 2.09, 2.94$ as L runs 1953 to 31250 — block sums growing like $L^{0.91}$, which is what a phase that is *smooth* on a block does — while the kernel's stays at $0.943, 0.948, 0.947, 0.963, 0.981$ across the same five lengths. The weight does not merely fail to hurt the sum:

it is what produces the cancellation, by destroying the smoothness that makes $e(n^{3/2})$ a stationary-phase object in the first place.

And the separation is not a feature of one P; it opens where the second-derivative test says it must. The control is the Weyl sum itself, since $e(\{n^{3/2}\}) = e(n^{3/2})$. Over odd n with step 2 its curvature in the term index is $\lambda = 3n^{-1/2} \sim 3P^{-1/2}$, so a block sum of length L is $\ll L\lambda^{1/2} + \lambda^{-1/2}$. The first term is *linear* in L, and it dominates once $L \gg 1/\lambda = \sqrt{P}/3$ — so the control's fitted exponent runs to 1, not to $\frac{1}{2}$. The five fitted lengths are $L \in [P/512, P/32]$, so the whole window clears the crossover once $\sqrt{P} \geq \frac{2}{3} \cdot 256$, i.e. $P > (512/3)^2 = 2.91 \cdot 10^4$.

Measured at $k = 1$ along that ladder (`decoration_budget.level1_control_trend`):

P	kernel	control	gap
10^4	0.3866	0.4902	0.104
$3 \cdot 10^4$	0.5081	0.5733	0.065
10^5	0.5178	0.7252	0.207
$3 \cdot 10^5$	0.4738	0.8181	0.344
10^6	0.5139	0.9099	0.396
$3 \cdot 10^6$	0.5563	0.9852	0.429

The kernel is flat at $\frac{1}{2}$ — mean 0.4928, spread 0.170 across two and a half decades and no trend — while the control climbs monotonically to 0.985 and the gap widens monotonically from $3 \cdot 10^4$ on. At $P = 10^4$ there is no separation at all: the control reads 0.4902, and that is the one row whose fitted window does not clear the crossover ($L \in [20, 312]$ against $1/\lambda = 33$). The contrast the previous paragraph reads off $P = 10^6$ is therefore not a coincidence of scale in the direction one would fear; it is absent below $2.91 \cdot 10^4$, appears exactly there, and grows.

The kernel has no such threshold, and it is worth saying why not. Its coefficient is $c(n) = \frac{27k}{32} n^{33/32}$, so $c'(n) = \frac{891k}{1024} n^{1/32}$ and, over odd n with step 2, the coefficient advances by $2c' = \frac{891k}{512} n^{1/32}$ per summand. "More than a whole period per step" is $2c' > 1$, and that holds from $n = (512/891k)^{32} \approx 2 \cdot 10^{-8}$ upward — below every P anyone would run. There is no threshold to cross, which is why the kernel column is flat where the control column is not. The condition is not comfortable, mind: $2c'$ runs from 2.32 at 10^4 to 2.77 at $3 \cdot 10^6$, and reaches 10 only near $2 \cdot 10^{24}$. It grows like $n^{1/32}$, which is the same $\frac{1}{32}$ the drift threshold turns on. One exponent does both jobs: it is what puts c past the drift-1 window and what decorrelates the summands, and it does the second from the start and the first only just.

So the one kernel reading outside the instrument's interval — 0.3866 at $P = 10^4$, $k = 1$ — is not a regime boundary. It is the estimator. Across $k = 1, \dots, 8$ at that P the mean is 0.4771 and there are two excursions, one low and one *high* ($k = 8$ reads 0.584), which is what a 90% interval predicts for eight draws. What moves with P is the spread — 0.062, 0.035, 0.024 at 10^4, $3 \cdot 10^4$, 10^5 — and that is a statement about the number of terms per block, not about the sum: the calibration was run at $N = 5000$, which is exactly the term count at $P = 10^4$ (`decoration_budget.level1_kernel_k_spread`).

So the drift threshold is not only an obstruction. It is the reason the shifted window has nothing to run on *and* the reason there is something for a method to find. This is an observation and nothing more: no bound on K_1 is claimed anywhere in this paper, a block-scaling exponent is a statistic, and 10^6 is far below any scale at which an asymptotic could be read. What it settles is which of the two the depth-seven deficit is — and it is the method.

What one mode is worth. The modes themselves are not the difficulty. For a phase of size $P^{81/32}$ on $n \sim P$ — which is what $e(rn^{3/2})$ has at $r \asymp kP^{33/32}$ — the van der Corput differencing and reflection processes, run from the trivial pair, take their best value at $(\frac{1}{11}, \frac{3}{4})$:

$$\sum_{n \sim P} e(rn^{3/2}) \ll P^{313/352},$$

a saving of $\frac{39}{352}$. The differencing chain quarters a saving, so it needs at least $\frac{1}{24}$ to reach $P^{1-1/96}$; this is 2.66 times that. (These are van der Corput pairs, not the ordered pairs drawn from E that Section 2 calls exponent pairs.)

The caveat is the whole of the difficulty, and it is the paragraph above. That bound is for one mode at one frequency. Assembling the modes means summing over r against coefficients depending on n through $c(n)$, and the device for that is the shifted window, whose window here holds no integer. So the arithmetic says the branch is not dead — a single mode carries more saving than the chain needs, with a factor of two and a half to spare — and says nothing whatever about how to collect the modes.

What one differencing does. The device is not a better shifted window. It is Weyl differencing — Step 1 of Theorem 5.3, applied once instead of twice — and it works because it attacks the weight rather than the defect. Write $\varphi(n) = c(n)\theta_1(n)$ with $\theta_1 = \{X\}$, $X = n^{3/2}$. Since $\lfloor X \rfloor$ is an integer, $\theta_1(n+h) = \{\theta_1(n) + \{\Delta_h X\}\}$, and as both summands lie in $[0,1)$ this is $\theta_1 + \{\Delta_h X\} - \kappa$ with $\kappa \in \{0,1\}$. Hence, exactly,

$$\Delta_h\varphi = (\Delta_h c)\,\theta_1(n) + c(n{+}h)(\{\Delta_h X\} - \kappa), \qquad \kappa = 1 \iff \theta_1(n) \geq 1 - \{\Delta_h X\}.$$

Both are identities, not approximations: verified to fifty-five digits at $n \sim 10^6$, and the characterisation of κ over four thousand samples without a mismatch.

Three terms, and each sits on the near side of a threshold this paper already crosses.

- $(\Delta_h c)\theta_1$. The weight is $\Delta_h c \asymp \frac{891}{1024} khn^{1/32}$, of coefficient exponent $\frac{1}{32}$. That is *below* the drift threshold, where $\frac{33}{32}$ was above it. One differencing carries the weight across the very threshold that made the shifted window unavailable, and this term is no longer drift-blocked.
- $c(n{+}h)\{\Delta_h X\}$. The large weight survives, $\asymp kP^{33/32}$, but the defect is now $\{\Delta_h X\}$, whose n-derivative is $\asymp hP^{-1/2}$: constant on runs of length $\asymp P^{1/2}/h$ — measured, five wraps over four thousand consecutive odd n at $P = 1.5 \cdot 10^6$, against five predicted. On a run the phase is the monomial $\frac{27k\beta}{32} n^{33/32}$ with β frozen. These are the b-runs of Lemma 5.1(iii), one level down.
- $-c(n{+}h)\kappa$. Since $\kappa \in \{0,1\}$, $e(-c\kappa) = 1 - \kappa(1 - e(-c))$, so the carry enters as an indicator multiplying the other two; and on a run β is frozen, so $\kappa = \mathbf{1}_{\theta_1 \in [1-\beta,1)}$ is the indicator of an equidistributing θ_1 in a *fixed* interval. That is a Vaaler expansion — Lemma 3.5 — and not a shifted window at all. It returns monomial waves $e(jn^{3/2})$ against $e(-\frac{27k}{32}(n{+}h)^{33/32})$, which is the shape Theorems 4.4 and 4.7 estimate.

So the reason the paragraph above finds no device is that it looks for one in the *undifferenced* sum, where the coefficient depends on n and the shifted window is the only tool available. After one differencing there is no coefficient problem left to solve: the exponent is $\frac{1}{32}$.

This is an accounting and not a proof, and the gap is the whole of Section 5's work at one level down. It does not choose the Vaaler truncation, bound the oscillation of the $1 - e(-c)$ weight, or handle the two terms that are not smooth. What the identity settles is that the route is Step 1 followed by Lemma 3.5, not a sharper Lemma 3.7, and that every ingredient it needs is already in Sections 3 to 5.

What the balance returns. One differencing halves a saving, so matching $P^{1-1/96}$ needs the differenced sum to reach $P^{1-1/48}$. The smooth term can be priced, and the price turns entirely on which curvature a b-run carries. It is not c''. What a run freezes is the *integer* $b = \lfloor \Delta_h X \rfloor$, not the fractional part, so the smooth phase on it is $c(n{+}h)(\Delta_h X - b)$, and all four terms of that product's second derivative — $c''\Delta_h X$, $2c'(\Delta_h X)'$, $c(\Delta_h X)''$ and $-c''b$ — are of one size,

$$\lambda \;\asymp\; kh\,P^{-15/32},$$

measured at 0.989 times it over ten samples at $n \sim 10^6$. Against $c'' \asymp kP^{-31/32}$ that is larger by $P^{1/2}/h$, which is exactly the run length: the factor the frozen integer is worth.

With it the stationary-point count per cell is $L\lambda \asymp kP^{1/32}$, comfortably above 1, so the second-derivative test is the right tool. Over $\asymp hP^{1/2}$ cells of length $P^{1/2}/h$ it gives $U(h) \ll (kh)^{1/2}P^{49/64}$ for the differenced sum $U(h)$, and the Weyl balance $|K_1|^2 \ll P^2/H + k^{1/2}H^{1/2}P^{113/64}$ is optimised at $H = P^{5/32}k^{-1/3}$, returning

$$K_1 \;\ll\; P^{59/64}k^{1/6} \;\ll\; P^{59/64+1/144} \quad \text{uniformly in } k \leq P^{1/24},$$

a saving of $\frac{41}{576}$ against the $\frac{1}{48} = \frac{12}{576}$ required — a factor $\frac{41}{12}$ of room. Run instead with c'' as the curvature, the same accounting returns $\frac{1}{256}$, short by $\frac{16}{3}$; the whole difference is the misidentified λ, whose $\lambda^{-1/2}$ term then charges for a stationary point the cell does not contain.

So the balance is not where this stands or falls. It has three times the room it needs on the term it can price, and prices only that term. What is unpriced is $(\Delta_h c)\theta_1$, where the shifted window now applies but has still to be run, and $-c(n+h)\kappa$, where the Vaaler expansion has still to be truncated against a weight of size $kP^{33/32}$. Those two are the problem; the arithmetic of H is not.

What the truncation costs. The carry is the one carrying the large weight, and its budget is also a closed formula. Vaaler at truncation J leaves a remainder $\asymp P/J$ and returns waves $e(jn^{3/2})$ against $e(-\frac{27k}{32}(n+h)^{33/32})$ with coefficients $|a_j| \ll 1/|j|$. Since $\frac{3}{2} > \frac{33}{32}$ the first monomial dominates the derivatives for every $j \geq 1$, so a van der Corput pair $(\varkappa, \ell)$ prices each at $(jP^{1/2})^{\varkappa}P^{\ell}$; the $1/j$ weights make the sum over $|j| \leq J$ of order $J^{\varkappa}P^{\varkappa/2+\ell}$, and balancing that against P/J gives

$$J = P^{\delta}, \qquad \delta = \frac{1 - \varkappa/2 - \ell}{\varkappa + 1}.$$

Over the pairs the two processes generate from the trivial one the best is $(\frac{1}{4}, \frac{13}{22})$, returning $J = P^{5/22}$ and a saving of $\frac{5}{22}$; the classical $(\frac{1}{2}, \frac{1}{2})$ already returns $\frac{1}{6}$. Against the $\frac{1}{48}$ required these are 10.9 and 8 times over.

Two checks that the range is the one it looks like. The $(\Delta_h c)\theta_1$ term, sawtooth-expanded in its turn, shifts j by at most $\asymp khP^{1/32} \leq P^{11/96}$, which $P^{5/22}$ dominates — so both unpriced terms reduce to the *same* family of two-monomial sums, over one range of j. And the interval endpoint $1-\beta$ moves, so Vaaler cannot be applied globally with frozen coefficients: the window on which β moves by less than $1/J$ has length $\asymp P^{1/2}/(Jh)$, and it holds integers exactly when $Jh \leq P^{1/2}$. At $J = P^{5/22}$ and $h \leq P^{1/24}$ that reads $P^{71/264}$ against $P^{1/2}$, a margin of $P^{61/264}$.

That last is the arc closing. The shifted window is not useless here; it was being asked of the wrong quantity. Applied to c, whose exponent is $+\frac{1}{32}$ above the drift threshold, it has no interval. Applied after differencing to β, whose exponent is $-\frac{1}{2}$ and far below it, it has one with a quarter of an order to spare.

And it is not needed. A window would have to be paid for. Across one of length $P^{1/2}/(Jh)$ the derivative of $e(jn^{3/2})$ moves by $\leq P^{-1/24}$, so f' is frozen there, no pair beats the trivial W, and $JhP^{1/2}$ windows of length $P^{1/2}/(Jh)$ reassemble to P: a per-window assembly returns nothing at all. The way past that is not a better assembly but the observation that the carry never needed an indicator. Since $\{X(n+h)\} = \{\{X(n)\} + \{\Delta_h X\}\}$,

$$\kappa = \lfloor \{X(n)\} + \{\Delta_h X\} \rfloor = \{X(n)\} + \{\Delta_h X\} - \{X(n+h)\} = \tfrac{1}{2} + \psi(X(n)) + \psi(\Delta_h X) - \psi(X(n+h)),$$

two lines of algebra, and Lemma 5.1(iii)'s gap identity $G_i = \lfloor \delta_{h_i} \rfloor + \kappa_i$ written in sawtooths. Verified in exact integer arithmetic — $\lfloor x^{3/2} \rfloor$ is isqrt(x^3) and the gap's floor is pinned by squaring — over ten thousand pairs (n, h), with $\kappa = 1$ in 50.3% of them.

Every argument there is a *smooth* function of n over the whole range: $n^{3/2}$, $(n+h)^{3/2}$, and their difference. So each sawtooth Vaaler-expands into pure monomial waves with absolute coefficients, and no endpoint moves. The two large families are $e(jn^{3/2})$ and $e(j(n+h)^{3/2})$, which are what the budget above prices; the third, $e(j\Delta_h X)$, has $F/P \asymp jhP^{-1/2} \leq P^{-61/264}$, so there $c(n+h)$ dominates the derivatives and the pair applies at $kP^{1/32}$, returning $k^{1/2}P^{33/64}$ — no constraint at all.

What is still not done is the two-monomial estimate itself, uniformly in j, k and h: every phase above is $e(jn^{3/2} \pm \frac{27k}{32}(n+h)^{33/32} + \cdots)$, and this paper proves no such bound. That, and not the assembly, is where a proof would start.

And it is a smaller thing than the leftover it resembles. The repository carries an open two-monomial exponent-pair question (`exponent_pair_two_monomial.md`), and it is tempting to read the two as one. They are not. Write $\psi(p, q)$ for the linear form a pair must satisfy. There the target is $\frac{5}{4}p + q < \frac{2}{3} = 0.667$ against a hull minimum of 0.861 over the closure under the two processes — $95/112 = 0.848$ once Huxley and Bourgain are admitted — so any solution lies *below* the hull and is a subconvexity result. Here the requirement $\delta \geq \frac{1}{48}$ reads

$$25p + 48q \ \leq \ 47,$$

against a hull minimum of 34.5: a margin of 27%, *inside* the hull. Of the fifty-six pairs the two processes generate, three fail — the trivial $(0, 1)$ at 48 against 47, and two of its neighbours, all three with $q \geq 0.984$ where every pair that clears has $q \leq 0.973$. The failures are the trivial pair and what crawls back towards it. Everything else clears: van der Corput's $(\frac{1}{6}, \frac{2}{3})$ at 36.17, Weyl's $(\frac{1}{2}, \frac{1}{2})$ at 36.5, Bourgain's at 35.30.

Nor is what separates them the second monomial. Both problems are dominated by one term: there $m^{9/4}$ leads $jm^{2/3}$ by $M^{71/60}$; here $jn^{3/2}$ leads $\frac{27k}{32}(n+h)^{33/32}$ by $P^{41/96}$ at the worst corner $j = 1$, $k = P^{1/24}$, and by $P^{691/1056}$ at the top of the j-range. What separates them is only where the target sits relative to the hull.

So the missing ingredient is not a new exponent pair. It is the verification that a pair's derivative hypotheses hold on the perturbed monomial uniformly in $j \leq P^{5/22}$, $k \leq P^{1/24}$, $h \leq P^{1/48}$ — routine by the margins above, and unwritten.

The three thresholds are independent, and they re-sort depth seven. Applying all of them to the three targets at once:

split	kernel	branches on	runs?	$> 9/4$
$OOO*$, letter 4	level 2, 3/2	3/2	yes	none
$OOOO*$, letter 5	level 3, 3/2	9/4	**no**	57/16, 45/16
$OOOEOEE$, letter 6	level 1, 3/2	1	yes	none
$OOEOOEE$, letter 6	level 3, $\surd$	9/4	**no**	none
$OOOOEEE$, letter 5	level 3, 3/2	9/4	**no**	57/16, 45/16

The two conditions in the last two columns are genuinely independent: over the words of length at most nine all four combinations occur, so neither implies the other. $OOEOOEE$ occupies the cell that is easiest to overlook — nothing above 9/4, yet no branch runs — and the reason is exact rather than analogous. Its deepest blocked defect is θ_3, so it would branch on J^2, which is the same v whose jump of $\asymp n^{5/4}$ per step is the obstruction Conjecture 7.3 names. It is not merely of a species the paper lacks a theorem for; it inherits, verbatim, the branching failure of the open conjecture.

And one of the three is not a linear object at all. Composing power maps composes to a single power, so letter t's wave is $(J^s)^E$ with $E = \prod_{q=s+1}^{t-1} p_q = e_{t-1}/e_s$, and the squared-defect term sits at $e_{t-1} - 2e_s$ — negligible exactly when $E < 2$. Every kernel this paper forms has $E = \frac{3}{2}$: its defect sits one letter below its wave, which is the shape of Lemma 5.1(i), $c\,\theta = \frac{k}{2}(Z^{3/2} - \lfloor Z \rfloor^{3/2})$. $OOOEOEE$ has $E = \frac{27}{16}$ — four steps, but three $\frac{3}{2}$'s against a $\frac{1}{2}$ — and is still sub-quadratic. $OOEOOEE$ has $E = \frac{9}{4}$, two $\frac{3}{2}$'s in succession, and its squared term sits at $P^{9/32}$. Measured on the orbit of $n = 1000057$, which realizes $OOEOO$, that term is 49.9 against $5.5 \cdot 10^{-4}$ for $OOOEOEE$'s: not a correction to a kernel but a second wave beside it, of a shape the paper never has to form. (Positive second-order exponents at defects a kernel keeps *exact* are harmless, and $OOO*$ and $OOOO*$ both have them at $s = 1$; what matters is the defect actually expanded.)

That leaves $OOOEOEE$ as the only one of the three whose deepest blocked defect admits a branch decomposition at all. The near-term reading of depth seven is therefore 1/128 rather than 2/128: $7/8 \to 113/128$ on $OOOEOEE$ alone, with $114/128 = 57/64$ available only if a square-root kernel is found for an object that branches no better than Conjecture 7.3's.

What one theorem would buy. Run the same screen over every contractor at every paying depth — each letter tested for a blocked defect, and each blocked defect for branch runs, for the 9/4 stop, and for $E < 2$ — and almost nothing survives:

depth	gain	unobstructed	word	profile
7	3/128	1 of 3	*OOOEOEE*	letter 4 at level 2, letter 6 at level 1
8	7/256	1 of 7	*OOOEOEOE*	letter 4 at level 2, letter 6 at level 1
10	3/256	0 of 12	—	—
12	15/2048	0 of 30	—	—
13	85/8192	0 of 85	—	—

The two survivors share their first six letters, so their profiles are not merely alike but identical. Letter 4 asks for $\frac{3k}{4}n^{9/8}$ at level 2 — Theorem 5.3's own monomial, already proved. Letter 6 asks for $\frac{27k}{32}n^{33/32}$ at level 1, which is not. One theorem, about one monomial, closes both, and it is the level-1 kernel already identified above.

What it buys is $\frac{1}{128}+\frac{1}{256}=\frac{3}{256}$, carrying the certified density from 7/8 to 227/256. After that the screen is empty: across depths 10, 12 and 13 not one of the 127 contractors is free of an obstruction, so 227/256 is where this line ends without new machinery. The screen is negative evidence only — it names the words carrying no obstruction this paper knows how to state, not the words that are provable.

OOOOEEE repeats the *OOOO∗* row exactly, so Conjecture 7.3 is necessary for that third of the depth-seven increment and not merely sufficient — and it is the only one of the three carrying anything above 9/4. *OOOEOEE* is blocked no deeper than level 1, the defect $\{n^{3/2}\}$ of a monomial base, which is one level *below* Theorem 5.3 rather than above it. *OOEOOEE* is blocked at level 3 on a square-root defect, a level-and-species combination for which this paper has no theorem and no conjecture; it is the harder of the two, not the easier, and the longest-odd-run reading had it the wrong way round.

None of this makes any of the three a corollary. Theorem 5.3 is a statement about one monomial and says so — its Step 5a cancellation uses the exact ratio of that monomial's derivatives, a two-sided scale $c^{(r)} \asymp kP^{9/8-r}$ being insufficient to determine the composite — so a kernel at 33/32 means recomputing the composites $\frac{243}{512}$ and $\frac{2187}{2048}$ there and showing they do not vanish. Each depth-seven word also needs six waves where the proved rows need four, every extra wave bringing its own Vaaler truncation, majorant and remainder into a balance that would have to be redone. What the table settles is the order of attack — *OOOEOEE* first, then *OOEOOEE* behind a square-root kernel that does not exist, then *OOOOEEE* behind Conjecture 7.3 — and that the first two, unlike the third, stay inside the 9/4 threshold throughout. Together they would carry the certified density to 57/64 with Conjecture 7.3 still open — though the branch condition below cuts that expectation back to the first of the two.

What "recomputing the composites" comes to. The requirement is discharged for 33/32, and how it is discharged says more than the answer. Each of the three sign-critical composites of Sections 5 and 6 is a quadratic in the weight exponent alone. Put $c(\nu)=a\nu^{\alpha}$ against the geometry the map fixes, $X=\nu^{3/2}$ and $F=\frac{3}{2}jm^{1/2}$; the algebra of (E6), of Step E and of Lemma 5.2b then runs unchanged with α for 9/8 and returns

$$\begin{aligned}
\lambda_a &= a\big[\tfrac{3}{2}(\alpha+\tfrac{3}{4})(\alpha-\tfrac{1}{4})-\tfrac{9}{16}\big]\,j\nu^{\alpha-5/4},\\
\lambda_a' &= a\big[\tfrac{3}{2}\alpha(\alpha-1)-\tfrac{27}{32}\big]\,j\nu^{\alpha-5/4},\\
\lambda_0 &= \tfrac{3}{4}a\,\big[\tfrac{21}{16}-\tfrac{3}{2}\alpha\big]\,\beta_1\beta_2\,\nu^{\alpha-11/4},
\end{aligned}$$

the first two anchor curvature minus window-centre mode — with $\frac{3}{2}(\alpha+\frac{3}{4})(\alpha-\frac{1}{4})$ the $(cF)''$ and $\frac{3}{2}\alpha(\alpha-1)$ the J_Fc'' — and the third the pair $2c'G_F'+cG_F''$. At $\alpha=\frac{9}{8}$, $a=\frac{3k}{4}$ they return $\frac{729}{512}$, $-\frac{243}{512}$ and $-\frac{27}{128}$, the last becoming $-\frac{243}{128}$ after $\beta_1\beta_2\to 9h_1h_2\nu$.

The third is the corrected constant of the erratum at Lemma 5.2b, not the printed one. Restoring the $c''G_F$ the anchor does not carry turns the linear form into the quadratic $(\alpha-\frac{3}{4})(\alpha-\frac{7}{4})$, whose value at $\frac{9}{8}$ is $-\frac{15}{64}$

and which returns the manuscript's $-\frac{135}{1024}$ and $-\frac{1215}{1024}$ exactly — so the α-form identifies which object each printed rational belongs to, and that is how the slip was found.

So the composites do have zeros, at $\alpha = \frac{\sqrt{10}-1}{4} = 0.5406$, $\alpha = \frac{2+\sqrt{13}}{4} = 1.4014$ and — exactly — $\alpha = \frac{7}{8}$ (and $\frac{3}{4}, \frac{7}{4}$ for the three-term form). The caveat is not a formality: there the architecture has no leading curvature for Lemma 3.3 to act on, whatever the rest of a proof does.

At 33/32 *all three survive.* With $a = \frac{27k}{32}$,

$$\lambda_a = \tfrac{84321}{65536} kj\nu^{-7/32}, \qquad \lambda_a' = -\tfrac{43983}{65536} kj\nu^{-7/32}, \qquad \lambda_0 = -\tfrac{1215}{8192} k\beta_1\beta_2\nu^{-55/32} = -\tfrac{10935}{8192} kh_1h_2\nu^{-23/32}.$$

What decides survival is not a composite's size but its *cancellation factor* κ, the sum of the absolute values of its terms over the absolute value of their sum: a relative perturbation of the terms inflates into a relative perturbation of the composite κ times as large, so the sign is determined exactly while that perturbation stays below $1/\kappa$. At $\frac{9}{8}$ the three factors are 1.59, 1.67, 8.00; at $\frac{33}{32}$ they are 1.74, 1.12, 12.20. The level-1 exponent is better than the proved one on the Step E composite and half again worse on the anchor; the two stay the same order, which is what the requirement asked.

And nothing the map produces is degenerate. Run the same three forms over every blocked coefficient exponent carried by a contractor of depth at most thirteen — 222 distinct values, from $\frac{4131}{4096}$ to $\frac{525297}{4096}$. Not one is a zero of any composite, and the extreme factors are

α	arises at	κ_{5a}	κ_E	κ_0
9/8	Theorem 5.3, proved	1.59	1.67	8.00
33/32	*OOOEOEE*, letter 6	1.74	1.12	12.20
27/16	Conjecture 7.3	1.24	2.88	3.15
45/32	*OOEOOEE*, letter 6	1.35	**129**	4.29
4131/4096	depth 13, the least	**1.78**	1.03	**14.10**

against ceilings the proofs already carry: the $(1 + O(P^{-1/4}))$ of (E6) allows $\kappa < 3071$ at P_0, and Lemma 5.2b's $O(hP^{-1})$ with $h \le P^{1/16}$ allows $\kappa < 1.2 \cdot 10^{13}$. Every frontier exponent clears both. The composite condition is therefore a genuine threshold with genuine zeros that the Juggler map never approaches — a further one beside drift-1, $e < 2$, the 9/4 stop and the linearisation bound $E < 2$, and the only one of them that never binds.

It is not idle, because it ranks, and it ranks on one axis only. By worst factor the depth-seven targets order 27/16 at 3.15, the proved 9/8 at 8.00, 33/32 at 12.20, and then 45/32 at 129 — the largest Step E factor anywhere on the frontier, 77 times the proved exponent's, and two orders clear of everything else in the column. The first three sit inside a factor of four of each other and say nothing; the fourth is the statement. It is a fourth reason *OOEOOEE* is the hard one, independent of its species, its branching and the 9/4 stop.

Three limits on that. 45/32 is a square-root defect, so its geometry is not the $X = \nu^{3/2}$, $F = \frac{3}{2}jm^{1/2}$ of Step E: the 129 is what this paper's own recipe returns at that exponent, not a composite the paper has ever formed. The κ_0 column is the *corrected* anchor of the erratum at Lemma 5.2b; read against the printed three-term form it would give 13.4, 14.3, 85.4, 17.7 and would rank the four differently — which is the sharpest evidence that a composite's identity matters before its size. And Step E's zero-offset $\frac{2187}{2048}$ is absent from the three for a different reason. It is now derived, in two halves, at Step E itself — $\frac{675}{2048}$ from $\Delta\Delta(\frac{k}{2}m^{9/4})$ less $\frac{432}{2048}$ from the corrected anchor — but the first half is not a function of the weight exponent at all: change α and the identity of Lemma 5.1(i) that produced $\frac{k}{2}m^{9/4}$ changes with it. Only the anchor half carries an α-form, and that is the one already in the table.

Sections 3–5 prove the hypothesis at every depth $d \le 4$, so the conclusion of Proposition 7.1 is unconditional for those depths. Corollary 6.4 raises the certified class to certificate density 7/8 without counting every depth-5 word. The first open counting case is the *OOOO∗* split. It has an exact shape, one nesting deeper than Theorem 5.3. Write $Z = v^{3/2}$, $z = \lfloor Z \rfloor$, $\theta_3 = Z - z$; after four odd letters the fifth is the parity of $\lfloor z^{3/2} \rfloor$.

Lemma 7.2 (level-3 kernel reformulation). For odd $n \ge 5$,

$$\tfrac{1}{2}(v^{9/4} - z^{3/2}) - \tfrac{3}{4}\, z^{1/2}\theta_3 = R_3, \qquad 0 \le R_3 \le \tfrac{3}{16}\, z^{-1/2}.$$

Consequently the fifth-letter phase is, up to $kR_3 \ll kn^{-27/16}$ per term, the exponential sum of the level-3 local floor defect, with coefficient $\varrho = \frac{3k}{4} z^{1/2} \asymp kn^{27/16}$: Lemma 5.1(i) with (m, v) replaced by (v, z).

Proof. Taylor of $(z + \theta_3)^{3/2}$ at z, with $Z^{3/2} = v^{9/4}$. ☐

For smooth ϱ with $\varrho \asymp kP^{27/16}$ and $\varrho' \asymp kP^{11/16}$ on $n \sim P$ (the $z^{1/2}$-shaped family), define

$$K_3(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e\big(\varrho(n)\, \{\lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor^{3/2}\}\big).$$

Conjecture 7.3 (level-3 kernel cancellation). $K_3(P) \ll P^{1-\delta}$ for some $\delta > 0$, uniformly over the family above with $k \le P^{\varepsilon}$.

The boundary between Theorem 5.3 and Conjecture 7.3 is quantitative and sharper than a derivative count. The smooth model iterates: where Theorem 5.3 used two Weyl differencings against $Y'' \asymp P^{1/4} \gg 1 > P^{-3/4} \asymp Y'''$, the level-3 model $n^{27/8}$ has $G''' \asymp P^{3/8} \gg 1 > P^{-5/8} \asymp G^{(4)}$ and predicts three. But the prediction does not descend to the nested floors. The kernel weight now has $\varrho' \asymp kP^{11/16} \gg 1$, so no drift-1 interval exists for any expansion of the weight itself; the branch decomposition of Lemma 5.1(iii) has no analogue at the v-level, because v jumps by $\asymp n^{5/4}$ per step and the branch set is a *product* of two carry lattices, not a copy of one; and the forced inner linearization $v^{3/2} = m^{9/4} - \frac{3}{2} m^{3/4}\theta_2 + E_2$ trades θ_3 for a sawtooth family at coefficient scale $kn^{45/16}$, *above* the 9/4 threshold where every method of this paper stops.

What survives at the frontier is a model problem and one theorem about it. Its objects carry script letters — $\mathcal{S}$ for the sum, $\mathcal{A}$ for the amplitude, $\mathcal{B}$ for the phase — to keep them clear of the S of Lemma 3.9 and the curvature triple A, B, C of the same lemma, which are unrelated. Stripping every problem-specific structure (carries, defects, nesting) from K_3 leaves the *amplitude-product* sums

$$\mathcal{S} = \sum_{t \le L} e\big(\mathcal{A}(t)\, \{\mathcal{B}(t)\}\big), \qquad 1 \ll \mathcal{A}' \ll \mathcal{A},$$

with smooth monomial-type $\mathcal{A}, \mathcal{B}$ (the instance above has $\mathcal{A} \asymp P^{27/16}$, $\mathcal{A}' \asymp P^{11/16}$). For $\mathcal{A}' \ll 1$ partial summation makes the amplitude a tame passenger and the classical single-floor machinery applies; for $\mathcal{A}' \gg 1$ we know of no nontrivial deterministic bound by any method.

That dichotomy is this section's drift threshold, written in the model's own notation. With $\mathcal{A} \asymp n^c$ one has $\mathcal{A}' \asymp n^{c-1}$, so $\mathcal{A}' \ll 1$ says $c < 1$ and $\mathcal{A}' \gg 1$ says $c > 1$ — precisely the condition deciding whether a defect coefficient admits a drift-1 window. The instance quoted, $\mathcal{A} \asymp P^{27/16}$, is Conjecture 7.3's own kernel weight $\frac{3k}{4} n^{27/16}$, and its $\mathcal{A}' \asymp P^{11/16}$ is that weight's derivative. Every coefficient in the table above sorts the same way: 3/16 and 9/16 windowed, 33/32 and 45/32 not. The model problem is the blocked case with the words taken out. What can be proved is an L^2 identity in the shift, a one-page computation that uses no harmonic analysis:

Proposition 7.4 (shift-averaged L^2 bound). Let $\mathcal{A}_1 < \cdots < \mathcal{A}_L$ be reals with $|\mathcal{A}_t - \mathcal{A}_{t'}| \ge \mathcal{A}'_{\min}|t - t'|$ for some $\mathcal{A}'_{\min} \ge 1$, and let $x_1, \ldots, x_L$ be arbitrary reals. For $\mathcal{S}_\lambda = \sum_{t \le L} e(\mathcal{A}_t\{x_t + \lambda\})$,

$$\Big|\int_0^1 |\mathcal{S}_\lambda|^2 \, d\lambda - L\Big| \le \frac{4}{\pi} \, \frac{L}{\mathcal{A}'_{\min}} \, (\log L + 1).$$

Consequently, for any $\eta \in (0,1)$, Markov's inequality gives

$$|\mathcal{S}_\lambda| \le \sqrt{\frac{L}{\eta}\Big(1 + \frac{4}{\pi}\,\frac{\log L + 1}{\mathcal{A}'_{\min}}\Big)}$$

outside a shift set of measure at most η. This is square-root cancellation times $\sqrt{\log L}$ in general, and genuine square-root cancellation when $\mathcal{A}'_{\min} \gg \log L$ — which holds in the instance above, where $\mathcal{A}'_{\min} \asymp P^{11/16}$. By a wide margin: at P_0 that is $3.9 \cdot 10^9$ against a $\log P_0$ of 32.

It is worth measuring what this does and does not give, since the two are easily conflated. Rearranging the display, $|\mathcal{S}_\lambda| \le P^{1-\delta}$ holds outside a shift set of measure $\eta \asymp P^{2\delta-1}$ — at $\delta = \frac{1}{96}$, the saving the level-2 kernel actually achieves, that is $P^{-47/48}$, about $2 \cdot 10^{-14}$ at P_0. And were the deterministic shift generic, the bound would read $P^{1/2}$, a saving of $\frac{1}{2}$ where Conjecture 7.3 asks only for some $\delta > 0$ — an overshoot of 48 over $\frac{1}{96}$. So the deficit at the frontier is not strength: it is the quantifier alone, and the strength available on the good side of it is excessive rather than marginal.

None of which is evidence. A single deterministic shift may lie in a set of any measure, including this one, and the smallness of $2 \cdot 10^{-14}$ is a fact about the proposition and not about the Juggler map.

Proof. Expand the square; the diagonal gives L. For $t \ne t'$, the function $\varphi(\lambda) = \mathcal{A}_t\{x_t + \lambda\} - \mathcal{A}_{t'}\{x_{t'} + \lambda\}$ is piecewise linear with *real* slope $\mathcal{A}_t - \mathcal{A}_{t'}$, with jumps at $1 - \{x_t\}$ and $1 - \{x_{t'}\}$. Those two points cut $[0,1)$ into three intervals, but the first and the last carry the same linear branch — their constants differ by exactly the slope — so on the circle there are two arcs, not three, and on each $|\int e(\varphi)\,d\lambda| \le 1/(\pi|\mathcal{A}_t - \mathcal{A}_{t'}|)$. Summing, $\sum_{t \ne t'} 2/(\pi \mathcal{A}'_{\min}|t - t'|) \le (4/\pi)(L/\mathcal{A}'_{\min})(\log L + 1)$. For the second display, the measure of $\{|\mathcal{S}_\lambda|^2 > \tau\}$ is at most $\tau^{-1}(L + (4/\pi)(L/\mathcal{A}'_{\min})(\log L + 1))$; take $\tau = (L/\eta)(1 + (4/\pi)(\log L + 1)/\mathcal{A}'_{\min})$. $\square$

Two cautions. The amplitude separation $\mathcal{A}' \gg 1$ — the very property that defeats every character expansion, since a Fourier window centred at the amplitude drifts by $\mathcal{A}'$ harmonics per step — is what makes the shift *average* trivial; but an average over shifts says nothing about the single shift $\lambda = 0$, which is the deterministic sum. Proposition 7.4 does not make the level-3 kernel "generically cancelling" in any sense that bears on Conjecture 7.3; it only locates the conjecture's difficulty as a specific-point problem inside a metric statement.

Conjecture 7.5 (the pure amplitude-product model). $|\mathcal{S}| \le L^{1-\delta}$ for some $\delta > 0$, for smooth monomial-type $\mathcal{A}, \mathcal{B}$ with $1 \ll \mathcal{A}' \ll \mathcal{A}$ as above.

Three structural facts locate the difficulty of the de-randomization. No second averaging variable exists: amplitude separation forces any two sample points with $|\mathcal{A}(p) - \mathcal{A}(q)| \le 1$ to coincide, and every family average available in the application re-enters either the differenced-kernel class or the amplitude-product class itself. The inverse theory is self-similar: any concentration or discrepancy inverse for $\mathcal{A}\{\mathcal{B}\}$ mod 1 is a statement about $\sum e(jA\{B\})$ — the same class with amplitude jA — so the class is closed under its own inverse theorems and no bootstrapping is possible. And the metric statement does not transfer: $|dS_\lambda/d\lambda| \le 2\pi \mathcal{A}_{\max} L$ almost everywhere, so $\mathcal{S}_\lambda$ decorrelates at shift scale $1/\mathcal{A}_{\max}$, and an almost-all-λ theorem leaves $\asymp \eta \mathcal{A}_{\max}$ exceptional cells among $\asymp \mathcal{A}_{\max}$ — no measure argument pins $\lambda = 0$. The deterministic content of Conjecture 7.5 is a specific-point-in-metric-theory problem: a class whose known successes all use special arithmetic.

Two analytic shortcuts around Conjecture 7.3 fail at recorded points, and we state them once: composing gap cells across two levels fails because on a cell where the first-level gap is constant the second-level gap still takes a new value at essentially every point, so no usable sub-cell survives; and reindexing by the image m strips one nesting level but introduces a fiber indicator of density $\asymp m^{-1/3}$, whose sawtooth requires savings beyond the exponent 1/3 while the engine saves only 1/24.

The obstruction ladder above blocks routes to a kernel bound *with a power saving.* It is worth recording that the density-one conclusion does not need one. Two weakenings of the hypothesis of Proposition 7.1 suffice, and both are elementary.

Proposition 7.6 (rate-free reduction). Suppose that for **each fixed** depth d, each of the N_d classes with no contracting prefix satisfies

$$\limsup_{N\to\infty} \frac{\#\{n \le N : \mathrm{word}_d(n) = w\}}{N} \;\le\; 2^{-d}$$

— an upper bound only, with no rate and no uniformity in d. Then the set of starts admitting a finite descent certificate has natural density 1.

Proof. Fix d and let $N \to \infty$ *before* $d \to \infty$. The hypothesis is Proposition 7.1's with $E_d(N) = o(N)$, so the starts with no contracting prefix of length $\le d$ have upper density at most $N_d/2^d$. This holds for every d, and $N_d/2^d \to 0$. □

Proposition 7.7 (biased-split reduction). Equidistribution is not needed either. Suppose that for each fixed d and every itinerary σ of length $< d$,

$$\limsup_{N\to\infty} \frac{\#(\sigma O)(N)}{\#\sigma(N)} \;\le\; 1-\beta \qquad \text{for a fixed } \beta \;\ge\; \beta_* := 1 - \frac{\log 2}{\log 3} = 0.36907\ldots$$

Then the same conclusion holds.

Proof. Write $\mu_N(\sigma) = \#\sigma(N)/N$ and let μ be a subsequential limit, which exists because each depth is finite; the hypothesis passes to $\mu(\sigma O) \le (1-\beta)\mu(\sigma)$. A node-wise bound on the conditional O-probability couples letter by letter, so the path $(o_t)_{t\le d}$ is stochastically dominated under μ by the i.i.d. Bernoulli$(1-\beta)$ walk. The never-contracting event $\{o_t \ge t\log 2/\log 3 \text{ for all } t \le d\}$ is increasing — turning an E into an O raises every later o_t and lowers none — so its μ-measure is at most its probability under that walk.

Under it, $o_t - t\log 2/\log 3$ is a random walk whose mean step is $(1-\beta) - \log 2/\log 3$, which is ≤ 0 exactly when $\beta \ge \beta_*$. If $\beta > \beta_*$ the drift is negative and the probability of staying nonnegative for d steps is $\le e^{-D(\log 2/\log 3\,\|\,1-\beta)d}$ by Chernoff. If $\beta = \beta_*$ the drift is exactly zero and Chernoff gives nothing, but the classical fluctuation estimate for a mean-zero walk still gives $\asymp d^{-1/2}$. Either way the measure tends to 0; certificates as before; let $d \to \infty$. □

The critical case is not a technicality, and the constant is visible. At $\beta = \beta_*$ the extremal measure — the one saturating $\mu(\sigma O) = (1-\beta)\mu(\sigma)$ at every node — makes each letter O with probability exactly $\log 2/\log 3$, so $o_t - t\log 2/\log 3$ has mean step 0 to machine precision, and the dynamic program gives never-contracting mass $9.46\cdot 10^{-2}$, $2.36\cdot 10^{-2}$, $1.18\cdot 10^{-2}$ at $d = 50, 800, 3200$: the products with $\sqrt{d}$ are 0.6689, 0.6675, 0.6675, and the ratio across each doubling is $0.7071 = 2^{-1/2}$ to four figures.

The threshold cannot be lowered. Below β_* the drift is positive and the extremal measure keeps a fixed positive mass on never-contracting words: at $\beta = 0.30$ it is 0.228 at $d = 200$ and still 0.228 at $d = 3200$. What the hypothesis $\beta \ge \beta_*$ asks is exactly that no node be asymptotically more than $\log 2/\log 3$ odd, which is the contraction line read as a branching rate rather than as an exponent.

The threshold is where it must be: β_* is the bias at which a node-wise O-share stops forcing the odd count below the contraction line $d\log 2/\log 3$, so the two constants in Proposition 7.7 are the same constant read from opposite sides. What the proposition buys is a change of species rather than of strength. Proposition 7.1 wants an analytic input with a rate; 7.6 wants a qualitative one; 7.7 wants no equidistribution at all, only that no node is asymptotically more than 63.09% odd. The obstruction ladder of this section — BB global, GG intra-block, JJ de-randomization — is a statement about power-saving mechanisms, and none of the three formally blocks a rate-free or ergodic route.

One caution, in the spirit of Section A.6. Just above β_* the Chernoff rate is useless and the finite-depth behaviour is entirely its prefactor. At $\beta = 0.37$, barely supercritical, the rate is $1.85\cdot 10^{-6}$ per letter, while the extremal measure — the one saturating $\mu(\sigma O) = (1-\beta)\mu(\sigma)$ at every node — has observed decay 0.0841, 0.0274 and 0.0154 per letter at $d = 24, 100, 200$. The same dynamic program at $\beta = \frac{1}{2}$ returns $N_d/2^d$ exactly, which is the check that the biased and unbiased accountings are one computation.

The open question, stated once:

It is open whether almost every odd-to-odd start has a finite descent certificate. By Proposition 7.1 that would follow from all-depth parity equidistribution, which is now a theorem through depth four for odd-rooted itineraries; by Proposition 7.6 a rate-free version of the same hypothesis suffices, and by Proposition 7.7 no equidistribution at all is needed, only that no node is asymptotically more than $\log 2/\log 3$ odd. The certified class through length five has certificate density 7/8. The first open counting case is the $OOOO*$ kernel of Conjecture 7.3, whose deterministic model instance is Conjecture 7.5 — but that line is the *power-saving* route, and Propositions 7.6 and 7.7 are what say the program does not require it.

8. Relation to the Juggler map

The companion paper [24] reduces the termination problem to one statement about parity words and makes precise which statistics it needs. That paper's results bear on the program of this one in two opposite directions, and we record both.

Two localization facts, stated so they cannot be confused.

1. Section 3.5 proves the depth-≤ 3 discrepancy estimates (Theorems 4.11 and 4.12) on sub-dyadic intervals of length $\geq P^{1/2}$ with a slow twist. Corollary 4.13 is the only localization in this paper that presently supplies a production to [24].
2. Theorem 5.3 is a dyadic-block statement; Theorem 5.5 localizes it to intervals of length $\geq P^{29/48+\delta}$, at the same exponent. The $P^{23/32}$ interval belongs to the already-treated $OOEEE$ word. The proposed $OOOEEE$ and $OOEOEE$ words land at exponent 27/64, so their broad inverse scale is $P^{37/64}$; since $37/64 < 29/48$, Theorem 5.5 as stated does not make those productions available. The threshold $29/48 > 1/2$ is also above the one in item 1, so the two localizations are not interchangeable.

What each new depth buys. Every new certificate class enters the contagion recursion of [24] as a production: a word w of fair probability P_w landing at scale x^{e_w} contributes $(P_w/e_w)\, g(e_w t)$, provided its cylinder is counted on the preimage intervals of the landing points — intervals of length x^{1-e_w}, which is why Section 3.5 localizes Theorems 4.4 and 4.7 to sub-dyadic intervals. Through Corollary 4.13 the $OOEEE$ production on even blocks raises the pairing-only baseline $\lambda_{\text{pair}} = 0.4480$ to $\lambda^{***} = 0.5392$, the corresponding rate threshold from 0.552 to 0.461, and the least depth constant from 20 to 18. The current elementary baseline, after the later V_k productions, is instead $\lambda^{**} = 0.4926$, with rate threshold 0.5074 and least depth constant 19. Those are distinct comparisons; the $OOEEE$ dividend is the analytic one supplied by the depth-≤ 3 localization. The localized kernel theorem (Theorem 5.5; the scaling architecture is indeed the same, but the per-window absolute costs reach $P^{25/48}$, not the $P^{7/16}$ of the depth-≤ 3 localization, which is what puts its threshold at $P^{29/48}$) adds no production at its stated range. Its two-term bookkeeping, evaluated formally at 37/64, has a residual saving 11/1536, but the theorem does not state that shorter estimate and no argument here transports the exact floor and parity restrictions of the production. This is a possible salvage proof obligation, not a theorem. If future work established both words with their fair weights, each would have probability 2^{-6} at landing scale $e_w = 27/64$, hence coefficient $\frac{1}{27}$, and the production rule above would return 0.6066. The quoted 0.5561 does not follow from that rule and is retracted; neither number is a proved dividend of Theorem 5.5. Each depth also raises the certificate density of Corollaries 4.9 and 6.4 and the constants of the Tao-type reduction. These are the quantitative dividends of the proved depth-≤ 3 production. The strongest short-interval kernel statement in this paper is Theorem 5.5, while the strongest localization currently supplied as a production to [24] remains Corollary 4.13.

What no depth can buy. The frontier statement of [24] is that the odd starts in $(y, 2y]$ whose orbit is still above a fixed floor after $C \log_2 \log y$ steps number at most $y(\log y)^{-e}$ for some $e > 0.508$, where $C \geq 19$. Its weakest sufficient condition is an exponential moment of the odd count on live starts — a statement about the tilted average of parity words of length $\asymp \log\log y$, which is insensitive to bias at any $o(\log\log y)$ initial depths and to any single cylinder unless it is over-populated by an exponential factor. Consequently: (i) no fixed-depth theorem, this paper's or any other, is necessary for that statement — the depth-5 split of Conjecture 7.3 is irrelevant to it; (ii) no cylinder statement of bounded depth is sufficient for it, because a word measure fair to depth k and all-O afterwards satisfies every depth-$\leq k$ statement and violates the bound; and (iii) along the per-cylinder route, an analytic method whose saving exponent loses a factor 2^c

per depth reaches the required depth only if $cC < 1$, i.e. $c < 1/19$, whereas Weyl differencing loses $c \geq 1$ (this paper's chain $\frac{1}{24} \to \frac{1}{96}$ from depth three to four is $c = 2$). The gap is worth naming as one number: the frontier needs $c < \frac{1}{19}$ and differencing delivers $c = 2$, a factor of 38 in the rate at which the saving is lost per depth. That is not a margin to be closed by sharpening constants. The differencing machinery of Sections 3–6 cannot therefore be iterated to the termination frontier, however far the kernel program is pushed; what would be needed is a saving uniform in the depth up to a factor $2^{d/19}$, or a direct treatment of the exponential moment, neither of which this paper offers.

(The pairing-only and *OOEEE* exponents above are roots of the same recursion and differ by exactly one term. $\lambda_{\text{pair}} = 0.4480$ solves $2^{-\lambda} + \frac{1}{9}(\frac{3}{8})^{\lambda} + \frac{2}{9}(\frac{3}{4})^{\lambda} = 1$, and $\lambda^{***} = 0.5392$ solves it with $\frac{1}{9}(\frac{9}{32})^{\lambda}$ added — that term *is* the *OOEEE* production, so the dividend of the depth-≤ 3 localization is attributable to a single summand.)

The honest summary is the one the abstract gives: this paper solves the first genuinely nested layers of the parity process of nested floor powers — depth four completely, depth five for the two contractors — with a bound whose critical piece is the level-2 wave; it identifies the level-3 kernel as the next analytic object; and through the depth-≤ 3 localization of Section 3.5 it supplies constants to the termination reduction of [24]. Theorem 5.5 does localize Theorem 5.3, but not far enough at the printed saving to supply the two proposed length-six productions, and it does not approach the infinite-depth problem that the termination question is.

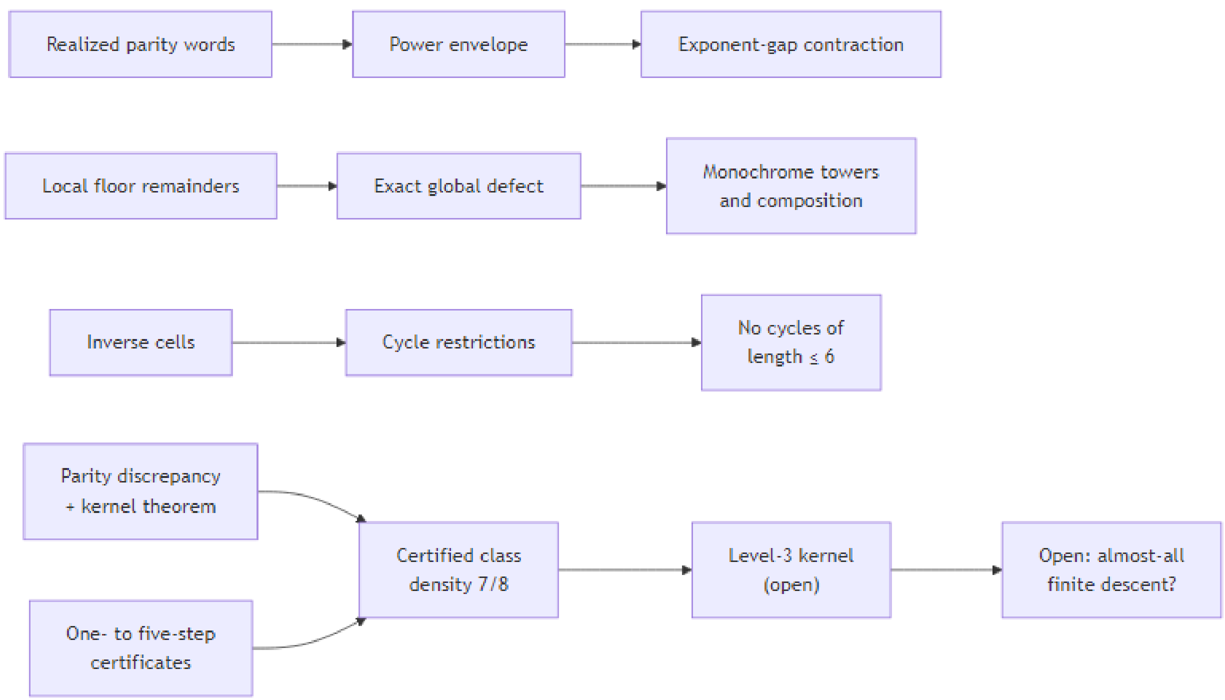

Figure 1: The theorem flow of the paper. The exact finite-itinerary calculus of the companion manuscript feeds the contraction certificates; the discrepancy calculus with the kernel theorem counts every O-rooted itinerary class through depth four and the two length-five contractors (certified-descent density 7/8), leaving the level-3 kernel — and with it almost-all descent — open.

A repository accompanies the paper (https://github.com/sneakyweasel/btlab/). It is not required to read or check any proof. Lean certificates cover the exact floor identities; the module `research.juggler_sequence.paper_b_audit` and the companion paper_b_audit_ledger.md record numerical and exponent checks. Those are not proofs and are not an independent verification of Lemma 5.2.

Appendix A. The threshold P_0

Every numerical margin of Sections 4–6 is claimed for $P \geq P_0$. This appendix makes P_0 effective. Each printed threshold inequality is solved separately for the least P beyond which it holds, and P_0 is the maximum. The computation is machine-checked in `src/research/juggler_sequence/p0_certificate.py`, which also generates the table below; the probe and the paper therefore cannot drift apart.

The probe solves each row by bisection in floating point, which was the only inexact step in an otherwise exact chain. It is no longer a link in that chain. Every exponent in the paper lies in $\frac{1}{96}\mathbb{Z}$, so the substitution $P = t^n$ turns each row into a *polynomial* inequality in t, with no real powers at all, and **all thirty-eight rows are proved that way** in `formal/Problems/Juggler/ThresholdCertificate.lean` (33 theorems, which is $38-7+2$: seven rows have no Lean theorem at all, and the window-boundary and λ_0-range rows each split in two). The seven are Claim D's shift range, the collision band, the Step 3(a) flat cost, the Step 5b(a) q'' ratio, Theorem 6.1's Step B discard, and the two depth-five sites of Theorem 6.3. None of them binds — their crossings run from $6.5 \cdot 10^5$ to $3.0 \cdot 10^{11}$, all below P_0 — but the Lean file certifies thirty-one of the thirty-eight rows and not the table. Each theorem it does carry states its substitution $P = t^k$ and a rational t_0 above the true crossing, so the certified thresholds are conservative: by under 10% in nineteen of the thirty that carry a loss figure (three hold from $P \geq 1$), and by 2.39 in the loosest, `row_5b_lam0_upper`, which certifies from $17^4 = 83521$ against a crossing of 35027. The largest is the binding row — `row_5b_binding`, at $t = 1.92$, i.e. $P \geq 1.92^{48} = 4.0 \cdot 10^{13}$ against the bisected $3.6 \cdot 10^{13}$, a loss of under 11%. Two irrational constants are replaced by rational bounds, both recorded: $\sqrt{0.56} \geq 0.7483$ in the boundary row, and $\frac{1}{12}\sqrt{0.56} \leq 0.06237$ in the binding one.

The file is regenerated against the corrected table of the erratum at Lemma 5.2b. Nine of the thirty-eight rows carry that anchor — the λ_0 range, the four $\rho_0(E)$ ratios, the wave remainder and the three balance comparisons — and every one of them still admits a rational witness, which was not guaranteed: the two balance rows are tight to four figures in the coefficient budget ($6.4969 + 1.6177 + 3.6895 = 11.8041$ against $7/5800 = 12.0690$, in units of 10^{-4}), and two of them first failed on a rounding of 1 part in $4 \cdot 10^5$. Two rows that look as though they carry the anchor do not: `row_s3s2_bdry_a` and the Step 5b(a) q'' ratio divide by Theorem 4.1's Stage-4 curvature $0.35\,uh\,P^{-3/4}$, so `sqrt_0_35_lower` stays and `sqrt_0_56_lower` is the new one beside it.

Constants that share a value. That last remark is an instance of the one systematic hazard in a paper carrying this many explicit constants: a numeral that names more than one quantity. Two rows looked as though they carried Lemma 5.2b's anchor because its pre-correction floor was 0.35 and Theorem 4.1's Stage-4 curvature is also 0.35. Reading one for the other cost an afternoon once, and it is not the only such pair. The collisions, so that a reader meeting a familiar numeral knows to check which quantity it is:

value	quantities it names
0.35	Thm 4.1's Stage-4 curvature $0.35uhP^{-3/4}$
	Lemma 5.2b's pre-correction λ_0 floor (now 0.56)
0.11	the smooth remnant $\lvert c''\rvert \leq 0.11kP^{-7/8}$, hence E's second term
	the collision band's lower edge $0.11uhP^{-1/4}$
	Step 5a's ratio $V/S \leq 0.11P^{-7/48}$ at $S \geq 0.60P^{-5/8}$
1.2	the Stage-4 curvature's upper end, $[0.35, 1.20]uhP^{-3/4}$
	the (s2) window length $\geq 1.2P^{3/4}$
	Step 5's cell sum $1.2R^{1/2}Y'P^{-1/4}$
	the cross-coefficient bound $\frac{63}{64} \leq 1.2$, in $k\nu^{-3/8}$
1.1	the (s2) window-boundary cost $1.1P^{17/32}(= 0.65/\sqrt{0.35})$
	Theorem 4.4's Lemma 3.3 sum $1.1uP^{3/4}$
	Step 5b's good pieces $(1.1C(E)S)^{1/2}$
1.5	the cell count $1.5hP^{1/2} + 1$
	the offset term's floor $\frac{3}{2}\lvert j\rvert P^{3/4}$

None of these is an error and none is avoidable by renaming — each constant is what its own derivation produces. What is avoidable is reading across them, and the two places this paper does distinguish them explicitly are this appendix's `sqrt_0_35_lower` and the erratum at Lemma 5.2b. `tools/manuscript_self_audit.py` keeps the list current: a numeral acquiring a new role has to be added to it.

The list is partly generated. Clustering every numeral that occurs in math mode by *what it multiplies* — $0.11kP^{-7/8}$ and $0.11uhP^{-1/4}$ fall into different clusters, and a numeral with two clusters is a candidate — flags fourteen of the three hundred and fifty-five numerals, of which 0.11, 1.1 and 1.2 survive inspection. That is how 1.1 and 1.2's fourth role were found; both had been missed by eye.

It flags 0.35 and 1.5 as well, but for the wrong reason, and the distinction is worth keeping. Their clusters are $0.35uhP^{-3/4}$ against $(0.35uh)^{-1/2}$, and $1.5hP^{1/2}$ against $1.5hY'$ — two notations for one quantity in each case, not two quantities. The roles that make them collisions are exactly the ones a check reading *what a numeral multiplies* cannot see: 0.35's is the endpoint of a bracket, $[0.35, 2.6]$, which multiplies nothing, and 1.5's is written $\frac{3}{2}$. So the agreement between the two methods on those two rows is a coincidence. Curation and clustering miss opposite things, and the list is kept by both.

A third check reads the manuscript as arithmetic. Every relation whose two sides are numeric literals — or a numeral times a symbolic factor that both sides carry, which is most of the displayed algebra — is evaluated and compared. There are eighty. Fifty-four are exact identities, and they include all of the fraction arithmetic of Section 5: $\frac{945}{512} - \frac{81}{512} = \frac{864}{512}$, $\frac{675}{2048} - \frac{432}{2048} = \frac{243}{2048}$, $\frac{11}{12} - \frac{29}{32} = \frac{1}{96}$, and the rest. Not one of them is off.

Twenty-three more are decimals correctly rounded to the precision they are printed at, and one is quoted in units of 10^{-4}. Two are neither: $(1.20)^{1/2} = 1.096$ at Stage 4, whose nearest four-figure decimal is 1.095, and $1.5 \cdot (0.35)^{-1/2} = 2.536$ in the same display, whose nearest is 2.535. Both are rounded up, and both feed an upper bound — $1.1\,(uh)^{1/2}P^{5/8}$ and $2.6\,(h/u)^{1/2}P^{7/8}$ — so in both the printed decimal is rounded *away* from the inequality it serves. That is the safe direction, and it is the paper's convention throughout.

The distinction matters because the unsafe direction is invisible to every other check here. A decimal rounded *into* its own bound is within a unit of the last place, agrees with the constant it names to the precision anyone would compare at, and is nonetheless a claim the line does not support. `tools/manuscript_self_audit.py` separates the two cases by direction, and the list of downward roundings is empty.

This check found one error. Claim D's shift range printed its threshold, 1.45^{36}, as $1.1 \cdot 10^6$; the value is 644537, and A.1's own row for that comparison read $6.4 \cdot 10^5$. The prose contradicted the table it was summarising. Both figures sit far below P_0, so nothing downstream moved, and that is the point: an error with no consequence is exactly the kind that survives reading.

Both were wrong in the same way, which the check as stated here could not see. A threshold is not a measurement; it is the left endpoint of the range over which a row holds. Printing $6.4 \cdot 10^5$ for 644537 is the nearest four-figure decimal and asserts the row over $[6.4 \cdot 10^5, 644537)$, where it fails. The rule for that column is therefore not "round to nearest" but "round up", and A.1 states it. Twenty of A.1's thirty-eight entries had been nearest-rounded below their crossings and have been raised; three were not roundings at all and are recorded there as errata.

A.1 The certificate

threshold	site	least P
P^(1/72-1/24) <= 1	Claim G	always
16 h1 P^(1/2) + 30 k h1 h2 P^(5/8) <= 46 P^(3/4)	Thm 5.3 St.3(a)	always
linearization remainder P^(43/96) <= P^(1-1/96)	Thm 6.3	always
P^(1/2) >= 8(1+|B|) with |B| < 1/2	Thm 4.1 St.3(s1)	144

threshold	site	least P
4.5 - 1.5/(h P^(1/2)) >= 4.4 at h = 1	Thm 4.1 St.5	225
P^(1/2) >= 8(1+6) = 56	Thm 5.3 St.5b (j=0)	3140
8(1+2.25P$^{(1/4))P}$(1/2) <= 19 P^(3/4)	Thm 4.1 St.3(s2)	4100
41 P^(5/36) <= P^(1/2)	Claim C	$2.93 \cdot 10^4$
[0.62,3.90] with its corrections inside [0.56,4.20]	Lemma 5.2b	$3.51 \cdot 10^4$
P^(7/72) >= 3	Claim C	$8.1 \cdot 10^4$
72 t^(-1) P^(-1/2) <= 1/4 at t = 1	Thm 5.3 St.6(D1)	$8.3 \cdot 10^4$
P^(1/2) >= 8(1 + 2.25 P^(1/4))	Thm 4.1 St.3(s2)	$1.16 \cdot 10^5$
window boundaries <= 1.1 P^(17/32) <= P^(5/8)	Thm 4.1 St.3(s2)	$1.51 \cdot 10^5$
0.6 P^(1/4) + 1 <= 0.65 P^(1/4)	Thm 4.1 St.3(s2)	$1.6 \cdot 10^5$
P^(1/2)/(2h1) >= 8(1+|B|): 0.5 P^(23/48) >= 15 P^(10/48)	Thm 5.3 St.3(a)	$2.85 \cdot 10^5$
P^(1/2)/(2h2) >= 8(1+|B|): 0.5 P^(22/48) >= 15 P^(9/48)	Thm 5.3 St.3(b)	$2.85 \cdot 10^5$
1.45 P^(7/72) <= P^(1/8): shift range of (i)	Claim D	$6.5 \cdot 10^5$
(3 pi k/4) P^(-1/8) <= 1 at k <= 2 P^(1/96)	Thm 6.1 St.B	$7.51 \cdot 10^5$
P^(1/2) >= 8(1 + 5 P^(1/4))	Thm 5.3 St.6(D1)	$2.62 \cdot 10^6$
wave remainder 300 P^(-35/24) vs S: 536 P^(-5/6) <= rho_0	Thm 5.3 St.5b	$1.58 \cdot 10^7$
beta-substitution error 2.3043 P^(-1/2) <= rho_0	Thm 5.3 St.5b	$1.83 \cdot 10^7$
cells + anchor runs + windows <= 3.5 P^(13/24)	Thm 5.3 St.5b	$5.14 \cdot 10^7$
mode/cell curvature ratio 0.39 P^(1/8) >= 4	Thm 4.1 St.2	$1.23 \cdot 10^8$
flat cost 23 P^(19/24) inside the P^(23/24) budget	Thm 5.3 St.3(a)	$1.49 \cdot 10^8$
every competitor ratio <= 1/4 (margin 4)	Thm 5.3 St.5a	$4.3 \cdot 10^8$
P^2|c''''/2|/S <= rho_0: (0.044/0.56) P^(-1/4)	Thm 5.3 St.5b	$4.53 \cdot 10^8$
P|c'''/2|/S <= rho_0: (0.047/0.56) P^(-1/4)	Thm 5.3 St.5b	$5.9 \cdot 10^8$
Lemma 3.7 window T = R_0 >= 8(1 + |C|)	Thm 6.3	$7.5 \cdot 10^8$
|c''/2|/S <= rho_0: (0.053/0.56) P^(-1/4)	Thm 5.3 St.5b	$9.6 \cdot 10^8$
3 R_0^(1/2) P^(3/4) = 3 P^(29/32) <= P^(23/24)	Thm 5.3 St.5	$1.45 \cdot 10^9$
96 P^(-5/24) <= 1	Claim G	$3.3 \cdot 10^9$
flat cost 8(1+|C|)/R_0 <= P^(-1/96) per point	Thm 6.3	$5.51 \cdot 10^9$
2.25 P^(-1/16) < 1/2	Thm 4.1 St.3(s1)	$2.83 \cdot 10^{10}$
widened |B_0| <= R_0: 5 P^(1/4) <= P^(5/16)	Thm 5.3 St.6(D1)	$1.53 \cdot 10^{11}$

<table>
<tr><th>threshold</th><th>site</th><th>least P</th></tr>
<tr><td>|q''| curvature ratio (1.85 P^(7/24) + R_0) 6 P^(-5/4) / (0.35 P^(-3/4)) <= 1/4</td><td>Thm 5.3 St.5b(a)</td><td>$3.0 \cdot 10^{11}$</td></tr>
<tr><td>E alone <= c_7 S/2 (the floor as kappa -> 0)</td><td>Thm 5.3 St.5b</td><td>$4.11 \cdot 10^{12}$</td></tr>
<tr><td>W = V + E <= c_7 S/2 at S >= 0.60 P^(-5/8)</td><td>Thm 5.3 St.5a</td><td>$2.92 \cdot 10^{13}$</td></tr>
<tr><td>W = V + E <= c_7 S/2 at S >= 0.56 P^(-5/8)</td><td>Thm 5.3 St.5b</td><td>$3.6 \cdot 10^{13}$</td></tr>
</table>

$$P_0 = 3.6 \cdot 10^{13},$$

attained at the Lemma 3.9 hypothesis $W \le c_7 S/2$ of Theorem 5.3, Step 5b. Three rows hold for every $P \ge 1$ and are listed for completeness rather than because they constrain anything.

The last column rounds up. It names a P from which the row holds, so an entry rounded to nearest can name a P at which it does not: Claim D's shift range crosses at 644537, and the nearest four-figure decimal, $6.4 \cdot 10^5$, asserts the row over $[6.4 \cdot 10^5, 644537)$, where it fails. Every entry is therefore the crossing rounded up, to as many figures as keeps the overshoot under one per cent — at most 0.85% anywhere in the column, and under 0.3% in twenty-three of the thirty-five printed rows. The other three hold for every $P \ge 1$ and are written *always*. `tools/manuscript_self_audit.py` checks each entry against `p0_certificate.thresholds`, and the check is that the printed value is at or above the computed crossing, never that it equals it.

Errata in this table. Three entries were not roundings. The Stage 3(s2) window-boundary row was listed at 403; its crossing is $1.50527 \cdot 10^5$, a factor of 373, and the printed value matches no sub-condition of the row — neither conjunct alone crosses above 3. The Step 5b(a) q'' row carried the constant 30.5, which appears nowhere else in this paper; the constant is 48.9 (= 17.1/0.35, fixed in Step 5b(a)), and with it the crossing is $2.98 \cdot 10^{11}$ rather than the listed $2.8 \cdot 10^{10}$. The Lemma 5.2b row carried $[0.62, 3.94]$ against the $[0.62, 3.90]$ of Lemma 5.2b itself, and with the correct endpoint the crossing falls from $6.1 \cdot 10^4$ to $3.51 \cdot 10^4$. All three sit far below P_0, and none is the binding row, so P_0 is unchanged at $3.5858 \cdot 10^{13}$.

P_0 from the constants printed above. The five figures of $3.5858 \cdot 10^{13}$ come from one row and five numbers, all of them printed in this paper: the interpolant error's $170.6\,P^{-25/24} + 0.11 P^{-5/6}$, the scale floor $0.56 P^{-5/8} \le S$, the $\frac{1}{12}$ of $V = \frac{1}{12} S^{1/2} P^{-11/24}$, and Lemma 3.9's $c_7 = \frac{1}{232}$. Solving $V + E \le c_7 S/2$ with those alone gives $3.58576 \cdot 10^{13}$, and `tools/manuscript_self_audit.py` does exactly that, reading each constant out of the text rather than from the certificate. A headline number quoted to five figures should be recoverable from the paper it appears in.

It was not, by one constant. E's definition printed $171\,P^{-25/24}$ while the derivation beside it gives $85.3 \cdot 2 = 170.6$ and Lean proves 170.6 (`interpolant_assembly`). Both are true — 171 is the rounder statement of a bound, and rounding a coefficient up in an upper bound is safe — but the crossing at 171 is $3.5969 \cdot 10^{13}$, so a reader recomputing P_0 from E as displayed could not reach the figure printed for it. E now carries 170.6 at every site, which is the value the derivation produces, the value Lean proves, and the value P_0 was always computed from. The erratum's list also read $106 \to 171$ while claiming a reader finds both ends in Lean; the corrected end in Lean is 170.6, and the list now says so.

A.2 The stratification

The thresholds are not spread out; they cluster and then jump. Thirty-three of the thirty-eight hold from $2.83 \cdot 10^{10}$ on, and that value is set by a single soft inequality ($2.25 P^{-1/16} < \frac{1}{2}$ in Stage 3(s1) of Theorem 4.1, which merely names the regime). Of the remaining five, two are the price of $R_0 = P^{5/16}$: the widened mode index of Lemma 5.2(iii) at $5^{16} = 1.53 \cdot 10^{11}$ and the q'' curvature ratio of Step 5b(a) at $3.0 \cdot 10^{11}$, both two orders below P_0 (A.6). The other three are the Lemma 3.9 balance comparisons of Steps 5a and 5b, which alone carry P_0 up by two orders of magnitude. The erratum at Lemma 5.2b moved only those three: it left the q'' row where it was, since that row divides by Theorem 4.1's Stage-4 curvature and not by λ_0, and it left the mode index where it was, since that row mentions no curvature at all.

The q'' row is the one place where the table's sentence and the certificate's predicate had drifted apart. The sentence read $48.9P^{-3/16} \le \frac{1}{4}$, the merged bound the prose uses; the predicate is the unmerged $(1.85P^{7/24} + R_0)$ form, sharper by $P^{1/48}$, and it is that form A.5's floor is measured at. The two clear at $2.98 \cdot 10^{11}$ and $1.66 \cdot 10^{12}$, so the row was false at its own printed threshold — $48.9\,(3.0 \cdot 10^{11})^{-3/16} = 0.345$, not $\le \frac{1}{4}$. A row must hold as written at the P written beside it, so the sentence now states the form that is checked. The $\tilde{\beta}$-substitution row had the same defect at one part in four hundred: it printed 2.31 where the derivation gives $9 \cdot 0.68/2.656 = 2.30422$, and 2.31 appears nowhere else in this paper. It now prints 2.3043.

The mode-index row is also the only one in the table whose printed threshold is *exact* rather than conservative. Every other row is certified at a rational t_0 at or just above the true crossing; this one substitutes $P = t^{16}$ and reads $5t^4 \le t^5$, whose crossing is $t = 5$ on the nose.

$W \le c_7 S/2$ is a *hypothesis* of Lemma 3.9, not an optimisation: it is what makes Ω_W empty on the $r = 2$ pieces. Its size is fixed by $c_7 = 1/\|M^{-1}\|_\infty = 1/232$ at the Step 5b exponent triple (Lean `step5b_curvature_norm`), which is exact and so not improvable there (A.5), and by the normalisation κ of $V = \kappa S^{1/2} P^{-11/24}$, which is free:

κ	P_0	P_1 (A.5)	boundary coefficient
$\frac{1}{3}$	$1.23 \cdot 10^{16}$	$2.82 \cdot 10^{20}$	7.01
$\frac{1}{8}$	$1.12 \cdot 10^{14}$	$1.3 \cdot 10^{19}$	11.5
$\frac{1}{12}$ (used here)	$3.6 \cdot 10^{13}$	$9.9 \cdot 10^{18}$	14.1
$\frac{1}{16}$	$2.05 \cdot 10^{13}$	$1.23 \cdot 10^{19}$	16.2
$\frac{1}{20}$	$1.47 \cdot 10^{13}$	$1.84 \cdot 10^{19}$	18.1

Both columns fall together until $\kappa = \frac{1}{12}$, where the piece-boundary term turns P_1 around; that is the operating point. The left column keeps falling below it, but only to the largest κ-free row: at $\kappa = \frac{1}{16}$ and below, P_0 is the gate by a shrinking margin, and it would stop at $5^{16} = 1.53 \cdot 10^{11}$. Read with the printed offset bound $|j'| \le 3$ it would have stopped at $3.3 \cdot 10^{13}$, i.e. one entry past the operating point. The turn is structural, not numerical: the $\frac{8}{5}$ correction to λ_0 moved every entry of both columns and left the minimum where it was. As $\kappa \to 0$ the *gate* tends to $4.1 \cdot 10^{12}$, the point at which the interpolant error alone satisfies $E \le c_7 S/2$, and P_0 follows it: every row below that is κ-free and smaller, the largest being the q'' ratio at $3.0 \cdot 10^{11}$. The exponent 89/96 does not depend on κ at all.

This table rounds up, for the reason A.1 does. Both columns are crossings — the left the least P at which the printed inequalities hold, the right the least at which the middle band beats the trivial bound — so an entry rounded to nearest can name a P at which the thing it promises has not happened yet. At $\kappa = \frac{1}{12}$ the crossing is $9.83914 \cdot 10^{18}$, and the nearest two-figure decimal, $9.8 \cdot 10^{18}$, asserts non-vacuity over $[9.8 \cdot 10^{18}, 9.83914 \cdot 10^{18})$, where the middle-band estimate is still the weaker of the two. Eleven of the fifteen entries were nearest-rounded below their crossings and have been raised; the overshoot is under one per cent throughout. The boundary coefficient is $3.5\,V^{-1/2}$ at $S = \lambda_0 P^{-5/8}$, i.e. $3.5(\kappa\sqrt{0.56})^{-1/2}$, and rounds up for the same reason: it is a cost.

P_1 itself is recoverable from the display above it. Solving $4PW/(c_7 S) + P(W/(c_7 S))^{1/2} + 3.5P^{13/24}V^{-1/2} \le P$ with the five constants of A.1 — 170.6, 0.11, $\lambda_0 = 0.56$, $\kappa = \frac{1}{12}$, $c_7 = \frac{1}{232}$ — and the 4, 3.5 and 13/24 printed in the display gives $9.83914 \cdot 10^{18}$. `tools/manuscript_self_audit.py` runs that solve, and the whole table, against the printed entries.

A near miss, recorded because it nearly cost a factor three. Claim D of Lemma 5.2(ii)→(i) must place every index of the Claim C sum inside the shift range of (i), i.e. $h_3 \le P^{1/8}$, from $h_3 \le t^{1/3} P^{1/12}$. The exponent gap $\frac{1}{8} - \frac{7}{72} = \frac{1}{36}$ is small, so whatever constant stands in front is paid at the thirty-sixth power, and the choice of bound on t decides the row:

$$|t| \le 16P^{1/24} \;\Rightarrow\; 16^{12} = 2.8\cdot 10^{14}, \qquad |t| \le 3P^{1/24} \;\Rightarrow\; 3^{12} = 5.3\cdot 10^{5}.$$

The first is what the individual bound $|q_d| \le 4P^{1/24}$ gives over the four elements of $\mathcal{D}$; the second is the total-frequency bound that Theorem 5.3, Step 4 — the only place (ii) is invoked — actually supplies, since the wave modes arrive one per expansion layer from three layers at truncation $J_2 = P^{1/24}$. Lemma 5.2(ii) therefore carries $|t| \le 3P^{1/24}$ as a hypothesis, and the row is $6\cdot 10^5$ rather than $2.8\cdot 10^{14}$. Had it been the latter it would have been the binding row of the whole paper, and P_0 would have been $2.8\cdot 10^{14}$ — larger by 3.15 than the value above, for no analytic reason whatever. This is the sharpest illustration in the paper of the rule that governs Appendix A: on a gap of g, a constant c costs $c^{1/g}$, so a constant is only harmless when the gap is wide.

A.3 P_0 does not depend on ε

No divisor sum, gcd sum or large-sieve average occurs anywhere in Sections 3–6. Every $\ll_\varepsilon$ in those sections is therefore a power of $\log P$, and the powers can be counted. The mode masses are $O(\log^3 P)$ over at most three expansion layers, so $|T_2| \ll P^{23/24}\log^3 P$; the two Weyl steps of Theorem 5.3 each halve the exponent of the log,

$$|T_1| \ll P^{1-1/48}\log^{3/2} P, \qquad K_c(P) \ll P^{1-1/96}\log^{3/4} P,$$

and Theorem 6.3 costs a further $\log^3 P$, giving $\#\mathrm{OOOEE}(N) = \frac{N}{32} + O(N^{1-1/96}(\log N)^{15/4})$. These forms carry no ε, and P_0 is the same for all of them.

The three logs in that last step are worth naming, because only two of them are Theorem 6.3's own. They are the Vaaler expansion of the fifth wave at $J_5 = 2P^{1/96}$; the Lemma 3.7 window at $T = R_0 = P^{5/16}$ against $|C| \le 1.30P^{19/96}$; and the first-letter expansion at $|i| \le 2P^{1/96}$, which Theorem 6.3 inherits from Theorem 6.1 when it merges the two indices into $|I_{\mathrm{tot}}| \le 2P^{5/16}$. That third one is not inside the $\log^{3/4}$ — that power is K_c's, and Theorem 6.1 is where K_c is applied rather than proved — so counting it here is right, and 15/4 stands. Only the phrase "its own" would be loose.

Either count is far outside the range where it could matter: $\log^A P \le P^{1/96}$ first holds at 10^{190} for $A = 3/4$, 10^{268} for $A = 1$, 10^{872} for $A = 11/4$ and 10^{1245} for $A = 15/4$ (`decoration_budget.log_power_ledger`). A larger A is the weaker claim, so the generous count is the safe one.

The threshold at which $\log^A P$ would be *absorbed* into P^ε is a different quantity and is not part of P_0: for $\varepsilon = 1/96$ it is $1.5\cdot 10^{190}$ at $A = 3/4$ and beyond 10^{300} at $A = 15/4$. Sections 4–6 carry the P^ε rather than spending it, so those numbers never enter. The same remark disposes of the $P^{15/16}$ reading in Step 5b: it would need $C(E)\log P \le P^{1/96}$, first true near 10^{274}, and Step 6 uses only $89/96 < 15/16$.

A.4 What the certificate is not

It is not a proof. It certifies that the inequalities the paper prints are true beyond the stated threshold; it does not certify that they are the right inequalities, and it inherits every modelling choice made in reducing a displayed estimate to a predicate in P. Two of its inputs are conservative substitutes for statements the paper leaves in $O(\cdot)$ form — the $\tilde{\beta}$-substitution error, bounded here by the ± 1 of the gap floor at $0.68P^{-1/2}$ rather than by the printed $O(hP^{-1})$, and the wave remainder $O(uh^2\nu^{-7/4})$, bounded at $200P^{-35/24}$ on the printed inventory. Both clear ρ_0 by more than nine orders of magnitude, so neither choice affects P_0.

Nor does P_0 answer every question one might ask of a threshold. It is the point beyond which the proof's inequalities hold; it is not the point beyond which the resulting bound is better than the trivial one. That second threshold is computed in A.5.

Nor is P_0 the threshold of the whole paper for every choice of R_0. Run Stage 2 of Theorem 5.3 at $R_0 = P^{1/4}$ and the depth-five Theorem 6.3 needs $1.8\cdot 10^{24}$, ten orders above P_0 — and Lemma 5.2(iii) needs $5P^{1/4} \le P^{1/4}$, which no P supplies at all. Section A.6 shows that $R_0 = P^{5/16}$ removes both gaps at no cost that binds, and that is the value carried throughout. With it, every threshold in the paper is P_0.

A.5 The two constants that carry the threshold

P_0 is carried entirely by $W \le c_7 S/2$, whose two ingredients are the curvature constant c_7 and the interpolant error E. Both were attacked; only one moved.

c_7: not by changing the exponent triple. $c_7 = 1/\|M^{-1}\|_\infty$ depends only on the exponents, through $\det M = \prod_{i<j}(x_j - x_i)$, and scales as the square of their gap. For an equally spaced triple of gap δ about x_0 the middle Lagrange row has denominator $-\delta^2$ and numerator entries $|x_0^2 - \delta^2|$, $|1 - 2x_0|$ and 1; when $|x_0| > \delta$ and $x_0 < \frac{1}{2}$ that row is the largest and the absolute values open, so

$$\frac{\delta^2}{c_7} = x_0^2 - 2x_0 + (2 - \delta^2), \qquad \text{i.e.} \qquad c_7 = \frac{\delta^2}{(x_0-1)^2 + 1 - \delta^2}.$$

The additive constant is therefore exactly $2 - \delta^2$, attaining $\frac{7}{4}$ at $\delta = \frac{1}{2}$ and approaching 2 as $\delta \to 0$ without reaching it. Step 5b's triple is $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$, and each entry is forced: $\frac{3}{2}$ is the level-1 wave $X = \nu^{3/2}$; $\frac{11}{8}$ is the frozen-shape global model; $\frac{5}{4}$ is the differenced-wave monomial after the frozen gap $G \sim 3h\nu^{1/2}$. In eighths they are $10, 11, 12$ — adjacent on the lattice $\frac{1}{8}\mathbb{Z}$ the whole paper lives on — so $\det M = \frac{1}{256}$ and $\|M^{-1}\|_\infty = 232$. Across all 165 triples of the paper's exponent inventory c_7 runs from $1/259$ to $144/287$, the good end being triples of gap $\frac{3}{2}$; these three lie within $\frac{1}{4}$ of each other by construction.

c_7: by dropping the uniform constant, under a factor ten, and not for free. The proof needs only $|M^{-1}|c \le 1$ for a vector $c = (c_2, c_3, c_4)$, and only c_2 gates $W \le c_2 S/2$. The middle row of $|M^{-1}|$ is $(24, 144, 64)$, so $c_3, c_4 \to 0$ gives $c_2 \le \frac{1}{24}$ (Lean `step5b_c2_ceiling`). But the uniform choice **saturates that row exactly**, $24+144+64 = 232$ (Lean `step5b_uniform_saturates`): every increase in c_2 is paid out of c_3, c_4, and those sit in C, through the $r = 3$ length $2PV/(c_3 S)$ and the $r = 4$ length $P(V/(c_4 S))^{1/2}$. Taking $c = (\frac{1}{27}, \frac{1}{1872}, \frac{1}{1872})$ — again exactly tight, $\frac{8}{9} + \frac{1}{9} = 1$ (Lean `step5b_c2_optimum_feasible`) — moves P_0 to $4.03 \cdot 10^{12}$ but P_1 below from $9.9 \cdot 10^{18}$ to $1.02 \cdot 10^{23}$. We keep the uniform constant.

c_7: and it saturates, at a value R_0 has already been tuned to. The lever has a ceiling, because only the sites that mention c_7 move when it does. Removing those, the largest threshold left is the Step 5b(a) q'' curvature ratio at $2.98 \cdot 10^{11}$, with the next below it the widened mode index of Lemma 5.2(iii) at $5^{16} = 1.53 \cdot 10^{11}$. Improving c_7 drives P_0 towards that value and no further: the Step 5b gate drops beneath it at $c_7 = 1/61$, so the whole of the lever is spent by then, and what it buys in total is a factor of 120, from $3.6 \cdot 10^{13}$ to $2.98 \cdot 10^{11}$. The vector trade above realises 8.9 of that 120, the rest being paid out of c_3 and c_4.

Two constants had to be right for that paragraph, and the second one was found late. The mode-index row is stated inside the proof of Lemma 5.2(iii) and was never collected into A.1, so the floor above was for a while computed from a table missing an entry. It matters which value that entry has. At the offset bound $|j'| \le 3$ the manuscript printed, the row is $7^{16} = 3.32 \cdot 10^{13}$: it would then be the largest c_7-free threshold, 8% under P_0, the lever would be worth 1.08 rather than 120, and the vector trade would return the P_0 it started from. At the corrected $|j'| \le 2$ — the erratum at Lemma 5.1(iii) — the row is 5^{16}, an order below the q'' ratio, and every figure in the paragraph above is the one printed. The conclusion was right; one of its two premises was not checked, and the other was not in the table.

The erratum at Lemma 5.2b moved neither figure: the floor divides by Theorem 4.1's Stage-4 curvature and is untouched, so the lever's worth fell from 300 to 120 simply because P_0 fell, and its saturation point eased from $c_7 = 1/54$ to $1/61$ because the gate sits lower at every c_7. The erratum at Lemma 5.1(iii) moves the mode-index row and nothing else here: 5^{16} is a pure power and mentions no curvature at all.

The floor is not an accident of this lever, and it is a function of R_0. Among the four exponents Appendix A.6 tabulates, $2.98 \cdot 10^{11}$ is the best: at $P^{9/32}$ the flat cost rises to $7.4 \cdot 10^{13}$ and at $P^{1/3}$ the q'' ratio rises to $1.6 \cdot 10^{12}$. Scanning the exponent continuously, the flat cost falls and the q'' ratio rises, and they cross at $a = 0.29919$, where the worst of the four is $1.40 \cdot 10^{11}$ — a factor 2.13 below what $P^{5/16}$ gives them.

That crossing is infeasible. At $a = 0.29919$ the mode index of Lemma 5.2(iii) needs $5^{1/(a-1/4)} = 1.6 \cdot 10^{14}$, a factor 4.5 above P_0. The four sites alone leave a free across $[0.2829, 0.3463]$; with the fifth the band is $[0.3016, 0.3463]$, and neither 0.29919 nor its nearest simple value $\frac{3}{10}$ — which needs $5^{20} = 9.5 \cdot 10^{13}$ — lies in it. That is the sharper reason $P^{1/4}$ will not do, too: at $a = \frac{1}{4}$ the mode index is not late, it is *unsatisfiable at every* P, since $5P^{1/4} \leq P^{1/4}$ is false.

Read as a minimax over all five, the optimum moves the other way and does not move far. The mode index falls in a while the q'' ratio rises, and they cross at $a^* = 0.3111$, where the worst site is $2.73 \cdot 10^{11}$ — so 5/16 is a factor 1.09 above the five-site optimum, against the factor 2.13 it costs on the four. Counting the site that pins the band makes 5/16 a *better* choice than A.6 rated it, not a worse one. None of this touches P_0, which is carried by the balance comparisons.

So the c_7 lever's floor is $2.98 \cdot 10^{11}$ at the truncation in force and would be $2.73 \cdot 10^{11}$ at the five-site crossing; the choice of R_0 is feasible rather than optimal, and the margin is now small enough that moving it is not worth the constants. P_0 itself is R_0-independent and stays $3.6 \cdot 10^{13}$. And the coincidence recorded at Lemma 5.2(i) — that $R_0 = 2(\frac{1}{24} + \frac{1}{8} - \frac{1}{96})$ — is a coincidence of a feasible choice and not the output of an optimisation, on a band whose left endpoint is 0.30157.

The band's width depends on a constant nobody sharpened. The collected 5 is not tight. It gathers $4P^{1/4}/h'$ and $20hP^{-1/4}$, and the second is not of the first's order: $4P^{1/4} + 20P^{-1/8} \leq (4+\delta)P^{1/4}$ from $P \geq (20/\delta)^{8/3}$, so 4.001 serves from $2.95 \cdot 10^{11}$ on, below P_0. At 4.001 the row falls from $1.53 \cdot 10^{11}$ to $4.3 \cdot 10^{9}$ and the band's left endpoint eases from 0.30157 to 0.29443 — which is below 0.29919, so at the sharp constant, and only there, A.6's four-site crossing and its nearest simple value $\frac{3}{10}$ both come back inside the band. We do not take it: the downstream constants of Lemma 5.2(iii) (8.5, 48, 5/0.6) are all stated at 5 and all sit in absorbed or dominated positions, and the floor is already the q'' row, so re-deriving them would move the band and not the threshold.

For the record, what it costs to be wrong about the offset. At $|j'| \leq 3$ the collected constant is 7, the row is $3.32 \cdot 10^{13}$, the band is $[0.3123, 0.3463]$ with 5/16 clearing its left endpoint by $1.5 \cdot 10^{-4}$, the five-site optimum is $a^* = 0.3218$ at $5.79 \cdot 10^{11}$ and 5/16 is a factor 57 above it, and the c_7 lever is worth 1.08 rather than 120. One integer in Lemma 5.1(iii) carries all of that.

The crossing itself is not at a nice exponent: each threshold has the form log(constant) over a linear function of a, so $a^* = 0.29919\ldots$ solves a transcendental equation. The nearest simple value is $\frac{3}{10}$, which lands within 4% of the optimum, against 5/16's factor 2.13. Neither figure moved with the erratum at Lemma 5.2b: both sites in the crossing divide by Theorem 4.1's Stage-4 curvature, not by λ_0.

Among those two the loss is steeply asymmetric, and that used to be what recommended staying high. Below the crossing the flat cost explodes — $\frac{2}{7}$ is worse by a factor 84 — while above it the q'' ratio rises gently, 5/16 costing only 2.13. And the crossing moves with the constants: $\frac{3}{10}$ sits $8 \cdot 10^{-4}$ above a^*, enough to survive a 25% shift in either site but not a doubling of the flat threshold, whereas 5/16 sits $1.3 \cdot 10^{-2}$ above and survives a factor of five.

The fifth site does not overturn that reading, and came close to. The band is defined, as A.6 defines it, by every R_0-sensitive site staying below the balance row; the mode index stays below it exactly when the collected widened constant is at most $P_0^{1/16} = 7.0333$. At the corrected $|j'| \leq 2$ that constant is 5, with 41% of room, and 5/16 clears the band's left endpoint by 0.0109 in the exponent — comfortable on both counts. So $P^{5/16}$ is the robust choice rather than the optimal one against all five sites, as it was against four. At the printed $|j'| \leq 3$ the constant is 7, the room is half of one percent, and 5/16 clears the left endpoint by $1.5 \cdot 10^{-4}$: a coefficient of 7.04 would have carried the row past the balance row and made $R_0 = P^{5/16}$ the paper's threshold. The margin that makes the truncation safe is the offset bound, and it was the one constant here nobody had checked.

E: yes, by a factor 1.28*; and the factor* 10 *beside it, by removing it.* The earlier $219 = 202.5 + 16$ opened the middle-band cap to 360 and carried 8 where step (ii) gives 0.907. Keeping the shifts visible lets the two error terms combine into $85.3\, k(h_1{+}h_2)P^{-9/8} \leq 170.6P^{-25/24}$. More consequential is the comparison beside it: the former $V \geq 10|f'' - \Lambda|$ is not needed at all. Running Lemma 3.9 at the raised threshold $W = V + E$ gives $|f''| \geq W - E = V$ off Ω_W directly, under the single hypothesis $W \leq c_7S/2$. The factor 10 had forced V to be large exactly where c_7 wanted it small; with it gone, κ can fall from $\frac{1}{3}$ to $\frac{1}{12}$ and P_0 with it.

The second threshold. A.1 certifies that the printed inequalities are true beyond P_0; it does not ask when the resulting bound is better than the trivial one. The middle band costs

$$\underbrace{4P\,\frac{W}{c_7S}}_{r=3,\ \asymp P^{41/48}} + \underbrace{P\Big(\frac{W}{c_7S}\Big)^{1/2}}_{r=4,\ \asymp P^{89/96}} + \underbrace{3.5\,P^{13/24}V^{-1/2}}_{\asymp P^{89/96}},$$

and P_1 is the least P at which that total is $\le P$. The three exponents are not equal, so they cannot be collected into a single coefficient of $P^{89/96}$; doing so over-counts the $r = 3$ term by $P^{7/96}$. Honestly computed, $P_1 = 9.9 \cdot 10^{18}$ at the operating point, against $2.82 \cdot 10^{20}$ at $\kappa = \frac{1}{3}$. Between P_0 and P_1 the middle-band estimate is true but weaker than the trivial bound; the theorem is asymptotic and its implied constant absorbs the difference.

What is left. At the operating point the binding comparison is still $W \le c_7S/2$, now with c_7 shown near its floor and E reduced by a factor two. The remaining slack is in E itself: the middle-band half-width 60, which enters 106 linearly through the cap $\frac{60\cdot 2.6}{0.84}$, and the range $[0.35, 2.6]$ for λ_0. Narrowing the band would reduce E but weakens the margin-20 domination claims in the anchor-dominant and mode-dominant regimes, which is a trade this draft has not made.

A.6 Why Stage 2 is truncated at $P^{5/16}$

Stage 2 of Theorem 5.3 Vaaler-expands the gap-cell indicator at a truncation $R_0 = P^a$. Four printed inequalities depend on a, two of them monotone each way, so a is not free.

Paid for by raising a.

- The **collision-band sum** of Stage 5 is $3R_0^{1/2}P^{3/4}\log P = 3P^{a/2+3/4}\log P$, which must stay inside the theorem's own $P^{23/24}$; this forces $a \le 5/12$.
- The q'' **curvature** of Step 5b(a) is $(1.85khP^{1/8} + R_0) \cdot 3|j|P^{-5/4}$, whose ratio to the main curvature is $\asymp P^{a-1/2}$ once R_0 is the larger term; it must clear the margin $\frac{1}{4}$.

Bought by raising a. Both come from the fifth-letter Lemma 3.7 window of Theorem 6.3, opened at $T = R_0$ against $|C| \le 1.2812\,P^{19/96}$:

- the **window hypothesis** $T \ge 8(1 + |C|)$, and
- the **flat cost** $8(1 + |C|)/T$ per point, which over a block must stay inside $P^{1-1/96}$.

Both have margin $P^{a-19/96}$, so both need $a > 19/96 = 0.198$, and they need it with room: the constant is $8 \cdot \frac{9}{8} \cdot 2^{3/16} = 10.25$, and the flat cost must beat not 1 but $P^{-1/96}$.

Solving each site for its least admissible P:

a	collision	q''	window	flat cost	worst
1/4	$5.32 \cdot 10^5$	$2.97 \cdot 10^{10}$	$2.55 \cdot 10^{19}$	$1.81 \cdot 10^{24}$	$1.81 \cdot 10^{24}$
9/32	$1.12 \cdot 10^7$	$6.7 \cdot 10^{10}$	$1.4 \cdot 10^{12}$	$7.4 \cdot 10^{13}$	$7.4 \cdot 10^{13}$
5/16	$1.45 \cdot 10^9$	$3.0 \cdot 10^{11}$	$7.5 \cdot 10^8$	$5.51 \cdot 10^9$	$3.0 \cdot 10^{11}$
1/3	$2.83 \cdot 10^{11}$	$1.59 \cdot 10^{12}$	$3.5 \cdot 10^7$	$1.42 \cdot 10^8$	$1.59 \cdot 10^{12}$
3/8	$8.0 \cdot 10^{22}$	$1.08 \cdot 10^{15}$	$6.9 \cdot 10^5$	$1.53 \cdot 10^6$	$8.0 \cdot 10^{22}$

The window is what makes the choice sharp. At $a = 1/4$ the two fifth-letter requirements are not merely inconvenient, they are the whole threshold of the depth-five theorem, and the binding one is the flat cost, which does not clear $P^{1-1/96}$ until $1.81 \cdot 10^{24}$ — eleven orders above P_0. At $a = 9/32$ the worst site is $7.4 \cdot 10^{13}$, which no longer sneaks under the threshold at all: it is a factor 2.1 *above* $P_0 = 3.6 \cdot 10^{13}$ and would set it. (Against the pre-correction $8.95 \cdot 10^{13}$ it sat just below, by less than 1.3 — too close to print then, and decided now.) At $a = 5/16$ the worst site is $3.0 \cdot 10^{11}$, a clear factor 120 below P_0, and it is the minimum of the last column over the admissible range. That is the value carried.

This table rounds up too, and its middle row is A.1's. Every entry is a least admissible P, so the convention of A.1 and A.5 applies here as well; twelve of the twenty-five were nearest-rounded below their crossings and have been raised, with the overshoot under one per cent throughout. The $a = 5/16$ row is not an independent computation: it is the collision-band, q'', window and flat-cost rows of A.1 at the exponent actually used, so the two appendices must print the same four numbers. They did not — A.1 read $1.45 \cdot 10^9$, $7.5 \cdot 10^8$ and $5.51 \cdot 10^9$ where this table read $1.4 \cdot 10^9$, $7.4 \cdot 10^8$ and $5.5 \cdot 10^9$, because A.1's column had been raised and this one had not. `tools/manuscript_self_audit.py` now checks both the rounding and the agreement, which is the check that would have caught the drift the day it appeared.

Two things this table settles. First, the choice costs the paper nothing that binds: the collision-band term moves from $3P^{7/8} \log P$ to $3P^{29/32} \log P$, still inside $P^{23/24}$ with $P^{5/96}$ to spare, and the q'' ratio moves from $52P^{-5/24}$ to $48.9P^{-3/16}$, which at P_0 is 0.14 against a margin of $\frac{1}{4}$. P_0 itself is unchanged. One exponent in a statement does move: the collision-band term is the fifth term of Lemma 5.2(i), which carries $R_0^{1/2} P^{3/4}$ explicitly for this reason, and reads $P^{7/8}$ at $a = 1/4$ and $P^{29/32}$ at $a = 5/16$. Second, the window's margin is genuinely narrow — $P^{a-19/96}$ with a constant above 10 — so this is a place where an asymptotic argument and an effective one diverge by twenty orders of magnitude, and the paper's claim of effectivity is what forced the choice.

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I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. The theorems of this paper are human proofs from the cited classical inequalities; the Lean certificates cover only the exact floor identities and the companion's finite-itinerary theorems. I take full responsibility for the contents.

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